ABET
Self-Study Report

for the

Chemical Engineering Program

at

The University of Illinois at Chicago

Chicago, IL 60607

UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

July 1, 2014

CONFIDENTIAL

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<u>Please note:</u>
Exhibits (Section 4.C) are not included in the pagination on the lower right of the pages.

Program Self-Study Report
for
EAC of ABET
Accreditation or Reaccreditation

BACKGROUND INFORMATION

A. Contact Information

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B. Program History

The College of Engineering was one of the original Colleges when University of Illinois at Chicago was founded in 1965 (originally named University of Illinois at Chicago Circle – renamed in 1982 after a merger with the University of Illinois at the Medical Center). Chemical Engineering started as a program within the Department of Energy Engineering. The other programs in the Department were Thermo-Mechanical Engineering and Fluids Engineering. In 1982 the College of Engineering was reorganized and Chemical Engineering became one of its six constituent Departments. In 1986 the Department moved to its current location at 810 S Clinton Street (Chemical Engineering Building, CEB), with offices, laboratories and classrooms located conveniently in the same building. In 2001 the department began offering Chemical Engineering as a minor to students not enrolled in the Chemical Engineering undergraduate program. In 2003 the Department initiated an area of concentration in Biochemical Engineering,

The last general review took place October 19-21, 2008. Since then there has been some turnover among the faculty. Two senior faculty members (including the department head) and one emeritus professor teaching part-time retired and one senior faculty member moved to another institution. Commensurately, four new faculty were hired: three assistant professors and one senior professor (who will start in August 2014). One adjunct professor teaching the senior design and laboratory courses was replaced with two new adjunct professors. Significant enhancements have been made to the coverage of computation in the curriculum. A new, sophomore-level course (CHE 205 – Computational Methods in Chemical Engineering) has been added as a bachelor's degree requirement, replacing 3 credit hours of free elective. A technical elective course on process simulation with Aspen Plus (CHE 433) was added. This provides in-depth coverage of process simulation and underlying chemical engineering concepts beyond the usage of Aspen in other courses (e.g., CHE 313 – Transport Phenomena III, ChE 397 – Senior Design II), and is a popular choice among seniors. The Senior Design II (ChE 397) project experience has been bolstered by a team of industrial mentors who meet regularly with their student groups.

C. *Options*

<u>BS in Chemical Engineering — Biochemical Engineering Concentration</u>

This concentration entails the completion of 8–10 semester hours distributed among CHE 422—Biochemical engineering (3 cr. hr.) as the technical elective and two elective courses in the non-major rubric category chosen from among BIOS 350—General Microbiology (3 cr. hr.), BIOS 351—Microbiology Laboratory (2 cr. hr.), CHEM 352—Introductory Biochemistry (3 cr. hr.) and CHEM 452—Biochemistry I (4 cr. hr.). (iii) Free elective (if needed) to make up 1 credit hour. Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 130 semester hours for the degree.

<u>Minor in Chemical Engineering (for non-ChE majors)</u>

Students outside the Department of Chemical Engineering who wish to minor in Chemical Engineering must complete a specified sequence of classes totaling 16-18 semester hours (excluding prerequisite courses). This sequence includes three courses in Chemical Engineering:

CHE 210—Material and Energy Balances (4 cr. hr.),
CHE 301—Chemical Engineering Thermodynamics (3. cr. hr.),
CHE 321—Chemical Reaction Engineering (3 cr. hr.),

One fluid mechanics course chosen from among

CHE 311—Transport Phenomena I (3 cr. hr.)
ME 211—Fluid Mechanics I (4 cr. hr.)

and one heat or mass transfer course chosen from among

CHE 312—Transport Phenomena II (3 cr. ht.)
ME 321—Heat Transfer (4 cr. hr.)
CHE 313—Transport Phenomena III (3 cr. hr.)

Minors appear on a student's transcript, but are not listed on the UIC diploma. Among the small number of Engineering minors, none have as yet been minors in Chemical Engineering since 2004.

D. Program Delivery Modes

The Bachelor of Science in Chemical Engineering is offered to interested students in conventional day classes (lecture / laboratory / recitation). The program can be completed by full time students in four years by taking 15-18 credits per semester.

E. Program Locations

The UIC Chemical Engineering program is offered only on campus.

F. Deficiencies, Weaknesses or Concerns from Previous Evaluation(s) and the Actions Taken to Address Them

For the 2008 accreditation cycle the ABET Final Statement (dated August 12, 2009) contained no Deficiencies, Weaknesses, or Concerns.

GENERAL CRITERIA

CRITERION 1. STUDENTS

The evaluation of Chemical Engineering students takes three forms:

1. Evaluation at the time of admission to the BSCE degree
2. Evaluation of progress towards the BSCE degree
3. Evaluation for completion of the BSCE degree

These evaluation processes are discussed in individual sections below for both incoming freshman and transfer students.

A. Student Admissions

Students are admitted directly into the Chemical Engineering Program in the College of Engineering as: a new freshman; an internal transfer student from another UIC college; or a transfer student from an institution other than UIC. The general evaluation procedures for new Freshman and internal transfer students are described below. Evaluation of transfer students from outside UIC is discussed later in this section.

Freshman students

The two primary factors used to determine the admissibility of a new freshman applying to the University of Illinois at Chicago are the high school percentile rank (HSPR) and the American College Test (ACT) composite score. The University will also accept Scholastic Aptitude Test (SAT) scores. SAT scores are converted to an ACT composite equivalent. Should an applicant submit more than one set of entrance exam scores, then the Office of Admissions will use the highest score when processing an application.

Before making the admission decision, a student's high school course work is checked to meet the specified subject pattern. All applicants must have successfully completed the following high school courses in order to be eligible for admission to UIC.

A. Four years of English
B. Three years of Mathematics (must include algebra, geometry, advanced algebra/trigonometry)
C. Three years of Science
D. Three years of Social Science
E. Two years of Language (other than English)
F. One year of an elective.

The College of Engineering provides the Office of Admissions the guidelines to be used in admitting freshmen to the College. Freshman applications are evaluated by a College of

Engineering Admission Committee consisting of the Director of Admission and Records, the Associate Dean for Undergraduate Affairs, and the Director of the Minority Engineering Recruiting and Retention Program. Admission is supported by high high school grades, particularly in STEM courses, high composite ACT scores, and high scores on the mathematics subscale of the ACT. Table 1.1 shows the history of College of Engineering admissions standards for freshmen admissions for the past five years and indicates a gradual increase in incoming student quality as measured by the composite ACT score.

All entering freshmen are admitted with the condition that they must take campus placement exams in math, chemistry and English composition. The results of these exams are critical in determining what courses students take during their first semester at UIC. If the results indicate the student needs a preparatory course in math, chemistry or English composition, then the student must register and pass the course.

Academic Year	Composite ACT		Composite SAT		Percentile Rank in High School		Number of New Students Enrolled
	MIN.	AVG.	MIN.	AVG.	MIN.	AVG.	
2009	21	25.4	n/a	n/a	n/a[1]	75.1	394
2010	22	26.2	n/a	n/a	n/a[1]	75.2	341
2011	23	26.5	n/a	n/a	n/a[1]	75.5	326
2012	23	27.1	n/a	n/a	n/a[1]	78.1	327
2013	23	27.3	n/a	n/a	n/a[1]	79.7	371

[1] Admission is based on unweighted High School GPA and not Percentile Rank in High School

TABLE 1-1. History of College of Engineering Admissions Standards for Freshmen Admissions for the Past Five Years

B. Evaluating Student Performance

Student progress is monitored both by the Office of the Associate Dean for Undergraduate Affairs, and by the Director of Undergraduate Studies (DUGS) of the Department of Chemical Engineering. The DUGS consults with the student's faculty advisor to discuss issues that may arise during such monitoring. During the ninth week of each semester each student must schedule an appointment with his/her adviser during the tenth week to plan an academic program for the following semester (see details on advising procedures in Section D), but especially to evaluate the student's progress during the past and current semester. During the advising sessions, students discuss with advisors their academic program and progress and may be advised to take remedial classes if needed, or rearrange their schedule to change their credit hour load to their advantage, or in extreme circumstances move to a program of study more suited to the student's academic strengths including changing their program of study. In the next two subsections, the processes for evaluating students' progress and probation procedures are described. These include departmental college and university policies.

1.B.1. Evaluation of Progress

Students are evaluated at the end of every semester during the College of Engineering Grade Review process. At that time, the Associate Dean for Undergraduate Administration (ADUA) reviews the file of each student who has either a term Grade Point Average (GPA) or cumulative UIC GPA below 2.00 (A=4.00).

In addition to monitoring the Grade Point Average of students, the College also uses the Deficit Point system to determine probation and drop decisions. The Deficit Point system has the advantage over the GPA system in that it clearly indicates the future academic performance that a student must achieve to return to clear status.

The following scale is on a per semester hour basis:

Letter grade	Grade points per credit hour	Deficit points
A	4	+2
B	3	+1
C	2 (2.0 minimum GPA at graduation)	0
D	1	-1
F	0	-2

If a student takes one 3-hour class and receives a grade of "D," then this student would have a GPA of 1.00 and –3 deficit points. If another student takes four 3-hour classes and receives all D's, then this student would also have a GPA of 1.00, but would have –12 deficit points. Although they both have the same GPA, the second student with –12 deficit points is having more serious academic difficulty. A student with negative deficit points must earn a positive number of deficit points in the future to bring the total back up to zero. Thus, a student with –12 deficit points must earn a combination of A's and B's for the number of hours necessary to achieve +12 deficit points to return to clear status.

See Appendix E for grade distributions over the past five years of all ChE courses.

Grade Forgiveness. A GPA of 2.0/4.0 is required overall and in a student's major courses. Students are advised each semester and their options are discussed when appropriate. Students who do not meet these graduation requirements may improve their GPA in accordance with the following policy for repeating course and GPA recalculation:

1. Students may repeat a course to increase their knowledge of the subject matter. There are circumstances under which repeating a course is advisable and to a student's advantage. There are also circumstances where repeating a course may disadvantage a student and narrow a student's options. Courses with A or B grades may not be repeated. Normally, courses with a C grade may not be repeated. Courses with D or F grades may be repeated once without written permission. In all cases, the original grade for the course and the grade for each repeat will appear on the transcript. The original grade will be calculated into the

grade point average, unless the student initiates a request for ***Repeating a Course with Grade Point Average Recalculation*** as described below. (In other words, absent the grade recalculation request, the effective GPA in the course becomes the average of the original and retaken grades – whether for the overall or major GPA.)

2. Grade point average recalculation for a repeated course **is not** automatic. The student must initiate a request in the college office. For the grade point average recalculation policy to apply, a student must declare to the college the intent to repeat a course for a change of grade. Students must submit this request to their college before the end of the official add/drop period, no later than the second Friday of the fall and spring semesters, the first Wednesday of Summer Session 1, or the first Friday of Summer Session 2. The course must be repeated within three semesters of the receipt of the original grade, and it must be taken at UIC. Only one registration for the course counts toward the total number of credits required for graduation.

3. Undergraduate students are allowed grade point average recalculation in up to four repeated courses. Under the course repeat policy, all courses taken and their grades appear on the transcript in the semester in which they were taken. Under the grade point average recalculation policy, the grade earned the first time the course is taken will be dropped from the calculation of the cumulative GPA and the grade(s) earned when the course is repeated will be used in the calculation. This rule holds, even if the second grade is lower than the first. If a course is repeated more than once, the first grade is not counted in the GPA, but all other grades for that course are calculated in the cumulative GPA. The original grade for the course and the grade for each repeat will appear on the transcript. University, college, or departmental honors will be awarded on the total cumulative GPA.

1.B.2. Probation Rules

Rule 1. A less serious probation level is called Probation Level 1 or 2.00 Pro. This probation is for any student whose Term GPA is below 2.00, but whose Cumulative GPA is above 2.00. In the next semester, the student is expected to earn no grade less than C to continue.

Rule 2. Any student whose UIC Cumulative GPA falls below 2.00 is placed on Probation Level 2 or 2.25 Pro. In the next semester, the student is expected to earn no grade less than a C and at least one B in order to continue. A student is not required to return to clear status in one semester. For example, if a student finishes the first semester with –10 deficit points, then finishes the second semester with 3 hours of B and 3 hours of C, this student will have reduced the total deficit points to –7. Although the student is still on probation, the student satisfied the probation conditions and is allowed to continue. At this rate of +3 deficit points per term, it will take the student 4 semesters to return to clear status, and the student is making progress towards that goal.

Rule 3. Graduation with a degree from the College of Engineering requires a minimum GPA of 2.00 in the major courses. Major courses are those required in the specific degree program as listed in the UIC Undergraduate Catalog. Rules 1 and 2 described above are also applied to the major courses.

1.B.3 Dismissal Rules

Rule 1. Any student who was on probation and did not satisfy the conditions of that probation may be dropped from the College of Engineering B.S. degree program.

Rule 2. A student who fails to make satisfactory progress toward a degree in the College of Engineering may be dropped. Examples of unsatisfactory progress are: -12 deficit points, excessive number of incomplete grades, failure to take courses required for the degree.

Rule 3. This rule applies to those students who had previously been dropped, were readmitted, and failed to meet the probation conditions necessary to continue in the College of Engineering. Students who have been dropped multiple times should pursue some other career goals. Only in rare cases will a student be readmitted after being dropped twice.

1.B.4. Enforcement of Prerequisites

The importance and enforcement of prerequisite course requirements is advertised frequently during Advising Week and during registration periods. Early each semester the Transfer Articulation and Degree Audit (TADA) office prepares a list of all students in the College that do not meet prerequisite course requirements for courses they are enrolled in. This list is sent to the Director of Engineering Admissions and Records, who then determines which students on the list have transfer credit that has not posted yet, or who have permission to take the course without the listed prerequisite courses (granted under rare circumstances via a General Petition form). The remaining students are notified via email (cc to DUGS, department head, course instructor, associate dean for undergraduate affairs) that they will be dropped from the course if they do not show proof of credit for the prerequisite course(s). The student generally has several days in which to respond to the email and/or make registration adjustments. In the third week, all remaining students are removed from the courses for which they are registered for improperly. Automatic, real-time prerequisite checking is implemented during registration for English, Math and Chemistry.

C. Transfer Students and Transfer Courses

1.C.1. Admissions

Transfer applicants are encouraged to complete 60 semester (90 quarter) hours of transferable work at the time of enrollment and are advised to complete English Composition I and II; Calculus I, II, III; Differential Equations; Physics I and II; Chemistry I. (Computer Science majors should have 12 semester hours (18 quarter hours) of lab science, which includes an 8 hour sequence). These are recommendations as opposed to rigid requirements, and a transfer student can be admitted student without these courses if their GPA meets college standards. This eventuality would occur most commonly for second-degree students with non-technical backgrounds.

Students must have earned at least a cumulative GPA of 2.50 (A=4.00), including all UIC and all transfer work.

More than half of new Chemical Engineering enrollees are now transfer students; see Table 1-2. The rising trend is more obviously monotonic at the College level; see Table 1-3.

Year (Fall)	Freshmen	Transfers	Total	% Transfers
2014				
2013	33	39	72	54.2
2012	39	24	63	38.1
2011	21	24	45	53.3
2010	25	22	47	46.8
2009	36	17	53	32.1

TABLE 1-2. New fall enrollments in the Chemical Engineering department.

Academic Year	Number of Transfer Students Enrolled
2009	287
2010	265
2011	233
2012	222
2013	169

TABLE 1-3. New fall enrollments of transfer students in the Engineering College.

1.C.2. Transfer credit evaluation criteria

The State of Illinois has formal articulation agreements between state community colleges and senior public institutions. Credit from a community college is limited in the sense that the last 60 hours of the degree must be completed at a senior institution. In addition, the University maintains a general provision that either the first 90 or the last 30 hours of the degree must be earned at UIC.

The College of Engineering maintains an extensive evaluation program for courses available at 32 Chicago area colleges that a majority of UIC transfer students come from. The articulation agreement transfer guide is available to prospective students on the COE website: http://www.uic.edu/depts/enga/prospective_students/transfer_guides.htm. The transfer information is housed on a website (Transferology.com) which conveniently allows students to review course equivalency information between institutions participating in the Illinois Articulation Initiative. Transfer credit is determined based on course equivalency.

Work successfully completed in other fully accredited institutions (either those approved by one of the regional accrediting associations or those approved by one of the agencies recognized by the National Commission of Accrediting) is generally accepted by the University on an hour-for-hour basis, as shown on the official transcripts received from those institutions. For consistency, credit awarded on the quarter hour basis is converted to the semester hour system by multiplying the number of quarter hours by 2/3.

Credit from institutions with provisional accreditation is accepted on a deferred basis until it is validated by satisfactory completion of additional work taken in residence at the University or in another fully accredited institution. Credit from unaccredited institutions is not accepted. However, knowledge in courses taken at such institutions may be awarded credit for the equivalent UIC course by successfully passing a UIC proficiency exam. No transfer credit is awarded for any course work completed at an institution external to UIC with a grade of D.

Credit evaluation is done by the Office of the Associate Dean for Undergraduate Administration in the College of Engineering. In many instances, descriptions, syllabi, etc. are sent to a faculty member in the department offering similar courses for their recommendation as to the equivalency to a UIC course and acceptance for transfer credit. The general principle used in accepting work from another college or university is that of "equivalency." Whether a course taken elsewhere is equivalent to a course at UIC is determined by the college office or in the appropriate department at UIC. Catalog descriptions that show course content and prerequisites for a course are the primary means for determining equivalency. If the catalog description is not sufficient, syllabi, exams, computer programs, homework, textbooks, etc. may be requested from the student. Table 1.2 shows the Transfer students for the past five academic years. The data show a substantial increase in the number of transfer students entering the program during this period. This increase is driven largely by the increase in tuition costs, which encourages students to use lower cost community colleges to complete their pre-engineering course work.

D. Advising and Career Guidance

1.D.1. Student Advising

The College of Engineering employs a mandatory advising system. Prior to the first semester of attendance, whether entering as freshmen or transferring from other schools, all students are advised by staff from the Office of the Dean for Undergraduate Administration. During their first semester, Chemical Engineering students are assigned to faculty advisors in the Department of Chemical Engineering, who will advise them every semester until graduation. No student is allowed to register without meeting with their advisor and receiving a signed registration form signed by their academic advisor. Registration holds are utilized to ensure that students meet with their advisors every semester prior to registering for the next term.

Advisor Evaluation Form

To the student: *In a continued effort by the college to recognize and reward faculty advising, we ask you that you fill out this form and drop-off the form in the ballot box in the student affairs office in your department (in order to maintain anonymity) along with your faculty advising form with your pre-registration information for the next semester.*

Please select between 1 and 5 or each of the statements: 1 for strongly disagree and 5 for strongly agree.

My Advisor is (Name_________________________)

Signature of Advisor __________________________)

	Strongly Disagree	Disagree	No Opinion	Agree	Strongly Agree
1. Shows genuine concern for my academic and professional development	1	2	3	4	5
2. Has detailed knowledge of curriculum and provides good advice about courses	1	2	3	4	5
3. Was available at posted advising time during advising week	1	2	3	4	5
4. Encourages students to come at other times beyond the required meeting during advising week	1	2	3	4	5

Comments:____

____

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Attach additional sheets if necessary

Advising Evaluations Criteria: 5 = Very Satisfied, 4 = Satisfied, 3 = Neutral, 2 = Unsatisfied, 1 = Very Unsatisfied.

FIGURE 1-1. Adviser evaluation form and evaluation criteria

Faculty advisors help students with curriculum matters, selection of electives, career path input, and other curriculum and career advice. The ChE DUGS acts as an intermediary between faculty

Spring 2014

Professor	Concern	Knowledge	Availability	Encouragement	Average
Akpa	4.67	4.71	4.71	4.43	4.63
Chaplin	4.26	4.00	4.68	3.89	4.21
Liu	4.56	4.39	4.83	4.61	4.60
Meyer	4.71	4.81	4.95	4.48	4.74
Murad	4.68	4.80	4.72	4.60	4.70
Nitsche	4.77	4.81	4.77	4.54	4.72
Sharma	4.72	4.72	4.62	4.14	4.55
Wedgewood	4.95	5.00	5.00	5.00	4.99

Fall 2013

Professor	Concern	Knowledge	Availability	Encouragement	Average
Akpa	4.90	4.90	4.85	4.85	4.88
Chaplin	4.37	4.21	4.95	4.21	4.43
Liu	4.57	4.62	4.76	4.33	4.57
Meyer	4.83	4.90	4.87	4.67	4.82
Murad	4.92	4.96	4.96	4.85	4.92
Nitsche	4.90	4.79	4.62	4.48	4.70
Sharma	4.69	4.66	4.94	3.66	4.48
Wedgewood	4.81	4.85	4.93	4.81	4.85

Advising Evaluations Criteria: 5 = Very Satisfied, 4 = Satisfied, 3 = Neutral, 2 = Unsatisfied, 1 = Very Unsatisfied.

TABLE 1-4. Student evaluations of faculty advising.

advisors and the Office of the Dean for Undergraduate Administration. Students first are advised to discuss curriculum issues with the DUGS for routine questions. The Office of the Dean for Undergraduate Administration maintains an academic counseling staff that helps students with questions related to degree requirements, transfer credits, course substitution, special programs and other related issues.

Based upon discussions at a "townhall" meeting between ChE students and the COE Dean, department head and ChE DUGS, the COE implemented an advisor evaluation process starting in Fall 2006 to provide feedback to advisors on their performance in the advising process. Advisees rate their faculty advisors on a 5 point scale (1 lowest and 5 highest) as to their: 1) concern for the student, 2) knowledge of the curriculum, 3) availability during advising week and 4) encouragement of the student. Scores for all faculty members in each department are compiled by the COE and sent to all faculty within the department for feedback. The COE has also implemented an advisor award program for the top-rated advisor in each department as an incentive to improve advisor performance.

Figure 1-1 shows the form that is used for advisor evaluation, and the pages that follow show the evaluations of all ChE faculty over the past five years. The efforts of the ChE faculty seem to be appreciated by our ChE students; see Table 1-4.

1.D.2. Career Guidance

We have four main activities to guide students for carrier that fit their skills:

- Career Fair
- Career Prep Day
- Engineering Career Center (ECC)
- Alumni Career Center

1.D.2.a. Career Fair.

The Engineering College holds a career fair once every year. The latest engineering career fair was on Tuesday, February 18, 2014 at Student Center East - Illinois Room from 12:00-4:00pm

The UIC Engineering Career Fair is an effective resource to match employers with engineering students, not only for internships and co-op positions, but also for full-time careers. It is sponsored by the UIC Engineering Council and the UIC College of Engineering. No prior student registration is necessary. This is not open to the general public and is only available to UIC students and UIC alumni. Example employers who participated in the career fair at UIC are listed in Table 1-4.

TABLE 1-4. Examples of Engineering Employers Registered for 2014

ORGANIZATION/COMPANY	MAJORS	POSITIONS	F1 Sponsorship Offered
Caterpillar	ChemE,Civil,Comp. Eng.CS,EE,IE, ME, Materials	FT, Internships	No
iLink Resources	Chemical,Civil,EE,ME, IE	FT	No
Integrys	Chemical,Civil,CS,EE,IE, ME	FT, Internship	No
LanzaTech	Chemical,Energy, ME	FT, Internship	Yes
Metropolitan Water Reclamation District of Greater Chicago	Chemical,Civil,EE, ME	FT	No
Michael Best & Friedrich LLP	Chemical (MS/Ph.D), Comp. Engr., EE,CS, ME	FT	No
Therapeutic Proteins International, LLC	BioE, Chemical, EE, IE, ME	FT	Yes
US Navy	All Majors	FT	No

1.D.2.b. Career Prep Day.

We hold a career prep day once every semester. Career preparation workshop and networking event is to connect with UIC Engineering Alumni. The latest career prep day event information is listed below:

What: Engineering Career Prep Day
When: Friday, February 7, 2014
Where: Student Center East - Illinois Room
Time: 9:00am-12:00pm
Who: Open to all undergraduate engineering students

Student Expectations: Students will be able to discuss some of the following career development topics:

- Catching the notice of a recruiter (resume, networking, etc.)
- Selling Yourself - What is an elevator pitch?
- Communication-of student's research, past jobs, internship experience, GPA, etc.
- Interviewing skills - How to prepare for an interview
- Business etiquette- presentation, dress, communication, manners, etc.

Alumni mentors from leading companies usually will lead interactive discussions covering these important topics!

1.D.2.c. The Engineering Career center (ECC).

ECC supports UIC engineering students with services to assist them in the opportunity to gain practical work experience while they are still students.

Participating employers are located throughout the Chicago area, the Midwest and the across the United States, providing employment opportunities in all fields of engineering. Small manufacturers, global corporations, and government agencies look to UIC students for expertise in solving their unique needs.

Below is some of the information we post on our website for students to be part of the ECC:

Why Should Students Make Internships and/or Co-ops Part Of Their Education Plan? ***According to hiring companies: (Survey – NACE Corp Recruitment Trends-Spring 2010)***

- *Students who have taken part in an internship are a better "risk" for companies. Within 1 year of hire, nearly 86% of those that did an internship at the hiring company (and 85% of those who did an internship elsewhere) are still on the job; compared to 81% who did not do an internship.*
- *They want students to have "hands on" experience in their field.*
- *Nearly all responding companies (86.5 %) have formal internships/co-op programs.*

- *Among respondents, the primary focuses of their programs is to feed their full-time hiring program. Approximately 83 % of respondents cited this as the primary focus of their internship program, while nearly 80% identified this as the primary focus of their co-op program.*

Fifty percent of employers said students should have at least one internship prior to graduation; and 50% responded that students should have a minimum of 2 internships on their resume!

Program Benefits:

- *Free service to all COE students*
- *Access to internships from local and national employers*
- *Resume Reviews, Mock Interviews, Workshops, Networking, Salary Negotiation, ENGR 289 (putting your practical experience on your permanent transcript) and more. Do not wait - build your resume NOW!!*

Eligibility:

- *All College of Engineering students are eligible to register with the Engineering Career Center (no special requirements).*

More information about the ECC center could be found on the following link: http://www.ecc.uic.edu/ECC/WebHome

1.D.2.d. Alumni Career Center.

The experienced Alumni Career Center staff has been serving University of Illinois Alumni Association members since 1987, giving alumni access to one of the most comprehensive employment and career management services in the country. Throughout their career, Illinois graduates gain the competitive edge in employment opportunities and career transition assistance with access to job postings, a resume referral program, networking contacts, workshops and career advising. Active connections with top employers who value the education, skills and prestige of University of Illinois graduates enhance our partnership with alumni.

Alumni can visit the Center at 200 S. Wacker Dr., Chicago from 8:30 a.m. to 5 p.m. Monday through Friday, and by appointment only on Wednesday evenings, 5 to 7:30 p.m.

E. Work in Lieu of Courses

The College of Engineering does not allow credit towards any degree based on work or life experience. However, students may establish credit toward an undergraduate degree through several examinations, the most common being Advanced Placement.

Advanced Placement (AP).
UIC will award course credit on the basis of scores earned on the Advanced Placement Examinations administered by the College Board. A list of applicable courses and corresponding scores required for credit is provided in the Undergraduate Catalog.

http://www.uic.edu/ucat/catalog/AS.shtml#g

Aside from liberal-arts courses satisfying General Education requirements, technical courses relevant to the Chemical Engineering major are as follows:

MATH 180
MATH 181
CHEM 112
CHEM 114
PHYS 141
PHYS 142

International Baccalaureate Exams (IB).
UIC will award credit on the basis of scores earned on the International Baccalaureate examinations. A list of applicable courses and corresponding scores required for credit is provided in the Undergraduate Catalog.

http://www.uic.edu/ucat/catalog/AS.shtml#g

Aside from liberal-arts courses satisfying General education requirements, technical courses relevant to the Chemical Engineering major are as follows:

CHEM 112
CHEM 114

ACT English/SAT Verbal.
UIC will award three hours of passing credit for ENGL 160 for an ACT English subscore of 27 or more or an SAT Verbal score of 610 or more.

College-Level Examination Program (CLEP).
In general, UIC may award credit on the basis of scores earned on the College Level Examination Program (CLEP). A maximum of 30 semester hours of credit on the basis of CLEP examination scores may be applied toward degree requirements, if approved by the Engineering College and Chemical Engineering Department. CLEP tests are administered on campus by the Office of Testing Services

Proficiency Examinations for Enrolled Students.

These are covered by general University rules, largely deferring to the department and college in question and delineated in detail in the Undergraduate Catalog.

http://www.uic.edu/ucat/catalog/AS.shtml#g

The Chemical Engineering department does not offer proficiency exams for any of its courses.

Students must submit official grade reports/examination results to the Office of Admissions before credit can be awarded. UIC will not award transfer course credit based on another institution's evaluation of test results. More information could be found on the university weblink: http://www.uic.edu/ucat/catalog/AS.shtml#g

F. Graduation Requirements

To earn a BS degree from the College of Engineering at UIC, students need to complete University, college, and department degree requirements. University and college degree requirements for all College of Engineering students are outlined below.

1.F.1. Semester Hour Requirement

Degree Program/ Department	Degree Conferred	Total Hours
Chemical Engineering	BS in Chemical Engineering	128

1.F.2. Course Requirements

General Education Core.

General Education at UIC is designed to serve as a foundation for lifelong learning. Students are required to complete a minimum of 24 semester hours in the General Education Core with at least one course from each of the following categories:

I. Analyzing the Natural World
II. Understanding the Individual and Society
III. Understanding the Past
IV. Understanding the Creative Arts
V. Exploring World Cultures
VI. Understanding U.S. Society

A description and list of courses for each General Education Core category is provided in the *General Education* section of the Undergraduate Catalog.

http://www.uic.edu/ucat/catalog/EG.shtml

Information on meeting the General Education requirements for each degree program is provided in the College of Engineering department sections. Chemical Engineering majors satisfy the General Education Core category I through other requirements of the major.

General Education Proficiencies—University Writing Requirement.
College of Engineering students meet the requirement by achieving a passing grade in English 160 and 161 (6 hours total). Credit for English 160 may be earned on the basis of a score of 4–5 on the AP English Language and Composition exam, an ACT English subscore of 27 or more, or an SAT Verbal score of 610 or more. Information on required scores is found in the *Standards Impacting Academic Performance and Progress* section of the *Undergraduate Catalog.*

http://www.uic.edu/ucat/catalog/AS.shtml

Orientation Course Requirement.
All incoming freshmen and transfer students must take an engineering orientation course ENGR 100 or ENGR 189, as appropriate, during the first or second term at UIC. Satisfactory completion of the engineering orientation course is a graduation requirement.

The ongoing evaluation of all Chemical Engineering students is aided by the Degree Audit Reporting System (DARS); an example of a DARS report will be provided to the site visit team. This appraises students of what courses they have taken and which courses remain to be taken for the degree. DARS communicates with the Course Registration System (CRS) software, which checks that course pre-requisites have been taken by the registering student. If the pre-requisite has not been taken, the student is not allowed to register for that class.

In addition to specific course requirements which are detailed in Section 5, the BS in Chemical Engineering also entails three activities.

1. Participation in the officially judged Senior Engineering EXPO in conjunction with CHE 397 (Senior Design II)
2. Completion of the Senior Exit Survey and Senior Exit Interview and Oral Exam in conjunction with CHE 499 (Professional development Seminar).

1.F.3. Residency Requirements

A minimum residency requirement stipulating the number of consecutive credits that must be taken at UIC is in place and must be satisfied for graduation. The policy (which was most recently revised in Fall 2007) states:

> "Either the first 90 semester hours or the last 30 semester hours of University work must be taken at UIC. In addition, at least one-half of the semester hours required in the student's major area of study must be completed at UIC. Concurrent attendance at the University of Illinois at Chicago and another collegiate institution or enrollment during the summer at another institution when approved by the student's college, does not interrupt the UIC enrollment residence requirement for graduation. Under exceptional circumstances, the enrollment residence requirement may be waived by the dean of the student's college upon petition of the student."

G. Transcripts of Recent Graduates

Student transcripts will be provided as requested by the EAC Team Chair.

CRITERION 2. PROGRAM EDUCATIONAL OBJECTIVES

A. *Mission Statement*

2.A.1. Scope and Mission Statement of the University of Illinois at Chicago (UIC)

"UIC provides the broadest access to the highest levels of intellectual excellence."

"UIC's mission is:

> *To create knowledge that transforms our views of the world and, through sharing and application, transforms the world.*
>
> *To provide a wide range of students with the educational opportunity only a leading research university can offer.*
>
> *To address the challenges and opportunities facing not only Chicago but all Great Cities of the 21st century, as expressed by our Great Cities Commitment.*
>
> *To foster scholarship and practices that reflect and respond to the increasing diversity of the U.S. in a rapidly globalizing world.*
>
> *To train professionals in a wide range of public service disciplines, serving Illinois as the principal educator of health science professionals and as a major healthcare provider to underserved communities."*

The UIC mission statement was most recently ratified by the faculty senate of the University of Illinois at Chicago on April 27, 2006. This statement is prominently displayed to all the UIC's constituencies at the following URL.
http://www.uic.edu/uic/about/scope.shtml

2.A.2. Mission Statement of the College of Engineering (COE)

> *"The mission of the College of Engineering at the University of Illinois at Chicago is to provide the opportunity for each student to become all that he or she is capable of becoming through excellence in education in the three areas of teaching, research, and service. In the area of teaching, the college provides academic excellence to its students through ten Bachelor of Science programs in six departments: Bioengineering; Chemical Engineering; Civil and Materials Engineering; Computer Science; Electrical and Computer Engineering; and Mechanical and Industrial Engineering. With the changing dynamics of society, the college continues to strive for excellence and innovation in both its instructional and research programs. In the area of community service and as part of the University's Great Cities Program related to economic development and environmental concerns, the college is continuously strengthening ties with the industrial community, especially the dynamic region of Illinois."*

The College of Engineering mission statement is available online to students and all other College constituencies at the following URL.

http://engineering.uic.edu/bin/view/COE/DeanMessage#MISSION

It also appears in the undergraduate catalog at the following URL.

http://www.uic.edu/ucat/catalog/EG.shtml#b

Consistent with this mission statement, the College of Engineering publishes the following Educational Objectives for students in the Undergraduate catalog at the following URL.

http://www.uic.edu/ucat/catalog/EG.shtml#d

"The UIC College of Engineering offers undergraduate and graduate students opportunities to join faculty in cutting-edge research. In the classroom, students become familiar with the fundamental mathematical and scientific principles that are common to engineering and computer science disciplines, and they learn to apply these principles to current engineering and computer science problems of analysis, design, and experimentation. Through individual and group projects, students make use of current techniques, instruments, equipment, and computers, and gain proficiency in communicating the results of their work. Study in other disciplines provides students with an understanding of the professional ethical responsibilities of practicing engineers. Students also have the opportunity to participate in a number of the many on-campus student chapters of national engineering professional organizations as a way to supplement their classroom experiences.

In the first two years, each student will be required to complete courses in mathematics, chemistry and physics (or other science requirements, for computer science majors), and University Writing. Beginning in the second year, the student will start course work in a particular major that represents the technical phase of the student's academic career and constitutes a cohesive program of advanced work in a chosen field. Although the course work in the major becomes progressively specialized in the junior and senior years, each student is also required to take engineering or computer science courses outside of his or her chosen field.

A student must also complete course work in the general fields of humanities and social sciences. Because engineers and computer scientists are no longer narrow specialists, they must recognize the effects of their work on the general welfare of society. The humanities/social sciences phase of their education helps them to become serious contributors to the quality of life. Requirements for the degrees often include free electives that introduce flexibility into the curricula."

2.A.3. Mission statement of the Department Chemical Engineering

> "The mission of the Department of Chemical Engineering is to serve society through excellence in education, research, and public service. We provide for our students an education in chemical engineering, and we aspire to instill in them the attitudes, values, and vision that will prepare them for lifetimes of continued learning and leadership in their chosen careers. Through scholarship, the Department strives to generate new knowledge and technology for the benefit of the State of Illinois, the nation, and beyond."

The departmental mission statement is available online to students and all other College constituencies at the following URL.

http://www.che.uic.edu/CHE/Welcome

B. Program Educational Objectives

PEO	Category	Description
1	Depth	In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.
2	Breadth	Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3	Professionalism and Service	In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.
4	Life-Long Learning and Leadership	Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.

The Program Educational Objectives are prominently listed in the "Objectives and Outcomes" link on the home page of the departmental website: www.che.uic.edu. Additionally, the PEOs are included in all undergraduate course syllabi, and class time is devoted to explaining how the knowledge, skills and principles developed by the course in question contribute to the professional capacities represented by the PEOs.

C. Consistency of the Program Educational Objectives with the Mission of the Institution

Component of UIC mission	Contributing PEO
To create knowledge that transforms our views of the world and, through sharing and application, transforms the world.	1
To provide a wide range of students with the educational opportunity only a leading research university can offer.	1, 4
To address the challenges and opportunities facing not only Chicago but all Great Cities of the 21st century, as expressed by our Great Cities Commitment.	2, 3
To foster scholarship and practices that reflect and respond to the increasing diversity of the U.S. in a rapidly globalizing world.	3
To train professionals in a wide range of public service disciplines, serving Illinois as the principal educator of health science professionals and as a major healthcare provider to underserved communities.	2

The healthcare component of UIC's mission (arising from the specific history of UIC and its medical school within the Chicago Metropolitan Area) is not directly served by the Chemical Engineering program, and would not be expected among the objectives of typical chemical engineering programs. However, to the extent that chemical engineering impacts on medical technology (e.g., artificial organs, instrumentation, biocompatible materials, pharmaceuticals, etc.) and that graduates could be employed in related sectors, the Chemical Engineering bachelor's degree program could contribute indirectly.

D. Program Constituencies

1.D.1. Program Constituencies.

Program Constituency	**Vehicles of Consultation**
Students	Course Update Surveys (each semester) Academic advising (each semester) Senior Exit Interview (semester of graduation)
Faculty	Faculty meetings (monthly)
Alumni	Alumni feedback surveys (ongoing) Alumni at large (in-person / telephone contact and guest lectures) Engineering Expo judging (each spring semester)
Industrial and Professional	External Advisory Board meeting (triennial) AIChE Chicago Section (ongoing) Industrial mentors in Senior Design II (CHE 397) class Engineering Expo judging (each spring semester)
University Administration	Educational Policy Committee meetings (monthly)

Our students are critically important stakeholders in the curriculum and policies of the Chemical Engineering department, and we consult them regularly as summarized above. However, students are not in as strong a position (as other constituents) to determine the appropriate aspects and levels of professional accomplishment several years after graduation. Surely their feedback forms part of the conceptual milieu in which the department formulates policy. But we do not regard students formally as "constituents" in the review and updating of PEOs.

1.D.2. Meeting the Needs of Program Constituents and Other Stakeholders

Students. The Chemical Engineering program exists to educate and promote young engineers and launch them on successful careers in industry, post-graduate education and government service. In order that students understand the broader professional context toward which they are being educated, the PEOs are listed on the syllabus of each undergraduate course and class discussion is devoted to putting the performance indicators of the course in the professional and career context of PEOs. With this motivation, PEOs guide the professional development of our students and set targets for their knowledge, skills and principles.

Faculty. PEOs represent the ultimate goals of undergraduate teaching and learning, and thereby serve to organize our efforts in design, development, implementation and improvement of the curriculum toward professional success and effectiveness of our graduates.

Alumni. PEOs represent the cornerstone for professional success, effectiveness and leadership of our alumni.

Industry. For local and National industry that seeks to hire, develop and promote professional engineers, PEOs represent the spectrum of professional abilities and capacities of UIC Chemical

Engineering graduates that make them desirable and enable them to contribute and lead. The industrial constituency is represented by the External Advisory Board (EAB) and also members of the AIChE Chicago Section.

The EAB has been assembled from among prominent industrial and corporate engineers from various sectors and localities, who complement each other's technical expertise.

Dr. Tarun Chandra
Managing Principal
EmpiriQA LLC
1580 Roanoke Court
Long Grove, IL 60047

Mr. Michael H. Goluszka
Technical Director
LA-CO Industries
Elk Grove Village, IL 60007

Dr. Diane Graziano
Deputy Division Director
Argonne National Laboratory
9700 S. Cass Avenue
Argonne, IL 60439

Dr. S. S. Kumaran
Sr. Research Engineer
Global Operations
Cabot Corporation
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Dr. Norman Li
President and CEO
NL Chemical Technology, Inc.
479 Business Center Dr., Suite 100
Mount Prospect, IL 60056
Member, National Academy of Engineering of USA

Dr. Paul Mathias
Fluor Corporation
3 Polaris Way
Aliso Viejo, CA 92698

Mr. Dennis O'Brien
Group Manager, Refining and Petrochemicals Division
Jacobs Engineering Group Inc.

525 West Monroe, Suite 1350
Chicago, Illinois 60661

Dr. Anil Oroskar
CTO, CFO
OroChem Technologies Inc.
331 Eisenhower Ln S
Lombard, IL 60148

Dr. Gavin Towler
Senior Manager of Process Design Development
Universal Oil Products LLC
25 E. Algonquin Road
Des Plaines IL 60017-5017

Dr. Samuel Wong, P.E.
Engineer, Kingwood Technology Center
Chevron Phillips Chemical Company
1862 Kingston Drive
Kingwood, TX 77339-3097

The EAB is helping to steer the department's direction, structure and mission.

A number of practicing chemical engineers, members of the AIChE Chicago Section, partner with the department on (i) improving pedagogy (industrial mentors and guest lecturers in the Senior Design sequence, CHE 396, 397 – Senior Design), (ii) outreach activities (high schools, middle schools), and (iii) interactions with the AIChE Student Chapter. These industrial constituents include

Alma de la Garza, UOP LLC
Shannon Brown, Ambitech
Dima Hart, UOP LLC
Annette Johnston, Abbott Labs
Adam Kanyuh, UOP LLC
Dan Rusinak, Middough
Pat Shannon, Middough
Alan Zagoria, UOP LLC

E. Process for Review of the Program Educational Objectives

Periodic review and updating of the PEOs involves all constituencies of the department. The PEOs are discussed by faculty and industrial constituents at the triennial meeting of the EAB. Our originally scheduled 2011 EAB meeting was postponed till May 8, 2012 because of scheduling constraints among the membership. Discussion of changes to the PEOs at this meeting led to the current formulation (Section 2.B). The most recent EAB meeting took place on April 25, 2014. The PEOs were considered again but no further changes were deemed necessary.

Alumni feedback is obtained where possible in person at return visits (judging at Engineering Expo, invited class lectures and student seminars, etc.) and other points of contact with the faculty. Limited success with paper-based surveys led us to develop an online survey system for the departmental Assessment website. This survey does not query the PEOs directly (as our experience has convinced us that the terminology is too technical for a survey brief enough to garner high response rates). Instead the survey poses open-ended questions from which answers can be mapped to feedback on the PEOs. This approach was discussed at a faculty meeting and approved on second reading at a subsequent faculty meeting.

Finally, input from University administration is obtained from the Office of the Associate Dean for Undergraduate Affairs at regular meetings of the COE Educational Policy Committee (EPC). This last avenue is an avenue for maintaining consistency between departmental PEOs and the UIC and COE mission statements.

For the ABET site visit we will present the following documentation on PEOs:

Results from the Alumni Survey.

Relevant agendas, notes and/or minutes from the EAB meetings.

Correspondence, notes and/or minutes on faculty and EPC discussion of PEOs.

CRITERION 3. STUDENT OUTCOMES

A. Student Outcomes

Student Outcomes are prominently listed in the "Objectives and Outcomes" link on the home page of the departmental website: www.che.uic.edu. Additionally, relevant SOs are included in all undergraduate course syllabi, and class time is devoted to explaining how the knowledge, skills and principles developed by the course in question contribute to the understanding and abilities represented by the SOs.

The processes (Section 2.E.) for revisiting and updating PEOs also afford opportunities to consider SOs and amend them as necessary. Since 2008 the departmental SO have coincided with the ABET outcomes (a)-(k) with one exception: division of the experimental outcome (b) into three parts. This is done to facilitate assessment of those elements of experimentation (experimental design, analysis/interpretation of data) that can be covered in non-laboratory lecture courses. The last update of SOs occurred in the faculty meeting of March 18, 2014. This was a minor change for conformity with the current ABET (a)-(k) and to formally articulate the above description as a policy that will track future changes in the ABET outcomes.

SO	Description	ABET	PEO
1	Ability to apply knowledge of mathematics, science and engineering	a	1
2	Ability to identify, formulate and solve engineering problems	e	1
3	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	k	1
4	Ability to design experiments	b	1
5	Ability to conduct experiments	b	1
6	Ability to analyze and interpret data	b	1

TABLE 3-1. Student Outcomes related to engineering fundamentals, and correspondence with ABET criteria (a) - (k) and Program Educational Objectives

SO	Description	ABET	PEO
7	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	c	1
8	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	h	2
9	Knowledge of contemporary issues	j	2

TABLE 3-2. Student Outcomes related to design and broader knowledge, and correspondence with ABET criteria (a) - (k) and Program Educational Objectives.

SO	Description	ABET	PEO
10	Recognition of the need for and ability to engage in life-long learning	i	4
11	Ability to function on multidisciplinary teams	d	3
12	Ability to communicate effectively	g	3
13	Understanding of professional and ethical responsibilities	f	3

TABLE 3-3. Student Outcomes related to professionalism, and correspondence with ABET criteria (a) - (k) and Program Educational Objectives.

B. Relationship of Student Outcomes to Program Educational Objectives

Tables 3-1 through 3-3 show how the SOs relate to the PEOs. SOs 1—7 represent the core of fundamental engineering knowledge, skills and tools that prepare graduates to enact PEO 1 (depth) in their subsequent careers. SOs 8 and 9 embody the social, economic and political context in which graduates advance their careers and impact society. SOs 11—13 establish the abilities of graduates to attain high professional and ethical standards in their careers and advance public safety and environmental protection. SO 10 reflects the capacity of graduates to renew their knowledge and skills, enhance their innovation and lead.

C. Coverage of Student Outcomes in Courses

Coverage of Student Outcomes in any given course is classified using the following four-point scale:

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge.
3 = course grade partially dependent on demonstrated knowledge on this topic.
2 = topic introduced in course, course grade not dependent on demonstrated knowledge.
1 = topic not covered specifically in course.

Tables 3-4 and 3-5 indicate the coverage of Student Outcomes in each of the Chemical Engineering courses. Overall coverage of SOs by ChE courses is summarized in Table 3-6: all SOs are covered at levels 3 or 4 in at least two ChE courses. Both SOs #1 and #2 receive this level of emphasis in seventeen courses. Tables 3-7 and 3-8 indicate the coverage of SOs in each of the technical courses outside the Chemical Engineering department.

Course	(1) Ability to apply knowledge of mathematics, science and engineering	(2) Ability to identify, formulate and solve engineering problems	(3) Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	(4) Ability to design experiments	(5) Ability to conduct experiments	(6) Ability to analyze and interpret data	(7) Ability to design a system, component, or process to meet desired needs within realistic constraints*
CHE 201	4	3	4	1	1	2	1
CHE 205	4	3	4	1	1	3	1
CHE 210	4	4	3	1	1	1	2
CHE 301	4	4	2	1	3	1	2
CHE 311	4	4	1	1	1	1	1
CHE 312	4	3	4	1	1	2	1
CHE 313	4	4	4	2	1	1	2
CHE 321	4	4	4	1	1	3	2
CHE 341	4	4	1	1	1	1	2
CHE 381	3	3	4	4	4	4	1
CHE 382	3	3	4	4	4	4	1
CHE 396	4	3	2	1	1	2	4
CHE 397	4	3	4	1	1	2	4
CHE 422	4	3	4	1	1	3	2
CHE 431	4	4	4	1	1	2	1
CHE 433	4	4	4	1	1	1	3
CHE 494	4	4	2	1	1	2	1
CHE 499	2	2	2	1	1	1	2

* Such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability.

TABLE 3-4. Coverage of Student Outcomes 1—7 by Chemical Engineering courses.

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;
3 = course grade partially dependent on demonstrated knowledge on this topic;
2 = topic introduced in course, course grade not dependent on demonstrated knowledge;
1 = topic not covered specifically in course.

Abbreviated, descriptive course titles:

CHE 201 = Intro Thermo
CHE 205 = Computational Methods
CHE 210 = Mass & Energy Balances
CHE 301 = Thermo
CHE 311 = Fluid Mechanics
CHE 312 = Heat and Mass Transfer
CHE 313 = Separations
CHE 321 = Reaction Engineering
CHE 341 = Process Control
CHE 381 = Unit Ops Lab I
CHE 382 = Unit Ops Lab II
ChE 396 = Senior Design I
CHE 397 = Senior Design II
CHE 422 = Biochemical Engineering
CHE 431 = Numerical Methods
CHE 433 = Aspen Process Simulation
CHE 494 = Selected Topics – Electrochemical
ChE 499 = Prof. Development Seminar

Course	(8) Broad education necessary to understand the impact of engineering solutions*	(9) Knowledge of contemporary issues	(10) Recognition of the need for and ability to engage in life-long learning	(11) Ability to function on multidisciplinary teams	(12) Ability to communicate effectively	(13) Understanding of professional and ethical responsibilities
CHE 201	1	1	1	1	1	2
CHE 205	1	1	1	1	1	2
CHE 210	1	1	1	1	1	1
CHE 301	1	1	1	1	1	1
CHE 311	1	1	1	1	1	1
CHE 312	1	1	1	1	1	2
CHE 313	2	2	2	2	3	2
CHE 321	2	1	1	1	1	1
CHE 341	2	1	1	1	1	1
CHE 381	1	1	1	3	4	3
CHE 382	1	1	1	3	4	3
CHE 396	3	3	3	2	2	3
CHE 397	3	3	3	4	4	3
CHE 422	1	1	1	1	1	1
CHE 431	1	1	1	1	1	1
CHE 433	3	3	1	1	1	1
CHE 494	1	1	1	1	4	2
CHE 499	2	2	2	2	2	2

[2] In a global, economic, environmental, and societal context.

TABLE 3-5. Coverage of Student Outcomes 8—13 by Chemical Engineering courses.

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;
3 = course grade partially dependent on demonstrated knowledge on this topic;
2 = topic introduced in course, course grade not dependent on demonstrated knowledge;
1 = topic not covered specifically in course.

Abbreviated, descriptive course titles:

CHE 201 = Intro Thermo
CHE 205 = Computational Methods
CHE 210 = Mass & Energy Balances
CHE 301 = Thermo
CHE 311 = Fluid Mechanics
CHE 312 = Heat and Mass Transfer
CHE 313 = Separations
CHE 321 = Reaction Engineering
CHE 341 = Process Control
CHE 381 = Unit Ops Lab I
CHE 382 = Unit Ops Lab II
ChE 396 = Senior Design I
CHE 397 = Senior Design II
CHE 422 = Biochemical Engineering
CHE 431 = Numerical Methods
CHE 433 = Aspen Process Simulation
CHE 494 = Selected Topics – Electrochemical
ChE 499 = Prof. Development Seminar

Student Outcomes		Number of courses at the indicated level of coverage			
#	Description	4 Topic is main focus of course, grade largely dependent on demonstrated knowledge;	3 Course grade partially dependent on demonstrated knowledge on this topic;	2 Topic introduced in course, course grade not dependent on demonstrated knowledge;	1 Topic not covered specifically in course.
1	Ability to apply knowledge of mathematics, science and engineering	15	2	1	0
2	Ability to identify, formulate and solve engineering problems	9	8	1	0
3	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	11	1	4	2
4	Ability to design experiments	2	0	1	15
5	Ability to conduct experiments	2	1	0	15
6	Ability to analyze and interpret data	2	3	6	7
7	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	2	1	7	8
8	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	0	3	4	11
9	Knowledge of contemporary issues	0	3	2	13
10	Recognition of the need for and ability to engage in life-long learning	0	2	2	14
11	Ability to function on multidisciplinary teams	1	2	3	12
12	Ability to communicate effectively	4	1	2	11
13	Understanding of professional and ethical responsibilities	0	4	6	8

TABLE 3-6. Summary of the Coverage of Student Outcomes in Chemical Engineering Courses.

Course	(1) Ability to apply knowledge of mathematics, science and engineering	(2) Ability to identify, formulate and solve engineering problems	(3) Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	(4) Ability to design experiments	(5) Ability to conduct experiments	(6) Ability to analyze and interpret data	(7) Ability to design a system, component, or process to meet desired needs within realistic constraints*
MATH 180	4	2	2	1	1	1	1
MATH 181	4	2	2	1	1	1	1
MATH 210	4	2	2	1	1	1	1
MATH 220	4	2	2	1	1	1	1
PHYS 141	4	2	2	3	3	3	1
PHYs 142	4	2	2	3	3	3	1
CHEM 112	4	2	1	3	3	3	1
CHEM 114	4	2	1	3	3	3	1
CHEM 222	4	2	1	3	3	3	1
CHEM 232	4	2	1	1	1	1	1
CHEM 233	4	2	1	4	4	4	1
CHEM 234	4	2	1	1	1	1	1
CHEM 342	4	2	1	1	1	1	1
CHEM 346	4	2	1	1	1	1	1
CS 109	2	2	4	1	1	1	1
CME 260	4	4	3	3	3	3	1
ECE 210	3	3	3	1	1	3	1
ENGR 100	1	1	2	1	1	1	1

* Such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability.

TABLE 3-7. Coverage of Student Outcomes 1—7 by non-ChE technical courses.

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;
3 = course grade partially dependent on demonstrated knowledge on this topic;
2 = topic introduced in course, course grade not dependent on demonstrated knowledge;
1 = topic not covered specifically in course.

Abbreviated, descriptive course titles:

MATH 180 = Calculus I
MATH 181 = Calculus II
MATH 210 = Calculus III
MATH 220 = Intro. Differential equations
PHYS 141 = General Physics I
PHYS 142 = General Physics II
CHEM 112 = General chemistry I
CHEM 114 = General Chemistry II
CHEM 222 = Analytical Chemistry
CHEM 232 = Organic Chemistry I
CHEM 233 = Organic Chem. Lab I
CHE 234 = Organic Chemistry II
ChEM 342 = Physical Chemistry I
CHE 346 = Physical Chemistry II
CS 109 = C/C++ Programming & MatLab
CME 260 = Properties of Materials
ECE 210 = Electrical Circuit Analysis
ENGR 100 = Engineering Orientation

Course	(8) Broad education necessary to understand the impact of engineering solutions*	(9) Knowledge of contemporary issues	(10) Recognition of the need for and ability to engage in life-long learning	(11) Ability to function on multidisciplinary teams	(12) Ability to communicate effectively	(13) Understanding of professional and ethical responsibilities
MATH 180	1	1	1	1	1	1
MATH 181	1	1	1	1	1	1
MATH 210	1	1	1	1	1	1
MATH 220	1	1	1	1	1	1
PHYS 141	1	1	1	1	1	1
PHYs 142	1	1	1	1	1	1
CHEM 112	1	1	1	1	1	1
CHEM 114	1	1	1	1	1	1
CHEM 222	1	1	1	1	2	1
CHEM 232	1	1	1	1	1	1
CHEM 233	1	1	1	1	2	1
CHEM 234	1	1	1	1	1	1
CHEM 342	1	1	1	1	1	1
CHEM 346	1	1	1	1	1	1
CS 109	1	1	1	1	1	1
CME 260	1	1	1	2	1	1
ECE 210	1	1	1	2	1	1
ENGR 100	2	2	2	2	1	2

[2] In a global, economic, environmental, and societal context.

TABLE 3-8. Coverage of Student Outcomes 8—13 by non-ChE technical courses.

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;
3 = course grade partially dependent on demonstrated knowledge on this topic;
2 = topic introduced in course, course grade not dependent on demonstrated knowledge;
1 = topic not covered specifically in course.

Abbreviated, descriptive course titles:

Abbreviated, descriptive course titles:

MATH 180 = Calculus I
MATH 210 = Calculus III
PHYS 141 = General Physics I
CHEM 112 = General chemistry I
CHEM 222 = Analytical Chemistry
CHEM 233 = Organic Chem. Lab I
ChEM 342 = Physical Chemistry I
CS 109 = C/C++ Programming & MatLab
ECE 210 = Electrical Circuit Analysis

MATH 181 = Calculus II
MATH 220 = Intro. Differential equations
PHYS 142 = General Physics II
CHEM 114 = General Chemistry II
CHEM 232 = Organic Chemistry I
CHE 234 = Organic Chemistry II
CHE 346 = Physical Chemistry II
CME 260 = Properties of Materials
ENGR 100 = Engineering Orientation

CRITERION 4. CONTINUOUS IMPROVEMENT

A. *Student Outcomes*

Oversight and conduction of assessment is ultimately the responsibility of the Undergraduate Committee (UGC) and its chair, the Director of Undergraduate Studies (DUGS). However, the overall assessment methodology has been developed in such a way that all faculty and some industrial representatives are involved.

4.A.1. Assessment Methods and Frequency of Administration.

The overall plan and schedule of SO assessment is summarized in Table 4-1. Direct assessment rests on a triad of methods, each of which probes the outcomes in different ways, each semester or annually, by different evaluators: (1) Grading rubrics in CHE courses, (2) Engineering Expo judging rubrics, (3) Senior Exit Interview and Oral Exam. Table 4-2 shows how these three assessment methods cover the various SOs. In summary, SOs 1-3, 8, 10-13 are covered by two direct assessment methods and SOs 4-6, 9 are covered by one assessment method. Indirect assessment of SOs is achieved via an online Exit Survey that is filled out by graduating seniors

#	Description (Abbreviation)	Category	Schedule	Responsibility
1	Grading rubrics in CHE courses (Course Rubrics)	Direct / Formative	Each semester	All faculty
2	Senior Exit Interview and Oral Exam (Exit Interview)	Direct / Summative	Spring semester	Undergraduate Committee
3	Engineering Expo judging rubrics (Expo Rubrics)	Direct / Summative	Spring semester	Technical Judges
4	Senior Exit Survey (Exit Survey)	Indirect / Summative	Spring semester	Undergraduate Committee
5	Course Update Surveys (Course Survey)	Feedback	Each semester	All faculty

TABLE 4-1. SO assessment methods and frequency of administration.

toward the end of each spring semester (Method #4). Finally, in each course the regular student evaluations of teaching (administered centrally by the Office of the Vice Provost for Faculty Affairs) are augmented by a much more detailed and penetrating online feedback survey that was developed in-house to facilitate continuous improvement of courses by individual faculty (Method #5). These assessment and feedback methods are now described in detail.

Assessment Method #1 (Direct / Formative): Rubric-based assessment of a major assignment in required courses. In each course the professor formulates a detailed rubric to assess specific performance indicators (PIs) among the students on a major assessment assignment (e.g., exam or portion of exam, term paper, lab report, milestone presentation). These PIs are mapped to some or all of the relevant SOs for that course as specified in the syllabus (see Appendix A). Table 4-3

SO	Description	ABET (a)-(k)	PEO	Method #1 Course Rubrics	Method #2 Exit Interview	Method #3 Expo Rubrics
1	Ability to apply knowledge of mathematics, science and engineering	a	1	X	X	
2	Ability to identify, formulate and solve engineering problems	e	1	X	X	
3	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	k	1	X	X	
4	Ability to design experiments	b	1	X		
5	Ability to conduct experiments	b	1	X		
6	Ability to analyze and interpret data	b	1	X		
7	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	c	1			X
8	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	h	2		X	X
9	Knowledge of contemporary issues	j	2		X	
10	Recognition of the need for and ability to engage in life-long learning	i	4		X	X
11	Ability to function on multidisciplinary teams	d	3		X	X
12	Ability to communicate effectively	g	3		X	X
13	Understanding of professional and ethical responsibilities	f	3	X	X	

TABLE 4-2. Coverage of SO by direct assessment methods.

indicates the coverage and schedule of SO assessment by this mechanism. We cannot claim to cover every relevant SO for every course every semester, because the coverage in any given course hinges on the major assessment assignment being formulated by the professor. But the partially overlapping "collage" of SO "footprints" in all the courses does ultimately provide good coverage of the SOs overall. Furthermore, this approach regularly engages the entire faculty in the process of assessment, and keeps assessment uppermost in our minds. The SO assessment rubrics certainly do require more time and effort from professors than does the typical academic grading process for the assignments involved, but at a manageable level that encourages buy-in among the faculty. Significantly, we thereby also avoid the burden of additional assessment activities for students. The assessment information compiled by the professor includes (1) major assessment assignment, (2) assessment rubric, (3) calculations (e.g., Excel spreadsheet, when appropriate), (4) summary of results. Exhibit 4-1 (Section 4.C) shows these documents for CHE 205 (Computational Methods in Chemical Engineering) in Fall 2013.

Assessment Method #2 (Direct / Summative): Senior Exit Interview and Oral Exam. After experimenting with written exit exams for seniors (2004, 2005), and discussing the results at meetings of the External Advisory Board, (May 2006) and ChE Alumni Association (February 2008), our department settled upon a compulsory exit interview administered to all graduating seniors by the Undergraduate Committee as part of CHE 499 (Professional Development Seminar) taken in the spring semester. The interview portion provides feedback from the student constituency whereas the oral exam portion provides an excellent vehicle for directly assessing proficiencies of soon-to-be graduates over a spectrum of skills. Although we reassure students that educational assessment results of their oral exams will not influence their course grades or graduation in any way, the face-to face, oral format provides a strong motivator for them to want to “do well” in front of us. There certainly seems to be bit of an intimidation factor because nearly all undergraduates have never had an old-fashioned oral exam before (which practice seems to be fading even from many graduate programs). Nevertheless, we have found the oral exam extremely valuable for assessing how students think and teasing out their abilities, especially in communication. We try to correct, at least partially, for on-the-spot nervousness by using a three-point scale and aligning this with the upper three ratings of the four-point scale that we use for the other two direct assessment methods. But especially for SOs 1—3, the oral exam has perhaps been a little too tough (i.e., too similar to a graduate qualifying exam). In the future we will probably detune our grading standards a little. Since the last ABET accreditation cycle in 2008, the oral exam has grown to encompass (1) numerical methods and computational/software proficiency, (2) engineering estimates and problem solving, (3) career planning and potential for life-long learning, (4) professional ethics and broader impacts of technology, (5) process and laboratory safety, (6) interdisciplinary teamwork. Past versions of the exit interview required 20-25 minutes with each student, whereas the most recent iteration in spring 2014 took about 40-60 minutes. This arduous work was shared between all three members of the Undergraduate Committee (Professors Nitsche, Meyer, Chaplin). Scoring is done with a detailed rubric that maps to the SOs and also anticipated subsequent achievement of PEOs. Two sample question sheets and the associated blank rubrics are included as Exhibit 4-2 (Section 4.C). <u>It should be emphasized that the oral exam portion constitutes a rigorous **direct** assessment of student knowledge, abilities and proficiencies by faculty</u>, and is not based on self-assessment by students (i.e., indirect assessment).

Assessment Method #3 (Direct / Summative): Judging rubrics from the annual Engineering Expo. Since before the previous ABET accreditation cycle in 2008, participation in the annual Engineering Expo has been a formal course requirement in CHE 397 (Senior Design II). At this event the design groups present posters on their projects to the UIC community, and their work and communication skills are evaluated by a panel of judges drawn from industry (including alumni) and faculty. To augment the College-wide, standardized judging sheets (used to award prizes), the Chemical Engineering department developed a rubric with rating categories that map to our specific SOs. There is also a general comments section so that free-format feedback can be cycled back to the students. Sample rubrics forms appear in Exhibit 4-3 (Section 4.C).

SO	Fall 2013	Spring 2013	Fall 2012	Spring 2012	Fall 2011	Spring 2011	Fall 2010	Spring 2010
1	205 301 381	321 341 382	311 381 433	321 341 382	301 311 381	321 341 382	381	321 341 433
2	205 301 311 382	321 341 382	311 381 433	321 341 382	301 311 381	321 341 382		321 341 433
3	205 301 311 381	321 341 382	311 381 433	321 341 382	301 311 382	321 341 382	381	321 341 433
4	381	382	381	382	381	382	381	
5	381	382	381	382	381	382	381	
6	205 301 381				301			
7		341	433					341
8								
9								
10	205 301 311 381	331 341 382	311 381 433	321 341 382	301 311 381	321 341 382	381	321 341 433
11								
12	381		381 433		381		381	433
13	205 301 311 382	321 341 382	311 382	321 341 382	301 311 381	321 341 382	381	321 341

TABLE 4-3. Schedule of direct assessment of SO by Method #1 (Course Rubrics).

Assessment Method #4 (Indirect / Summative): Senior Exit Survey. A survey was developed by the Undergraduate Committee in consultation with the department faculty to obtain self-assessment of graduating seniors on the SOs as well as feedback for improvement of the curriculum. This survey methodology has progressed from a paper form to an online system programmed in-house in PHP. Upon submission of the filled-in survey, each student receives an automatically generated, anonymous code number that he/she is required to bring to the Exit

Interview (Assessment Method #2) as proof of completion. The system automatically compiles a summary report that (i) calculates averages and (E,S,D,U) vectors for the SOs, and (ii) aggregates student responses for each of the feedback questions. (In previous years some of the calculations had to be done with Excel.) A sample printout of the survey is provided in Exhibit 4-4 (Section 4.C.

SO	2014	2013	2012	2011	2010	2009
1	97.7	80.6	81.5	83.3	92.6	
2	97.7	76.7	70.4	75.0	77.8	
3	97.7	64.5	74.1	66.7	70.4	
4	84.1	87.1	85.2	75.0	88.8	
5	95.5	87.1	85.2	75.0	88.8	
6	97.7	87.1	85.2	75.0	88.9	
7	88.4	67.6	76.9	41.7	77.8	
8	95.5	67.7	77.8	66.7	81.5	
9	90.9	54.8	66.7	50.0	63.0	
10	100.0	74.2	81.5	66.7	84.6	
11	97.7	71.0	84.6	75.0	84.6	
12	97.7	67.7	85.2	58.3	92.6	
13	100.0	63.3	77.8	75.0	81.5	

TABLE 4-4. Attainment of SOs as determined by indirect assessment (Method #4 – Senior Exit Survey). In this Level 3 analysis we tabulate the percentage of students self-rating at E (Exemplary) or S (Satisfactory).

Assessment Method #5 (Feedback): Course Update Survey. A survey was developed by the Undergraduate Committee to enable student feedback toward professors improving courses from semester to semester. After earlier paper versions, this survey has been implemented in online form (programmed in-house in PHP) on the departmental assessment website. The system automatically generates summary reports (that aggregate student responses to each question) and emails each professor the report(s) for his/her course(s). An interesting and useful feature is the labeling of each response with the number of hours per week that the student spent on the course (ascertained in a preliminary multiple-choice question). This puts the feedback in the context of student effort. A sample printout of the survey is provided in Exhibit 4-5 (Section 4.C).

SO	2014	2013	2012	2011	2010	2009
1	73.5	53.3	53.0	66.4	57.8	
2	76.2	57.3	51.5	63.1	52.7	
3	71.2	45.1	40.8	74.5	56.4	
4	100.0	100.0	100.0	100.0		
5	100.0	100.0	100.0	100.0		
6	83.5	63.0	63.6	69.0	42.5	
7	67.9	82.6	75.0		58.3	
8	82.1	71.4	88.0	65.7	72.8	85.7
9	76.8	71.4	88.0	65.7	72.8	85.7
10	86.9	69.6	72.5	73.3	72.8	100.0
11	94.8	83.0	88.2	75.0	74.4	100.0
12	97.8	67.2	76.2	75.4	75.0	100.0
13	87.1	76.4	70.6	71.0	64.8	75.0

TABLE 4-5. Attainment of SOs as determined by direct assessment (aggregate of Method #1 – Course Rubrics, Method #2 – Exit Interview, and Method #3 – Expo Rubrics). In this Level 4 analysis we tabulate the percentage of students attaining rating of E (Exemplary) or S (Satisfactory).

The improvement in student attainment of SOs #1 and #2 in 2014 is discussed in Section 4.B.4. The improvement in attainment of SO #3 explained in Section 4.B.2.

4.A.2. Expected levels of Attainment for the Student Outcomes.

Based upon our formulation of rubrics and the very stringent nature of the Senior Exit Interview and Oral Exam (Assessment Method #2) particularly, we would align the rubric assessment levels of E (Exemplary), S (Satisfactory), D (Developing) and U (Unsatisfactory) roughly with letter grades A, B, C and D, respectively. Thus, we are looking to achieve about 70% or better attainment of E or S for each Student Outcome among out students.

4.A.3. Evaluation of the Attainment of Student Outcomes.

Reduction of assessment data and their synthesis into higher-level information on attainment of SOs proceeds via four levels of analysis. The calculations are carried out by the faculty member(s) conducting the assessment and/or the Undergraduate Committee (assisted by an assessment TA).

> Level 1 analysis:
> Reduction of data from assessment Rubrics (Excel workbooks).
> For each of the three direct assessment methods (Table 4-1), ratings of students on individual rubric items (performance indicators, exams or portions of exams, lab reports, milestone presentations, etc.) are averaged and also binned into 3-dimensional or 4-

dimensional vectors indicating percentage of students attaining the various levels: (E, S, U) or (E,S,D,U). Here E = “Exemplary”, S = “Satisfactory”, “D = “developing” and U = “Unsatisfactory”. These results for individual rubric items are then mapped onto the applicable SOs. One rubric item may (but need not) relate to more than one SO, and each SO may have contributions from several rubric items. This procedure produces results on attainment of SOs for (i) students in a particular course in a given semester (Assessment Method #1 – Course Rubrics) or (ii) a graduating senior class (Assessment Method #2 – Exit Interview; Assessment Method #3 – Exit Interview; Assessment Method #4 – Exit Survey).

<u>Level 2 analysis:</u>
<u>Average and (E,S,D,U) attainment assessed with 4 methods by Student Outcome.</u>
For each SO we combine the (E,S,D,U) results of all Level 1 analysis year by year into a one-page summary report. This allows tracking of trends for each assessment method over time and also exposes similarities and differences between the methods. The summary reports for all SOs appear in Exhibit 4-6 (Section 4.C).

<u>Level 3 analysis:</u>
<u>Average and satisfactory attainment for all Student Outcomes by method.</u>
For each method we combine the average attainment rating and the percentage of students attaining at least “Satisfactory” for all SOs year by year into a one-page summary report. This facilitates tracking SOs over time and also comparisons between SOs to determine which aspects of the curriculum are most in need of improvement. The summary reports for all methods appear in Exhibit 4-7 (Section 4.C.). The results of indirect assessment are shown in Table 4-4.

<u>Level 4 analysis:</u>
<u>Satisfactory attainment for all Student Outcomes.</u>
For each SO we aggregate the Level 3 results from all three direct assessment methods into an overall percentage of students attaining at least “Satisfactory”. The Level 2 results from indirect assessment (Method #4) are carried over from the Level 3 analysis. This summary (Table 4-5) gives us an overall, “bird’s eye” view of SO attainment and which outcomes need improvement.

Table 4-5 shows that, by and large, UIC Chemical Engineering undergraduates are sufficiently attaining the departmental Student Outcomes. There have been some issues with computational and problem-solving skills (covered by SOs #1, #2, #3) and in some practical aspects of design (covered by SO #7). These are addressed by previous and ongoing improvements to the curriculum, as is discussed in the next section.

B. Continuous Improvement

The loop of assessment, evaluation and feedback from constituencies has been closed in various ways to improve the curriculum and student learning experience. These cases are now described in detail.

4.B.1. Ongoing Feedback-Based Improvement of Courses.

The Course Update Surveys (Assessment Method #5) are very detailed and give useful information to individual professors for improving their courses. Since the last ABET cycle in 2008, this survey procedure has been upgraded from paper and hand calculation to an automated online system programmed in-house with PHP.

We maintain on ongoing feedback loop for continuous improvement of existing courses by the following procedure. At the end of every semester, each professor administers the Course Update Survey in his applicable courses toward the end of the semester. (This is in addition to the standard student evaluations of teaching conducted centrally at the university level.) Within two weeks or so the professor receives by email the compiled survey results. The professor reflects on this student feedback and formulates a brief self-study statement on how he/she intends to improve that course for the next year (or semester). These statements are filed with the DUGS and maintained as PDF copies/scans in the Assessment Database. As needed, they are discussed when faculty meetings turn to curriculum matters. The Course Update Surveys represent a regular and extensive consultation with our student constituency with tangibly positive effects on their courses and learning experiences.

Sample self-study statements for CHE 205 (Computational Methods in Chemical Engineering) in Exhibit 4-8 (section 4.C) show how this course has evolved since its introduction in spring 2012. A total of 22 self-study statements written by 5 faculty members will be available for inspection during the upcoming ABET site visit.

4.B.2. Introduction of CHE 205 (Computational Methods in Chemical Engineering).

The origin of CHE 205 and the program change that made it a bachelor's degree requirement lies in an alumni focus group on February 5, 2008 and a meeting of the departmental External Advisory Board (EAB) on May 8, 2008. Guidance thereby obtained from both alumni and industrial constituencies convinced us of the importance and career relevance of problem solving with Excel, mathematical packages like MATLAB and process simulation software like Aspen Plus. We also noted opinions on the fading relevance of traditional programming languages (although other consultation experiences made this less clear-cut). So we began considering changes to the computation requirement in the ChE program – which at that time was a sophomore-level C++ class (CS 109). Discussion at a faculty meeting led to this issue being delegated to the Undergraduate Committee, which in due course produced a plan to develop a new sophomore-level computation course and replace CS 109 with this. This was approved at a subsequent faculty meeting, and then the process for College level approval was started by filing forms for a new course proposal (CRS Outline) and a Program Change to replace the computation requirement for the bachelor’s degree. The Educational Policy Committee (EPC) of the Engineering College reviewed both proposals during spring semester 2011 and approved the new CHE 205. The committee members applauded the introduction of Excel, MATLAB and Aspen plus for sophomores, but did not agree that this material should replace C++ programming. They felt that programming in a high-level computer language remained a relevant skill for students and also a thought process worth cultivating. After reconsideration at the department level, the final proposal was changed to add the CHE 205 as a requirement while keeping CS 109 in place as well. Room in the 128 credit hour program was found by eliminating a free elective. This was the version of the plan that was approved by the EPC and subsequently affirmed by the Senate Committee on Educational Policy (SCEP). CHE 205 was offered for the first time in spring 2012. After spring 2013 it was decided to offer CHE 205 in both semesters (like the other sophomore courses, CHE 201 and 210) to facilitate the study programs of students transferring to UIC from community colleges.

Assessment Method	2014	2013	2012	2011	2010
#2 (Exit interview)	58.0	16.7	9.1	65.8	50.0
#1 & #2 (Aggregate)	71.2	45.1	40.8	74.5	56.4

TABLE 4-6. Assessment results on computational proficiency (Student SO #3).). Here we tabulate the percentage of students attaining a rating of E (Exemplary) or S (Satisfactory).

Although the original alumni and EAB discussions on computational course material were not actually precipitated by results of assessment, we did decide because of them to start assessing computational proficiency (relating to SO #3) in the Senior Exit Interview and Oral Exam, and have done so regularly since spring 2010. The starting attainment in 2010 was less than our desired 70%, and we observed a dip in student computational proficiency in 2012 and 2013. But there was a marked improvement this last spring 2014, when the first cohort of student to have taken CHE 205 as sophomores (in spring 2012) graduated. So this is a case of assessment verifying the improvement achieved by a curriculum change suggested in feedback from alumni and industrial constituencies, filtered through discussion among departmental faculty, faculty from other departments and the College level EPC.

For the ABET site visit we will present the following documentation on the continuous improvement cycle described above:

- Minutes of departmental faculty meetings.
- Minutes of College EPC meetings.
- Notes/minutes from the alumni focus group in 2008.
- Notes/minutes from the EAB meeting in 2008.
- CRS course outline and program change proposal form.
- PDF scans of all scoring rubrics from the Exit Interview and Oral Exam.

4.B.3. Safety Content of the Curriculum.

Discussion at the EAB meetings in 2008 and 2012 emphasized the importance of safety training for students. In line with the introduction into the Chemical Engineering Program Criteria (since the last ABET cycle in 2008) of education on “hazards associated with [chemical] processes”, the Undergraduate Committee was charged to implement the incorporation of safety-related material into the Chemical Engineering curriculum. The recommended plan that was approved by faculty vote was to make the completion of specific SaChE modules mandatory in CHE 210, 312, 321, 381, 382, 396 and 397. The official CRS outlines of these courses were amended to codify these changes, after being approved by the College EPC in spring 2013. Starting in spring 2013 we added questions on lab safety and risk assessment to the Senior Exit Interview and Oral Exam (Assessment Method #2), and are tracking this rubric item and the associated SO #7.

For the ABET site visit we will present the following documentation on the continuous improvement cycle described above:

Recommendations to the department faculty from the Undergraduate Committee.
Official CRS outlines for in CHE 210, 312, 321, 381, 382, 396 and 397.
Minutes of the relevant College EPC meeting in spring 2013.
PDF scans of all scoring rubrics from the Exit Interview and Oral Exam in 2010 –2014.

4.B.4. Feedback-Driven Improvement of CHE 210 (Material and Energy Balances).

At the start of academic year 2012 we had a conference involving five faculty members (Professors Akpa, Liu, Meyer, Murad, Perl) and a member of the External Advisory Board (Dennis O’Brien) who regularly serves as an industrial mentor in Senior Design II (CHE 397). The purpose was to address an apparent disconnect between sophomores’ exposure to material and energy balances in CHE 210 and their abilities in this area in senior year. As one outcome of this discussion, Mr. O’Brien furnished industrially relevant flow sheets with which Professor Akpa could motivate the concepts and problem-solving strategies in CHE 210, so that students would appreciate where in the subsequent curriculum and their professional careers they would need this material. She has also shifted increasingly toward active learning in the CHE 210 course. These pedagogic strategies, together with the problem-solving emphasis of the new CHE 205 course (Computational Methods in Chemical Engineering), which started in spring 2012, may explain the recent upturn of SOs #1 and #2 evident in Table 4-5.

For the ABET site visit we will present the following documentation on the above course improvement:

Minutes of the curriculum meeting on August 23, 2012.
Peer observation letter on Professor Akpa’s teaching of CHE 210, dated November 1, 2012.

4.B.5. Ongoing Improvements of the Curriculum and Teaching Infrastructure.

Discussion at the 2014 EAB meeting, an earlier industrial consultation involving both professors who have taught CHE 341 (Chemical Process Control), and feedback from the industrial mentors in CHE 397 (Senior Design II) have all pointed to the incorporation into the undergraduate curriculum of further practical material on process control to augment the more theoretical emphasis of CHE 341. This has also been a theme of student feedback uncovered by the Senior Exit Interview and Oral Exam (Assessment Method #2). Current faculty discussion centers on appropriately revamping the content of CHE 396 (Senior Design I) and introducing a process control experiment in the Unit Operations Laboratory (CHE 381 and/or CHE 382).

Results from the Senior Exit Interview and Oral Exam (Assessment Method #2) and also the Senior Exit Survey (Assessment Method #4) have shown us the interest of students in hands-on exposure to the concepts of the junior core courses. (Years ago we had moved the Unit Operations Lab from the junior year to the senior year precisely because data analysis required concepts that students had by then not completed.) The Armfield tabletop demonstration units that we recently examined seem to fill this pedagogic niche, and we are now considering incorporating them into the junior Transport Phenomena sequence (CHE 311 and 312).

The Dean's office has been very supportive of upgrades to our teaching infrastructure. We have had an equipment demonstration visit from a representative of the Armfield company along with correspondence on relevant cost quotes. At the time of writing of this report, we are working with the Dean's office on a new teaching-equipment procurement budget. For the upcoming ABET site visit (October 5-7), we will document new developments on the aforementioned curriculum and teaching-infrastructure initiatives.

C. Additional Information

The following exhibits document the assessment, evaluation and continuous improvement procedures described in Sections 4.A and 4.B.

Exhibit 4-1.
Assessment Method #1 (Direct / Formative):
Rubric-based assessment of a major assignment in required courses.
Sample Dates: Fall 2013
Sample Description: Assessment assignment, rubric and results for CHE 205.

Exhibit 4-2.
Assessment Method #2 (Direct / Summative):
Senior Exit Interview and Oral Exam.
Sample Dates: Spring 2014 and 2013.
Sample Description: Question sheets and rubrics.

Exhibit 4-3.
Assessment Method #3 (Direct / Summative):
Judging rubrics from the annual Engineering Expo.
Sample Date: Spring 2013.
Sample Description: Assessment rubric for Senior Design (CHE 397) project.

Exhibit 4-4.
Assessment Method #4 (Indirect / Summative):
Senior Exit Survey.
Sample Date: Spring 2014.
Sample Description: PDF printout of online survey (PHP/CSS file).

Exhibit 4-5.
Assessment Method #5 (Feedback):
Course Update Survey.
Sample Date: Spring 2014.
Sample Description: PDF printout of online survey (PHP/CSS file).

Exhibit 4-6.
Level 2 analysis of assessment data

Exhibit 4-7.
Level 3 analysis of assessment data

Exhibit 4-8.
Continuous improvement – closing the feedback loop on assessment:
Sample Dates: Fall 2013, Spring 2013, Spring 2012.
Sample Description: Faculty self-studies for CHE 205.

Exhibit 4-1.

Assessment Method #1 (Direct / Formative):
Rubric-based assessment of a major assignment in required courses.
Sample Dates: Fall 2013
Sample Description: Assessment assignment, rubric and results for CHE 205.

ChE 205 Final Exam

Student Name____

For this exam you should prepare an Excel workbook and save it under your name (i.e., "lastname_firstname.xlsx") in the middle of the desktop of your computer. There is no provision for crediting lost work, so please be sure to save frequently as you proceed, in order to avoid accidentally losing your work. Put your work on each problem in a separate, appropriately named worksheet within your workbook. Any formulas or derivations should be typed into your workbook, as handwritten answers will not be collected. **Do NOT write on this test sheet.** It is important to distinguish between absolute versus relative cell references, as appropriate, in the course of your computations. You need not use VBA on this exam, but you can if you wish to. Be sure to sign and return this exam sheet upon finishing. **Exams not accompanied by this signed exam sheet will not be graded and thereby receive a score of zero.**

PROBLEM 1 (10 points, 10 minutes). Ethics — no crib sheet allowed.

Write the first 4 provisions of the AIChE Code of Ethics.

PROBLEM 2 (44 points).

Consider the steady-state, radial temperature profile $T(r)$ inside a spherical fuel pellet in a nuclear reactor. To obtain $T(r)$ you must solve a second-order ODE that represents conduction (left-hand side) of heat produced by fission (right-hand side).

$$k\left[\frac{\mathrm{d}^2T}{\mathrm{d}r^2} + \frac{2}{r}\frac{\mathrm{d}T}{\mathrm{d}r}\right] = -S\left[1 + b\left(\frac{r}{R}\right)^2\right] \qquad (0 < r < R)$$

For boundary conditions we have vanishing slope at the center

$$\left.\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{r=0} = 0$$

and a surface temperature prescribed by the surrounding cooling water.

$$T(R) = T_\mathrm{c}$$

In this problem the parameters are as follows:

$R = 0.6$ cm
$S = 270$ W cm^{-3}

$k = 0.032$ W cm^{-1} C^{-1}
$b = 18$
$T_c = 100$ C

Since these parameters are given in consistent units, you can use them directly as numerical values in the ODE, without explicitly stating the units.

Solve the above boundary-value problem using least-squares collocation with boundary constraints (as opposed to weighted boundary conditions[1]). You will be using 5 power-law basis functions

$$\phi_n(r) \quad = \quad r^n \qquad\qquad (n = 0, 1, 2, 3, 4)$$

The 11 collocation points at which you should impose the differential equation (in a least-squares sense) are $r = 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 0.55$. Please note that you should not use the point $r = 0$ because the coefficient $2/r$ on the left-hand side of the ODE blows up there. You can save yourself considerable algebraic work by first tabulating columns of the functions $\phi_n(r)$, $\phi_n'(r)$ and $\phi_n''(r)$ before generating the columns associated with the least-squares collocation. Tabulate the solution $T(r)$ and plot it up.

PROBLEM 3 (40 points).

A 30-liter storage tank with a bursting pressure rating of 3600 psi is filled with chlorine gas ($T_c = 417.2$ K, $P_c = 7.71 \times 10^6$ J m^{-3}, MW $= 35.453$ g mol^{-1}). Use the Redlich-Kwong (RK) equation of state to estimate the gas pressure (psi) from the temperature T (K) and molar volume V (m^3 mol^{-1}).

$$P \quad = \quad \frac{RT}{V - b} - \frac{a}{T^{0.5}V(V + b)}$$

The constants a and b can be obtained from critical data as follows.

$$a \; = \; 0.4278 R^2 T_c^{2.5} P_c^{-1}, \qquad\qquad b \; = \; 0.0867 R T_c P_c^{-1}$$

The universal gas constant is $R = 8.3145$ J mol^{-1} K^{-1}. Unit conversions: 1 bar = 100,000 J m^{-3}. 1 bar = 14.5038 psi. The following temperature readings (Celsius) were recorded over a 48-hour period in the room in which the tank is to be stored:

28.7, 27.4, 25.2, 23.9, 24.5, 29.4, 28.4, 26.1, 25.5, 27.6, 26.8

Calculate the mean temperature along with the 95% confidence interval. Based upon this uncertainty in the temperature and a 10% safety (over-design) factor for the tank bursting pressure, what is the maximum mass (kg) of chlorine that can safely be stored in the tank?

No credit will be given for work carried out using the "AVERAGE" or "STDEV" functions in Excel. You may, however, use "SUMSQ" or "SUMPRODUCT".

[1] Using the method of weighted boundary conditions will cut very heavily into partial credit points.

ChE 205 – Fall 2013
Computational Methods in Chemical Engineering
Assessment rubric for final exam

LEVEL CRITERION	**(1)** Unsatisfactory	**(2)** Developing	**(3)** Satisfactory	**(4)** Exemplary
PO (13) AIChE Code of Ethics ***Problem #1***	Cannot remember more than one provision.	Writes two provisions correctly or with minor errors.	Writes three provisions correctly or with minor errors.	Writes all four provisions correctly.
PO (1), (2), (3) Collocation solution of conduction-generation problem ***Problem #2***	Unable to start or cannot set up rectangular matrix equation ODE.	Sets up rectangular matrix equation but not the boundary conditions.	Sets up boundary constraints correctly or with minor errors.	Correct solution for temperature profile and plot.
PO (6) 95% confidence interval for temperature ***Problem #3***	Incorrect mean or standard deviation.	Calculates substantially correct mean and standard deviation but does not start on Student t distribution	Shows s/he knows how to use t distribution but makes mistake in using it.	Correct 95% confidence interval for temperature.
PO (1), (2), (3) Estimate of tank operating limit. ***Problem #3***	Only able to calculate constants in RK equation of state and P_{safe}.	Enters RK formula into Excel and obtains V from T and P_{safe} using Goal seek or Solver.	Calculates propagation of temperature uncertainty into pressure uncertainty.	Successfully iterates at least once on V and determines safe mass of chlorine.

CHE 205 - Fall 2013
Final Exam (2013-12-10-TUE)
Grading Rubric

NAME_________________________________ _____ / **100**

PROBLEM 1 _____ / **10**

Provision 1 _____ / 2.5

Provision 2 _____ / 3.5

Provision 3 _____ / 2.5

Provision 4 _____ / 1.5

PROBLEM 2 _____ / **44**

Column of r values _____ / 1

5 columns of ODE left-hand-side matrix _____ / 15

ODE right-hand-side column vector _____ / 3

Two rows for boundary conditions _____ / 4

Sub-matrix $A^t A$ _____ / 3

Sub-matrix Q _____ / 1

Sub-matrix Q^t _____ / 2

Sub-matrix 0 _____ / 1

Right hand side sub-vector $A^t b$	_____ / 3
Right hand side sub-vector w (boundary conditions)	_____ / 2
Inverse matrix	_____ / 2
Coefficients	_____ / 2
Tabulated values of $T(r)$	_____ / 3
Plot of $T(r)$	_____ / 2

PROBLEM 3 _____ / **40**

Sample average temperature	_____ / 2
Sample standard deviation for T	_____ / 3
Calculate δT for 95% confidence interval on T	_____ / 4
Calculate a and b in RK equation from critical data	_____ / 2
Safe upper limit on pressure P_{safe} = 3273 psi	_____ / 2
RK formula for $P(T,V)$ entered into cell	_____ / 3
Calculate V from T and P_{safe} (Goal Seek or Solver)	_____ / 3
Correct formula $\partial P/\partial T$ entered into cell	_____ / 4

Correct formula for $\delta P = (\partial P/\partial T)\,\delta T$	_____ / 4
Calculate δP at V and T obtained earlier	_____ / 3
Recalculate P upper bound based on P_{safe} and δP	_____ / 3
Recalculate V and confirm convergence of 1st iteration	_____ / 3
Calculate # moles given molar volume V and tank volume	_____ / 2
Calculate mass of chlorine # moles and molecular weight	_____ / 2

ChE 205 – Fall 2013
Computational Methods in Chemical Engineering

Assessment results for final exam
Prepared by: Ludwig Nitsche

Number of students: 36

SO	**1**	**2**	**3**	**6**	**13**		
% (4 = Exemplary)	49	49	49	78	97		
% (3 = Satisfactory)	36	36	36	0	3		
% (2 = Developing)	15	15	15	0	0		
% (1 = Unsatisfactory)	0	0	0	22	0		
Average score	3.33	3.33	3.33	3.33	3.97		
Standard deviation	0.56	0.56	0.56	1.26	0.17		

SO	**Description**
1	Ability to apply knowledge of mathematics, science and engineering
2	Ability to identify, formulate and solve engineering problems
3	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice
6	Ability to analyze and interpret data
13	Understanding of professional and ethical responsibilities

Conclusions:

Introduction to AIChE code of ethics was successful.

Students' achievement of basic computational and error-analysis skills is better than satisfactory.

Exhibit 4-2.

Assessment Method #2 (Direct / Summative):
Senior Exit Interview and Oral Exam.
Sample Dates: Spring 2014 and 2012.
Sample Description: Question sheets and rubrics.

BSChE Graduate Exit Interview and Oral Examination – Spring 2014
Oral discussions conducted with graduating seniors in Chemical Engineering

Question 1

Numerical methods and computational proficiency (Excel, MATLAB or Maple): choose one problem.

(a) Generate a graph of the solution of the following ODE initial value problem.

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{\tau}\Big[x_{\mathrm{in}}(t) - x(t)\Big] - kx(t)^2, \qquad x(0) = 2$$
$$x_{\mathrm{in}}(t) = 1 + \varepsilon \sin(\omega t)$$

with the constants $\tau = 3.4$, $k = 1.2$, $\varepsilon = 0.1$ and $\omega = 5$. Can you think of a chemical engineering problem (or, at least, a course) in which such an ODE would arise?

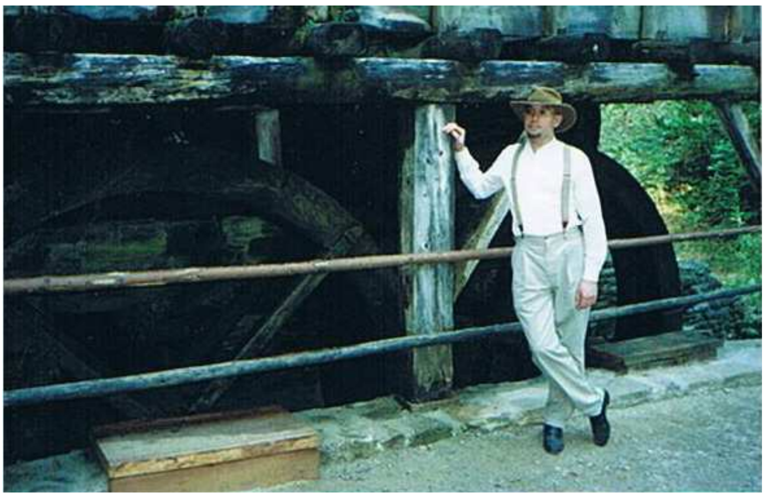
(b) Find a numerical value for the following definite integral.

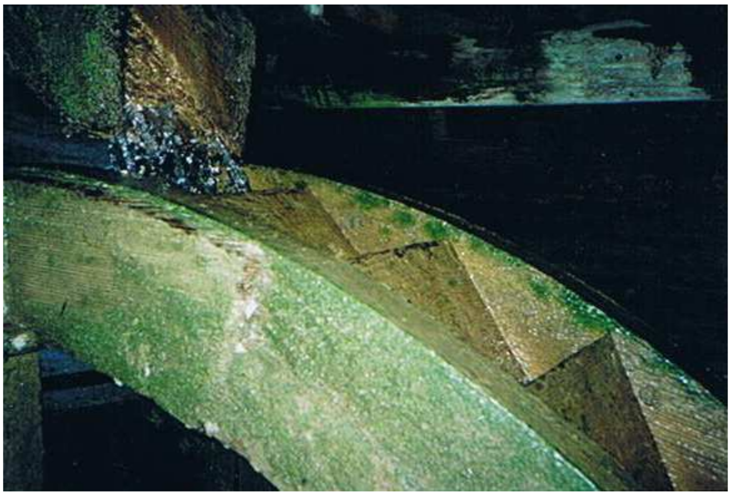
$$\int_1^7 \frac{\mathrm{d}x}{1 + \cos^3 x + \ln x}$$

Question 2

Engineering estimate.

The waterwheel shown in the following three pictures was observed to make one complete revolution every 15 seconds. Estimate the power output (in units of horsepower)?



Question 3

Career plans. (Scoring rubric was designed to assess potential for life-long learning.)

Where do you see yourself in 5-10 years, and what steps will you take to get there? How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)? Does your prospective employer sponsor master's degree or MBA? Under which terms?

Question 4

Professional ethics and ethical ramifications of technology.

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it?

Which courses at UIC have contributed most to your views on professional ethics? How?

Question 5

Safety

Name four lab safety rules that apply to the Unit Ops lab.

Which steps would you take to carry out a risk assessment protocol for one of the experiments in the Unit Ops lab? What information do you need? Where can you find it? What is a MSDS?

Which Unit Ops experiment is potentially the most dangerous? Why? Do an impromptu risk assessment for this experiment, and indicate how the risk can be mitigated.

Question 6

Working in multidisciplinary teams

Can you name two other types of engineers that you might find yourself working with?

In what context or application would you be collaborating with them?

Scoring Rubrics Mapped to Student Outcomes and Program Educational Objectives

GRADUATING SENIOR ______________________ **FACULTY EVALUATOR** ______________________

Date ______________________ Start time ______________ End time __________

Rubric pertaining to engineering fundamentals

#	**LEVEL DESCRIPTION**	**(1)** Unsatisfactory	**(2)** Satisfactory	**(3)** Exemplary	Score
1	***Question 1*** **SO: 1, 3** **PEO: 1** Description of numerical method and formulation of algorithm	Does not know how to proceed and/or unable to get started.	Describes most of numerical method but shows some gaps in understanding or is cannot formulate algorithm.	Describes numerical method correctly and successfully formulates algorithm.	
2	***Question 1*** **SO: 3** **PEO: 1** Proficiency with Excel or MATLAB	Unfamiliar with Excel, MATLAB (and any other math package). Unable to get started or needs extensive consultation of on-line manual to get started. Incomplete or incorrect solution.	Able to do significant portion or most of problem with some quick consultation of on-line help.	Fluent in finishing problem completely with little or no consultation of on-line help.	
3	***Question 2*** **SO: 1, 2, 6** **PEO: 1** Identification of system, surroundings and applicable flow streams	Does not mention system or surroundings or falters seriously in implementation.	Provides many details of system, surroundings and flow streams correctly	Complete and correct formulation of engineering system.	
4	***Question 2*** **SO: 1, 2, 6** **PEO: 1** Identification and application of mass, momentum and energy balances	Unable to identify applicable balance law(s) or applies incorrect balance.	Applies correct balance law, and progresses substantially toward mathematical formulation.	Correctly and completely formulates applicable balance law into mathematical formulation.	
5	***Question 2*** **SO: 1** **PEO: 1** Effective problem solving skills	Unable to translate physical ideas into mathematical form. Ineffective solution.	Formulates mathematical problem substantially and makes substantial progress toward correct solution	Correct mathematical formulation and completion of correct solution, including units	

GRADUATING SENIOR **FACULTY EVALUATOR**

Rubric pertaining to professionalism and broader knowledge

#	LEVEL DESCRIPTION	**(1)** Unsatisfactory	**(2)** Satisfactory	**(3)** Exemplary	Score
6	***Question 3*** **SO: 10** **PEO: 4** Background research on career options and plans	No significant research. Unfamiliar with application deadlines, interview procedures and employer policies.	Has checked some information on specific universities or companies on the internet. Actively seeking job interviews or knows relevant deadlines for graduate schools.	Can quote employer policies (e.g., career tracks, promotions, sponsorship of advanced degrees). Has submitted applications, has job interviews scheduled, or already accepted for job or graduate school.	
7	***Question 4*** **SO: 13** **PEO: 3** AIChE Code of Ethics	Unfamiliar with existence or content of code.	Knows where to find code and is familiar with some provisions.	Has read code more than once and can quote main provisions.	
8	***Question 4*** **SO: 8, 9, 13** **PEO: 3** Ethical and societal ramifications of technology	Unable to think of an example or cannot identify ethical ramifications of supplied technological topic.	Can independently come up with an example or identify an ethical ramification of supplied example.	Clearly explains ethical impact or social issues related to technological topic.	
9	***Question 4*** **SO: 13** **PEO: 3** Approach to ethical dilemmas	Unable to recognize ethical dilemma in hypothetical scenario. Purely *ad hoc* or “gut feel” approach toward ethical decisions. Does not know where or to whom to turn for guidance. Confounds fact-finding with ethical consideration.	Recognizes ethical dilemma in hypothetical scenario (or own experience). Logical considerations toward resolution. Can formulate steps to seek guidance.	Articulates resolution of dilemma as outcome of systematic thought process or ethical framework. Familiar with channels to air ethical concerns and/or cognizant officers.	

GRADUATING SENIOR **FACULTY EVALUATOR**

Rubric pertaining to professionalism and broader knowledge

#	LEVEL DESCRIPTION	**(1)** Unsatisfactory	**(2)** Satisfactory	**(3)** Exemplary	Score
10	***Question 5*** **SO: 7** **PEO: 3** Safety rules in the Unit Ops laboratory	Correctly names 1 or 2 safety rules.	Correctly names 3 safety rules.	Correctly names 4 or more safety rules.	
11	***Question 5*** **SO: 7** **PEO: 3** Familiarity with MSDS	Does not know what as MSDS or does not know where to find them.	Knows what an MSDS is and can obtain them as necessary.	Is familiar with at least several specific MSDS and can quote hazards associated with chemicals used in the Unit Ops lab.	
12	***Question 5*** **SO: 7** **PEO: 3** Risk assessment	Not familiar with risk assessment or its constituent steps.	Knows the 3 steps (hazard identification, assessment of risk of exposure, control of risk) and can relate one or two of these to some extent to the experiment in question.	Clearly explains all three steps of risk assessment in relation to the experiment and formulates a significant subset of appropriate procedures, precautions, controls, evacuation and first aid. Distinguishes between material hazards versus equipment and experimental-design hazards.	
13	***Question 6*** **SO: 11** **PEO: 3** Multidisciplinary teamwork	Cannot name any collaborating engineering fields or explain context of collaboration.	Names one other type of engineer and articulates some aspects of collaboration.	Names two other types of engineer and clearly articulates example or specific contexts in which his/her work could intersect.	
14	***Oral communication*** **SO: 12** **PEO: 3** Overall rating among items 1, 3, 4, 6, 8, 9, 10, 12, 13	Inarticulate or struggles to make him/herself understood. Unclear explanations require extensive probing by the interviewer.	Able to get across most technical and other ideas but interviewer must ask some follow-up questions to fully understand the student.	Student presents ideas clearly and concisely and his/her explanations are complete and easy to understand.	

GRADUATING SENIOR **FACULTY EVALUATOR**

Notes and comments for Question 1.
Numerical methods and computational proficiency.

______________________________ ______________________________
GRADUATING SENIOR **FACULTY EVALUATOR**

Notes and comments for Question 2.
Engineering estimate.

______________________________ ______________________________

GRADUATING SENIOR **FACULTY EVALUATOR**

Notes and comments for Question 3
Career plans.

GRADUATING SENIOR **FACULTY EVALUATOR**

Notes and comments for Question 4
Professional ethics and ethical ramifications of technology.

GRADUATING SENIOR **FACULTY EVALUATOR**

Notes and comments for Question 5
Laboratory and process safety.

______________________________ ______________________________
GRADUATING SENIOR **FACULTY EVALUATOR**

Notes and comments for Question 6
Multidisciplinary teamwork.

______________________________ ______________________________
GRADUATING SENIOR **FACULTY EVALUATOR**

Feedback on department, curriculum and/or program.
Positives.

GRADUATING SENIOR **FACULTY EVALUATOR**

Feedback on department, curriculum and/or program.
Negatives – suggestions for improvement.

BSChE Graduate Exit Interview and Oral Examination – Spring 2012
30-minute oral discussions conducted with ChE graduates by Professor L. C. Nitsche or Randall J. Meyer

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#1.) [<u>PEO (i) Depth</u>]
Maple (mathematics package) proficiency. Consider the function

$$f(x) \quad = \quad \frac{1}{4x^2 - 8x + 25}$$

and use Maple or Matlab to carry out the following mathematical tasks.

(a) Plot $f(x)$ for $0 \le x \le 1$.

(b) Find a formula for $\int_0^1 f(x)\,\mathrm{d}x$.

(c) Find the 6^{th} order term in the Taylor expansion of $f(x)$.

#5.) [<u>PEO (iv) Life-Long Learning</u>]
Career plans. Scoring rubric was designed to assess potential for life-long learning.

Where do you see yourself in 5-10 years, and what steps will you take to get there? How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)? Does your prospective employer sponsor master's degree or MBA? Under which terms?

#6-8.) [<u>PEO (ii) Breadth, (iii) Professionalism</u>]
Professional ethics and ethical ramifications of technology.

Can you name a recent significant item in current events or a piece of legislation impacting on technology or engineering, and comment on its ethical ramifications?

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

Are you familiar with the AIChE Code of Ethics?

Which courses at UIC have contributed most to your views on professional ethics? How?

#2-4.) [PEO (i) Depth]
Engineering estimates.

The *Warner Lawrence* is an innovative fireboat owned and operated by the Los Angeles Fire Department (LAFD) in Los Angeles. Designed by Robert Allan in the early 2000s, the Warner Lawrence was built in Washington and delivered to San Pedro on May 21, 2003. It was dedicated on April 12 of that year. The boat is driven primarily by pilot James Horimoto, a 30-year veteran of the LAFD. It was built by Nichols Boats of Freeland, Washington according to the LAFD. **Abilities/Description.** The *Warner Lawrence* is one of the most technologically advanced fireboats in the world. It is an omni-directional vessel driven by large propulsion devices similar to huge egg beaters. The Warner Lawrence has the capability to pump up to 38,000 US gallons per minute (2 m^3/s) up to 400 feet (120 m) in the air.

1.) What is the exit velocity of the nozzle(s)? Do you need to know the cross sectional area of the nozzle to answer this question? How far from shore (feet) can the Warner Lawrence put out a fire?

2.) What is the thrust (lb_f or N) produced by the 38,000 US gallons per minute?

3.) What is the minimum horsepower requirement for the pumping system?

NAME OF GRADUATING SENIOR ________________________________

NAME OF FACULTY EVALUATOR ________________________________

BSChE Graduate Exit Interview and Oral Exam – Spring 2012

Rubric for Program Educational Objectives and Program Outcomes

TECHNICAL PEO's & PO's

#	LEVEL PEO	**(1)** Unsatisfactory	**(2)** Satisfactory	**(3)** Exemplary	Score
1	**PEO (i) Depth** Maple / Matlab proficiency **PO: 3**	Unfamiliar with Maple (or any other math package). Unable to get started or needs extensive consultation of on-line manual to get started. Incomplete or incorrect solution.	Able to do significant portion or most of problem with some quick consultation of on-line help.	Fluent in finishing problem completely without consulting on-line help.	
2	**PEO (i) Depth** Identification of system, surroundings and applicable flow streams **PO: 1, 2, 6, 13**	Does not mention system or surroundings or falters seriously in implementation	Provides many details of system, surroundings and flow streams correctly	Complete and correct formulation of engineering system	
3	**PEO (i) Depth** Identification and application of mass, momentum and energy balances **PO: 1, 2, 6, 13**	Unable to identify applicable balance law(s) or applies incorrect balance	Applies correct balance law, and progresses substantially toward mathematical formulation	Correctly and completely formulates applicable balance law into mathematical formulation	
4	**PEO (i) Depth** Effective problem solving skills **PO: 1, 13**	Unable to translate physical ideas into mathematical form. Ineffective solution.	Formulates mathematical problem substantially and makes substantial progress toward correct solution	Correct mathematical formulation and completion of correct solution, including units	

NAME OF GRADUATING SENIOR ________________________________

NAME OF FACULTY EVALUATOR ________________________________

BSChE Graduate Exit Interview and Oral Exam – Spring 20112

Rubric for Program Educational Objectives and Program Outcomes

NON_TECHNICAL PEO's & PO's

#	LEVEL PEO	**(1)** Unsatisfactory	**(2)** Satisfactory	**(3)** Exemplary	Score
5	**PEO (iv) Life-long learning** Background research on career options and plans **PO: 10**	No significant research. Unfamiliar with application deadlines, interview procedures and employer policies.	Has checked some information on specific universities or companies on the internet. Actively seeking job interviews or knows relevant deadlines for graduate schools.	Can quote employer policies (e.g., career tracks, promotions, sponsorship of advanced degrees). Has submitted applications, has job interviews scheduled, or already accepted for job or graduate school.	
6	**PEO (iii) Professionalism** AIChE Code of Ethics **PO: 14**	Unfamiliar with existence or content of code.	Knows where to find code and is familiar with some provisions.	Has read code more than once and can quote main provisions.	
7	**PEO (ii) Breadth** **PEO (iii) Professionalism** Ethical and societal ramifications of technology **PO: 9, 10, 13, 14**	Unable to think of an example or cannot identify ethical ramifications of supplied technological topic.	Can independently come up with an example or identify an ethical ramification of supplied example.	Clearly explains ethical impact or social issues related to technological topic.	
8	**PEO (iii) Professionalism** Approach to ethical dilemmas **PO: 13, 14**	Unable to recognize ethical dilemma in hypothetical scenario. Purely *ad hoc* or "gut feel" approach toward ethical decisions. Does not know where or to whom to turn for guidance. Confounds fact-finding with ethical consideration.	Recognizes ethical dilemma in hypothetical scenario (or own experience). Logical considerations toward resolution. Can formulate steps to seek guidance.	Articulates resolution of dilemma as outcome of systematic thought process or ethical framework. Familiar with channels to air ethical concerns and/or cognizant officers.	

NAME OF GRADUATING SENIOR ____________________________________

NAME OF FACULTY EVALUATOR ____________________________________

Notes and Comments

Maple / Matlab

Technical question

NAME OF GRADUATING SENIOR ______________________________

NAME OF FACULTY EVALUATOR ______________________________

Notes and Comments

Ethics

Job search, graduate school, PE and/or career research and preparation

NAME OF GRADUATING SENIOR ___________________________________

NAME OF FACULTY EVALUATOR ___________________________________

Feedback on department, curriculum and/or program

Positives

Negatives

Exhibit 4-3.

Assessment Method #3 (Direct / Summative):
Judging rubrics from the annual Engineering Expo.
Sample Date: Spring 2013.
Sample Description: Assessment rubric for Senior Design (CHE 397) project.

UIC Engineering Expo 2013 **SUPPLEMENTARY PROJECT ASSESSMENT RUBRIC**

Judge's Name______________________________ ____
Print Signature

Check only one: ____UIC Judge ____UIC Alumni Judge ____Corporate Judge ____Design Mentor

Project Number:__________ Category:__________ Department: **Chemical Engineering**

Project Title: ____

____

SCORE CATEGORY	(1) Unsatisfactory	(2) Developing	(3) Satisfactory	(4) Exemplary	SCORE
(A.) Completeness of bibliography and sufficiency of citations.	Few or no references provided. Little evidence of solid background research.	Minimal bibliography that is not extensively referred to in the poster presentation. Some evidence of background research.	Sufficient bibliography and support of some points of presentation by specific literature citations.	Extensive bibliography with specific citations to support most main items of background, methodology and conclusions.	
(B.) Vetting of technical sources.	Mostly unedited web sources (e.g., Wikipedia) and unpublished material	Heavy reliance to general news media, magazines, and non-technical or semi-technical literature.	Some citations of handbooks, standard references, refereed journal articles and/or patents.	Extensive citations of handbooks, standard references, refereed journals and/or patents.	
(C.) Effectiveness of visual aides and graphic content.	Missing, poorly prepared, and/or confusing graphics. Insufficient detail or distracting ornamentation	Graphics are reasonably clear and help somewhat to understand the project.	Clearly presented graphics help substantially to understand the project.	Sophisticated and creative graphics (diagrams, graphs) provide conceptual organization. Miniature (solid) prototype or model.	
(D.) Attention to process safety.	Little or no consideration	Specific mention confined to qualitative aspects	Discussion of process safety and evidence of some impact on process design.	Thorough discussion of process safety and inclusion of relevant aspects in process flow diagram	
(E.) Attention to environmental concerns.	Little or no consideration	Specific mention confined to qualitative aspects	Discussion of environmental aspects and evidence of some impact on process design.	Thorough discussion of environmental aspects and inclusion of relevant aspects in process flow diagram	
(F.) Teamwork.	Oral presentation made mainly by one team member. Indications of uneven distribution of effort on project.	Oral presentation made by mainly by two team members. Can discern (possibly by asking directly) contributions of at least three members on project.	Oral presentation made by at least three team members. Discernible contributions of all members on completed project.	Oral presentation shared equally among team members. Documentation of equitable work distribution on project.	
(G.) Effectiveness of oral presentation.	Inarticulate, confusing or disorganized presentation. Inability to answer technically substantive questions.	Understandable presentation with some significant points presented out of order or as afterthoughts.	Team shows substantial mastery of the material and presents the project mostly in logical order. Most questions are answered sufficiently.	Team shows mastery of the material. Technical details are clearly explained in logical order. Questions are answered competently and concisely.	

The feedback you include below will be shared **confidentially** with teams and departmental project coordinators (faculty):

Project's strong points: ____

Areas for improvement: ____

Exhibit 4-4.

Assessment Method #4 (Indirect / Summative):
Senior Exit Survey.
Sample Date: Spring 2014.
Sample Description: PDF printout of online survey (PHP/CSS file).

UIC Department of Chemical Engineering
Anonymous Senior Exit Survey
Spring 2014

(1.) Based upon your experiences as a Chemical Engineering student at UIC, please rate your skills in the following categories, using the following criteria:

	Unsatisfactory	Minimal	Adequate	Excellent
(SO 1) Ability to apply knowledge of mathematics, science and engineering	○	○	○	○
(SO 2) Ability to identify, formulate and solve engineering problems	○	○	○	○
(SO 3) Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	○	○	○	○
(SO 4) Ability to design experiments	○	○	○	○
(SO 5) Ability to conduct experiments	○	○	○	○
(SO 6) Ability to analyze and interpret data	○	○	○	○
(SO 7) Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	○	○	○	○
(SO 8) Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	○	○	○	○
(SO 9) Knowledge of contemporary issues	○	○	○	○
(SO 10) Recognition of the need for and ability to engage in life-long learning	○	○	○	○
(SO 11) Ability to function on multidisciplinary teams	○	○	○	○
(SO 12) Ability to communicate effectively	○	○	○	○
(SO 13) Understanding of professional and ethical responsibilities	○	○	○	○
Overall, do you feel that your undergraduate degree will prepare you for your career?	○	○	○	○

(2.) Which courses or topics were the most interesting or useful?

(3.) Which courses or topics were the least interesting or useful?

(4.) What courses/topics should we consider deleting or adding to the curriculum?

(5.) Do you have any additional comments or suggestions?

(6.) How many years/months did it take for you to complete your B.S. degree in chemical engineering?

(7.) Did you work part-time while you were a student? If yes, then approximately how many hours per week?

(8.) On average, how much many hours did you spend on each ChE course outside of class?

(9.) Did you participate in any summer research/internship program at UIC or at other schools? Did you complete an internship at a company? What company? Was this a useful experience? What specifically did you learn that was most useful?

(10.) Have you found a job? If so, in which company and industry? Please indicate your job title or specialization.

(11.) Are you planning to attend a graduate or a professional school? If so please list the name of the school and the specific program (M.S./Ph.D/M.D.) that you will enroll in, or those programs that you have looked into.

(12.) Do you have a strategy in mind for continued professional development and/or continue learning? For example, do you intend to take the FE/PE exam at some point in your career? If so, when?

Submit Survey

Exhibit 4-5.

Assessment Method #5 (Feedback):
Course Update Survey.
Sample Date: Spring 2014.
Sample Description: PDF printout of online survey (PHP/CSS file).

UIC Department of Chemical Engineering
Anonymous Student Feedback Survey for Updating and Improving Courses
Spring 2014

ChE Course

201	205	210	312	313	321	341	382	397	494-Energy	494-Electrochem	51
○	○	○	○	○	○	○	○	○	○	○	○

(1.) How many courses did you take this term?

1	2	3	4	5	6
○	○	○	○	○	○

(2.) Approximately how many hours per week did you devote to **all of your courses** outside of lectures?

1-5	6-10	11-15	16-20	31-30	31-40
○	○	○	○	○	○

(3.) Approximately how many hours per week did you devote to **this course** outside of lectures?

1-3	4-6	7-9	10-12	13-15	15-20
○	○	○	○	○	○

(4.) Please rate usage, relevance and/or applicability of the following subjects to this course.

	None	Minor	Significant	Extensive
Calculus and vector calculus	○	○	○	○
Differential equations	○	○	○	○
Chemistry	○	○	○	○
Excel spreadsheets	○	○	○	○
Math packages (Maple / Matlab / Mathcad)	○	○	○	○
Computer programming (Fortran / C++)	○	○	○	○
Aspen Plus	○	○	○	○
Social/societal and geopolitical issues	○	○	○	○
Environmental, regulatory, health and safety issues	○	○	○	○
Energy and natural resource issues	○	○	○	○
Professional ethics and responsibilities	○	○	○	○
Analysis and interpretation of data	○	○	○	○
Design of system, component or process to meet desired needs	○	○	○	○
Application of engineering principles to solve engineering problems	○	○	○	○
Formulation of engineering problems	○	○	○	○

(6.) Please comment on the balance of coverage between theoretical derivations versus practical problems and examples.

(7.) Were there any problematic or unfamiliar mathematical concepts utilized in this course that would have benefited from preparation or exposure in previous classes?

(8.) Were there any computational methods utilized in this course that would have benefited from preparation or exposure in previous classes?

(9.) How useful was the class text? Would you prefer an alternative textbook (suggestions and comments)?

(10.) How well did the class follow the text? Can you make any suggestions in this regard?

(11.) Was the pace of lectures appropriate? Too fast? Too slow?

(12.) Was the difficulty and quantity of homework appropriate to the material?

(13.) Comment on the difficulty and applicability of exams to the course material.

(14.) Please list three most beneficial aspects of this course?

(15.) Please list three aspects of this course that you would like to see improved? Offer specific suggestions if you can.

(16.) Would you like to see any particular material added to the course? Was any specific topic less useful than the others?

(17.) Additional comments.

Did this course have a teaching assistant?

Yes	No
○	○

If yes, please give his/her name.

If yes, rate the teaching assistant with regard to the following aspects.

	Excellent	Very Good	Good	Fair	Poor
Approachability	○	○	○	○	○
Helpfulness in office hours and/or with questions in general	○	○	○	○	○
Knowledge of and background in the course material	○	○	○	○	○
Grading of homeworks (helpful comments on papers, fairness) - if applicable	○	○	○	○	○
Grading of exams and/or quizzes (helpful comments on papers, fairness) - if applicable	○	○	○	○	○
Overall effectiveness	○	○	○	○	○

Submit Course Feedback Survey

Exhibit 4-6.

Level 2 analysis of assessment data.

Attainment of Student Outcome #1
Ability to apply knowledge of mathematics, science and engineering

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	**2014**	**2013**	**2012**	**2011**	**2010**	
% (4 = Exemplary)	60.04	28.51	33.12	32.90	36.36	
% (3 = Satisfactory)	24.39	45.26	38.84	50.97	26.70	
% (2 = Developing)	12.25	19.63	19.88	11.61	23.30	
% (1 = Unsatisfactory)	3.31	6.60	8.16	4.51	13.64	
% ≥ Satisfactory	84.44	73.77	71.96	83.87	63.07	
Average Rating	3.41	2.96	2.97	3.12	2.86	

Assessment Method #2 (Direct / Summative): Exit Interview

Year	**2014**	**2013**	**2012**	**2011**	**2010**	
% (4 = Exemplary)	25.9	6.3	9.8	15.9	26.3	
% (3 = Satisfactory)	36.6	26.6	24.2	33.1	26.3	
% (2 = Developing)	37.5	67.1	65.9	51.0	47.5	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	0.0	
% ≥ Satisfactory	62.5	32.9	34.0	49.0	52.6	
Average Rating	2.88	2.39	2.44	2.65	2.79	

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year						
% (4 = Exemplary)						
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% ≥ Satisfactory						
Average Rating						

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	**2014**	**2013**	**2012**	**2011**	**2010**	
% (4 = Exemplary)	54.5	35.5	29.6	50.0	22.2	
% (3 = Satisfactory)	43.2	45.2	51.9	33.3	70.4	
% (2 = Developing)	2.3	19.4	18.5	16.7	3.7	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	3.7	
% ≥ Satisfactory	97.7	80.6	81.5	83.3	92.6	
Average Rating	3.52	3.16	3.11	3.33	3.11	

Attainment of Student Outcome #2
Ability to identify, formulate and solve engineering problems

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	58.61	31.64	35.48	40.00	35.61	
% (3 = Satisfactory)	25.83	45.01	35.77	43.66	27.27	
% (2 = Developing)	12.25	17.26	19.88	12.47	23.48	
% (1 = Unsatisfactory)	3.31	6.09	8.87	3.87	13.64	
% ≥ Satisfactory	84.44	76.65	71.25	83.66	62.88	
Average Rating	3.40	3.02	2.98	3.20	2.85	

Assessment Method #2 (Direct / Summative): Exit Interview

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	30.4	7.0	9.1	13.3	15.0	
% (3 = Satisfactory)	37.5	31.0	22.7	29.3	27.5	
% (2 = Developing)	32.1	62.0	68.2	57.3	57.5	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	0.0	
% ≥ Satisfactory	67.9	38.0	31.8	42.6	42.5	
Average Rating	2.98	2.45	2.41	2.56	2.58	

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year						
% (4 = Exemplary)						
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% ≥ Satisfactory						
Average Rating						

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	38.6	33.3	22.2	16.7	33.3	
% (3 = Satisfactory)	59.1	43.3	48.1	58.3	44.4	
% (2 = Developing)	2.3	23.3	29.6	25.0	14.8	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	7.4	
% ≥ Satisfactory	97.7	76.7	70.4	75.0	77.8	
Average Rating	3.36	3.10	2.93	2.92	3.04	

Attainment of Student Outcome #3
Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	60.93	32.28	35.75	37.74	35.99	
% (3 = Satisfactory)	23.51	41.24	36.84	45.48	26.89	
% (2 = Developing)	12.25	19.78	19.35	12.26	23.48	
% (1 = Unsatisfactory)	3.31	6.70	7.91	4.51	13.64	
% ≥ Satisfactory	84.44	73.52	72.59	83.22	62.88	
Average Rating	3.42	2.99	3.00	3.16	2.85	

Assessment Method #2 (Direct / Summative): Exit Interview

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	23.2	4.2	3.0	15.8	30.0	
% (3 = Satisfactory)	34.8	12.5	6.1	50.0	20.0	
% (2 = Developing)	42.0	83.3	90.9	34.2	50.0	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	0.0	
% ≥ Satisfactory	58.0	16.7	9.1	65.8	50.0	
Average Rating	2.81	2.21	1.42	2.82	2.80	

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year						
% (4 = Exemplary)						
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% ≥ Satisfactory						
Average Rating						

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	34.1	25.8	22.2	33.3	33.3	
% (3 = Satisfactory)	63.6	38.7	51.9	33.3	37.0	
% (2 = Developing)	2.3	32.3	22.2	33.3	25.9	
% (1 = Unsatisfactory)	0.0	3.2	3.7	0.0	3.7	
% ≥ Satisfactory	97.7	64.5	74.1	66.7	70.4	
Average Rating	3.32	2.87	2.93	3.00	3.00	

Attainment of Student Outcome #4
Ability to design experiments

Assessment Method #1 (Direct / Formative): Course Rubrics

Year		2014	2013	2012	2011	2010
% (4 = Exemplary)		84.0	100.0	100.0	92.5	
% (3 = Satisfactory)		16.0	0.0	0.0	7.5	
% (2 = Developing)		0.0	0.0	0.0	0.0	
% (1 = Unsatisfactory)		0.0	0.0	0.0	0.0	
% ≥ Satisfactory		100.0	100.0	100.0	100.0	
Average Rating		3.93	4.00	4.00	3.93	

Assessment Method #2 (Direct / Summative): Exit Interview

Year		2014	2013	2012	2011	2010
% (4 = Exemplary)						
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% ≥ Satisfactory						
Average Rating						

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year						
% (4 = Exemplary)						
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% ≥ Satisfactory						
Average Rating						

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	45.5					
% (3 = Satisfactory)	38.6					
% (2 = Developing)	13.6					
% (1 = Unsatisfactory)	2.3					
% ≥ Satisfactory	84.1					
Average Rating	3.27					

Attainment of Student Outcome #5
Ability to conduct experiments

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2014	2013	2012	2011		
% (4 = Exemplary)	84.0	100.0	100.0	92.5		
% (3 = Satisfactory)	16.0	0.0	0.0	7.5		
% (2 = Developing)	0.0	0.0	0.0	0.0		
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0		
% ≥ Satisfactory	100.0	100.0	100.0	100.0		
Average Rating	3.93	4.00	4.00	3.93		

Assessment Method #2 (Direct / Summative): Exit Interview

Year						
% (4 = Exemplary)						
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% ≥ Satisfactory						
Average Rating						

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year						
% (4 = Exemplary)						
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% ≥ Satisfactory						
Average Rating						

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	56.8					
% (3 = Satisfactory)	38.6					
% (2 = Developing)	2.3					
% (1 = Unsatisfactory)	2.3					
% ≥ Satisfactory	95.5					
Average Rating	3.50					

Attainment of Student Outcome #6
Ability to analyze and interpret data

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2014	2013	2012	2011		
% (4 = Exemplary)	74.0	44.5	56.0	44.0		
% (3 = Satisfactory)	26.0	43.5	39.5	51.5		
% (2 = Developing)	0.0	10.0	4.5	3.0		
% (1 = Unsatisfactory)	0.0	1.5	0.0	1.5		
% ≥ Satisfactory	100.0	88.0	95.5	95.5		
Average Rating	3.74	3.30	3.52	3.38		

Assessment Method #2 (Direct / Summative): Exit Interview

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	31.3	7.0	9.1	13.3	15.0	
% (3 = Satisfactory)	35.7	31.0	22.7	29.3	27.5	
% (2 = Developing)	33.0	62.0	68.2	57.3	57.5	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	0.0	
% ≥ Satisfactory	67.0	38.0	31.8	42.6	42.5	
Average Rating	2.98	2.45	2.41	2.56	2.58	

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year						
% (4 = Exemplary)						
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% ≥ Satisfactory						
Average Rating						

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	61.4	41.9	29.6	41.7	29.6	
% (3 = Satisfactory)	36.4	45.2	55.6	33.3	59.3	
% (2 = Developing)	2.3	12.9	14.8	25.0	7.4	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	3.7	
% ≥ Satisfactory	97.7	87.1	85.2	75.0	88.9	
Average Rating	3.59	3.29	3.15	3.17	3.15	

Attainment of Student Outcome #7
Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability

Assessment Method #1 (Direct / Formative): Course Rubrics

Year		**2013**			**2010**	
% (4 = Exemplary)		21.31			30.08	
% (3 = Satisfactory)		71.31			36.59	
% (2 = Developing)		5.73			19.52	
% (1 = Unsatisfactory)		1.64			14.63	
% ≥ Satisfactory		92.62			66.67	
Average Rating		3.12			2.84	

Assessment Method #2 (Direct / Summative): Exit Interview

Year	**2014**	**2013**				
% (4 = Exemplary)	24.1	38.5				
% (3 = Satisfactory)	43.8	51.0				
% (2 = Developing)	32.1	10.4				
% (1 = Unsatisfactory)	0.0	0.0				
% ≥ Satisfactory	67.9	89.5				
Average Rating	2.92	3.28				

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year		**2013**	**2012**		**2010**	**2009**
% (4 = Exemplary)		20.6	22.9		23.3	35.7
% (3 = Satisfactory)		45.3	52.1		26.7	39.3
% (2 = Developing)		16.6	20.8		36.7	25.0
% (1 = Unsatisfactory)		17.6	4.2		13.3	0.0
% ≥ Satisfactory		65.8	75.0		50.0	75.0
Average Rating		2.46	2.81		2.54	3.25

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	**2014**	**2013**	**2012**	**2011**	**2010**	
% (4 = Exemplary)	44.2	22.6	19.2	25.0	29.6	
% (3 = Satisfactory)	44.2	45.2	57.7	16.7	48.1	
% (2 = Developing)	11.6	29.0	23.1	58.3	18.5	
% (1 = Unsatisfactory)	0.0	3.2	0.0	0.0	3.7	
% ≥ Satisfactory	88.4	67.6	76.9	41.7	77.8	
Average Rating	3.33	2.87	2.96	2.67	3.04	

Attainment of Student Outcome #8
Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)						
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% ≥ Satisfactory						
Average Rating						

Assessment Method #2 (Direct / Summative): Exit Interview

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	47.3	11.4	34.4	14.3	31.6	
% (3 = Satisfactory)	34.8	51.4	50.0	51.4	47.4	
% (2 = Developing)	17.9	10.4	15.6	34.3	21.1	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	0.0	
% ≥ Satisfactory	82.1	62.8	84.4	65.7	79.0	
Average Rating	3.29	2.74	3.19	2.80	3.11	

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year		2013	2012		2010	2009
% (4 = Exemplary)		31.4	25.0		33.3	50.0
% (3 = Satisfactory)		48.6	66.7		33.3	35.7
% (2 = Developing)		17.1	8.3		33.3	14.3
=% (1 = Unsatisfactory)		2.9	0.0		0.0	0.0
% ≥ Satisfactory		80.0	91.7		66.7	85.7
Average Rating		2.93	3.06		2.90	3.50

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	40.9	25.8	25.9	33.3	29.6	
% (3 = Satisfactory)	54.5	41.9	51.9	33.3	51.9	
% (2 = Developing)	4.5	19.4	22.2	33.3	11.1	
% (1 = Unsatisfactory)	0.0	12.9	0.0	0.0	7.4	
% ≥ Satisfactory	95.5	67.7	77.8	66.7	81.5	
Average Rating	3.36	2.81	3.04	3.00	3.04	

Attainment of Student Outcome #9
Knowledge of contemporary issues

Assessment Method #1 (Direct / Formative): Course Rubrics

Year					2010	
% (4 = Exemplary)					0.00	
% (3 = Satisfactory)					28.57	
% (2 = Developing)					42.86	
% (1 = Unsatisfactory)					28.57	
% ≥ Satisfactory					28.57	
Average Rating					2.00	

Assessment Method #2 (Direct / Summative): Exit Interview

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	39.3	11.4	34.4	14.3	31.6	
% (3 = Satisfactory)	37.5	51.4	50.0	51.4	47.4	
% (2 = Developing)	23.2	37.1	15.6	34.3	21.1	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	0.0	
% ≥ Satisfactory	76.8	62.8	84.4	65.7	79.0	
Average Rating	3.16	2.74	3.19	2.80	3.11	

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year		2013	2012		2010	2009
% (4 = Exemplary)		31.4	25.0		33.3	50.0
% (3 = Satisfactory)		48.6	66.7		33.3	35.7
% (2 = Developing)		17.1	8.3		33,3	14.3
% (1 = Unsatisfactory)		2.9	0.0		0.0	0.0
% ≥ Satisfactory		80.0	91.7		66.7	85.7
Average Rating		2.93	3.06		2.90	3.50

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	25.0	22.6	14.8	41.7	22.2	
% (3 = Satisfactory)	65.9	32.3	51.9	8.3	40.7	
% (2 = Developing)	9.1	35.5	22.2	50.0	33.3	
% (1 = Unsatisfactory)	0.0	9.7	11.1	0.0	3.7	
% ≥ Satisfactory	90.9	54.8	66.7	50.0	63.0	
Average Rating	3.16	2.68	2.70	2.92	2.81	

Attainment of Student Outcome #10
Recognition of the need for and ability to engage in life-long learning

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	60.04	28.51	33.12	32.90	36.36	
% (3 = Satisfactory)	24.39	45.26	38.84	50.97	26.70	
% (2 = Developing)	12.25	19.63	19.88	11.61	23.30	
% (1 = Unsatisfactory)	3.31	6.60	8.16	4.51	13.64	
% ≥ Satisfactory	84.44	73.77	71.96	83.87	63.07	
Average Rating	3.41	2.96	2.97	3.12	2.86	

Assessment Method #2 (Direct / Summative): Exit Interview

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	54.5	38.9	31.0	17.1	28.6	
% (3 = Satisfactory)	34.8	38.9	55.2	45.7	64.3	
% (2 = Developing)	10.7	22.2	13.8	37.1	7.1	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	0.0	
% ≥ Satisfactory	89.3	77.8	86.2	62.8	92.9	
Average Rating	3.44	3.17	3.17	2.80	3.22	

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year		2013	2012	2011	2010	2009
% (4 = Exemplary)		1.6	8.4		8.3	64.3
% (3 = Satisfactory)		55.6	51.0		54.2	35.7
% (2 = Developing)		28.1	38.4		37.5	0.0
% (1 = Unsatisfactory)		14.7	2.2		0.0	0.0
% ≥ Satisfactory		57.2	59.4		62.5	100.0
Average Rating		2.08	2.46		2.38	3.75

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	61.4	45.2	37.0	50.0	53.8	
% (3 = Satisfactory)	38.6	29.0	44.4	16.7	30.8	
% (2 = Developing)	0.0	19.4	11.1	33.3	11.5	
% (1 = Unsatisfactory)	0.0	6.5	7.4	0.0	3.8	
% ≥ Satisfactory	100.0	74.2	81.5	66.7	84.6	
Average Rating	3.61	3.13	3.11	3.17	3.35	

Attainment of Student Outcome #11
Ability to function on multidisciplinary teams

Assessment Method #1 (Direct / Formative): Course Rubrics

Year						
% (4 = Exemplary)						
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% ≥ Satisfactory						
Average Rating						

Assessment Method #2 (Direct / Summative): Exit Interview

Year	2014					
% (4 = Exemplary)	46.4					
% (3 = Satisfactory)	45.5					
% (2 = Developing)	8.0					
% (1 = Unsatisfactory)	0.0					
% ≥ Satisfactory	91.9					
Average Rating	3.38					

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year		2013	2012		2010	2009
% (4 = Exemplary)		44.4	58.3		7.1	57.1
% (3 = Satisfactory)		30.6	33.3		57.1	42.9
% (2 = Developing)		25.0	8.3		35.7	0.0
% (1 = Unsatisfactory)		0.0	0.0		0.0	0.0
% ≥ Satisfactory		75.0	91.7		64.3	100.0
Average Rating		3.19	3.57		2.73	3.00

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	59.1	41.9	46.2	41.7	42.3	
% (3 = Satisfactory)	38.6	29.0	38.5	33.3	42.3	
% (2 = Developing)	2.3	22.6	11.5	25.0	11.5	
% (1 = Unsatisfactory)	0.0	6.5	3.8	0.0	3.8	
% ≥ Satisfactory	97.7	71.0	84.6	75.0	84.6	
Average Rating	3.57	3.06	3.27	3.17	3.23	

Attainment of Student Outcome #12
Ability to communicate effectively

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	57.21	14.42	43.64	37.44	71.43	
% (3 = Satisfactory)	42.79	73.16	56.36	62.56	19.05	
% (2 = Developing)	0.00	11.58	0.00	0.00	9.52	
% (1 = Unsatisfactory)	0.00	0.83	0.00	0.00	0.00	
% ≥ Satisfactory	100.00	87.58	100.00	100.00	90.48	
Average Rating	3.57	3.01	3.44	3.37	3.62	

Assessment Method #2 (Direct / Summative): Exit Interview

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	57.1	10.9	19.6	15.8	24.5	
% (3 = Satisfactory)	38.4	36.2	35.6	35.0	37.8	
% (2 = Developing)	4.5	52.9	44.8	49.2	37.8	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	0.0	
% ≥ Satisfactory	95.5	47.1	55.2	50.8	62.3	
Average Rating	3.53	2.58	2.75	2.67	2.87	

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year		2013	2012		2010	2009
% (4 = Exemplary)		7.6	21.9		11.3	65.0
% (3 = Satisfactory)		59.3	51.5		60.9	35.0
% (2 = Developing)		23.4	24.4		27.8	0.0
% (1 = Unsatisfactory)		9.8	2.1		0.0	0.0
% ≥ Satisfactory		66.8	73.5		72.2	100.0
Average Rating		2.61	2.77		2.72	3.75

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	59.1	29.0	44.4	50.0	51.9	
% (3 = Satisfactory)	38.6	38.7	40.7	8.3	40.7	
% (2 = Developing)	2.3	25.8	14.8	41.7	3.7	
% (1 = Unsatisfactory)	0.0	6.5	0.0	0.0	3.7	
% ≥ Satisfactory	97.7	67.7	85.2	58.3	92.6	
Average Rating	3.57	2.90	3.30	3.08	3.41	

Attainment of Student Outcome #13
Understanding of professional and ethical responsibilities

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	49.01	55.06	40.69	37.81	38.27	
% (3 = Satisfactory)	31.13	36.44	33.56	50.97	23.46	
% (2 = Developing)	16.56	6.21	17.59	7.36	24.69	
% (1 = Unsatisfactory)	3.31	2.30	8.15	3.87	13.58	
% ≥ Satisfactory	80.14	91.50	74.25	88.77	61.73	
Average Rating	3.26	3.44	3.07	3.23	2.86	

Assessment Method #2 (Direct / Summative): Exit Interview

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	56.0	26.3	20.8	12.1	24.1	
% (3 = Satisfactory)	38.1	45.5	41.7	41.1	58.6	
% (2 = Developing)	6.0	28.3	37.5	46.7	17.2	
% (1 = Unsatisfactory)	0.0	0.0	0.0	0.0	0.0	
% ≥ Satisfactory	94.1	71.8	62.5	53.2	82.7	
Average Rating	3.50	2.98	2.83	2.65	3.07	

Assessment Method #3 (Direct / Summative): Expo Rubrics

Year		2013	2012	2011	2010	2009
% (4 = Exemplary)		20.6	22.9		23.3	35.7
% (3 = Satisfactory)		45.3	52.1		26.7	39.3
% (2 = Developing)		16.6	20.8		36.7	25.0
% (1 = Unsatisfactory)		17.6	4.2		13.3	0.0
% ≥ Satisfactory		65.8	75.0		50.0	75.0
Average Rating		2.46	2.81		2.54	3.25

Assessment Method #4 (Indirect / Summative): Exit Survey

Year	2014	2013	2012	2011	2010	
% (4 = Exemplary)	58.1	36.7	29.6	41.7	44.4	
% (3 = Satisfactory)	41.9	26/7	48.1	33.3	37.0	
% (2 = Developing)	0.0	30.0	18.5	25.0	14.8	
% (1 = Unsatisfactory)	0.0	6.7	3.7	0.0	3.7	
% ≥ Satisfactory	100.0	63.3	77.8	75.0	81.5	
Average Rating	3.58	2.93	3.04	3.17	3.22	

Exhibit 4-7.

Level 3 analysis of assessment data.

Direct Assessment Method #1: Direct / Formative
Course Rubrics

4 = E = Exemplary
3 = S = Satisfactory
2 = D = Developing
1 = U = Unsatsifactory

Percentage of students attaining E or S

SO	2014	2013	2012	2011	2010	2009
1	84.4	73.8	72.0	83.9	63.1	
2	84.4	76.6	71.2	83.7	62.9	
3	84.4	73.5	72.6	83.2	62.9	
4	100.0	100.0	100.0	100.0		
5	100.0	100.0	100.0	100.0		
6	100.0	88.0	95.5	95.5		
7		92.3			66.7	
8						
9						
10	84.4	73.8	72.0	83.9	63.1	
11						
12	100.0	87.6	100.0	100.0	90.48	
13	80.14	91.5	74.2	88.8	61.7	

Average rating among students

SO	2014	2013	2012	2011	2010	2009
1	3.41	2.96	2.97	3.12	2.86	
2	3.40	3.02	2.98	3.20	2.85	
3	3.42	2.99	3.00	3.16	2.85	
4	3.93	4.00	4.00	3.93		
5	3.93	4.00	4.00	3.93		
6	3.74	3.30	3.52	3.38		
7		3.12			2.84	
8						
9						
10	3.41	2.96	2.97	3.12	2.86	
11						
12	3.57	3.01	3.44	3.37	3.62	
13	3.26	3.44	3.07	3.23	2.86	

Direct Assessment Method #2: Direct / Summative

Exit Interview

4 = E = Exemplary
3 = S = Satisfactory
2 = D = Developing
1 = U = Unsatisfactory

Percentage of students attaining E or S

SO	2014	2013	2012	2011	2010	2009
1	62.5	32.9	34.0	49.0	52.6	
2	67.9	38.0	31.8	42.6	42.5	
3	58.0	16.7	9.1	65.8	50.0	
4						
5						
6	67.0	38.0	31.8	42.6	42.5	
7	67.9	89.5				
8	82.1	62.8	84.4	65.7	79.0	
9	76.8	62.8	84.4	65.7	79.0	
10	89.3	77.8	86.2	62.8	92.9	
11	91.9					
12	95.5	47.1	55.2	50.8	62.3	
13	94.1	71.8	62.5	53.2	82.7	

Average rating among students

SO	2014	2013	2012	2011	2010	2009
1	2.88	2.39	2.44	2.65	2.79	
2	2.98	2.45	2.41	2.56	2.58	
3	2.81	2.21	1.42	2.82	2.80	
4						
5						
6	2.98	2.45	2.41	2.56	2.58	
7	2.92	3.28				
8	3.29	2.74	3.19	2.80	3.11	
9	3.16	2.74	3.19	2.80	3.11	
10	3.44	3.17	3.17	2.80	3.22	
11	3.38					
12	3.53	2.58	2.75	2.67	2.87	
13	3.50	2.98	2.83	2.65	3.07	

Assessment Method #3 – Direct / Summative
Expo Rubrics

4 = E = Exemplary
3 = S = Satisfactory
2 = D = Developing
1 = U = Unsatisfactory

Percentage of students attaining E or S

SO	2014	2013	2012	2011	2010	2009
1						
2						
3						
4						
5						
6						
7		65.8	75.0		50.0	75.0
8		80.0	91.7		66.7	85.7
9		80.0	91.7		66.7	85.7
10		57.2	59.4		62.5	100.0
11		75.0	91.7		64.3	100.0
12		66.8	73.5		72.2	100.0
13		65.8	75.0		50.0	75.0

Average rating among students

SO	2014	2013	2012	2011	2010	2009
1						
2						
3						
4						
5						
6						
7		2.46	2.81		2.54	3.25
8		2.93	3.06		2.90	3.50
9		2.93	3.06		2.90	3.50
10		2.08	2.46		2.38	3.75
11		3.19	3.57		2.73	3.00
12		2.61	2.77		2.72	3.75
13		2.46	2.81		2.54	3.25

Assessment Method #4 – Indirect / Summative

Exit Survey

4 = E = Exemplary
3 = S = Satisfactory
2 = D = Developing
1 = U = Unsatsifactory

Percentage of students attaining E or S

SO	2014	2013	2012	2011	2010	2009
1	97.7	80.6	81.5	83.3	92.6	
2	97.7	76.7	70.4	75.0	77.8	
3	97.7	64.5	74.1	66.7	70.4	
4	84.1	87.1	85.2	75.0	88.8	
5	95.5	87.1	85.2	75.0	88.8	
6	97.7	87.1	85.2	75.0	88.9	
7	88.4	67.6	76.9	41.7	77.8	
8	95.5	67.7	77.8	66.7	81.5	
9	90.9	54.8	66.7	50.0	63.0	
10	100.0	74.2	81.5	66.7	84.6	
11	97.7	71.0	84.6	75.0	84.6	
12	97.7	67.7	85.2	58.3	92.6	
13	100.0	63.3	77.8	75.0	81.5	

Average rating among students

SO	2014	2013	2012	2011	2010	2009
1	3.52	3.16	3.11	3.33	3.11	
2	3.36	3.10	2.93	2.92	3.04	
3	3.32	2.87	2.93	3.00	3.00	
4	3.27	3.29	3.15	3.17	3.15	
5	3.50	3.29	3.15	3.17	3.15	
6	3.59	3.29	3.15	3.17	3.15	
7	3.33	2.87	2.96	2.67	3.04	
8	3.36	2.81	3.04	3.00	3.04	
9	3.16	2.68	2.70	2.92	2.81	
10	3.61	3.13	3.11	3.17	3.35	
11	3.57	3.06	3.27	3.17	3.23	
12	3.57	2.90	3.30	3.08	3.41	
13	3.58	2.93	3.04	3.17	3.22	

Exhibit 4-8.

Continuous improvement – closing the feedback loop on assessment:
Sample Dates: Fall 2013, Spring 2013, Spring 2012.
Sample Description: Faculty self-studies for CHE 205.

ChE 205 – Fall 2013
Self-study based upon results of Course Update Survey
L. C. Nitsche
2014-02-04-TUE

The issue from spring 2013 regarding administration of quizzes has been resolved by using CEB 214 and SCE 408 simultaneously: there were no further comments. Replacement of Sturm-Liouville problems with a week on statistics seems to have been a good change. One student did not relate well to the lectures on error analysis, so perhaps this material and its presentation could be revisited.

Some students would have appreciated more time on the quizzes (or less problems for the class-hour duration). On the other hand, a good number of students regarded the quizzes as appropriately difficult, and the final grade distribution already matched the nominal 90/80/70/60/50 grade lines. So I hesitate to make the quizzes or exams easier or shorter. There was still the sentiment that lectures were a bit too abstract vis-à-vis applications: I can go through all the lectures and make sure that every topic or derivation is introduced with specific reference to a concrete chemical engineering problem. A purely practical concern is that Aspen shows up too small on the projector screen for students sitting in the back of the room.

Some themes emerged fairly clearly. First, we might consider assigning a textbook for the course (instead of the posted readings), although one covering this collection of material will be difficult to find. Especially the least-squares collocation is unlikely to show up on an available textbook, so I could at least prepare a dedicated write-up (like the Lecture Notes download on reaction-diffusion problems). Second, students felt that MATLAB got short-shrifted on the quizzes, as I have so far been able to directly test only Excel/VBA. I should integrate MATLAB into the quizzes; this will require advance planning with the ACCC. Also, the TA seemed less able to help with MATLAB than Excel. Third, the students would like the full VBA codes for homework solutions instead of just the correct answers. This I really don't see how I can accommodate without compromising some really good problems that took substantial time and effort to come up with. (And reading through VBA code is anyway not going to lead to the active learning that I am trying to foster). But something can be changed to help: given the continuing unpopularity of Fortran/C++, that material could stand to be replaced with more class time devoted to VBA. Better that the students really master one programming language than remain shaky on three. Fourth, several students wanted solutions posted to the quizzes and also more practice problems. I had resisted posting quiz solutions because the quizzes were so similar to the homeworks anyway (for which they got posted solutions) and I had to protect some material for future exams. But now is probably the time to prepare a new generation of quiz problems and post the old ones as practice problems.

ChE 205 – Spring 2013
Self-study based upon results of Course Update Survey
L. C. Nitsche
2013-08-22-THU

Two main, actionable points emerged from student feedback. First, dividing the class into three sections for rotating weekly quizzes (a practical necessary to hold quizzes in CEB 214) was distinctly unpopular, and opened the specter of non-uniform grading requirements. I think the resolution that I came up with for the final exam (using CEB 214 and SCE 408 simultaneously to accommodate all students) is the way to go for quizzes in the future, even if it is a little inconvenient for transit and proctoring. Second, students still did seem to favor more practical examples and less theory – despite the reorganization of and adjustments to course content that I made since last spring. I had included the Sturm-Liouville material because I thought students would really want to know where complete sets of basis functions come from. My approach to making this material accessible were partly successful, but after two attempts I now think that eigen-problems are a little out of reach for sophomores. So for the next offering of ChE 205 I intend to consolidate and compress weeks 8 and 9 into one week on more directly computational aspects least-squares collocation for ODEs and simply present the basis functions as given. That will free up one week for statistics and error analysis which I think can be more immediately useful and phrased in terms of practical problems.

The homework problems using Fortran were unpopular, and could have benefitted from more comprehensive, step-by-step instruction in class. I do want students to gain some facility with multiple computational platforms, so I'm not yet ready to excise Fortran entirely. I'll focus on an improved presentation in class and see how next semester's student feedback comes out.

Some students thought the online readings were a little too long or broad, so I should review them and see where they can be condensed and/or focused them down on the actual problems being covered in homeworks.

One category of feedback is harder to address, purely for practical reasons. Students would have liked to be able to follow along with the class on computers (as in Spring 2012), but we now have more than twice as many students as fit in CEB 214. On the longer term this should be discussed among the faculty.

ChE 205 – Spring 2012
Self-study based upon results of Course Update Survey
L. C. Nitsche
2012-05-18-FRI

Student feedback and my own experience teaching this course for the first time speak in favor of reorganizing the content. I think I started out the semester too quickly and my expectation outpaced sophomores' abilities. Introducing vectors, matrices and interpolation in addition to the Shomate correlation and numerical differentiation and integration in the first week was too much, and I had to spend the 6th week reviewing the linear algebra. Some students did not understand why converting between orthogonal and non-orthogonal bases is important: the analogies between basis vectors and basis functions (in solving ODEs) needs to be clearer. Continuity may benefit from moving the three weeks of Aspen Plus material to the very end instead of returning to nonlinear algebraic equations in the last week. I will have to reorder the content and make sure that basic linear algebra gets at least two weeks of coverage before starting on least squares fitting and collocation, which should get three full weeks. Also, a lecture devoted completely to homogenous ODEs and the characteristic polynomial would address the math background related to MATH 220.

CRITERION 5. CURRICULUM

A. Program Curriculum

5.A.1. Study Plan.

Details of our Bachelors of Science in Chemical Engineering curriculum can be found in Table 5-1. The program is a semester based program that requires 8 semesters (4 years) to complete. A typical degree plan is shown in Figure 5-1.

5.A.2. Alignment of the Program Curriculum with Program Educational Objectives.

The Chemical Engineering PEOs (Section 2.B) describe career success of our graduates based upon depth, breadth, professionalism and skills in life-long learning — acquired in a rigorous and innovative learning environment.

Depth. Table 5-1 demonstrates that students are required to take 42 credit hours in advanced chemical engineering core courses plus 6 credit hours of supporting engineering courses. These core courses cover a wide range of topics beginning with an Introduction to Thermodynamics (CHE 201, 3 hours) typically taken by the fall of the sophomore year. This is followed by Computational Methods in Chemical Engineering (CHE 205, 3 hours) and the Material and Energy Balances course (CHE 210, 4 hours). CHE 205 is a new course in our curriculum, the outcome of a feedback and assessment based continuous improvement process detailed in Section 4.B.2. This course standardizes the computational tools that the students will use throughout the curriculum (Excel, Matlab, ASPEN). The introduction of these tools in the sophomore year allows the students to build a base of computational expertise that they will further develop as they advance through the junior level courses and ultimately apply during their senior year in the Unit Operations Laboratory (CHE 381, 382) and Senior Design sequence (CHHE 396, 397). The requirement of CHE 205 eliminated 3 hours of free elective credit, but the early results indicate that our students are much better prepared as they exit the program. It must be emphasized that CHE 205 is a distinctly <u>chemical engineering</u> mathematics and computation course, carefully formulated to preview key concepts from the core courses and develop the relevant problem-solving skills. CHE 210 introduces the critical concepts of control-volume balances as well as basic problems-solving skills for problems that may contain a large number of variables and unknowns. In the junior year, students take the first two courses in Transport Phenomena (CHE 311 and 312, 6 hours), which cover fluid mechanics (momentum transfer), heat transfer, and mass transfer. In addition, students take an advanced course in Chemical Engineering Thermodynamics (CHE 301, 3 hours) in which students learn to apply equilibrium criteria to multiphase systems. The course in Chemical Reaction Engineering (CHE 321, 3 hours) which deals with the design of reactors and a third course in Transport phenomena (CHE 313, 3 hours) that covers separation processes complete the junior-level core in Chemical Engineering.

Senior-level courses complete the major in chemical engineering subject areas and complement the engineering science courses which dominate the junior year. These courses are: a two course sequence of Unit Operations Laboratory (CHE 381 and 382, 4 hours) in which transport and separation experiments are performed, data are analyzed and laboratory reports are written. The chemical process control course (CHE 341, 3 hours) introduces control theory and uses examples from all of the previous courses. Finally, the capstone two course Senior Design sequence (CHE 396 and 397, 7 hours) focuses on design of chemical processes and plants. In these courses, students demonstrate their ability to independently apply chemical engineering principles, cost estimation, and economic evaluation methodologies, flow sheet synthesis and process simulation, optimization techniques, and report writing and oral presentation skill in order to address contemporary chemical engineering design problems. The Senior Design II course (CHE 397, 3 hours) involves a team oriented design project which is supervised by an industrial mentor. Senior Design and Unit Operations Laboratory require our students to demonstrate and synthesize a depth of knowledge in many important chemical engineering areas and build their communication and presentation skills. In addition, students are required to take a technical elective course (3 hours) which may be any 400-level chemical engineering graduate course or an independent research course (CHE 392, 3 hours) in which the student can pursue in more depth a particular topic of personal interest under the direct supervision of a departmental faculty member. These topics range from Electrochemical Engineering, Renewable Energy Technologies, Advanced Transport Phenomena, to a host of other similar topics. In addition, students may pursue a special concentration in biotechnology which requires CHE 422 (3 hours) and supporting courses from outside of Chemical Engineering.

The rigorous sequence of course described above attests to the depth of knowledge with which Chemical Engineering emerge upon graduation and upon which they build their professional careers.

Breadth. The curriculum also requires students to demonstrate a breadth of knowledge. This includes a thorough grounding in chemistry and a working knowledge of advanced topics in chemistry such as organic, physical, and analytical chemistry. The courses that address the Program Educational Objective of breadth begin in the freshman year with an introduction to General Chemistry (CHEM 112 and 114, 10 hours). In the sophomore year students take a sequence of three courses in organic chemistry (CHEM 232, 233, and 234, 9 hours) which consist of two lecture courses and one laboratory course. In the junior year the chemistry courses are completed with a combination lecture/laboratory course in Analytical Chemistry (CHEM 222, 4 hours) and a two-course sequence in physical chemistry (CHEM 342 and 346, 6 hours). Students must also demonstrate a high level of mathematical skills in a four course sequence (MATH 180, 181, 210, and 220: 16 hours) taken during the freshman and sophomore years that involves differential and integral calculus, vector and multivariable calculus, differential equations and other advanced topics in mathematics. These mathematics skills are used throughout the curriculum. Also during the freshman and sophomore years a two-course sequence in physics (PHYS 141 and 142: 8 hours) requires our students to demonstrate a basic grounding in mechanics and electromagnetism.

In addition to the basic science courses described above, our students must also demonstrate their breadth of technical knowledge by taking engineering courses outside of Chemical Engineering. These are a course in Properties of Materials (CME 260: 3 hours), a course in Computer Programming (CS 109: 3 hours) and a course in Electrical Circuit Analysis (ECE 210: 3 hours).

These courses add breadth by giving our students insight into other related engineering disciplines and prepares them for the multidisciplinary career environment in which they will work.

Professionalism. In the freshman year our students take a general Engineering Orientation course (ENGR 100: 0 hours) course that gives an introduction to the development of a career in engineering. The students learn about engineering careers of all varieties and also learn good professional habits. The design course sequence described above (CHE 396 and 397: 7 hours) is also integral to our students' development of professionalism. The design courses bring in speakers from industry who discuss current issues and describe the requirements of a professional career in chemical engineering. Topics concerning safety, professional ethics, design strategies, business strategies, etc. are discussed. In addition, the students are required to take a professional development seminar (CHE 499, 0 hours) which focuses on career paths, job hunting and a variety of issues such as engineering ethics, writing resumes and effective cover letters, and concludes with an exit interview in the form of an oral examination. Finally, all of our students are required to participate in the College of Engineering's Engineering Expo where the final design projects are presented and judged by engineers from industry. This requires our students to connect their design projects and the basic engineering science and design principles applied to the projects to the complex modern world and the many issues that must be addressed in any real world application of engineering.

Life-Long Learning. We address this Program Educational Objective with a curriculum that forces students to develop the knowledge, skills and mental orientation that will equip them to be life-long learners. For example, in the Senior Design sequence, students learn to write a literature review, which requires that they begin to explore the vast research literature available. They must synthesize knowledge from the literature and apply it to their specific design project. During the course of the semester, the students are asked to give presentations detailing the progress on their designs and thereby develop their oral communication skills. Finally they receive feedback not only from the course instructor and industrial mentor, but also from professional engineers who participate as judges at the Engineering Expo. Also, students discuss many current issues in chemical engineering including development of new energy sources and products. These topics often become the major theme for many of the design projects. Finally, our students are required to participate in extensive one-on-one advising with a faculty member every semester. During the advising sessions, students are asked to plan their career or academic objectives and to consider how to best prepare to achieve the goals. This one-on-one mentoring is a valuable part of providing a learning environment that is suited to the individual student and to promote the skills and knowledge needed to pursue life-long learning. The Professional Development Seminar (CHE 499) brings in industrial and professional speakers to open up our students' vistas on unexpected career paths, real-life experience in subfields of chemical engineering, interacting with other types of engineers.

Finally, career options and advanced degrees relevant to pharmaceutical and medically relevant fields is enhanced by the Biochemical Engineering Concentration. To fulfill this option (which appears on the transcript), students are required to complete 9–10 semester hours in elective courses by taking the following courses: (i) CHE 422—Biochemical Engineering, (ii) two electives in nonmajor rubric category (i.e., outside Chemical Engineering) from among BIOS 350—General Microbiology (3 cr. hr.), BIOS 351—Microbiology Laboratory (2 cr. hr.), CHEM 352—

Introductory Biochemistry (3 cr. hr.) or CHEM 452—Biochemistry I (4 cr. hr.), (iii) Free elective (if needed) to make up 1 credit hour.

5.A.3. Support of the Attainment of Student Outcomes by the Curriculum and Prerequisites.

Section 3.C has detailed how attainment of SOs is facilitated by both Chemical Engineering courses (Tables 3-4, 3-5 and 3-6) and technical courses outside our department (Tables 3-7 and 3-8). Within the ChE program each SO receives significant coverage by multiple courses (between 2 and 17), and this is bolstered by further treatment in the non-ChE portion of the curriculum.

5.A.4. Prerequisite Structure of Required Courses.

Figure 5-2 shows the prerequisite structure of required courses within the ChE curriculum.

5.A.5. Meeting the Requirements of Depth of Study for General and Program Criteria.

It is shown in Table 5.1 that our programs course requirements meet or exceed the minimum credit hours and distribution. How the Chemical Engineering program stands in relation to ABET requirements is detailed as follows.

(a) ***ABET Requirement.*** ***"One year (32 semester hours) of a combination of college level mathematics and basic sciences (some with experimental experience) appropriate to the discipline. Basic sciences are defined as biological, chemical, and physical sciences."***

Our program features 56 semester hours of college level mathematics and basic sciences. This is distributed among mathematics (4 courses, 16 hours), physics (2 courses, 8 hours), chemistry (8 courses, 29 hours) and computer science (1 course, 3 hours). Laboratory experience is integrated into the General Chemistry (CHEM 112, 114, 10 hours) and General Physics (PHYS 141, 142, 8 hours) and accounts for half of Analytical Chemistry (CHEM 222, 4 hours). In addition, CHEM 233 (1 hour) is a dedicated Organic Chemistry Lab course.

(b) ***ABET Requirement.*** ***"One and one-half years (48 semester hours) of engineering topics, consisting of engineering sciences and engineering design appropriate to the student's field of study. The engineering sciences have their roots in mathematics and basic sciences but carry knowledge further toward creative application. These studies provide a bridge between mathematics and basic sciences on the one hand and engineering practice on the other. Engineering design is the process of devising a system, component, or process to meet desired needs. It is a decision-making process (often iterative), in which the basic sciences, mathematics, and the engineering sciences are applied to convert resources optimally to meet these stated needs."***

Our program has 48 hours of engineering topics, distributed as follows.
Sophomore Engineering courses outside major (6 hours): ECE 210, CME 260.
Sophomore Chemical Engineering courses (10 hours): CHE 201, 205, 210.
Junior Chemical Engineering Courses (15 hours): CHE 301, 311, 312, 313, 321.

Senior Chemical Engineering Courses (17 hours): CHE 341, 381, 382, 396, 397, 4XX.

(c) ***ABET Requirement. "A general education component that complements the technical content of the curriculum and is consistent with the program and institution objectives."***

Our program features 6 hours of English composition, 15 hours of General Education courses divided amongst five broad categories, and 3 additional hours of elective credit that can also be used for general education. Many students use the latter courses to study a foreign language or take business courses. We also have many students who opt for a minor in Mathematics since only two additional courses (one in the form of the free elective) are necessary to complete that requirement.

With 56 hours of basic mathematics and science, 48 hours of engineering topics and 21 hours of general education with an additional 3 hours of free electives, UIC's Chemical Engineering program satisfies ABET requirements.

5.A.6. Major Design Experience.

The major design experience in our program is composed of CHE 396 and 397, which are the two capstone design courses. The first course (CHE 396) in the sequence focuses on the design and economic analysis of chemical processes. This includes cost estimation and economic evaluation methodologies and well as design of individual processing units. Students become more familiar with process flow simulation, although they have already had experience with the Aspen Plus in several previous courses (at a minimum in CHE 205, CHE 301 and CHE 313). The design of unit operations relies on the Chemical Engineering core and basic engineering science courses from the junior year. These include chemical engineering thermodynamics where the students are taught the basics of equation of state models and activity coefficient models upon which to base their design simulations. In addition student utilize the transport phenomena sequence (CHE 311, 312, and 313) which supports the design of heat exchangers, piping systems, and separation processes by supplying the estimates of the transport properties and the insight provide by the engineering analysis of these unit operations. The reactor design course (CHE 321) furnishes an understanding of basic reactor modeling which is relevant for many process units. CHE 396 begins the conceptual transition from theory to practice, from "ideal" to "real". Operational consequences of connecting process units together are explored, as are pumping and piping, fouling, instrumentation, controls, process safety, occupational health, sustainability, optimization with regard to energy and raw materials, and engineering ethics. Since the last ABET site visit in 2008, we have moved to define the spring semester design projects late in CHE 396, in order to enable better time flow in CHE 397.

In the second design course (CHE 397) students are assigned to work on a major design project in small teams of three or four students each. Each team is assigned an industrial mentor, a practicing design engineer who provides feedback and guidance in mandatory weekly meetings carried out on-site. Within a well-structured framework of report milestones, the teams give multiple presentations describing their coalescing designs throughout the semester in front of their classmates and the team mentors to trace their progress and build detail into the final design. The overall design of a process flow sheet and the control and optimization of the process is supported by the previous design course (CHE 396), the mass and energy balance course (CHE 210) and the process control course (CHE 341). The milestones culminate in the final written and oral reports (co-judged by the industrial mentors) as well as presentation of the completed project at the annual Engineering Expo, where it is judged by practicing industrial engineers. Projects are reviewed by judges and the instructor to be such that the designs meet modern and realistic design constraints and standards. It is essential that projects are reviewed by practicing engineers. Throughout the semester, guest speakers are invited to give presentations on several aspects of the chemical industry and career development. Students also have an opportunity to meet informally with visitors to discuss their own interests. One advantage of extensive industrial contact and observation of individual students is the opportunity for networking. Senior Design often leads to job offers for our graduating seniors.

5.A.7. Cooperative Education

Although it is not required, many of our students do participate in co-op and internship programs. Our location in a major metropolitan center with extensive chemical, materials, energy, food, healthcare, consulting, electronics industries, to name a few, offers our students many opportunities for a meaningful industrial experience during their undergraduate education. However, course credit is not offered for these activities.

5.A.8. Material to be Available for Review at the Site Visit.

The Program Evaluator will have access to an extensive sampling of course materials (syllabi, handouts, copies of textbooks, homework assignments, quizzes, exams) and corresponding student work. We will also provide extensive documentation of assessment methods/data, evaluation procedures, feedback and continuous improvement, along with detail information on how course content supports attainment of Student Outcomes. This material will be organized in a web format that facilitates efficient examination and cross-checking of the material.

B. Course Syllabi

Appendix A provides the syllabi of all courses used to satisfy the mathematics, science, and chemical engineering discipline-specific requirements required by Criterion 5.

Table 5-1 Curriculum

Chemical Engineering

Course (Department, Number, Title) List all courses in the program by term starting with the first term of the first year and ending with the last term of the final year.	Indicate Whether Course is Required, Elective or a Selected Elective by an R, an E or an SE.[1]	*Subject Area (Credit Hours)* Math & Basic Sciences	Engineering Topics Check if Contains Significant Design (√)	General Education	Other	Last Two Terms the Course was Offered: Year and, Semester, or Quarter	Maximum Section Enrollment for the Last Two Terms the Course was Offered[2]
Freshman Year, First Semester							
MATH 180 – Calculus I	R	5					
CHEM 112 – General Chemistry I	R	5					
ENGL160 – English Composition I	R			3			
General Education Core: II. Understanding the Individual and Society	SE			3			
ENGR 100 – Engineering Orientation	R						
Freshman Year, Second Semester							
MATH 181 – Calculus II	R	5					
PHYS 141 – General Physics I (Mechanics)	R	4					
ENGL161 – English Composition II	R			3			
CHEM 114 – General Chemistry II	R	5					
Sophomore Year, First Semester							
MATH 210 – Calculus III	R	3					
PHYS 142 – General Physics II (Electricity and Magnetism)	R	4					
CS 109 – C/C++ Programming	R	3					
CHEM 232 – Organic Chemistry I	R	4					

CHE 201 – Intro to Thermodynamics	R		3			Fall 2013/ Spring 2014	59
Sophomore Year, Second Semester							
MATH 220 – Differential Equations I	R	3					
CHEM 234 – Organic Chemistry II	R	4					
CHEM 233 – Organic Chemistry Lab I	R	1					
CHE 210 – Material & Energy Balances	R		4√			Fall 2013/ Spring 2014	53
CHE 205 -Computational Methods in Chemical Engineering	R		3			Fall 2013/ Spring 2014	64
CME 260 – Properties of Materials	R		3				
Junior Year, First Semester							
ECE 210 – Electrical Circuit Analysis	R		3				
CHE 311 – Transport Phenomena I	R		3√			Fall 2012/ Fall 2013	65
CHE 301 – Chemical Engineering Thermodynamics	R		3			Fall 2012/ Fall 2013	63
CHEM 222 – Analytical Chemistry	R	4					
General Education Core: III. Understanding the Past	SE			3			
Junior Year, Second Semester							
CHEM 342 – Physical Chemistry I	R	3					
CHE 312 – Transport Phenomena II	R		3√			Spring 2013/ Spring 2014	57
CHE 313 – Transport Phenomena III	R		3√			Spring 2013/ Spring 2014	56
CHE 321 – Chemical Reaction Eng.	R		3√			Spring 2013/ Spring 2014	57
General Education Core: IV. Understanding the Creative Arts	SE			3			
Senior Year, First Semester							
CHE 381 – Chem. Eng. Lab I	R		2			Fall 2012/ Fall 2013	43

CHE 396 – Senior Design I	R		4√			Fall 2012/ Fall 2013	44
Technical Elective/. For example CHE 494 – Selected Topics in Chemical Engineering or CHE 422 – Biochemical Engineering, etc.	R		3√			Fall 2013/ Spring 2014	23
CHEM 346 – Physical Chemistry II	R	3					
General Education Core: V. Exploring World Cultures	SE			3			
Senior Year, Second Semester							
CHE 382 – Chem. Eng. Lab II	R		2			Spring 2013/ Spring 2014	43
CHE 341 – Process Control	R		3√			Spring 2013/ Spring 2014	42
CHE 397 – Senior Design II	R		3√			Spring 2013/ Spring 2014	43
CHE 499 – Professional Development Seminar	R		0			Spring 2013/ Spring 2014	42
Elective outside the Major Rubric					3		
General Education Core: VI. Understanding U.S. Society	SE			3			
TOTALS-ABET BASIC-LEVEL REQUIREMENTS		56	48	21	3		
OVERALL TOTAL CREDIT HOURS FOR COMPLETION OF THE PROGRAM	128						
PERCENT OF TOTAL		43.8	37.5				
Total must satisfy either credit hours or percentage	Minimum Semester Credit Hours	32 Hours	48 Hours				
	Minimum Percentage	25%	37.5 %				

1. **Required** courses are required of all students in the program, **elective** courses (often referred to as open or free electives) are optional for students, and **selected elective** courses are those for which students must take one or more courses from a specified group.
2. For courses that include multiple elements (lecture, laboratory, recitation, etc.), indicate the maximum enrollment in each element. For selected elective courses, indicate the maximum enrollment for each option.

Instructional materials and student work verifying compliance with ABET criteria for the categories indicated above will be required during the campus visit.

FRESHMAN		Total Hours = 16			Total Hours = 17
ENGL 160	3	English Composition I	**ENGL 161**	3	English Composition II
MATH 180	5	Calculus I	**MATH 181**	5	Calculus II
CHEM 112	5	General Chemistry I	**CHEM 114**	5	Gen. Chemistry II
ENGR 100	0	Engineering Orientation	**PHYS 141**	4	Gen. Physics - Mech.
HUM/SOC	3	General Education Course			
SOPHOMORE		Total Hours = 17			Total Hours = 18
MATH 210	3	Calculus III	**MATH 220**	3	Differential Equation I
CHEM 232	4	Organic Chemistry I	**ChE 205 (=CHE 201 & MATH 210)**	3	Comput. Methods in ChE
PHYS 142	4	Gen. Physics II-Elec. & Mag	**CHE 210 (= CHE 201 & 205)**	4	Material & En Balances
CHE 201 (MATH 181)	3	Intro to Thermodynamics	**CME 260 (MATH 181)**	3	Properties of Materials
CS 109	3	Fortran Prog. for Engineers	**CHEM 233 (=CHEM 232)**	1	Organic Chemistry Lab
			CHEM 234	4	Organic Chemistry II
JUNIOR		Total Hours = 16			Total Hours = 15 or 18
CHE 311 (=CHE 210)	3	Transport Phenomena	**CHE 312 (CHE 311)**	3	Transport Phenomena II
CHE 301 (CHE 201 & 205)	3	ChE Thermodynamics	**CHE 313 (CHE 301)**	3	Transport Phenomena III
CHEM 222	4	Analytical Chemistry	**CHE 321 (CHE 301, CHE 210)**	3	Chem. Reaction Eng.
ECE 210	3	Electrical Circuit Analysis	**CHEM 342**	3	Physical Chemistry I
HUM/SOC	3	General Education Course	***HUM/SOC***	3	General Education Course.
			HUM/SOC (option 1)	3	General Education Course.
SENIOR		Total Hours = 15			Total Hours = 14 or 17
CHE 396 (CHE 321 & 313)	4	Senior Design I	**CHE 397 (CHE 396)**	3	Senior Design II
CHE 381 (CHE 312)	2	Chemical Eng. Lab I	**CHE 341 (CHE 312, 313, 321 & MATH 220)**	3	Chemical Process Control
CHEM 346	3	PhysicalChemistry II	**CHE 382 (CHE 381)**	2	Chemical Eng. Lab II
HUM/SOC	3	General Education Course	**TECH. ELECTIVE**	3	400 Level ChE or Equiv.
ELECTIVE	3	Outside Major	**ELECTIVE**	3	Outside Major
			CHE 499	0	Prof. Development Seminar
			HUM/SOC (option 2)	3	General Education Course
Total Hours in Major = 128					

Legend:

[grey box] Milestone Courses: Must be taken in the specified semester to be able to graduate in 4 yrs.

( ) Prerequisites (=) Concurrent registration

FIGURE 5-1. Typical degree plan for the Chemical Engineering program.

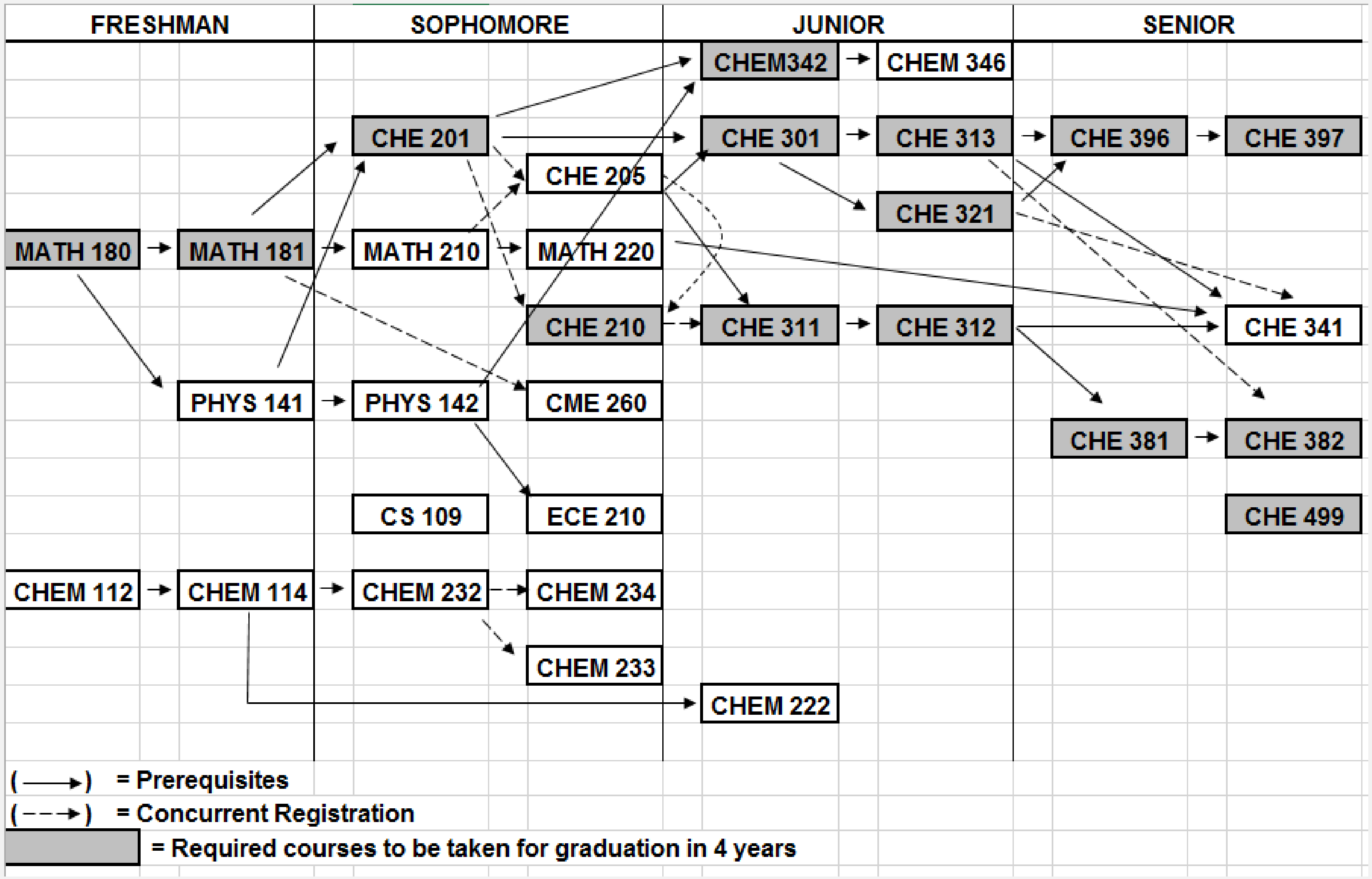

FIGURE 5-2. Prerequisite structure of required courses within the ChE curriculum.

CRITERION 6. FACULTY

A. Faculty Qualifications

All professors have PhD's in Chemical Engineering and two have industrial experience. Several have had post-doctoral experience outside the US, and are also very familiar with the educational system in Europe and Asia. In addition, during the past five years the Department has had three Adjunct Professors, who teach on a part time basis (from 50% to 100% each semester – 1 or two classes). All have had industrial experience. Two have full time Industrial positions, and one has spent most of his career in Industry. In addition one Emeritus faculty member who does not participate in teaching actively participates in advising undergraduate students. The normal course load of a research active faculty member is four classes per year. However, under College rules faculty who actively participate in the training of PhD students get a reduced course load of three classes and in some cases of two classes. Faculty is allowed to under very special circumstances "buy out" a class, but in the past five years it has only been done once in the Department.

B. Faculty Workload

Expertise among the faculty is sufficiently diverse that any course in the undergraduate curriculum could be taught by at least two of the current faculty members. This teaching versatility has been fostered by the diverse educational backgrounds of faculty members: both American and European educated; both in traditional chemical engineering and related fields, in addition to both long and short teaching experiences. Teaching commitment and accomplishment are evidenced by the high number of internal teaching awards received by faculty members (currently most eligible faculty members in the department have won at least one College or University Teaching Award; see Section 6.C.6). Research in support of teaching is also encouraged and pursued by most of the faculty through the NSF REU program and through the University Honors College. A summary of the distribution of faculty workload summary and Activity Analysis are given in Tables 6.1 and 6.2. We would like to point out that while Table 6.2 shows the actual time being spent by the faculty in different activities as per university policies, Table 6.1 also takes into consideration the actual impact of the activity in judging its level, so there may not necessarily be a direct correspondence between the two Tables.

C. Faculty Size

The Department currently has 252 undergraduate students. In academic year 2013-2014 there were ten tenured/tenure-track faculty (two among them 25% in the Chemical Engineering department, with the Bioengineering department being their main appointment) and three adjunct professors. Tables 6.1 lists these faculty members and their teaching and other responsibilities.

The current faculty workload as described earlier in Section 6.B permits more than adequate advising and educational interactions with students. It also provides enough depth of expertise to permit faculty sabbatical leaves for professional development without adverse effects on teaching.

6.C.1. Assistant Professors.

Dr. Belinda Akpa is an active member of the American Institute of Chemical Engineers and the American Chemical Society. She is a member of AIChE's Societal Impact Operating Council and a symposium organizer for ACS's Division of Colloids and Surface Science. She also attends meetings of the Biomedical Engineering Society and international conferences in the field of magnetic resonance. Dr Akpa has served as a reviewer for the the Mathematical and Physical Sciences and Engineering directorates of the National Science Foundation. She currently holds grants from NSF and NIH that entail regular collaboration with physician-scientists and experimental biologists in the College of Medicine. She is also active in leading educational outreach efforts to K-12 students in the Chicago region.

Dr. Brian Chaplin is an active member of the American Chemical Society, and organizes research symposia annually. He frequently attends the North American Membrane Society (NAMS) conference. He has also developed several teaching modules that were targeted at increasing scientific literacy of middle school students. Dr. Chaplin has served on proposal review panels for the National Science Foundation and Environmental Protection Agency, and is a frequent reviewer of several professional journals.

Dr. Ying Liu is an executive member of the Sino-American Pharmaceutical Professionals Association (SAPA) Midwest. She regularly attends the national meetings for the American Institute of Chemical Engineering, the American Chemical Society, and the American Physical Society. She has close collaborations with researchers in the medical school at UIC as well as researchers in the nation and out of the country, and Argonne National Laboratory. Dr. Liu's research is currently funded by National Science Foundation (NSF)-CAREER award and Department of Defense (DoD)-Army grant. Dr. Liu has been an active reviewer for more than 20 journals (including Nature Physics, JACS and ACS Nano) and panel reviewer for NSF proposals.

Dr. Vivek Sharma is active member of Society of Rheology (SOR), American Institute of Chemical Engineers (AIChE), American Physical Society (APS) and American Chemical Society (ACS). He has organized (or will be organizing) sessions and chairing sessions at national meetings for SOR (2013), AIChE (2013 & 2014), APS (2014), ACS (2014) as well as for U. S. National Congress on Theoretical and Applied Mechanics (2014). Dr. Vivek Sharma regularly attends the meetings for these professional societies, and presents papers and posters at these conferences. In addition, Dr. Sharma has participated in Gordon Research Conference on Protein Unfolding and a conference on Biological and Pharmaceutical Complex Fluids. Dr. Sharma has served on several panels for the National Science Foundation (NSF), and actively reviews papers for over a dozen scientific journals.

6.C.2. Associate Professors.

Dr. Randall J. Meyer is an active member in the Catalysis Club of Chicago, where he has formerly held the offices of Program Chair and President. He regularly attends the national meeting for the North American Catalysis Society as well as the annual meetings of the American Institute of Chemical Engineers, and the American Chemical Society. He collaborates extensively with researchers at Argonne National Laboratory. He also has initiated active research collaborations with universities in Europe as well as in the US. Dr. Meyer has been a panelist for the National Science Foundation for review of proposals and is an active reviewer for many professional journals.

Ludwig C. Nitsche is active in the local chapter of the American Institute of Chemical Engineers (AIChE) and has co-organized outreach symposia for high school and middle school students at Midwest AIChE conferences hosted at UIC (2007, 2008 and 2013). He is an active participant in the WISEST (Women in Science and Engineering System Transformation) Committee and the College level Diversity Committee. Dr. Nitsche has an active NSF GOALI research grant on skin permeation of chemicals and drugs and is PI on an NSF S-STEM grant that provides need-based scholarships to undergraduates. He has served on proposal review panels for the National Science Foundation and frequently reviews manuscripts for a number of archival journals.

Dr. Lewis E. Wedgewood is an active member of the American Institute of Chemical Engineers (AIChE) and the Society of Rheology (SOR) and has participated in the Midwest AIChE conferences. He is currently a member of the college Executive Committee which reviews administrators and participates in tenure and promotion and college Awards Committee which awards student scholarships and faculty teaching awards. He is a co-PI on an NSF S-STEM grant that provides need-based scholarships to undergraduates. He has been a panelist for the National Science Foundation for review of proposals and is an active reviewer for journals such as the Journal of Fluid Mechanics and Applied Physics Letters.

6.C.3. Full Professors.

Dr. Sohail Murad is Professor and Department Head of Chemical Engineering at University of Illinois at Chicago, where he joined the faculty in 1979 after receiving a PhD from Cornell University, Ithaca, NY. He spent 1981-82 at Exxon Research and Engineering Company at Florham Park, New Jersey, while on a leave of absence from the university. He was an ARO Research Fellow at the Ballistics Research Laboratory in 1985. He is the author of over 100 archival research publications and book chapters. He is/has been a member of the Editorial Advisory Board of Computer Applications in Engineering Education, Scientific Journals International and Research Letters in Chemical Engineering. His research is focused on alternate energy and its efficient utilization, computational molecular modeling of fluids on membrane surfaces and pores and on heat and mass flows in nanosystems. It has been funded by the US National Science Foundation, US Department of Energy, US Army Research Office, American Chemical Society, IBM, Dow Chemical Company, Sun Microsystems and other private and public funding agencies.

Dr. Christos Takoudis (25%, Home Department: BioEngineering) is an active participant in meetings of the American Institute of Chemical Engineers, American Chemical Society, and the Materials Research Society. Has active research projects with several local industrial organizations (Motorola, Air-Liquide, etc), in addition to several federal agencies. Is active in consulting activities, and as a panelist for funding agencies and reviewer for professional journals..

Dr. Ali Mansoori (25%, Home Department: BioEngineering) is active in the American Institute of Chemical Engineers, and has held several key offices in the organization. Is very active in industrial consulting, and has served as a panelist for funding agencies, and is an active reviewer of professional journals. Serves on the editorial boards of several professional journals.

6.C.4. Adjunct Professors.

Dr. Said Al-Hallaj is active in corporate research on alternative energy and battery technology. He serves as chairman and CEO of AllCell Technologies, LLC.

Dr. Jeffery P. Perl is an active member of the American Institute of Chemical Engineers (AIChE) and past chair of the aicHe Chicago section. He is also a member of the American Chemical Society (ACS), a registered Professional Engineer in Illinois and Kentucky, and a Certified Hazardous Materials Manager. His contributions to engineering education and certification include service on the National council of Examiners of Engineers and Surveyors, Chemical Exam.

Dr. Alan D. Zdunek is an active member of the American Institute of Chemical Engineers (AIChE), American Chemical Society (ACS) and The Electrochemical Society (ECS). He is co-PI on an active NSF grant studying the synthesis and characterization of intermediate temperature solid oxide fuel cells, and co-PI on an active UIC OTM Proof of Concept grant developing an electric-field enhanced desalination process. Dr. Zdunek is a reviewer for the Journal of the Electrochemical Society, has served as a technical advisory board member for the IIT Chemical Engineering Department (2004-2007) and was the keynote speaker at the Yellow River Delta High-efficiency Economic Zone International Talents Exchange and Projects Conference in Dongying City, China (2011).

6.C.5. Current/Past National Science Foundation CAREER/NYI Awardees Among the Faculty.

Dr. Ying Liu, 2014-2019.
Dr. Randall J. Meyer, 2008-2013.
Dr. Ludwig C. Nitsche, 1994-1999.

6.C.6.Faculty Teaching and student Advising Awards.

- Dr. Belinda Akpa
    - COE Faculty Teaching Award (2012, 2013)
    - COE Faculty Advising Award (2012)

- Dr. Sohail Murad
  - UIC Award for Excellence in Teaching (1999)
  - UIC Teaching Recognition Program Award (1997)
  - COE Harold A. Simon Award for Excellence in Teaching (2002)
  - COE Faculty Teaching Award (1998, 2004, 2006)
  - Department of Chemical Engineering Outstanding Teacher of the Year (1995)
- Dr. Ludwig Nitsche
  - UIC Award for Excellence in Teaching (2010-2011)
  - UIC Teaching Recognition Program Award (2006-2007)
  - COE Harold A. Simon Award for Excellence in Teaching (2012-2013)
  - COE Faculty Teaching Award (2006, 2008)
- Dr. Lewis Wedgewood
  - UIC Award for Excellence in Teaching (2013-2014)
  - UIC Teaching Recognition Program Award (2000-2001)
  - COE Harold A. Simon Award for Excellence in Teaching (2003-2004)
  - COE Faculty Teaching Award (2009, 2007)
  - COE Faculty Advising Award (2007, 2008)

6.C.7. Changes in Faculty.

Effective the end of May 2014, Dr. Murad has retired from UIC and now continues as an Adjunct Professor. Dr. Nitsche was named Interim Head effective June 1. Dr. Vikas Berry (formerly of Kansas State University and an NSF CAREER awardee, 2011-2016) has recently been hired as a tenured Associate Professor to start in fall 2014. In the intervening time he is an Adjunct Professor. Currently it is planned for Dr. Meyer to be on a leave of absence in academic year 2014-2015.

D. Professional Development.

The department places much emphasis on the development of all faculty members, especially junior and mid-career faculty. A formal mentor is appointed for each Assistant Professor appointed in the Department. The mentor works closely with the faculty member in developing better and more effective teaching strategies, as well as developing exciting new research ideas and provides continuous feedback on the faculty member's progress towards tenure. For mid-career faculty, although a formal mentor is not appointed, the Department Head holds regular meetings with them to enhance both their teaching and research effectiveness. This includes arranging meetings with representatives of both industrial and governmental funding agencies, and alerting them when funding is available for these activities through university wide-programs as well as from external agencies. The department also provides seed funding to both early career and mid-career faculty to initiate research projects. For example for two academic years, the College/Department and University have cost-shared a new research collaboration of an Associate Professor (Dr. Meyer) with Argonne National Laboratory. In addition the Department provides funding to faculty to visit funding agencies and present talks at professional meeting. As part of its contribution to Faculty Development, the University has provided a new personal computer to all faculty on a 3/4 year cycle as well as to new faculty members when they first join. The University additionally shares the cost of startup costs of new faculty with the College and the Department through their various programs. The department encourages faculty members to be active in professional societies, and supplements the cost of attending their meetings when they do not have travel funds available in grants. In addition the department provides funds to faculty members to visit granting agencies, and covers other related costs of grant proposal development. Finally to ensure that all faculty have adequate computing resources to ensure their professional development, the Department provides software needed to fulfill their teaching and professional responsibilities. A report of the professional development activities of faculty in the Department is given below. For faculty members with 25% appointments in the department, the primary responsibility of faculty development is that of their home departments.

E. Authority and Responsibility of Faculty

Every new course developed (or modified) is initiated by an individual faculty member in the Department. The faculty member in consultation with other faculty with similar expertise generally develops a new outline, which after discussion is approved by the Undergraduate Committee and presented to the entire faculty for approval. After such approval the faculty member prepares a detailed outline of the course with hours to be spent on each subject/topic, suggests an appropriate text book, and submits such a detailed proposal to the Undergraduate Committee. Once approved by the Undergraduate Committee it is discussed by the entire faculty during either a regularly scheduled or special faculty meeting. After approval by the Department it is sent to the College of Engineering, where it is considered by the College Education Policy Committee (which has one member from each department and is chaired by an elected faculty member). The Associate Dean of Undergraduate Affairs is an ex-officio member of this Committee. After approval by the College Educational Policy Committee it is considered and approved by the entire faculty of the College of Engineering. After such College approval it is forwarded to the Office of the Provost and Vice-Chancellor for Academic Affairs. That office seeks advice of the Senate Educational Policy Committee. After such approval it is sent to the University Senate for final approval for the following academic year

Table 6-1. Faculty Qualifications

Name of Program

Faculty Name	Highest Degree Earned- Field and Year	Rank [1]	Type of Academic Appointment[2] T, TT, NTT	FT or PT[3]	Years of Experience			Professional Registration/ Certification	Level of Activity[4] H, M, or L		
					Govt./Ind. Practice	Teaching	This Institution		Professional Organizations	Professional Development	Consulting/summer work in industry
B. S. Akpa	PhD (ChE), 2007	AST	TT	FT	0	5	5		M	H	M
S. Al-Hallaj	PhD (ChE), 1999	A	NTT	PT	15	3	3		M	M	H
B. Chaplin	PhD (ChE), 2007	AST	TT	FT	0	4	1		M	H	M
Y. Liu	PhD (ChE), 2007	AST	TT	FT	0	6	6		M	H	M
A. Mansoori	PhD (ChE), 1969	P	T	FT	0	46	46	IL	H	M	H
R. J, Meyer	PhD (ChE), 2001	ASC	T	FT	0	11	11		H	H	M
S. Murad	PhD (ChE), 1979	P	T	FT	2	36	36		M	H	M
L. C. Nitsche	PhD (ChE), 1989	ASC	T	FT		23	23		M	H	L
J. Perl	PhD (ChE), 1984	A	NTT	PT	28	7	6	IL, KY	H	M	H
V. Sharma	PhD (Polym. Sci. & Eng.), 2008	AST	TT	FT	0.5	2	2		H	H	M
C. Takoudis	PhD (ChE), 1982	P	T	FT	0	29	18		M	H	H
L. E. Wedgewood	PhD (ChE), 1988	ASC	T	FT	0	24	16		M	H	M
A. Zdunek	PhD (ChE), 1990	A	NTT	PT	17	6	4		M	M	H

Instructions: Complete table for each member of the faculty in the program. Add additional rows or use additional sheets if necessary. Updated information is to be provided at the time of the visit.

1. Code: P = Professor ASC = Associate Professor AST = Assistant Professor I = Instructor A = Adjunct O = Other
2. Code: TT = Tenure Track T = Tenured NTT = Non Tenure Track
3. At the institution
4. The level of activity, high, medium or low, should reflect an average over the year prior to the visit plus the two previous years.

Table 6-2. Faculty Workload Summary

Name of Program

Faculty Member (name)	PT or FT[1]	Classes Taught (Course No./Credit Hrs.) Term and Year[2]	Program Activity Distribution[3]			% of Time Devoted to the Program[5]
			Teaching	Research or Scholarship	Other[4]	
B. S. Akpa	FT	CHE 210 / 3 – fall 2013 and spring 2014 CHE 422 / 3 – fall 2013	50	40	10	100
S. Al-Hallaj	PT	CHE 494 / 3– spring 2014	100			100
B. Chaplin	FT	CHE 494 / 4 – spring 2014	50	40	10	100
Y. Liu	FT	CHE 311 / 3 – fall 2013 CHE 341 / 3 – spring 2014 CHE 410 / 4 – fall 2013				100
A. Mansoori	PT	CHE 502 / 4 – fall 2013	50	30	20	25
R. J. Meyer	FT	CHE 301 / 3 – fall 2013 CHE 321 / 3 – spring 2014 CHE 527 / 4 – spring 2014	50	40	10	100
L. C. Nitsche	FT	CHE 205 / 3 – fall 2013 CHE 494 / 3 – fall 2013 (new course number is 433) CHE 512 / 4 – spring 2014	40	40	20 (DUGS)	100
S. Murad	FT	CHE 201 / 3 – spring 2014	20	50	30 (Head)	100
J. Perl	PT	CHE 396 / 4 – fall 2013 CHE 397 / 3 – spring 2013	100			100
V. Sharma	FT	CHE 312 / 3 – spring 2013 CHE 494 / 3 – fall 2013	50	40	10	100
C. Takoudis	PT	CHE 313 / 3 – spring 2014	50	30	20	25
L. E. Wedgewood	FT	CHE 201 / 3 – fall 2013 CHE 205 / 3 – spring 2014	40	40	20 (DGS)	100

		CHE 445 / 4 – fall 2013				
A. Zdunek	PT	CHE 381 / 2 – fall 2013 CHE 382 / 2 – spring 2014	100			100

1. FT = Full Time Faculty or PT = Part Time Faculty, at the institution
2. For the academic year for which the Self-Study Report is being prepared.
3. Program activity distribution should be in percent of effort in the program and should total 100%.
4. Indicate sabbatical leave, etc., under "Other."
5. Out of the total time employed at the institution.

CRITERION 7. FACILITIES[1]

A. Offices, Classrooms and Laboratories

Here follows a summary of each of our program's facilities in terms of their ability to support the attainment of the student outcomes and to provide an atmosphere conducive to learning.

7.A.1. Offices.

The Chemical Engineering department is housed in a self-contained building (Chemical Engineering Building, CEB) containing 14 offices for faculty, 1 office for teaching assistants, and 1 central office for the administrative staff.

7.A.2. Classrooms.

There are three classrooms in CEB (Rooms 214, 218 and 230). All three are equipped with high-definition video projectors and computers for presentation. Room 214 is a state-of-the-art Computer Teaching Lab equipped with 21 Versatable Revolution Flip Top Computer Consoles for enhanced instruction and problem-based learning in numerical and computer courses.

7.A.3. Laboratory Facilities.

Laboratory facilities including those containing computers (describe available hardware and software) and the associated tools and equipment that support instruction. Include those facilities used by students in the program even if they are not dedicated to the program and state the times they are available to students. Complete Appendix C containing a listing of the major pieces of equipment used by the program in support of instruction.

The ground-floor high-bay houses the Unit Operations Laboratory for CHE 381 and 382. This contains 17 functional chemical engineering process demonstrators that support metrology, digital scales, stirrers, isothermal bath and other equipment. The laboratory is furnished with 11 computers, two laser printers for generating lab reports and includes laptop computer friendly counter running the length of the lab for student use. The second floor computer laboratory features 23 desks and 20 computers, configured as described in the next section.

7.A.4. Additional Facilities.

A former student lounge has been renovated and upgraded into multifunctional and inviting Student Collaboratoreum workspace. This features work/lunch tables, a laptop-friendly "information bar" and a meeting space centered around an Epson Brightlink Pro interactive whiteboard. A library/reading room and a meeting/teleconferencing room are also used by our students. Computers and laboratories are served by small machine and electronics shops on site.

B. Computing Resources

The Chemical Engineering department has 50 computers running Windows 7 Professional for student/teaching use and six laser printers are provided for student use. Additionally, our Faculty Research Laboratories include 39 computer workstations and PC clusters. These machines range from desktop PCs to Unix Workstations to a pair of 8 node LINUX clusters. These computers are utilized for both data analysis as well as simulations.

All student/ teaching computers have installed Aspen, Matlab, Microsoft Office Suite as well as other software specific to classes. The department of Chemical Engineering funds and maintains an exclusive 150 user license for worldwide standard Aspen Chemical Simulation Software. The two computer labs have 16 workstations and computer room accessible to all students has 9 workstations. The state-of -the-art Computer Teaching Lab includes 21 Versatable Revolution Flip Top Computer Consoles. In addition to the above, the students have access to an Epson Brightlink Pro interactive whiteboard and laptop-friendly "Collaboratoreum" workspace.

For advanced computing needs and high speed computing, the students have access to two clusters. Argo is a cluster with 57 computers connected together to perform as a single system with supercomputer-like performance. The department students also have access to the UIC Extreme Computing cluster recently acquired by the university. The UIC Extreme Computing facilities includes 2560 cores and with 30.5TB memory. The individual nodes have Intel(R) Xeon(R) CPU E5-2670 0 @ 2.60GHz, 16 Cores with a cache size of 20MB, 128GB RAM, 1TB on-board hard drive and the installed operating system is CentOS 6.3. The cluster includes 288TB fast scratch communicating with nodes over QDR infiniband, with 1.14PB of raw persistent storage, and large memory compute nodes, each with 32 cores has 1TB RAM giving 31.25GB per core and thus the total adds upto 96 cores and 3TB of memory.

In the Unit Operations Laboratory, there are 17 Functional Chemical Engineering Process demonstrators that support metrology, digital scales, stirrers, isothermal bath and other equipment. The laboratory is furnished with 11 computers, two laser printers for generating lab reports and includes laptop computer friendly counter running the length of the lab for student use.

Aside from computers dedicated to student use in the Chemical Engineering Building (CEB), computer labs are distributed throughout the campus – although the vast majority are concentrated on the east campus (within easy walking distance of CEB and also served by the campus shuttle buses). General information and a schedules for specifc computer labs can be found in the following websites:

http://accc.uic.edu/service/computer-labs

http://accc.uic.edu/lab/all?campus=All&building=All&reserve=1&printers=All&accessible=All&projector=All

Al of these computers are installed with Excel and MATLAB and Microsoft Office 2010/2013, which are the main software packages that Chemical Engineering students would need.

C. Guidance

As part of the College-wide Freshman Engineering Success Program, participating students conduct an 8-week group project that includes usage of library resources for literature research and bibliographic formatting. Similar material is covered during mentoring of ChemE Scholars (recipients of the NSF funded S-STEM scholarship, which started in Fall 2012).

7.C.1. Support Policies.

All computing support policies for computer account usage, network access, are other support policies could be found at http://accc.uic.edu/policy/all

All ChE students, faculty and staff should read and agree with all of these policies before using computing resources.

7.D.2. Accessibility.

ACCC also work closely with the Disability Resource Center to ensure accessibility resources are available to the UIC community. More information can be found following this link:

http://accc.uic.edu/service/learning-environments/accessibility

D. Maintenance and Upgrading of Facilities

7.D.1. Maintenance.

Routine maintenance and an ongoing improvement program for the Chemical Engineering Unit Operations Laboratories (ChE 381 and 382) are conducted on a continuous basis throughout the year. Suggestions for improvement are taken from student assessment (e.g., Senior exit Interview and Oral Exam) and faculty input, and are funded by the student laboratory fee and an internal College grant. It has included the following:

7.D.2. Ongoing Improvements.

Year	Experiment	Status
2010	Fluid flow in pipes	Replaced differential pressure transmitters with new units

2011	Filtration	Refurbished filter unit; machined and installed a new screw top/handle, added downstream pressure gauge
2011	Facilities	Purchased a new digital balance with 0.01 g resolution or more accurate weighing
2011	Ultrafiltration	Purchase handheld portable pH meters and calibration standards for acid/base cleaning procedure
2011	Membrane Separation	Replaced non-functional O2 sensors with new units
2011	Facilities	Purchased a new laser tachometer for accurately measuring the mixer speeds in the heat transfer and liquid-liquid extraction experiments
2012	Fluid flow in pipes	Installed two manual pressure gauges on unit for comparison with differential pressure transmitter readings
2012	Fluid flow in pipes	Replaced faulty main flow valve on unit for better flow control
2012	Liquid extraction	Constructed and installed custom fit acrylic covers on feed/product tanks to minimize acetic acid evaporation
2012	Liquid extraction	Refurbished middle stage impeller motor, tightened control dials on all impeller motors
2013	Distillation	Purchased refractive index standards for refractometer calibration
2013	Humidification	Purchased portable digital hygrometer for measuring humidity of exit air flow out of humidification column and to compare with wet/dry bulb measurements
2013	Packed Bed	Installed new air flowmeter with a smaller flowrate range for more accuracy in air flowrate readings
2013	Filtration	Installed new pressure gauges and gauge guards upstream and downstream of sealed disc filtration unit
2014	Filtration	Purchased a new filtration pump to replace existing worn-out pump; will allow better filtration performance and higher pressure operation (Summer 2014)

7.D.3. Routine and Periodic Maintenance.

Weekly: Students rinse, clean and dry all labware (beakers, flasks, graduated cylinders, buckets, pipettes, test tubes, racks, trays). Students clean up common lab bench areas, especially around scales/balances and drying oven, and put away any equipment used during lab period.
The students shutdown their experiments, clean up the apparatus and bench area, and leave the experiment ready for the next lab team. Water, air, steam, pumps and individual electric power boxes are turned off. Water bath is drained and left open to dry. Lockout is performed on the

distillation column electric power box. Instructor/TA performs an inspection at the end of each lab period to make sure all experiments are shutdown correctly, chemicals are properly stored and equipment is put away.

<u>Semester</u>: Faculty/staff perform a run-through of the lab experiments at the beginning of the semester before the first lab period to make sure the experiments are working properly. An inventory is performed at the end of the semester to determine if lab supplies/chemicals/gases need to be ordered for the next semester. Equipment is periodically cleaned, recalibrated, refilled or replaced by faculty/staff during the semester. For example, capillary tubes in the viscometer experiment are cleaned out with a solvent to remove Carbopol polymer residue left inside. Manometers are checked to make sure they are filled correctly. The boiler is turned on in the morning before the lab commences and is checked periodically to make sure it is working correctly. Safety equipment is maintained in good working order; safety glasses are checked for breakage and new glasses ordered if necessary.

7.D.4. Upgrading Facilities.

Classrooms: All three classrooms in the Chemical Engineering Building are equipped with computers and projectors to facilitate the use of software demonstrations and computer visuals during instruction. We are currently in the process of upgrading the technology in the classrooms to enhance the student's experience. In all three classrooms we are planning to add *Smart Board* technology that will enable instructors to directly annotate their presentations and create animations during lecture. This new technology will allow instructors to illustrate complex and abstract concepts much easier than can be accomplished using the white board. We anticipate that instructors will be able to achieve the desired learning outcomes on a wider group of students, as some students learn most effectively through visualization.

7.D.5. Laboratory Development Plan.

Over the past several years we have had numerous discussions with the Engineering Advisory Board (EAB) and the *Industrial Mentors* in order to solicit input on ways we can improve the laboratory experiments that we offer our students. Feedback from these groups and students suggest that a new laboratory focused on *Process Control* is needed in the Unit Ops Laboratory course. We are actively developing this new laboratory, and project to offer it in the 2014-15 academic year. Funds and space have already been allocated for this laboratory. Central to the *Process Control* laboratory is the purchase of a process control technology. We anticipate acquiring the PCT40 from Armfield. This unit is a multi-function process control teaching system, and is capable of level, flow, pressure, and temperature control loops. Experiments utilizing the PCT40 have not been finalized, but the specifications of the unit are shown below.

Process principles • Calibration of sensors (via PC software) • Inflow control • Outflow control • Direct heating • Indirect heating • Batch and continuous operation • Effect of sensor lag (e.g. thermometer pocket) • Effect of system time constant (e.g. volume change) •

Effect of dead time (e.g. holding tube) • Effect of mixing/stirring (requires PCT41)

Measured variables • Level - on/off switch (fixed hysteresis) • Level - differential on/off switch (adjustable hysteresis) • Level - proportional pressure sensor (slow system response) • Flow - Proportional turbine sensor • Temperature - on/off switch (fixed hysteresis) • Temperature - proportional thermocouple sensors • Static pressure - proportional sensor (fast system response) • Differential pressure - proportional pressure (fast system response) • pH - proportional sensor (requires PCT41 and PCT42)

Controller types • Manual control • Effect of reverse/direct action • On/off output with hysteresis • Time proportioned output with adjustable P, I & D terms and cycle time • PID proportional output with adjustable P, I & D terms • Remote set point control (requires PCT41) • Ratio, cascade and feedforward with feedback loops (requires PCT41)

Controlled variables • On/off flow control via solenoid valves • Time proportioned flow control via solenoid valves • Proportional flow control via pump speed • Proportional flow control via electrical proportioning valve • Proportional flow control via pneumatic proportioning valve (requires PCT44) • On/off heater power control via solid state relay • Time proportioned heater power control via solid state relay • Proportional heater power control via solid state relay

E. Library Services

A public research facility, the University Library at the University of Illinois at Chicago serves an urban campus of more than 25,000 students and 11,000 faculty and staff members, as well as the Chicago metropolitan area and the State of Illinois.

The Richard J. Daley Library roots located at 801 South Morgan on UIC's East Campus is the main library. The Richard J. Daley Library Library serves nine colleges, including Engineering. Patrons have access to more than 2.2 million volumes and 30,000 current journal titles, more than half of which are electronic. 381 journals are included in the category of Chemical Engineering, such as AIChE journal, Chemical Engineering Science, Colloid and Polymer Science, Colloids and Surfaces, Combustion and Flame, Journal o Catalysis.

The Library of the Health Sciences (LHS) located at 1750 West Polk is one of the largest health sciences libraries in the nation and continues to support education, research, and clinical practice. With more than 725,000 volumes and 5,200 journals.

The availability of the books and journals can be checked on the website of the library. The available books can be borrowed from the library and the journal articles can be either copied in the library or downloaded if the e-copies available. If an item is not available from UIC library, it may be accessed by requesting interlibrary loan. The items can be renewed in the library or online after signing up an account.

The faculty members can submit course reserves and recommend a purchase of books or subscription of journals by contacting the reserves staff or filling up the online form.

Besides UIC libraries, the Chemical Engineering has a small library located in our building. The reference books can be used in the library. The library can also be reserved as a study room.

Overall Comments on Facilities

Aside from the above specifics on facilities dedicated to the Chemical engineering program, it is worth mentioning the unique circumstance of the Chemical Engineering department occupying its own, dedicated building. Faculty are literally never more than a few feet away from classrooms, labs and the student lounge, and our student community benefits from a significant esprit de corps.

The departmental Safety Committee (which includes the Professor who teaches the Unit Operations Laboratory and the electronics technician and machinist on staff) regularly assess laboratories and equipment, keep the faculty and departmental administration apprised, and make updates as necessary.

CRITERION 8. INSTITUTIONAL SUPPORT

A. Leadership

The Undergraduate Program in Chemical Engineering is administered by a three member Undergraduate Committee chaired by a faculty member designated as the Director of Undergraduate Studies (DUGS). The DUGS is responsible for the monitoring of the effectiveness of the curriculum to meet the declared objectives and outcomes of each course as well as the overall curriculum. This responsibility also entails oversight of all assessment procedures and coordination The department has implemented a four year cycle to update every course taught in our undergraduate program. Each year a subset of the courses taught is chosen by this committee for evaluation. The committee then conducts student surveys as well as obtains input from other stake holders such as alumni, recruiters, external reviewers (members of the Industrial Advisory Board, Judges of our annual Engineering Expo) and suggests changes in the contents of a class chosen for such evaluation. These changes are then discussed during a Departmental Faculty Meeting, and after discussion and sometimes modifications to the recommendation, changes are approved and become effective from the following academic year. The committee also approves minor modifications in the curriculum of individual students under special circumstances. In addition the committee also approves new courses developed by individual faculty members, before they are submitted to the entire faculty for their approval.

The entire curriculum of the Department is in addition discussed as a whole during the biannual meetings of the Industrial Advisory Board, and regular meetings with Alumni Groups (there were two such formal meetings and several informal meetings in the AY 2013-2014), as well as representatives of organizations that hire significant number of our graduates. If issues concerning either courses or the overall curriculum are raised in these meetings, these issues are handled by the Undergraduate Committee outside the two-year cycle mentioned above. For example, in the AY 2010-2011 during one of such Assessment meetings a suggestion was made to develop an elective course to provide Junior and Senior students extensive exposure and practice in using software packages such as ASPEN. This course has been offered annually the last three years. Another suggestion was to introduce a required class that included an examination and introduction to many software packages such as ASPEN, Microsoft Excel, Maple, Matlab, etc. This course (CHE 205) has now been included in the curriculum as a required class.

B. Program Budget and Financial Support

The institutional budget for the Department of Chemical Engineering consists of two parts, personnel and the operating costs. The personnel costs are those for salaries of faculty, union staff members, academic professionals and teaching assistants. The budgeted amount for these salaries increases each year by an average percentage mandated by the University. These increases have so far been to retain qualified personnel in the department. The budget of the College is currently in part tied to the enrolment in Engineering and the tuition revenue generated by the enrolled students. While the Department budget is not tied directly to enrolments, it is heavily influenced by it. The department has had a healthy increase in student enrolments in the last few years. This has resulted in healthy increases in our operating budgets, as well as allocations for Teaching Assistants (TAs) in the department. In contrast, the budget for the operating costs has not increased for the last few years and due to poor statewide economic conditions and has in fact decreased slightly in the last fiscal year. A fraction of the overhead funds available from research grants and contracts are returned to the Department as discretionary funds. These additional funds have therefore also been used to ensure that the program objectives are met. Some examples of how these overhead funds have been used in meeting program objectives are given below:

(i) Augmenting the Teaching Assistants (TA) Budget in the Department: The allocated TA Budget is often not sufficient to cover all courses requiring TAs. The Department has routinely used overhead funds/gifts to hire additional TAs to maintain the quality of teachings mandated by our program objectives.

(ii) Audio-Visual Resources: The department has used overhead funds to obtain such resources for the classrooms. The Department now has three projectors as well as associated desktop/laptop computers available for presentations, which is sufficient to cover our current and anticipated teaching needs.

(iii) Updating Departmental Computing Services: Overhead funds have been used to obtain LCD monitors for our Computer Rooms as well as high performance computers that can be used efficiently with all the software packages our students are expected to use in their classroom activities (ASPEN, CHEMCAD. Etc.). In addition overhead funds have also been used to purchase site licenses for software needed by our students and faculty.

The current sources of financial support for the Department are the State of Illinois with additional funds available from gifts and overhead charges on research grants. State funding covers generally covers faculty and staff salary, as well as some additional non-permanent positions such as Teaching Assistants and part-time student workers. This portion of the budget is fixed each year by the University -- changes made are generally consistent with the support provided to the University by the State of Illinois. In addition the Department receives a portion of the Indirect Cost Recovery (ICR) from research grants housed in the department, or those in which the department faculty participate. Currently the University returns 50% of the overhead cost to the College of Engineering, and this is then divided equally between the College and the Department. The Department actively works with the Development Office of the College and University to raise gifts from alumni and local and national businesses with an interest in our Department and University. These gifts when they are discretionary are also used to fund special projects in the department including sometimes emergency repairs to equipment in the UG Laboratories that has not been budgeted. This year our development activities have resulted in one major gift and several smaller gifts.

While there have been state budget cuts over a period of time, alternate revenue streams have seen an increase driven by a steady increase in student enrollments and our international programs. This together with careful budgetary management in general by the Dean's office has allowed us to tide over the budget cutbacks and persist in providing a high quality education for the undergraduate students. Indeed, at the time of writing of this report, we are working with the Dean's Office to finalize an equipment procurement package totaling approximately $250K. This will be directed toward upgrading the Unit Operations Laboratory (CHE 381, 382) and enhancing classroom and learning technology. College alumni donor funding for the recently inaugurated Student Collaboratoreum workspace is another example of support for our program. Strong support enables our effectiveness in teaching and learning. The College conducts surveys of teaching and advising every semester, and the Chemical Engineering department usually ranks the highest in these surveys. Another measure of the quality of instruction we have provided to our students is the Annual Engineering Expo. This year as in the past years, our students have won a disproportionately higher share of awards. As an example, in 2014 Chemical Engineering students won the first prize in the Chemical Processes and Chemical Production Methods and Facilities categories.

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C. Staffing

The support staff in the department consists of an Assistant to the Head/Office Manager, an Undergraduate and Graduate Services Coordinator, a Computer Engineer and Workshop Engineer/Supervisor. In addition the department generally hires a graduate student to help the Computer Engineer in maintaining our Student Computer Laboratories. By way of professional development, the Assistant to the Head attends approximately 6-8 university workshops per year and this number is roughly 3-4 for the Undergraduate/Graduate Coordinator. Our staffers have won a number of service and recognition awards.

D. Faculty Hiring and Retention

The Department has been successful with support from the College and the University to recruit several excellent additions to our Faculty. In 2009 the Department hired Professor Belinda Akpa, who has since received an NSF BRIGE Award as well as an NIH award. In 2012 the Department hired Prof Vivek Sharma who is currently active in our Teaching Program and developing his research program. In 2013 the Department hired Professor Brian Chaplin who came to us from Vanderbilt University after a few years of teaching experience. He currently is the recipient of an NSF grant. Professor Vikas Berry, formerly from Kansas State University, will be joining the Department in Fall 2014 as a tenured Associate Professor. In the transitional period he is an Adjunct Professor. Professor Berry is also an NSF CAREER awardee. The department as a result of this support been successful in broadening the research expertise of the department, as well as broadening its teaching programs.

E. Support of Faculty Professional Development

8.E.1. Promotion and Tenure Seminars.

These professional development seminars are hosted by the Office of Faculty Affairs, and coordinated by the vice-provost for faculty affairs (Professor Renee Taylor).

They occur regularly, at several times each year. As an example the schedule for the months of April and May 2014 is listed below.

April 11, 2014
1:00 – 2:25
Junior Faculty
Location: Student Center East 603
This seminar focuses on the concerns of assistant professors and those on Q-contracts, whether tenure or non-tenure track (research, clinical).

April 18, 2014
1:30 – 3:30
Promotion & Tenure Seminar for Underrepresented Faculty
Location: Student Center East 603
This workshop will provide Underrepresented Faculty with an opportunity to discuss specific issues with senior faculty members experienced in the process. This workshop is designed to work in conjunction the April 11th and May 2nd campus seminars, which faculty are also encouraged to attend in order to get general information about the P&T process. Small groups will be facilitated by senior faculty panel members who will share their experience and insight with junior faculty. Junior faculty members are asked to come with prepared questions for the small group discussion.

May 2, 2014
12:00 – 1:00
Promotion & Tenure Seminar for Post Mid-Probation Pre-Tenure Faculty
Location: University Hall 401
This workshop is specifically designed for faculty members who are in T3 or T4, those faculty who will be going through mid-probationary review, and those who recently completed it. Senior faculty will help prepare faculty for upcoming promotion and tenure applications and answer questions. This workshop is designed to work in conjunction the April 11 campus seminar which faculty are also encouraged to attend in order to get general information about the P&T process.

8.E.2. Mentoring Program.

There are university-wide mentoring programs and department-wide mentoring as well. Mentoring will help UIC retain more of the faculty it hires and improve morale and performance generally. The university wide mentoring program supplements the efforts of individual departments to provide assistance and guidance to their probationary faculty members. The Office of Faculty Affairs coordinates the process by which a junior faculty may select a mentor from a list of senior faculty who have volunteered their services. In general junior faculty can ask for and receive guidance on how to get things done within the university and how to chart a productive research path.

The Chemical Engineering department appoints a faculty mentor for all new Assistant Professors as part of the support program. However, this is considered an informal program; the mentor does not have any extraordinary influence on personnel matters concerning the faculty member. The mentor is expected to help the faculty member in writing proposals, papers and any other help he may wish to seek. For mid career faculty a mentor is generally not appointed, but senior faculty members have open discussions with faculty to suggest research topics and upcoming research funding opportunities, to enhance their professional development.

8.E.3. Workshops on increasing teaching effectiveness.

Several workshops are offered each year for professional development in teaching through the Council for Excellence in Teaching and Learning. The Council for Excellence in Teaching and Learning (CETL), composed of faculty members who have been recognized for their demonstrated teaching excellence, provides a mechanism by which faculty, staff, and administrators can work collaboratively toward the improvement of instruction and the advancement of learning in UIC's culturally pluralistic community. Through the development of programs, policies, resources and rewards, CETL strives to encourage every member of the academic community to develop interests, abilities, and achievements as both teachers and learners. CETL offers various programs and resources to support excellent teaching and learning throughout the UIC community including:

- Teaching Recognition Program (TRP) is a program to recognize and reward documented excellence of UIC faculty in instruction.
- Curriculum and Instruction Grants (CIG) is a seed fund to support faculty initiatives designed to enhance the quality of teaching and learning.
- Departmental Teaching Excellence Award (DTEA) recognizes the commitment to teaching and documented instructional excellence of departments and other academic units.

- Resources: provides publications, reports, links, and other instructional resources for faculty, staff, and administrators This Council also provides awards for teaching excellence. The Council seminars are coordinated by the Vice-Provost for Faculty Affairs. Registration is free for these seminars and they are widely publicized. In most cases faculty members decide on their own which seminars would enhance their teaching abilities, while often a colleague will suggest one of these to a junior colleague.

8.E.4. Sabbatical leaves for professional development.

Faculty may apply and expect to receive sabbatical leaves for the purpose of study or research, as recommended by the head of the department and with the concurrence of the Dean of the College of Engineering.

8.E.5. WISEST (Women in Science and Engineering System Transformation) program .

WISEST supports the professional development of women faculty. It provides networking and mentoring venues for women faculty in science, technology, engineering, and mathematics through regularly scheduled lunches, leadership seminars, meetings with Visiting Scholars, and a WISER fund for some bridging funds to maintain continuity of research effort during maternity/infant care periods.

PROGRAM CRITERIA

Describe how the program satisfies any applicable program criteria. If already covered elsewhere in the self-study report, provide appropriate references.

The Chemical Engineering Program Criteria require that the curriculum must:

a.) Provide a thorough grounding in the basic sciences including chemistry, physics, and/or biology, with some content at an advanced level, appropriate to the objectives of the program.

b.) Include the engineering application of these basic sciences to the design, analysis, and control of chemical, physical, and/or biological processes, including the hazards associated with these processes.

Coverage of these areas in the Chemical engineering curriculum was discussed above under Criterion 5. The course work that addresses the program criteria are briefly recapped in this section. Examples of student work will be available during the on-site visit that will demonstrate a working knowledge of these topics. Chemical engineering students begin their chemistry courses during their freshman year with general chemistry (two lecture courses: Chem 112 and 114: 10 credit hours). Organic chemistry (two lecture courses: Chem. 232 and 234: 8 credit hours, and one laboratory course: Chem 233: 1 credit) is typically taken in the sophomore year, while physical chemistry (two lecture courses: Chem 342 and 346: 6 credit hours) and analytical chemistry (one lecture course: Chem 222: 4 credit hours) fall in the junior year.

Chemical engineering courses begin with an introduction to thermodynamics (ChE 201: 3 credit hours) in the fall semester of the sophomore year. This is followed by a computational methods course (ChE 205: 3 credit hours) and the material and energy balance course (ChE 210: 4 credit hours). The former develops mathematical background specifically addressing chemical engineering applications and general problem-solving skills using Excel, VBA, MATLAB and Aspen Plus process simulation. The letter introduces the critical concept of balances as well as basic problem-solving skills to be applied and further developed throughout the curriculum. In the junior year, students take the first two transport phenomena courses (ChE 311 and 312: 6 credit hours) which cover fluid mechanics, and heat and mass transfer; the first chemical engineering laboratory course (ChE 381: 2 credit hours), and chemical engineering thermodynamics (ChE 301: 3 credit hours). Senior level courses complete the major chemical engineering subject areas and complement the design experience. These courses are: chemical reaction engineering (ChE 321: 3 credit hours); the third transport course (ChE 313: 3 credit hours), dealing with separation processes, mostly stage-wise; chemical process control (ChE 341: 3 credit hours), and the second laboratory course (ChE 382). Preparation for engineering practice culminates in the design course sequence (ChE 396 and 397: 7 credit hours) taken in the senior year. In these courses the students demonstrate their ability to independently apply chemical engineering principles, cost estimation and economic evaluation methodologies, flow sheet synthesis and process simulation, optimization techniques, and report writing and oral presentation skills, to contemporary chemical engineering design problems. Content in the above courses is further augmented by a host of technical electives such as Catalytic Reaction Engineering, Pollution Control Engineering, etc. Syllabi for all required and elective courses is given as additional information.

Coverage of hazards associated with chemical processes was discussed in Section 4.B.3. Mandatory student completion of SaChE and relevant class discussion are integrated throughout the Chemical Engineering curriculum – namely in CHE 210, 312, 321, 381, 382, 396 and 397. Student knowledge and proficiency in laboratory and chemical process safety are assessed directly in the Senior Exit Interview and Oral Exam (Assessment Method #2).

APPENDICES

Appendix A – Course Syllabi

A.1. Chemical Engineering Courses.

ABET Course Syllabus – CHE 201
INTRODUCTION TO THERMODYNAMICS

1. Course number and name
   CHE 201. Introduction to Thermodynamics

2. Credits and contact hours
   Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name
   Lewis E Wedgewood

4. Text book, title, author, publisher, year

Fundamentals of Engineering Thermodynamics: 7th Edition (2011)
Moran/Shapiro/Boettner/Bailey

5. Specific course information
   a. Brief description of the content of the course (catalog description)

Work and energy; conversion of energy; theory of gases and other states of matter; applications to energy conversion devices. second law of thermodynamics, entropy, and equilibrium, with applications.

   b. Prerequisites or co-requisites
      Math 181, Calculus II and Phys 141, General Physics I (Mechanics).
   c. Indicate whether a required, elective, or selected elective.
      Required.

6. Specific goals for the course
   a. Performance indicators of student learning.

1	Introduce the basic concepts of thermodynamics.
2	A thorough understanding of the first and second laws of thermodynamics, and their applications to the design of heat engines, refrigerators and heat pumps. Calculation of thermodynamic properties and exposure to data sources for these properties.
3	Understand the application of thermodynamics to evaluate the efficiencies of various processes using the second law.
4	Application of thermodynamics to developing alternate energy sources such as Geothermal Energy, Wind Energy, Solar Energy. Ethical dilemmas and professional principles to address those.
5	Discussion of contemporary environmental issues include green house effect, global warming, carbon dioxide sequestration, etc. and their thermodynamics implications.
6	Non-ideal gas behavior and the calculation of thermodynamic properties for nonideal gases. Introduction to Phase Equilibrium and

   b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level

1	1-6	Ability to apply knowledge of mathematics, science and engineering	4
2	1-4	Ability to identify, formulate and solve engineering problems	3
3	2-6	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
6	2	Ability to analyze and interpret data	2
13	4	Understanding of professional and ethical responsibilities	2

7. Brief list of topics to be covered

Introduction to Work and energy — Week 1
Conservative force, electric, and magnetic fields and potential energy — Week 2
Introduction to the Kinetic theory of the ideal gas, temperature the ideal gas scale — Week 3
Principles of energy transfer as work and heat — Week 4
First law of thermodynamics, Thermodynamic properties and state — Week 5
The concept of equilibrium and its application to thermodynamics — Week 6
Equation of state, phases of matter. Algebraic, graphical, and tabulated equations of state — Week 7
Application of the first law to closed and open systems. Control volume analysis — Week 8
Second law of thermodynamics — Weeks 9/10
The entropy as a state property. Statistical interpretation of entropy.
Thermal and mechanical equilibrium. Gibbs equations — Week 11
Thermodynamics of state. Maxwell relations
First and second law analysis of engineering problems — Weeks 12/13
Energy issues and environmentally related problems — Week 14
Applications with electrical and mechanical systems
Spreadsheet solution of problems — Week 15

ABET Course Syllabus – CHE 205
Computational Methods in Chemical Engineering

1. Course number and name

CHE 205. Computational Methods in Chemical Engineering

2. Credits and contact hours

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Ludwig C. Nitsche

4. Text book, title, author, publisher, year

Readings and/or chapters from numerous sources are provided as PDF downloads each week on the course website. Major sources are listed below.

   a. Other supplemental materials

      Excel Scientific and Engineering Cookbook, D. M. Bourg, O'Reilly, 2006.
      Introduction to Linear Algebra, 4^{th} Ed., Wellesley Cambridge Press, 2009.
      Numerical Meth. for Engineers, 6th Ed., S. Chapra & R. Canale, McGraw-Hill, 2009.
      Elementary Numerical Analysis: Algorithmic Approach, S. D. Conte & C. de Boor, McGraw-Hill, 1980.

5. Specific course information

   a. Brief description of the content of the course (catalog description)

      Computational methods and software relevant to unit operations. Excel spreadsheets (curve fitting, heat conduction), Matlab, Aspen Plus (process simulation), algorithms and object oriented concepts in chemical engineering.

   b. Prerequisites or co-requisites

      Credit or concurrent registration in CHE 201 and MATH 210.

   c. Indicate whether a required, elective, or selected elective.

      Required.

6. Specific goals for the course

   a. Performance indicators of student learning.

1	Use basic Excel cell functionality to tabulate thermophysical property correlations (e.g., Shomate equation) and equations of state (e.g., Redlioch-Kwong) and numerically differentiate and integrate the results.
2	Carry out 1d (linear) and 2d (bilinear) interpolation from data tables (e.g., reduced viscosity chart, Lee-Kessler tables) by hand and with VBA macros programmed from scratch.
3	Carry out basic operations involving vectors and linear algebra (dot and cross products for vectors, matrix-vector and matric-matrix products, determinant and trace of a matrix) by hand and with Excel and MATLAB.
4	Calculate vector coordinates for changes of basis vectors by hand and with Excel.
5	Calculate and plot the Fourier series of an arbitrary function with Excel/VBA and Matlab
6	Formulate a dimensionless, 1d boundary value problem describing reaction and diffusion in a catalyst slab and calculate the Thiele modulus.
7	Solve a constant-coefficient, linear ODE using the characteristic polynomial.
8	Solve ODEs and coupled systems of ODEs With Euler's method and midpoint method.

9	Solve single and coupled nonlinear, algebraic equations using the Goal Seek and Solver functionalities of Excel.
10	Carry out error analysis of data.
11	Solve square and rectangular matrix equations arising in engineering problems with Excel and MATLAB. (Fit data by linear least squares; solve ODEs with least-squares collocation and weighted or constrained boundary conditions).
11	List (by memory) at least 5 provisions in the AIChE code of ethics.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	1-5, 8-10	Ability to apply knowledge of mathematics, science and engineering	4
2	7	Ability to identify, formulate and solve engineering problem	3
3	1-5, 8-10, 12-14	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
6		Ability to analyze and interpret data	2
13	11	Understanding of professional and ethical responsibilities	2

7. Brief list of topics to be covered

Week #1. Data correlations, numerical differentiation, numerical integration, basic operations in Excel.
Week #2. Corresponding states principle. Linear and bilinear interpolation.
Week #3. Introduce VBA and syntax. Automate Lee-Kessler interpolation.
Week #4. Introduction to linear algebra. Vectors: norm, dot product and cross product. Introduce MATLAB.
Week #5. Matrix operations. Transformation of bases.
Week #6. Linear least-squares fitting. Basis functions. MATLAB.
Week #7. Rectangular matrix equations. Fourier series and Fourier projection formula for coefficients
Week #8. Constant coefficient homogeneous ODEs (characteristic polynomial). Least-squares collocation for solving ODE boundary value problems (weighted BCs)
Week #9. Least squares collocation with boundary constraints.
Week #10. ODEs. Euler's and midpoint method. Coupled ODEs. Higher order ODEs. MATLAP ODE solvers.
Week #11. Nonoinear algebraic equations. Goal Seek and Solver functions on Excel. Shooting method for ODEs. Binary chop and VBA.
Week #12. Basic statistics. Student t distribution and confidence intervals. Error Propagation
Week #13. Introduction to Aspen Plus.
Week #14. Pipe flow and friction factor. Inferring the friction factor correlation inside Aspen plus
Week #15. Heat exchangers. HeatX model in Aspen.

ABET Course Syllabus – CHE 210
Material and Energy Balances

1. Course number and name
CHE 210. Material and Energy Balances

2. Credits and contact hours
Credit hours: 4. Contact hours: 3 lecture hours and 1 discussion session per week.

3. Instructor's or course coordinator's name
Belinda S Akpa

4. Text book, title, author, publisher, year
Introduction to Chemical Processes: Principles, Analysis, Synthesis by Regina M Murphy, McGraw-Hill, 1st edition, New York, 2007

5. Specific course information
   a. Brief description of the content of the course (catalog description)
   Material and energy balances applied to chemical systems. Introduction to chemical and physical properties. Introduction to the use of computers for chemical process calculations.
   b. Prerequisites or co-requisites
   Credit or concurrent registration in CHE 201 and CHE 205.
   c. Indicate whether a required, elective, or selected elective.
   Required.

6. Specific goals for the course
   a. Performance indicators of student learning.

1	Synthesize a block flow diagram from a process description
2	Identify and modify reaction pathways to achieve specific production while minimizing waste and occurrence of hazardous compounds
3	Perform a degree of freedom analysis to determine degree of process specification
4	Perform component, element, and total mass (or molar) material balances
5	Quantify and make appropriate use of system performance specifications for reactors, splitters, separators
6	Correctly identify modes of operation for process units (steady-state, transient, batch, continuous) and simplify the general material balance equation appropriately.
7	Perform unit conversions as necessary.
8	Calculate reaction equilibrium constants and determine fractional conversion and selectivity in equilibrium reactions.
9	Use model equations and graphical data to determine composition in vapor-liquid phase equilibria
10	Formulate process analysis problems as a system of linear equations

11	Use tabulated data or model equations to evaluate changes in state properties
12	Perform energy balances to determine basic heating/cooling requirements of a reactive process

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	2-5, 7-12	Ability to apply knowledge of mathematics, science and engineering	4
2	1-6, 10, 12	Ability to identify, formulate and solve engineering problems	4
3	2, 9, 10	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	3
7	1-3, 5, 12	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	2

7. Brief list of topics to be covered
   - Raw materials, reaction path synthesis
   - Generation-consumption analysis
   - Material balance equations
   - Process flowsheeting
   - Degree of freedom analysis
   - Reactor materials balance
   - Stream composition and system performance (conversion, selectivity, yield)
   - Recycle, purge
   - Chemical reaction equilibrium and chemical kinetics
   - Separation technologies
   - Phase equilibria
   - Energy balances, Energy flows
   - Energy data and model equations
   - Energy balances on reactive processes

ABET Course Syllabus – CHE 301
Chemical Engineering Thermodynamics

1. Course number and name
CHE 301. Chemical Engineering Thermodynamics

2. Credits and contact hours
Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name
Randall J. Meyer

4. Text book, title, author, publisher, year
Required text: *Introduction to Chemical Engineering Thermodynamic,* 7th Ed., J. M. Smith, H. C. Van Ness, M. M. Abbott, McGraw-Hill, 2005.

5. Specific course information
    a. Brief description of the content of the course (catalog description)
    Review of first and second laws of thermodynamics; Equations of state; Multi-component systems and phase equilibria; Chemical reaction thermodynamics and reaction equilibria.
    b. Prerequisites or co-requisites
    Credit for CHE 201.
    c. Indicate whether a required, elective, or selected elective.
    Required.

6. Specific goals for the course
    a. Performance indicators of student learning.

1	Formulate an energy balance to describe both open and closed systems for ideal gases using the First law of Thermodynamics
2	Formulate an entropy balance to describe both open and closed systems for ideal gases using a combination of the first and second laws of thermodynamics
3	Choose an appropriate equation of state to examine thermodynamic properties of a non-ideal gas.
4	Calculate enthalpy and entropy changes of simple processes using an appropriate equation of state
5	Use Mollier charts to evaluate thermodynamic properties at a given thermodynamic condition (T, P)
6	Use two component vapor –liquid equilibrium diagrams to understand how vapor phase composition and liquid phase composition are connected
7	Solve Bubblepoint and dewpoint pressure and temperature problems using modified Raoult's law for a two component vapor-liquid system.
8	Identify azeotropic behavior
9	Explain how the chemical properties of a molecule relate to its mixing behavior in solution
10	Use a combination of an equation of state and an activity coefficient model to solve for the partitioning with regard to phase and composition of a two component mixture in vapor/liquid equilibrium
11	Use ASPEN to determine the partitioning with regard to phase and composition of a two component system in vapor-liquid equilibrium.
12	Solve for the extent of reaction at a particular temperature and pressure condition for a single reaction.
13	Use the method of Lagrangian multipliers to evaluate the product distribution in multi-reaction equilibrium.
14	Identify the if a product in a chemical, reaction is the result of thermodynamic or kinetic considerations.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	1-10,12-14	Ability to apply knowledge of mathematics, science and engineering	4
2	1-8,10-13	Ability to identify, formulate and solve engineering problems	4
3	11	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	2
5	3,5,6,11	Ability to conduct experiments	3
7	9,14	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	2

7. Brief list of topics to be covered

Week #1. First Law of Thermodynamics. Definition of thermodynamic properties.
Week #2. Open system and closed system energy Balances. Phase diagrams. Mollier charts.
Week #3. Reversibility. Elementary processes for ideal gases: Isothermal, Isobaric, Isochoric and Adiabatic. Microscopic definition of Entropy
Week #4. Macroscopic definition of Entropy. 2nd Law of Thermodynamics. Entropy changes for elementary processes for ideal gases. Carnot Cycles. Lost Work.
Week #5. Maxwell Relationships. Equations of State.: Van der Waals gases, other cubic equations, Lee-Kesler tables. Theory of Corresponding States.
Week #6. Evaluation of Enthalpy and Entropy changes for processes involving non-ideal gases. Residual properties.
Week #7. Phase Changes. Clausius Clapeyron Equation. Phase Rule.
Week #8. Two Component Vapor Liquid Equilibrium. Raoult's Law. Introduction to Activity Coefficients. Henry's Law.
Week #9. Azeotropes. Flash Separations. Multi-component Vapor Liquid Equilibrium. Partial Molar Properties.
Week #10. Definition of Chemical Potential and Fugacity. Fugacity of a Component in a Mixture. Ideal Solutions. Lewis-Randall Rule. Excess Properties.
Week #11. Relationship between chemical properties and free energy of mixing. Infinite Dilution Activity Coefficients. Complex Activity Coefficient Models (Margules, UNIFAC).
Week #12. Complete solution of vapor-liquid equilibrium using both an equation of state model and an activity coefficient model. Introduction to Aspen as a tool for solving problems in Vapor-liquid equilibrium.
Week #13. Liquid-Liquid equilibrium. Vapor-Liquid-Liquid Equilibrium. Reaction Enthalpy. Introduction to Reaction Equilibrium.
Week #14. Effect of temperature on the free energy of reaction. Entropy vs. Enthalpy driven reactions.
Week #15. Multi-reaction equilibrium. Lagrangian Multipliers. Heterogeneous reaction equilibrium.

ABET Course Syllabus - CHE 311
Transport Phenomena I

1. Course number and name
   CHE 311. Transport Phenomena I

2. Credits and contact hours
   Credit hours: 3.
   Contract Hours: Lectures: MWF, 10:00 - 10:50 am in CEB RM218; Instructor Office Hours: MWF 11:00am – 12:00pm CEB RM211; TA Office Hours: TTh 2-3pm CEB RM228

3. Instructor's or course coordinator's name
   Ying Liu, Office: CEB RM211, Telephone: 312-996-8249, Email: liuying@uic.edu

4. Text book, title, author, publisher, year
   R.B.Bird, W.E. Stewart & E. N. Lightfoot, Transport Phenomena, 2ed Ed., Wiley (2002)

5. Specific course information
   a. Brief description of the content of the course (catalog description)
      Momentum transport phenomena in chemical engineering. Fluid statics. Fluid mechanics; laminar and turbulent flow; boundary layers; flow over immersed bodies.
   b. Prerequisites or co-requisites
      CHE 201; and credit or concurrent registration in CHE 210
   c. Indicate whether a required, elective, or selected elective.
      Required.

6. Specific goals for the course
   a. Performance indicators of student learning.

1	Use the mathematical language to calculate scalar, vector, and tensor operations.
2	Understand the essential physical principles of momentum, heat, and mass transport and definition of viscosity, heat conductivity, and binary diffusivity.
3	Set up differential momentum balance and analyze boundary conditions.
4	Solve constant-coefficient ODEs and apply boundary conditions to determine steady state one-dimensional flow velocity and stress profiles.
5	Solve PDEs by using the methods of separation of variables and combination of variables to result the velocity profiles of the flow flied which has two independent variables.
6	Formulate and solve the mathematical models to show how viscosity of a Newtonian fluid can be measured by various types of viscometers. Analyze the advantage and disadvantage of each viscometer.
7	Apply time-smoothed equations of change for turbulent flow.
8	Analyze the time-smoothed velocity profile near a wall.
9	Understand the physical meaning and mathematical expression of the empirical expressions for the turbulent momentum flux.

10	Understand the definition of friction factor. Use Moody diagram to solve various pipe-design problems.
11	Set up macroscopic mass, momentum, and mechanical energy balances for isothermal flow systems.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	2-8, 11	Ability to apply knowledge of mathematics, science and engineering	4
2	2, 3, 6, 9, 10	Ability to identify, formulate and solve engineering problems	4

7. Brief list of topics to be covered

Week #1: Introduction and mathematical preliminaries
    Scalars, vectors, and tensors—operations involving them
Week #2: Transport coefficients (Chap. 1, 9, 17)
    Fourier's law of heat conduction
    Fick's law of binary diffusivity
    Newton's law of viscosity
Week #3: Shell Momentum balances and velocity distribution in laminar flow
Week #4: Apply shell momentum balances on more representative laminar flows
Week #5: Conservation equations
    Differential equations for mass and linear momentum
Week #6: Apply differential mass and linear momentum balance on representative flow problems
Week #7: Conservation equations for angular momentum* and energy balance
Week #8: Macroscopic balances for mass and momentum
Week #9: Macroscopic balances for mechanical energy
Week #10: Velocity distributions with more than one independent variable
    Time dependence
    Stream function
Week #11: Similarity analysis
Week #12: Introduction of turbulent flow
Week #13: Time-smoothed velocity profile for turbulent flow near a wall
Week #14: Friction factors for flow in a circular tube
Week #15: Friction factors for flow around a sphere

ABET Course Syllabus – CHE 312
TRANSPORT PHENOMENA II

1. Course number and name
   CHE 312. Transport phenomena II
2. Credits and contact hours
   Credit hours: 3. Contact hours: 3 lecture hours per week.
3. Instructor's or course coordinator's name
   Vivek Sharma
4. Text book, title, author, publisher, year
   Required –Transport Phenomena, Revised 2nd Edition by R. B. Bird, W. E. Stewart and E. N. Lightfoot. John Wiley & Sons, Inc (2006).
5. Specific course information
   a. Brief description of the content of the course (catalog description)
      Heat and mass transport phenomena. Heat conduction, convection and radiation. Heat exchanger design. Diffusion. Mass transfer coefficients.
   b. Prerequisites or co-requisites
      CHE 201, CHE 210, CHE205, CHE 311
   c. Indicate whether a required, elective, or selected elective.
      Required.
6. Specific goals for the course
   a. Performance indicators of student learning.

1	Demonstrate knowledge of heat transfer [conduction, convection and radiation] and mass transfer [diffusion, convection]
2	Apply the mass and energy conservation principles and flux laws to model transport processes central to chemical sciences and engineering
3	Solve 1-d, 2-d and 3-d steady state and transient problems involving heat and mass transfer
4	Use the physical and mathematical similarities between the processes of heat and mass transfer to solve new problems "by analogy"
5	Design heat and mass transfer processes and equipment
6	Recognize the significant role played by transport phenomena in daily life and in typical chemical engineering practice

   b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	1-6	Ability to apply knowledge of mathematics, science and engineering	4
2	1-6	Ability to identify, formulate and solve engineering problems	3
3	2-6	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
6	4, 6	Ability to analyze and interpret data	2
13	5-6	Understanding of professional and ethical responsibilities	2

7. Brief list of topics to be covered

I. Heat and mass transfer fundamentals

- A. Introduction including examples from daily life and an overview. Week 1
- B. Flux Expressions Weeks 2-3
  1. Heat conduction (Fourier's Law) Ch 9
  2. Mass diffusion (Fick’s Law) Ch 17
  3. Convection transport, section 9.7
  4. Work by flow, section 9.8
  5. Mass and molar transport, section 17.7
- C. Shell Balances Weeks 4-7
  1. Energy and species balances, Sections 10.1 and 18.1
  2. Boundary Conditions, Sections 10.1 and 18.1
  3. Heat Conduction with a Heat Source, Section 10.2
  4. Diffusion Through a Stagnant Gas Film, Section 18.2
  5. Diffusion with Heterogeneous Reaction, Section 18.3
  6. Diffusion with Homogeneous Reaction, Section 18.4
  7. Cooling Fins, Section 10.7
  8. Heat Conduction through Composite Walls, Section 10.6
  9. Forced Convection, Section 10.8
  10. Free Convection, Section 10.9
  11. Diffusion into a Falling Film, Section 18.5
  12. Falling Film with Solid Dissolution, Section 18.6

II. Solving Problems with the General equations of Change Weeks 8-11

- A. Derivation of the Diffusion Equation, Section 19.1
- B. Derivation of the Energy Equation, Section 11.1
- C. Special Forms of the Energy Eq., Section 11.2
- D. Boussinesq Equation, Section 11.3
- E. Problem Solving, Section 11.4
- F. Nonisothermal, Multicomponent Equations of Change, Section 19.2
- G. Nonisothermal, Multicomponent Fluxes, Section 19.3
- H. Dimensional Analysis, Sections 11.5 and 19.5
- I. Transient Conduction and Diffusion, Section 12.1 and 20.1
- J. Steady Potential Flow of Heat in Solids, Section 12.3

III. Convection Weeks 12-13

- A. Introduction to Convection; define heat transfer coefficient and mass transfer coefficient & the energy equation (14.1-14.5; 15.2, 15.4, 22.1-22.4)
- B. External flows with forced convection (14.2-14.7)
- C. Free (or Natural) Convection
- D. Boiling and Condensation
- E. Design of Heat Exchangers
- F. Macrobalances for multicomponent systems (23.1-23.4)

IV. Radiation (Chapter 16) Week 14-15

ABET Course Syllabus – CHE 313
Transport Phenomena III

1. Course number and name
CHE 313. Transport Phenomena III

2. Credits and contact hours
Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name
Christos Takoudis

4. Text book, title, author, publisher, year

Separation Process Engineering, 3rd edition P. C. Wankat, Prentice Hall 2012.

5. Specific course information
   a. Brief description of the content of the course (catalog description)
   Mass transfer and phase equilibria. Multistage separations; applications in distillation; extraction; absorption and drying.
   b. Prerequisites or co-requisites
   Credit or concurrent registration in CHE 301.
   c. Indicate whether a required, elective, or selected elective.
   Required.

6. Specific goals for the course
   a. Performance indicators of student learning.

1	Design flash distillation by hand and computer calculations.
2	Design distillation systems by hand and computer calculations.
3	Design absorbers and strippers by hand and computer calculations.
4	Design extraction systems by hand and computer calculations.
5	Design membrane separation systems.

   b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	1-5	Ability to apply knowledge of mathematics, science and engineering	4
2	1-5	Ability to identify, formulate and solve engineering problem	4
3	1-5	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
4	1-5	Ability to design experiments	2

7	1-5	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	2
8	1-5	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	2
9	1-5	Knowledge of contemporary issues	2
10	1-5	Recognition of the need for and ability to engage in life-long learning	2
11	1-5	Ability to function on multidisciplinary teams	2
12	1-5	Ability to communicate effectively	3
13	11	Understanding of professional and ethical responsibilities	2

7. Brief list of topics to be covered

Week #1. Introductory Material.
Week #2. Flash Distillation.
Week #3. Binary Distillation.
Week #4. Binary Distillation.
Week #5. Multicomponent Distillation.
Week #6. Multicomponent Distillation.
Week #7. Multicomponent Distillation.
Week #8. Complex Distillation.
Week #9. Complex Distillation.
Week #10. Batch Distillation.
Week #11. Distillation Design.
Week #12. Absorption.
Week #13. Stripping and Extraction.
Week #14. Extraction.
Week #15. Membrane Separation.

ABET Course Syllabus – CHE 321
Chemical Reaction Engineering

1. <u>Course number and name</u>
   CHE 321. Chemical Reaction Engineering

2. <u>Credits and contact hours</u>
   Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name
   Randall J. Meyer

4. <u>Text book, title, author, publisher, year</u>
   Required text: *Chemical Reactor Analysis and Design Fundamentals*, 2nd Ed., J. B. Rawlings and J. G. Ekerdt, Nob Hill Publishing, 2012.

5. <u>Specific course information</u>
   a. <u>Brief description of the content of the course (catalog description)</u>
      Ideal Reactors: batch, stirred tank and plug flow systems; Conversion and yield in multi-reaction networks; Kinetics of homogeneous reactions and surface reactions; Catalysis and Catalytic Reactors; Non-isothermal reactors; Non-ideal reactors and residence time distribution functions.
   b. <u>Prerequisites or co-requisites</u>
      Credit for CHE 210 and CHE 301.
   c. <u>Indicate whether a required, elective, or selected elective.</u>
      Required.

6. Specific goals for the course
   a. Performance indicators of student learning.

1	Write appropriate material balance for each type of ideal reactor: Batch, CSTR, PFR.
2	Compare reactor types and decide which type of ideal reactor is to be used for a given process
3	Perform calculations with regard to reactor size, achievable conversion and selectivity
4	Write a rate expression based upon a reaction mechanism
5	Analyze reactor data to determine the rate law for a reaction
6	Write a MATLAB code to determine the effluent concentrations of products for a multi-reaction system in a ideal reactor
7	Postulate a set of elementary steps that could take place for a given reaction
8	Calculate the Thiele modulus for a catalytic pellet including the effect of shape and reaction order
9	Determine the size of a catalytic reactor where mass transfer may be important.
10	Evaluate the pressure drop for a packed bed reactor
11	Write an appropriate energy balance for each type of ideal reactor: Batch, CSTR, PFR
12	Determine if multiple steady state behavior exists for a CSTR including heat effects

13	Write a MATLAB code that simultaneously solves the energy balance and all relevant material balances to determine the product distribution and temperature as a function of reactor size (or time in the reactor).
14	Explain all relevant safety considerations in the design of a reactor for a highly exothermic reaction
15	Analyze data from an inert pulse tracer to determine the expected reactor performance for a given (non-ideal) reactor.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	1-15	Ability to apply knowledge of mathematics, science and engineering	4
2	1-6,8-15	Ability to identify, formulate and solve engineering problems	4
3	5,6,13	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
6	5,15	Ability to analyze and interpret data	3
7	14	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	2
8	7	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	2

7. Brief list of topics to be covered

Week #1. Reaction Stoichiometry. Definition of Reactor Types. Definition of Conversion, Selectivity and Yield. General Material Balance
Week #2. 1st order and 2nd order Batch Reactors. Constant Pressure vs. Constant Volume Batch Reactors. Reversible Reaction in a Batch Reactor.
Week #3. Reactions in Series in Batch Reactor. Introduction to Continuously Stirred Tank Reactors. Semi-Batch Reactors. Variable Density.
Week #4. Introduce MATLAB. Introduction to Plug Flow Reactors. CSTRs in Series
Week #5. Detailed Comparison of Reactor Types. Recycle Reactors
Week #6. Introduction to Analysis of Kinetic Data. Least Squares parameter fitting. Introduction to rate laws.
Week #7. Introduction to reaction mechanisms. Equilibrium Assumption
Week #8. Quasi-steady state assumption. Langmuir Hinshelwood Mechanisms.
Week #9. Complex Mechanisms. Catalysis. Introduction to Mass Transfer Effects
Week #10. Reaction and Diffusion in a Catalytic Pellet. Thiele Modulus. Effectiveness Factors. Sizing Catalytic Packed Bed Reactors. Effect of Pellet Shape.
Week #11. Catalytic Pellets with Higher order Reactions. Pressure Drop. Generalized Energy Balance for Chemical Reactors.
Week #12. Energy Balance for Non-isothermal Batch Reactors. Energy balance for Non-isothermal Plug Flow Reactors.
Week #13. Energy Balance for CSTRs. Multiple Steady States. Reactor Runaway. Reactor Design Safety Module. Non-Isothermal Catalytic Pellet.
Week #14. Introduction to Non-Ideal Reactors. Dispersion Model for Steady State.
Week #15. Dispersion Model for Pulse Input. Residence Time Distribution Functions.

ABET Course Syllabus – CHE 341
Chemical Process Control

1. Course number and name
CHE 341. Chemical Process Control

2. Credits and contact hours
Credit hours: 3.
Contract Hours: Lectures: MW, 11:00 am – 12:15 pm in CEB RM218; Instructor Office Hours: T Th 11:00 am – 12:00 pm CEB RM211; TA Office Hours: MWF 2-3pm CEB RM228

3. Instructor's or course coordinator's name
Ying Liu, Office: CEB RM211, Telephone: 312-996-8249, Email: liuying@uic.edu

4. Text book, title, author, publisher, year
Process Dynamics and Control (3rd Edition), Seborg, Edgar, Mellichamp, and Doyle, Wiley (2011), ISBN 978-0-470-12867-1

8. Specific course information
   a. Brief description of the content of the course (catalog description)
   Analysis and design of chemical processes and control systems. Feedback controllers. Stability, tuning, and simulation of PID controller bodies.
   b. Prerequisites or co-requisites
   MATH 220 and CHE 312 and CHE 313 and CHE 321.
   c. Indicate whether a required, elective, or selected elective
   Required.

9. Specific goals for the course
   a. Performance indicators of student learning.

1	Develop the ability to mathematically model process temporal response using linearization of ODE's (also touching on PDE's), Laplace transforms, transfer functions and information block diagrams.
2	Recognize the unifying abstract mathematical structure of classes of process units (integrating, first order, second order, etc.) and formulate this in general terms using the formalism of linear algebra and linear ODE systems.
3	Emerge with the ability to calculate time evolution of process units and networks of units by analytical means and numerical programming.
4	Identify and classify various feedback (and some feedforward) control strategies (proportional, integral, differential), recognize the characteristic features of each in closed-loop response, design control strategies and tune controllers toward quantitative specifications of performance (disturbance rejection and set point tracking) and robustness.

5	Emerge with qualitative understanding of frequency response of prototypical process and controller transfer functions, and the quantitative ability to fit transfer functions to time-evolution data using frequency-response methods and linear/nonlinear regression techniques.
6	Analyze the process dynamics consequences of societally relevant technologies.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	1-3	Apply knowledge of mathematics and science in engineering problems	4
2	4, 5	Identify and formulate engineering problems	4
7	4, 6	Ability to design a system, component, or process to meet desired needs within realistic constrains such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	2
8	6	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	2

10. Brief list of topics to be covered

Week #1: Introduction to chemical process control
Week #2: Theoretical models of unsteady state chemical processes
Week #3 and #4: Laplace Transforms and transfer functions
Week #5: Linearization of nonlinear models
Week #6 and #7: Solutions using Laplace transforms – dynamic response
Week #8: Definition of controllers
Week #9: Closed-loop dynamics and stability
Week #10 and #11: PID controller design and trouble shooting
Week #12 and #13: Instrumentation and control system design
Week #14: Frequency response analysis and control system design
Week #15: Problems related to broader societal context

ABET Course Syllabus – CHE 381
Chemical Engineering Laboratory I

1. Course number and name
CHE 381: Chemical Engineering Laboratory I

2. Credits and contact hours
Credit hours: 2. Contact hours: 4 lab hours per week.

3. Instructor's or course coordinator's name
Alan D. Zdunek

4. Text book, title, author, publisher, year
Unit Operations of Chemical Engineering, W.L. McCabe, J.C. Smith, P. Harriott, 7th Edition, McGraw-Hill, 2005.
Other supplemental materials: Provided online in the UIC Blackboard course folder to help aid in successfully preparing and carrying out the experiments. Material includes; Lab report templates, lab presentation templates, laboratory safety information, technical writing guidelines, error analysis methodology, specific articles and files related to the various experiments including vendor information on the experimental apparatus and components, theory and empirical correlations, engineering charts, and material and physical properties.

5. Specific course information
   a. Brief description of the content of the course (catalog description) Heat and momentum transfer operations associated with chemical processes. These include heat exchangers, fluid properties and fluid flow. Technical report writing, computer calculations.
   b. Prerequisites or co-requisites Credit in CHE 312.
   c. Indicate whether a required, elective, or selected elective.
   Required.

6. Specific goals for the course
Provide students with a hands-on opportunity to work with physical equipment in a laboratory environment.
   a. Performance indicators of student learning.

1	Solve complex, open-ended engineering problems creatively, in a team setting.
2	Integrate knowledge and skills from prior chemical engineering courses.
3	Design and conduct realistic experiments, analyze data, draw concrete conclusions and make relevant recommendations.
4	Communicate effort and findings effectively through the written and oral word.
5	Master and apply safety analysis and safety practices in a laboratory environment.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	1-3	Ability to apply knowledge of mathematics, science and engineering	3
2	1, 3	Ability to identify, formulate and solve engineering problems	3
3	1-4	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
4	1-3	Ability to design experiments	4
5	1-3	Ability to conduct experiments	4
6	1, 3-4	Ability to analyze and interpret data	4
11	2-5	Ability to function on multidisciplinary teams	3
12	3-4	Ability to communicate effectively	4
13	5	Understanding of professional and ethical responsibilities	3

7. Brief list of topics to be covered

Specific lab experiments performed are:
1) Viscosity: Non-Newtonian fluids
2) Evaporation
3) Fluid Flow in Pipe Systems
4) Packed Beds
5) Filtration
6) Membrane Separation

List of Topics:
Experimental verification of theoretical principles, familiarization with engineering equipment and exposure to an engineering sense, teamwork skills; participation, leadership and team member assessment, time value; experimental planning, implementation, and completion, experimental skills; design, process safety analysis, data collection, error analysis, written and oral technical communication, integrated problem-solving.
Other topics:
Week #1: General lab safety information, technical writing and technical presentation tips and guidelines.
Week # 2: Industrial Chemical Hygiene Plan review
Week #12-14: SaChE Module: Chemical Reactivity Hazards

ABET Course Syllabus – CHE 382
Chemical Engineering Laboratory II

1. Course number and name
CHE 382: Chemical Engineering Laboratory II

2. Credits and contact hours
Credit hours: 2. Contact hours: 4 lab hours per week.

3. Instructor's or course coordinator's name
Alan D. Zdunek

4. Text book, title, author, publisher, year
   Unit Operations of Chemical Engineering, W.L. McCabe, J.C. Smith, P. Harriott, 7th Edition, McGraw-Hill, 2005.
   Other supplemental materials: Provided online in the UIC Blackboard course folder to help aid in successfully preparing and carrying out the experiments. Material includes; Lab report templates, lab presentation templates, laboratory safety information, technical writing guidelines, error analysis methodology, specific articles and files related to the various experiments including vendor information on the experimental apparatus and components, theory and empirical correlations, engineering charts, and material and physical properties.
5. Specific course information
   a. Brief description of the content of the course (catalog description) Heat and momentum transfer operations associated with chemical processes. These include heat exchangers, fluid properties and fluid flow. Technical report writing, computer calculations.
   b. Prerequisites or co-requisites CHE 381 and concurrent registration in CHE 313.
   c. Indicate whether a required, elective, or selected elective.
      Required.

6. Specific goals for the course
   Provide students with a hands-on opportunity to work with physical equipment in a laboratory environment.
   a. Performance indicators of student learning.

1	Solve complex, open-ended engineering problems creatively, in a team setting.
2	Integrate knowledge and skills from prior chemical engineering courses.
3	Design and conduct realistic experiments, analyze data, draw concrete conclusions and make relevant recommendations.
4	Communicate effort and findings effectively through the written and oral word.
5	Master and apply safety analysis and safety practices in a laboratory environment.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	1-3	Ability to apply knowledge of mathematics, science and engineering	3
2	1, 3	Ability to identify, formulate and solve engineering problems	3
3	1-4	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
4	1-3	Ability to design experiments	4
5	1-3	Ability to conduct experiments	4
6	1, 3-4	Ability to analyze and interpret data	4
11	2-5	Ability to function on multidisciplinary teams	3
12	3-4	Ability to communicate effectively	4
13	5	Understanding of professional and ethical responsibilities	3

7. Brief list of topics to be covered

Specific lab experiments performed are:
1) Distillation
2) Liquid-Liquid Extraction
3) Fluid Flow in Fluidized Beds
4) Ultrafiltration
5) Humidification
6) Heat Transfer: CSTR

List of Topics:
Experimental verification of theoretical principles, familiarization with engineering equipment and exposure to an engineering sense, teamwork skills; participation, leadership and team member assessment, time value; experimental planning, implementation, and completion, experimental skills; design, process safety analysis, data collection, error analysis, written and oral technical communication, integrated problem-solving.
Other topics:
Week #1: General lab safety information, technical writing and technical presentation tips and guidelines.
Week # 2: JT Baker Hazardous Chemical Safety Course; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards, Compressed Gases
Week #9: SaChE Module: Dust Explosion Control

ABET Course Syllabus – CHE 396
Senior Design I

1. Course number and name
CHE 396. Senior Design I

2. Credits and contact hours
Credit hours: 4. Contact hours: 4 lecture/discussion hours per week.

3. Instructor's or course coordinator's name
Jeffery P. Perl

4. Text book, title, author, publisher, year
The adopted text for first and second semesters is *Chemical Engineering Design*, Towler and Sinnott, Elsevier, 2013. Readings and/or chapters from numerous sources are provided as PDF downloads from time to time on the course Blackboard Website. Major sources are listed below.
    a. Other supplemental materials
Engineering Economics, Grant and Ireson, Ronald, 1970
Plant Design and Economics for Chemical Engineers, Peters, Timmerhaus and West, McGraw Hill, 2003
Flow of Fluids through Valves, Fittings and Pipes Pub 410, Crane, 1970
Specialty Presentations on Economics, power point, Aspen Plus, General Engineering Design by numerous external design engineers to support course objectives.
Lectures and material posted to Blackboard when available/allowed

5. Specific course information
    a. Brief description of the content of the course (catalog description)
    Introduction to modern, process design and development, engineering economics, and report writing. Design and cost of equipment relating to materials handling to heat transfer, mass transfer, and reactors

    b. Prerequisites or co-requisites
    CHE 312 and CHE 313 and CHE 321

    c. Indicate whether a required, elective, or selected elective.
    Required.

6. Specific goals for the course
    a. Performance indicators of student learning.

1	State the problem at hand (design basis).
2	Draw a process flow sheet (PFD).
3	Perform material and energy balances.
4	Carry out equipment design calculations.

5	Identify conceptual process control schemes.
6	Conduct an economic, environmental, safety, health and sustainability process review, always in an ethical manner.
7	Determine whether your primary conceptual process design should go to the next level.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	3, 4	Ability to apply knowledge of mathematics, science and engineering	4
2	1, 2	Ability to identify, formulate and solve engineering problems	3
3	3, 4	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	2
6	4	Ability to analyze and interpret data	2
7	5, 6, 7	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	4
8	6	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	3
9	6	Knowledge of contemporary issues	3
10		Recognition of the need for and ability to engage in life-long learning	3
11		Ability to function on multidisciplinary teams	2
12		Ability to communicate effectively	2
13	6	Understanding of professional and ethical responsibilities	3

7. <u>Brief list of topics to be covered</u>

Approach / Purpose / Scoping.
Material and energy balances.
Equipment calculations: unit operations, reaction engineering, hand calculations, computational design methods.
Engineering economics: net present value, internal rate of return, depreciation, equipment sizing, 6/10 rules, total installed cost, factored estimates, simple breakeven, stage gates.
Utilties
Environment, safety and occupational health.
Instrumentation, controls and process safety.
Pilot studies: scale-up, identifying problems.
Theory versus practice: efficiency, fouling, scaling, plugins, mixing, mass transfer.
Sustainable engineering: design for environment, energy optimization, raw material optimization.
All-discipline approach to detail design engineering.
Final design engineering and drawings.
Construction, start-up and operations.
Engineering ethics.

ABET Course Syllabus – CHE 397
Chemical Engineering Design II

1. Course number and name

CHE 397: Chemical Engineering Design II

2. Credits and contact hours

Credit hours: 3. Contact hours: 3 lecture hours per week on separate two days.

3. Instructor's or course coordinator's name

Jeffery P. Perl

4. Text book, title, author, publisher, year

The adopted text for first and second semesters is *Chemical Engineering Design*, Towler and Sinnott, Elsevier, 2013. Readings and/or chapters from numerous sources are provided as PDF downloads from time to time on the course Blackboard Website. Major sources are listed below.

a. Other supplemental materials

Engineering Economics, Grant and Ireson, Ronald, 1970

Plant Design and Economics for Chemical Engineers, Peters, Timmerhaus and West, McGraw Hill, 2003

Flow of Fluids through Valves, Fittings and Pipes Pub 410, Crane, 1970

Specialty Presentations on Economics, power point, Aspen Plus, General Engineering Design by numerous external design engineers to support course objectives.

Lectures and material posted to Blackboard when available/allowed

5. Specific course information
    a. Brief description of the content of the course (catalog description)
    Objective is to develop methods and procedures for mounting preliminary stage gate I process design initiatives including economic evaluation of alternatives
    b. Prerequisites or co-requisites
    Credit in CHE 396
    c. Indicate whether a required, elective, or selected elective.
    Required
6. Specific goals for the course
    a. Performance indicators of student learning.

1	Effective coordination with other sub-block teams of overall class design and intra-group collaboration.
2	Effective interaction with outside industry mentor with professional design experience
4	Formulate and articulate a Process Design Statement and Basis for the Team Block
5	Develop Block and Process Flow Diagrams to support stated Design Basis
6	Perform fully developed and closed material and energy balances around assigned elements
7	Conduct a Net Present Value engineering economic assessment to support all designs
8	Environment, Safety and Occupational Health review
9	Intermediate design including conceptual process control, plant layouts for operation/safety, all supporting equipment size and cost calculations
10	Present intermediate process design project reviews as well as final design and recommendations
11	Successful Completion of AIChE SAChE Modules assigned for ChE 397
12	Final Oral and Written Presentation before class, faculty and external industry mentors

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	6	Apply knowledge of mathematics and science in engineering problems	4
2	4, 5	Identify and formulate engineering problems	3
3	6	Use techniques, skills, and modern chemical engineering tools	4
6		Analyze and interpret data	2
7	4—9, 11	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	4
8	8	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	3
9	8	Knowledge of contemporary issues	3
10	8, 11	Recognition of the need for and ability to engage in life-long learning	3
11		Ability to function on multidisciplinary teams	4
12	10, 12	Ability to communicate effectively	4
13	8, 11, 12	Understanding of professional and ethical responsibilities	3

7. Brief list of topics to be covered

At the end of each of the following 5 periods, student teams all make a 30 minute presentation revolving around the stated periodical assignments:

Week #1-3 **Review Projects & Course Objectives**
- Develop Design Basis
- Examine Competing Processes
- Commence Conceptual Proc Block Flow
- Report Outlines Established

Week #4-6 **Review and Realign Process Units**
- Develop Flow Sheets
- Commence Material and Energy Balances
- Process Flow Sheet Frozen
- Compile Supporting Design Data
- Use Initial Hand Calculations
- Start Rough Economics

Week #7—9 **Preliminary Design Presentation**
- Improved Flow Sheeting
- Identify all Energy Sinks/Loads and Streams
  ID Stream #'s & components etc.
- Start Equipment Sizing
- Initial Costs using ASPEN Process Estimator

Week #10-12 **Presentations in Aspen Plus-Initial**
- PFD
- Initial Control Scheme
- Plant Layout (General Arrangements)
- Calculations
- Refined Economics
- Draft Final Written Report

Week #13-15 **Final Report Preparation and Presentation**
- Final Report
- Summary of Stage Gate recommendations

ABET Course Syllabus – CHE 422
Biochemical Engineering

1. Course number and name
   CHE 422. Biochemical Engineering

2. Credits and contact hours
   Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name
   Belinda S Akpa

4. Text book, title, author, publisher, year
   Introduction to Bioprocess Engineering: Basic Concepts by Michael L Shuler and Fikret Kargi, Prentice Hall, 2nd edition, Jersey, 2002

5. Specific course information
   a. Brief description of the content of the course (catalog description)
      Enzyme-catalyzed and microbially-mediated processes. Free and immobilized enzymes. Batch and continuous cell cultures. Transport phenomena in microbial systems and fermentation processes. Design of biological reactors.
   b. Prerequisites or co-requisites
      Consent of the instructor.
   c. Indicate whether a required, elective, or selected elective.
      Elective.

6. Specific goals for the course
   a. Performance indicators of student learning.

1	Identify structure and function of important biomacromolecules
2	Identify and distinguish between types of cells
3	Derive kinetic expressions from proposed enzymatic reaction mechanisms
4	Determine kinetic parameters from experimental data
5	Identify and characterize modes of enzyme inhibition
6	Integrate cell growth models with reactor operating equations
7	Analyze performance of bioreactors operating in different modes and configurations
8	Use MATLAB to simulate operation of bioreactors. Identify optimal operating conditions and limits on operating parameters.
9	Apply bioprocess engineering concepts in a critical reading of the biochemical engineering literature

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	3-8	Ability to apply knowledge of mathematics, science and engineering	4
2	7-8	Ability to identify, formulate and solve engineering problems	3
3	4, 5, 7, 8	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
6	4, 7, 8, 9	Ability to analyze and interpret data	3
7	8	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	2

7. Brief list of topics to be covered

- Biology basics
  - Structure and function of key biomacromolecules
  - Molecular genetics and protein synthesis
  - Recombinant DNA and genetic engineering
  - Basic microbiology and structure of microorganisms
- Principles of enzyme catalysis
  - Kinetics of single substrate reactions
  - Enzyme inhibition
  - Enzyme denaturation and inactivation
- Methods of enzyme and cell immobilization
  - External and internal mass transfer effects on immobilized enzyme kinetics
- Unstructured models of microbial growth
- Bioreactors
  - Continuous stirred tank, plug flow, packed bed, and fed-batch
  - Chemostats in series

ABET Course Syllabus – CHE 431
Numerical Methods in Chemical Engineering

1. Course number and name

CHE 431. Numerical Methods in Chemical Engineering

2. Credits and contact hours
   Credit hours: 3 (undergraduate) / 4 (graduate). Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name
   Ludwig C. Nitsche

4. Text book, title, author, publisher, year
   Numerical Computation in Science and Engineering, 2nd Ed., C. Pozrikidis, Oxford University Press (2008).
   a. Other supplemental materials
      Numerical Recipes, 3rd Ed., W. H. Press, S. A. Teukolsky, W. T. Vetterling, B. Flannery, Cambridge University Press (2007).
      Selected readings and papers drawn from the numerical and engineering literature.

5. Specific course information
   a. Brief description of the content of the course (catalog description)
   b. Introduction to the application of numerical methods to the solution of complex and often non-linear mathematical problems in chemical engineering. Includes methods for the solution of problems arising in phase and chemical reaction equilibria, chemical kinetics, and transport.
   c. Prerequisites or co-requisites
   d. Graduate or advanced undergradate standing.
   e. Indicate whether a required, elective, or technical elective.
      Technical elective.

6. Specific goals for the course
   a. Performance indicators of student learning.

1	Explain the concept and benefits of object-oriented programming (OOP).
2	Formulate and program a Fortran object (type with bound procedures) to evaluate a property correlation or root-finding formula for a cubic equation of state.
3	Write basic routines for linear algebra: (i) Gaussian elimination for a full mtrix; (ii) iterative solution using compacted storage for a sparse matrix.
4	Solve 1d and 2d thermal conduction problems using finite differences and successive over-relaxation.
5	Program 1d and 2d interpolation routines for arbitrary data: (i) linear/bilinear/trilinear interpolation; (ii) cubic slines.
6	Program numerical differentiation and integration by various methods: (i) uneven data; (ii) trapezoid/Simpson's rule; (iii) Monte Carlo method.
7	Solve coupled ODEs by Euler's method, midpoint method and Runge-Kutta methods and apply them to chemical engineering problems such as chemical kinetics and reaction networks.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	2, 3, 5, 6, 7	Ability to apply knowledge of mathematics, science and engineering	4
2	4, 7	Ability to identify, formulate and solve engineering problems	4
3	1	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
6	5	Ability to analyze and interpret data	2

7. Brief list of topics to be covered

Week #1. Compiling and running Fortran codes; physical property correlations; introduction to object-oriented programming.
Week #2. Object-oriented programming concepts; arrays in Fortran; linear (bilinear, trilinear) interpolation.
Week #3. Linear algebra; matrix operations; determinants; matrix inverse; Gramm-Schmidt orthogonalization; banded matrices; operations with sparse matrices (packed versus rolled storage).
Week #4. LU decomposition; QR decomposition.
Week #5. Linear algebraic systems, null space, eigenproblems; principal vectors and Jordan normal form.
Week #6. Applications of linear algebraic systems; thermal conduction problems and finite differences; tridiagonal systems; nearly easy systems; row operations and Gaussian elimination.
Week #7. Iterative methods for linear equations; Jacobi method; Gauss-Seidel method; Successive overrelaxation.
Week #8. Minimization and residual methods for linear systems; steepest-descent search; conjugate gradients.
Week #9. Interpolation (Newton, Lagrange, Hermite); cubic splines; numerical differentiation; Interpolation in two variables; numerical partial differentiation.
Week 10. Numerical integration; error function; elliptic integrals; trapezoid and Simpson's rules; Gauss-Legendre quadrature; area integrals; Monte Carlo methods.
Week #11. Numerical algebraic equations #1 of 3: mathematical and physical context; formalism and general procedures; binary chop method.
Week #12. Numerical algebraic equations #2 of 3: fixed point iteration; Newton's method; secant method; Muller's method; regula falsi.
Week #13. Numerical algebraic equations #3 of 3: Newton's and related methods for systems of Equations; roots of polynomials; continuation for nonlinear equations with a parameter; applications in process control.
Week #14. Numerical solution of ODEs; problem formulation and physical context; initial-value problems; Euler's method, 2nd, 3rd and 4th order Runge-Kutta methods; implicit methods and stability.
Week #15. Optimization; relation to nonlinear algebraic systems; Newton's method; equality Constraints and Lagrange multipliers; inequality constraints and slack variables.

ABET Course Syllabus – CHE 433
Process Simulation with Aspen Plus

1. Course number and name

CHE 433. Process Simulation with Aspen Plus

2. Credits and contact hours
   Credit hours: 3 (undergraduate) / 4 (graduate). Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name
   Ludwig C. Nitsche

4. Text book, title, author, publisher, year
   Readings and/or chapters from numerous sources are provided as PDF downloads each week on the course website. Major sources are listed below.
   a. Other supplemental materials
      Sections from the *Aspen Plus User's Manual* and tutorials.
      Conceptual Design of Chemical Processes (Chapter 8: Heat-Exchanger Networks), J. M. Douglas, McGraw-Hill (1988).
      Distillation Design and Control Using Aspen Simulation, W. L. Luyben, Wiley (2006).

5. Specific course information
   a. Brief description of the content of the course (catalog description)
      Application of Aspen Plus to design, modeling and simulation of process flow sheets. Property models, unit operations, heat integration and pinch analysis, electrolytes, nonconventional solids (e.g., coal), computational aspects.
   b. Prerequisites or co-requisites
      CHE 312 and CHE 313 and CHE 321; or consent of the instructor.
   c. Indicate whether a required, elective, or technical elective.
      Technical elective.

6. Specific goals for the course
   a. Performance indicators of student learning.

1	Select and justify choices for property models required for accurate flow sheet predictions.
2	Design and model unit operations such as chemical reactors, heat exchangers, distillation and other separation columns, pumping and piping networks, etc.
3	Model electrolyte systems such as acid-base reactions and salt precipitation.
4	Students will be able to apply pinch analysis and systematic methods of heat exchanger network design to heat integration.
5	Model nonconventional solids, as are involved in drying, desulfurization and combustion of coal.
6	Students will be able to make quantitative estimates of carbon footprint and other environmental ramifications of chemical process flow sheets, and design for sustainability.
7	Gain facility with computational aspects such as design specs, calculator blocks, sensitivity analyses, equation-oriented modeling and flow sheet convergence.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	1, 3	Ability to apply knowledge of mathematics, science and engineering	4
2	2	Ability to identify, formulate and solve engineering problems	4
3	1-5, 7	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	4
7	2, 4	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	3
8	6	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	3
9	6	Knowledge of contemporary issues	3

7. Brief list of topics to be covered

Week #1. Introduction to Aspen GUI, basic features and input, instructional meta-language.
Week #2. Survey of uit operation models, flow sheet sections and property methods, Refrigeration and heat pump cycles, environmental impact of common Refrigerants.
Week #3. Aspen property database and thermodynamic methods, corresponding states Principle, corelations for vapor pressure and heat capacity, thermodynamics of vapor-liquid equilibrium, Raoult's law and Henry's law.
Week #4. Design specifications, calculator blocks.
Week #5. Vapor-liquid equilibrium of ternary mixtures, distillation, McCabe-Thiele diagram, Underwood equation.
Week #6. Aspen RadFrac column model, key components.
Week #7. RGibbs and REquil reactor models, introduction to kinetic reactors.
Week #8. Processes with solids (conventional inert versus nonconventional solids), drying and combustion of coal.
Week #9. Nonconventional solids, problems involving proximate, ultimate and sulfur analyses.
Week #10. Pinch technology, problem table algorithm, composite curves, path diagrams.
Week #11. Design of heat exchanger networks
Week #12. Software for design of heat exchanger networks, HEN with heat pumps.
Week #13. Detailed design of heat exchangers, HeatX model in Aspen Plus
Week #14. Equation-oriented modeling, convergence.
Week #15. Modeling electrolyte systems, sweetening of sour water and sour gas.

ABET Course Syllabus – CHE 494
Special Topics: Fundamentals of Electrochemistry

1. Course number and name

CHE 494. Special Topics: Fundamentals of Electrochemistry

2. Credits and contact hours

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Brian P. Chaplin

4. Text book, title, author, publisher, year

(Required) Bard, A. J.; Faulkner, L. R., *Electrochemical Methods: Fundamentals and Applications*. Second ed.; John Wiley and Sons: 2001; p 833.

   a. Other supplemental materials

   Bockris, J. O. M.; Reddy, A. K.; Gambra-Aldeco, M., *Modern Electrochemistry*. 2nd ed.; Plenum Press: New York, 2000.

   Orazem, M. E.; Tribollet, B., *Electrochemical Impedance Spectroscopy*. John Wiley and Sons: 2008; p 525.

   A. Lasia, Electrochemical Impedance Spectroscopy and Its Applications, In: *Modern Aspects of Electrochemistry*, B. E. Conway, J. Bockris, and R.E. White, Eds., Kluwer Academic/Plenum Publishers, New York, 1999, Vol. 32, p. 143-248.

5. Specific course information

   a. Brief description of the content of the course (catalog description)

   Introduction to the fundamentals of electrochemistry and its application in a variety of technologies (i.e., batteries, fuel cells, electrolysis cells). Includes methods for the analysis of cells using electrochemical techniques.

   b. Prerequisites or co-requisites

   Graduate or advanced undergraduate standing.

   c. Indicate whether a required, elective, or selected elective.

   Elective.

6. Specific goals for the course

   a. Performance indicators of student learning.

1	Use basic Excel cell functionality to tabulate and graph electrochemical data derived from cyclic voltammetry and electrochemical impedance spectroscopy experiments, and interpret with mathematical models
2	Solving differential equations by the Laplace transform technique, with application to the diffusion equation
3	Applying Stirling's formula to derive the diffusional chemical potential of the intercalation of Li ions at a cathode using a lattice gas model

4	Using complex variable notation to derive equivalent circuit model fits of experimental electrochemical impedance spectroscopy data and compare to computer generated fits using Gamry software
5	Prepare a research proposal that identifies a key limitation in an electrochemical technology, reviews the "state of knowledge" related to the electrochemical technology, and proposes electrochemical and/or theoretical experiments that address this key limitation
6	Prepare an oral presentation that summarizes and defends #5
7	List (by memory) at least 5 provisions in the AIChE code of ethics.

b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	1-6	Ability to apply knowledge of mathematics, science and engineering	4
2	5,6	Ability to identify, formulate and solve engineering problems	4
3	1,4	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	2
6	1, 4	Ability to analyze and interpret data	2
12	5,6	Ability to communicate effectively	4
13	7	Understanding of professional and ethical responsibilities	2

7. Brief list of topics to be covered

Week #1. Introduction and overview of electrochemical cells and electrode processes

Week #2-3. Thermodynamics of electrochemical cells

Week #4-6. Kinetics of electrochemical reactions

Week #7. Mass transfer in electrochemical systems

Week #8. Adsorption and double-layer structure

Week #9-11. Electrochemical methods including potential step and sweep methods and controlled current techniques

Week #12-13. Electrochemical impedance spectroscopy of electrochemical cells

Week #14. Industrial electrochemical processes including the chlor-alkali process, electrowinning, adiponitrile process, and water treatment

Week #15. Student presentations

ABET Course Syllabus – CHE 499
Professional Development Seminar

1. Course number and name

CHE 499. Professional Development Seminar

2. Credits and contact hours

Credit hours: 0.

Contact hours: 4 lecture/discussion/seminar hours during the semester plus Senior Exit Interview and Oral Exam.

3. Instructor's or course coordinator's name
Ludwig C. Nitsche

4. Text book, title, author, publisher, year

None.

a. Other supplemental materials

Downloads from course website.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Students are provided general information about their roles as UIC Chemical Engineering alumni in society and the role of the University in their future careers. Students provide evaluations of their educational experience in the Chemical Engineering Department.

b. Prerequisites or co-requisites

Open only to seniors; and approval of the department. Must be taken in the student's last semester of study.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

   a. Performance indicators of student learning.

1	Effective researching of career prospects.
2	Reflection on educational experiences at UIC and completion of Senior Exit Survey
3	Assessment of Student Outcomes and feedback on Senior Exit Interview and Oral Exam.

   b. Student Outcomes covered by course and measured by Performance Indicators.

SO	PI	Description	Level
1	3	Ability to apply knowledge of mathematics, science and engineering	2
2	3	Ability to identify, formulate and solve engineering problems	2
3	3	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	2
7	3	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	2
8	3	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	2
9	3	Knowledge of contemporary issues	2
10	1—3	Recognition of the need for and ability to engage in life-long learning	2
11	3	Ability to function on multidisciplinary teams	2
12	3	Ability to communicate effectively	2
13	3	Understanding of professional and ethical responsibilities	2

7. Brief list of topics to be covered

Managing technical information, relevant software, and basics of web communication and publishing.

Intellectual property and related career opportunities (Ashley Romano, UIC Office of Technology Management).

Engineer career preparation (Michel Reeve, Kraft Foods Group).

Careers in instrumentation and process control (Dima Hart, UOP LLC).

Senior Exit Survey and Senior Exit Interview and Oral Exam.

A.2. Other Engineering Courses.

1. Course number and name	**CME 260 Properties of Materials**
2. Credits and contact hours	3 credits, 3 - 1 hour lectures per week
3. Instructor's or course coordinator's name	J. Ernesto Indacochea, PhD **Prepared:** September 23, 2013
4. Textbook title, author, yr	*Materials Science and Engineering – An Introduction, 9th Edition;* William D. Callister and David G. Rethwisch.
5. Specific course information	
a. Brief description of the content of the course (catalog description)	Introduction to the relationships between composition and microstructure; correlation with physical and mechanical behavior of metals, ceramics, and polymers. Manufacturing methods. Service performance. Materials selection.
b. Prerequisites or co-requisites	CHEM 112 (General Chemistry I); MATH 181 (Calculus II); and PHYS 141 (General Physics I).
c. Required/elective/selective	Required
6. Specific goals of the course	
a. Specific outcomes of instruction.	**Course Objectives:** Familiarize the student with the multiplicity of materials, their corresponding structures and properties. Introduce them to the processing methods and their correlation to the product/material characteristics and performance. **Educational Outcomes:** Students will recognize the performance and properties of the different materials based structure and chemical make-up; students will be able to select the proper sustainable material based on application needs and environment. **Assessment criteria:** Using a distribution of skills and learning outcomes: reading comprehension, homework/quizzes to develop skills and applications, examinations (testing both concept breadth and depth). Quizzes 22%; Midterm exams 51%; Final 27%.
b. Student outcomes in Criterion 3 addressed by course.	See Accompanying Table
7. Brief list of topics covered	1. Atomic and Crystalline Structure of Materials 2. Imperfections in Solids and Dislocations 3. Mechanical Properties of Metals and Failure 4. Diffusion and Thermal Processing. 5. Phase Diagrams and Phase Transformations. 6. Structures and Properties of Ceramics and Polymers 7. Composite Materials; Corrosion of Materials.

Important Notes to Instructors: CME 260

The CME BSCE undergraduate curriculum process is dependent on each required course covering Student Outcomes 1-12 at the levels indicated. CME faculty members have agreed to these levels for all instructors. Student learning in these areas will be assessed as described in the CME Departmental Self Study.

Program Student Outcomes:	**Coverage**	**Comments to Instructors**
1. *Apply knowledge of mathematics and science in engineering problems*	**EH**	Homework, quizzes and exams emphasize problem solving
2. *Design and conduct experiments, analyze and interpret data*	**H**	Requires evaluating materials properties and experimental data to solve design problems
3. *Design Civil Engineering systems within realistic constraints*	**L**	Some design problems look at materials selection for some Civil Engineering systems
4. *Function effectively in multidisciplinary design teams*	**L**	
5. *Identify, formulate and solve engineering problems*	**EH**	Equations are provided to identify its use in homework, quizzes and exams
6. *Understand their professional and ethical responsibilities*	**L**	
7. *Communicate in a professional and effective manner*	**L**	
8. *Understand the economic, environmental and societal impact of engineering solutions*	**M**	Students are exposed to the impact of materials processing on the environment
9. *Recognize the importance and need to engage in life-long learning*	**L**	Not specifically covered
10. *Comprehend the significance of contemporary issues*	**M**	Green engineering discussed in the context of the use of alternate recyclable materials.
11. *Use techniques, skills, and modern engineering tools for efficient practice of Civil Engineering*	**M**	Use internet to explain lab tests and materials processes; textbook multimedia resources available for active learning
12. *A majority of the graduates should pass the FE exam*	**H**	Topics covered in FE exam are discussed tested

ECE 210

1. Course number and name
ECE 210, Electrical Circuit Analysis

2. Credits and contact hours
3 credit hours, 5 contact hours/week (2 lecture, 3 laboratory)

3. Instructor's or course coordinator's name
Kimberly Fitzgerald, Department of Electrical and Computer Engineering

4. Text book, title, author, and year
Fundamentals of Electric Circuits, 5th Edition, by Charles K. Alexander and Matthew N. O. Sadidu , McGraw-Hill Companies, Inc., 2013.

4a. Other supplemental materials
None.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Linear circuit analysis: networks, network theorems, dependent sources, operational amplifiers, energy storage elements, transient analysis, sinusoidal analysis, frequency response, filters.

5b. Prerequisites or co-requisites
PHYS 142, General Physics II, Electricity and Magnetism; credit or concurrent registration in MATH 220, Introduction to Differential Equations.

5c. Indicate whether a required, elective, or selected elective
Required for some non-ECE undergraduate students.

6. Specific goals for the course
6a. Specific outcomes of instruction
This course has two main objectives: to introduce fundamental circuit theorems and to introduce circuit analysis methods and strategies in the time and frequency domains. Laboratory experiments, in addition to introducing electronic measurement techniques, have been designed to verify theorems and circuit analysis methods.

6b. Explicitly indicate which of the student outcomes are addressed by the course
a. an ability to apply knowledge of mathematics, science, and engineering - Use of complex numbers, linear algebra, calculus, differential equations; above math applied with fundamental physical laws and theorems to solve real electric circuit problems.

b. an ability to design and conduct experiments, as well as to analyze and interpret data - Experiments are performed to obtain data used to verify theorems and analysis methods.

e. an ability to identify, formulate, and solve engineering problems - Major point of the course is to solve electrical engineering circuits.
f. an understanding of professional and ethical responsibility - Solutions must be accurate to have confidence in system behavior.
g. an ability to communicate effectively - Students must each write a number of well-organized and informative lab reports.
h. the broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
i. a recognition of the need for, and an ability to engage in life-long learning - Solution strategy techniques will apply to study of new technological developments.
j. a knowledge of contemporary issues - Some discussion of electronics permeating contemporary life.
k. an ability to use the techniques, skills, and modern engineering tools necessary for engineering practice - (1) Students learn how to use software for: (i) circuit simulation, (ii) solutions of linear systems with real and complex numbers (2) Students must develop skill at using scientific calculators for the solution of linear systems with real and complex number solutions.

7. Brief list of topics to be covered

1. Basic electric circuit variables and elements, Ohm's Law, current vs. voltage characteristics.
2. DC power and power balance circuit solution strategy.
3. Kirchhoff’s Laws and basic analysis strategy.
4. Transformations to equivalent circuits (series, parallel resistors and sources).
5. Voltage and current division rules, double dividers.
6. Circuits with dependent sources.
7. Superposition principle.
8. Thevenin's and Norton's Theorems, source transformations, maximum power transfer.
9. Formal Methods (mesh current and node voltage).
10. Capacitor and Inductor time domain characteristics, RC and LR transient analysis.
11. Review of signals (sinusoidal, exponential, etc.).
12. Review of complex numbers and complex arithmetic.
13. AC steady state, phasors, impedances, frequency characteristics of circuits.

1. Course number and name	**ENGR 100 Engineering Orientation**
2. Credits and contact hours	1 hour seminar/discussion/lecture per week
3. Instructor's or course coordinator's name	Chris Kuypers **Prepared:** June 25, 2014
4. Textbook title, author, yr	None
a. Supplemental materials	GYST Sheets – Course work organizational schedule to enhance studying for best performance-adaptable for computer; COURSE PLANNER-semester-by-semester fill in chart to complement Degree Audit Report ENGINERERING 100 Facebook Page for enhanced communication BLACKBOARD – announcements, Power point presentations, sample resumes
5. Specific course information	
a. Brief description of the content of the course (catalog description)	A general orientation course on careers in the engineering profession. Discussion of college advising procedures. Required of all engineering students. Satisfactory/Unsatisfactory grading only. No graduation credit. Should be taken in the first semester after acceptance into the College of Engineering.
b. Prerequisites or co-requisites	Admission to the College of Engineering
c. Required/elective/selective	Required
6. Specific goals of the course	
a. Specific outcomes of instruction.	**Course Objectives:** 1) Develop baseline knowledge of the engineering profession; 2) Establish a critical and ongoing connection with the College of Engineering and departments; 3) Guide incoming freshmen and transfer students through their first semester in the College of Engineering; 4) Provide enhanced communication utilizing social media and networking opportunities via guest speakers. **Educational outcome:** Students become familiar with Engineering education, the engineering profession, the various fields of engineering covered by the COE, engineering professional societies, interviews and resumes, ethics, the role of alumni, academic advising and the senior design competition **Assessment criteria:** Attendance; resume assignment
b. Student outcomes in Criterion 3 addressed by course.	See accompanying table
7. Brief list of topics covered	1. General Information on class/syllabus discussion 2. Role of Engineering in Society 3. Navigating the Engineering College rules 4. Grading systems (deficit points, probation) and more 5. The Fields of Engineering presentation 6. Your degree and the work world: 7. The Professional Engineers License & procedure 8. Resume discussion 9. Guest: Rotating guest from industry 10. Interviewing skills for Career Fair 11. Engineering Ethics overview

	12. ATTEND ENGR CAREER FAIR 13. Experience the employers 14. Students and Internships (Part 1) 15. Alumna – Rotating UIC alumni 16. Internships and their role in job acquisition 17. Transitioning to the professional employee 18. Engineering Advising expectations 19. Academic plans and preparation 20. Engineering Ethics Discussion and ethics cases 21. Ethics from classroom to professionalism 22. ALUMNI (open panel discussion and advice) 23. Civil, Mechanical, ECE breakout sessions 24. Senior Design Competition 25. Robotics demonstration

Important Notes to Instructors: ENGR 100

The CME BSCE undergraduate curriculum process is dependent on each required course covering Student Outcomes 1-12 at the levels indicated. CME faculty members have agreed to these levels for all instructors. Student learning in these areas will be assessed as described in the CME Self Study.

Program Student Outcomes:	Coverage	Comments to Instructors
1. *Apply knowledge of mathematics and science in engineering problems*	**L**	
2. *Design and conduct experiments, analyze and interpret data*	**L**	
3. *Design Civil Engineering systems within realistic constraints*	**L**	
4. *Function effectively in multidisciplinary design teams*	**L**	
5. *Identify, formulate and solve engineering problems*	**L**	
6. *Understand their professional and ethical responsibilities*	**M**	Covered in great detail, but grade does not depend on demonstrated knowledge
7. *Communicate in a professional and effective manner*	**M**	Covered in great detail, but grade does not depend on demonstrated knowledge
8. *Understand the economic, environmental and societal impact of engineering solutions*	**M**	
9. *Recognize the importance and need to engage in life-long learning*	**M**	
10. *Comprehend the significance of contemporary issues*	**M**	
11. *Use techniques, skills, and modern engineering tools for efficient practice of Civil Engineering*	**M**	
12. *A majority of the graduates should pass the FE exam*	**M**	Not graded, but the importance of passing the FE and professional licensure are covered in detail.

A.3. Mathematics and Basic Science Courses.

MATH 180

1. Course number and name
MATH 180, Calculus I

2. Credits and contact hours
5 credit hours, 5 contact hours/week (3 lecture, 2 discussion)

3. Instructor's or course coordinator's name
Department of Mathematics staff

4. Text book, title, author, and year
Hughes-Hallet and Gleason, Calculus, Single Variable, 2nd Ed. John Wiley.
4a. Other supplemental materials
None.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Differentiation, curve sketching, maximum-minimum problems, related rates, mean-value theorem, antiderivative, Riemann integral, logarithm, and exponential functions.

5b. Prerequisites or co-requisites
Grade of C or better in Math 121 (Precalculus Mathematics) or appropriate performance on the department placement test or a Math ACT subscore of 28.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction
An introduction to the basic concepts of Calculus. Students will develop advanced mathematical skills and will be able to apply calculus to other areas.

6b. Explicitly indicate which of the student outcomes are addressed by the course
a. (SO #1) an ability to apply knowledge of mathematics, science, and engineering
e. (SO #2) an ability to identify, formulate, and solve engineering problems

7. Brief list of topics to be covered

Functions, Limits and Continuity	10
Derivative, Techniques of differential Calculus	15
Applications: Mean Value Theorem, graphing Max-Min Problems, Related rates	20
The Integral: Antiderivative, fundamental Theorem of Calculus, Riemann Sum	20
Logarithm and exponential functions, applications	10
Total	75

MATH 181

1. Course number and name
MATH 181, Calculus II

2. Credits and contact hours
5 credit hours, 5 contact hours/week (3 lecture, 2 discussion)

3. Instructor's or course coordinator's name
Department of Mathematics staff

4. Text book, title, author, and year
L.Loomis, Calculus 3rd Ed. Addison-Wesley.

4a. Other supplemental materials
None.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Techniques of integration, arc length, solids of revolution, applications, polar coordinates, parametric equations, infinite sequences and series, power series.

5b. Prerequisites or co-requisites
Grade of C or better in Math 180 (Calculus I).

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction
This course provides students with a knowledge of calculus.

6b. Explicitly indicate which of the student outcomes are addressed by the course
a. (SO #1) an ability to apply knowledge of mathematics, science, and engineering
e. (SO #2) an ability to identify, formulate, and solve engineering problems

7. Brief list of topics to be covered

Techniques of Integration	10
Applications of the Integral	20
Polar Coordinates, parametric equations	15
Infinite Series	15
Power Series	15
Total	75

MATH 210

1. Course number and name
MATH 210, Calculus III

2. Credits and contact hours
3 credit hours, 4 contact hours/week (3 lecture, 1 discussion)

3. Instructor's or course coordinator's name
Department of Mathematics staff

4. Text book, title, author, and year
L.Loomis, Calculus 3rd Ed. Addison-Wesley.

4a. Other supplemental materials
None.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Vectors in the plane and space, vector valued functions, functions of several variables, partial differentiation, maximum-minimum problems, double and triple integrals, applications, Green's theorem.

5b. Prerequisites or co-requisites
Grade of C or better in Math 181 (Calculus II).

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction
This course provides students with a knowledge of multivariate calculus.

6b. Explicitly indicate which of the student outcomes are addressed by the course
a. (SO #1) an ability to apply knowledge of mathematics, science, and engineering
e. (SO #2) an ability to identify, formulate, and solve engineering problems

7. Brief list of topics to be covered

Vectors in the plane and space	5
Vector valued functions	10
Functions of several variables; partial derivatives, max-min problems	15
Double and Triple integrals	10
Green's Theorem	5
One hour of exercises in the computer laboratory each week. Exercises are based on the same topics that are covered each week in the lecture-discussion	15
Total:	60

MATH 220

1. Course number and name

MATH 220, Introduction to Differential Equations

2. Credits and contact hours

3 credit hours, 4 contact hours/week (3 lecture, 1 discussion)

3. Instructor's or course coordinator's name

Department of Mathematics staff

4. Text book, title, author, and year

Fundamentals of Differential Equations, 4th ed. Nagle and Saff, Addison-Wesley.

4a. Other supplemental materials

None.

5. Specific course information

5a. Brief description of the content of the course (catalog description)

Techniques and applications of differential equations. First order equations: separable and linear. Linear second order equations, Laplace transforms, and series solutions. Graphical and numerical methods. Fourier series and partial differential equations.

5b. Prerequisites or co-requisites

Grade of C or better in Math 181 (Calculus II).

5c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

6a. Specific outcomes of instruction

Student will be able to solve differential equations.

6b. Explicitly indicate which of the student outcomes are addressed by the course

a. (SO #1) an ability to apply knowledge of mathematics, science, and engineering
e. (SO #2) an ability to identify, formulate, and solve engineering problems

7. Brief list of topics to be covered

Introduction, definitions, direction fields	3
Using Maple and computer labs	3
First order equations: separable, linear,	5
applications, numerical methodsSecond Order linear equations:	
general solutions, applications	9
Systems of equations and numerical methods	3
Laplace transforms and applications	9
Series approximations and special functions	4
Partial differential equations and Fourier series	9
Total	45

CHEM 112

1. Course number and name
CHEM 112, General College Chemistry I

2. Credits and contact hours
5 credit hours, 7 contact hours/week (3 lecture, 3 laboratory, 1 discussion)

3. Instructor's or course coordinator's name
Department of Chemistry staff

4. Text book, title, author, and year
"Chemistry: The Central Science (9th edition)" by Brown, LeMay & Bursten, Prentice Hall.

4a. Other supplemental materials
None.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Topics in general chemistry, including stoichiometry, periodicity, reaction types, the gaseous state, solution stoichiometry, chemical equilibria, acid-base equilibria, dissolution-precipitation equilibria. Includes a weekly three-hour laboratory.

5b. Prerequisites or co-requisites
Grade of C or better in Chem 101 (Preparatory Chemistry) or adequate performance on the UIC chemistry placement examination. Students with a course equivalent to Chem 101 from another institution must take the UIC chemistry placement examination.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction
Will obtain a comprehensive introduction to modern general chemistry.

6b. Explicitly indicate which of the student outcomes are addressed by the course
SO #1. Ability to apply knowledge of mathematics, science, and engineering
SO #5. Ability to conduct experiments
SO #6. Ability to analyze and interpret data

7. Brief list of topics to be covered
1. Review
2. Atomic Nature of Matter
3. Chemical Formulas, Equations, Stoichiometry
4. Chemical Periodicity

5. Types of Chemical Reactions
6. The Gaseous State
7. Solutions and Stoichiometry in Solution
8. Chemical Equilibrium
9. Acids and Bases
10. Dissolution and Precipitation Equilibria
11. Dissolution & Precipitation Equilibria
12. Thermo-chemistry
13. Spontaneous Change & Equilibrium
14. Exams and holiday
15. Laboratory

CHEM 114

1. **Course number and name**
   CHEM 114, General College Chemistry II

2. **Credits and contact hours**
   5 credit hours, 4 contact hours/week

3. **Instructor's or course coordinator's name**
   Dr. Gregory Jursich

4. **Text book, title, author, and year**
   Oxtoby, Freeman, and Block. Chemistry SCIENCE OF CHANGE, 4^{th} edition.
   4a. Other supplemental materials
   Laboratory Manual with blank lab notebook, lab safety goggles, student solutions manual (optional), and scientific calculator.

5. **Specific course information**
   5a. Brief description of the content of the course (catalog description)
   Topics in general chemistry including phase transitions, thermochemistry, spontaneity/equilibrium, electrochemistry, kinetics, bonding, order/symmetry in condensed phases, coordination compounds, descriptive chemistry. Includes a weekly 3-hour lab. Credit is not given for CHEM 114 if the student has credit for CHEM 118. *Prerequisite(s)*: Grade of C or better in CHEM 112 or the equivalent. Students with an equivalent course from another institution must take the chemistry placement examination. *Natural World - With Lab course.*
   5b. Prerequisite or co-requisites
   Prerequisite(s): Grade of C or better in CHEM 112 or the equivalent.
   5c. Indicate whether a required, elective, or selected elective course in the program

6. **Specific goals for the course**
   6a. Specific outcomes of instruction

   6b. Explicitly indicate which of the student outcomes are addressed by the course
   SO #1. Ability to apply knowledge of mathematics, science, and engineering
   SO #5. Ability to conduct experiments
   SO #6. Ability to analyze and interpret data

7. **Brief list of topics to be covered**
   General Chemistry:
   - phase transitions
   - thermochemistry
   - spontaneity/equilibrium

- electrochemistry
- kinetics, bonding
- order/symmetry in condensed phases
- coordination compounds
- descriptive chemistry

CHEM 222

1. Course number and name
CHEM 222, Analytical Chemistry

2. Credits and contact hours
4 credit hours
Lecture: 1 hr, twice a week
Lab: once a week

3. Instructor's or course coordinator's name
Scott Shippy

4. Text book, title, author, and year
Harris. Quantitative Chemical Analysis. 8th Edition. Freeman, 2010.

4a. Other supplemental materials
Sapling Online Homework (www.saplinglearning.com)

BOUND Lab Notebook

Splash goggles

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Theory and application of chemical equilibria and instrumentation in quantitative analysis. Includes two weekly three-hour laboratories.

5b. Prerequisites or co-requisites
Grade of C or better in CHEM 114 or grade of C or better in CHEM 118 or the equivalent.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction

7. Brief list of topics to be covered
- Statistics
- Calibration Methods
- Chemical Equilibria
- Activity and Equilibria
- Monoprotic Acid/Base
- Polyprotic Acid/Base
- Acid/Base Titrations

- EDTA Titrations
- Electrochemistry
- Electrodes and Potentiometry
- Redox Titrations
- Spectrophotometry and its Applications
- Atomic Spectroscopy
- Analytical Sep'ns

Chromatographic methods and electrophoresis

CHEM 232

1. Course number and name
CHEM 232, Organic Chemistry I

2. Credits and contact hours
4 credit hours
Lecture: 1 hr 15 min, twice a week
Discussion: 1 hr, once a week

3. Instructor's or course coordinator's name
Dr. Lindsey McQuade

4. Text book, title, author, and year
Carey and Giuliano. Organic Chemistry. 9th Edition (8th Edition is ok too)

Recommended:
Allison, Giuliano, Atkins, and Carey. Student Solutions Manual (for above text)

4a. Other supplemental materials
Sapling Online Homework (www.saplinglearning.com)

iClicker2 (www.iclicker.com)

Molecular Model Set

5. Specific course information
5a. Brief description of the content of the course (catalog description)
First semester of a one-year sequence. Structure, reactivity, and synthesis of organic molecules.

5b. Prerequisites or co-requisites
Grade of C or better in CHEM 114 or grade of C or better in CHEM 118. Recommended background: Concurrent registration in CHEM 233.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction

7. Brief list of topics to be covered

- Alkanes and Cycloalkanes
- Stereochemistry
- Alcohols and Alkyl Halides
- Nucleophilic Substitution
- Structure and Preparation of Alkenes: Elimination Reactions
- Reactions of Alkenes
- Alkynes
- Conjugation and Reactions of Dienes and Allylic Compounds
- Arenes and Aromaticity
- Reactions of Arenes
- Spectroscopy

CHEM 233

1. Course number and name
CHEM 233, Organic Chemistry Laboratory I

2. Credits and contact hours
1 credit hour
Lab: 4 hours, once a week

3. Instructor's or course coordinator's name
Dr. Lindsey McQuade

4. Text book, title, author, and year
Gilbert and Martin. Experimental Organic Chemistry: A Miniscale and Macroscale Approach. 5th Edition (but 4th Edition may also be used).

Landrie and McQuade. Organic Chemistry Laboratory Manual. 3rd Edition.

4a. Other Supplemental Materials
Chemistry Lab Notebook – 100 pages

Chemistry Splash Goggles

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Introductory organic chemistry laboratory. Basic organic techniques (distillation, crystallization), reactions (esterification, oxidation, addition, substitution, elimination), instruments (gas and liquid chromatography).

5b. Prerequisites or co-requisites
Credit or concurrent registration in CHEM 232

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction

6b. Explicitly indicate which of the student outcomes are addressed by the course
SO #5. Ability to conduct experiments
SO #6. Ability to analyze and interpret data

7. Brief list of topics to be covered

CHEM 233 is divided into two complementary parts. In Part One, students will learn the theories and principles of isolation, purification, characterization, and compositional analysis of organic molecules through techniques such as thin-layer and column chromatography, gas chromatography, infrared spectroscopy, melting point, and distillation. These laboratories provide the foundation for Part Two of the course, in which students will learn the techniques of various chemical reactions that are relevant to organic synthesis. Each experiment in Part Two typically involves one functional group transformation.

CHEM 234

1. Course number and name
CHEM 234, Organic Chemistry II

2. Credits and contact hours
4 credit hours
Lecture: 1hr 15min, thrice a week
Discussion: 1 hr, once a week

3. Instructor's or course coordinator's name
Duncan J. Wardrop, Ph.D.

4. Text book, title, author, and year
Francis A. Carey. Organic Chemistry. 8^{th} Edition.

Atkins and Carey. Solutions Manual.

4a. Other supplemental materials
A chemistry molecular model set is strongly recommended.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Continuation of CHEM 232. Structure, reactivity, and synthesis of organic molecules.

5b. Prerequisites or co-requisites
Grade of C or better in CHEM 232 or the equivalent.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction

7. Brief list of topics to be covered

- Organometallic Compounds
- Alcohols, Diols and Thiols
- Ethers, Epoxides and Sulfides
- Aldehydes and Ketones
- Enols and Enolates
- Carboxylic Acids
- Carboxylic Acid Derivatives
- Ester Enolates
- Amines
- Aryl Halides
- Phenols
- Carbohydrates
- Lipids
- Amino acids, Peptides and Proteins
- Nucleosides, Nucleotides and Nucleic Acids

CHEM 342

1. Course number and name
CHEM 342, Physical Chemistry I

2. Credits and contact hours
3 credit hours
Lecture: 1 hr, thrice a week
Discussion: 1 hr, once a week

3. Instructor's or course coordinator's name
Dr. George A. Papadantonakis

4. Text book, title, author, and year
Thomas Engel and Philip Reid. Physical Chemistry. 3rd Edition with MasteringChemistry online homework (required).

James R. Barrante. Applied Mathematics for Physical Chemistry.

Any Calculus and General Physics with Calculus textbook

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Thermodynamics of gases, solutions, reaction equilibria, and phase transitions.

5b. Prerequisites or co-requisites
Grade of C or better in MATH 181; and Grade of C or better in PHYS 142; C or better in or concurrent registration in MATH 210.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction

7. Brief list of topics to be covered
Covers chapters 1-11 of the textbook. Handouts posted in the course website will complement the material that will normally be covered in a typical Physical Chemistry course.

CHEM 346

1. Course number and name
CHEM 346, Physical Chemistry II

2. Credits and contact hours
3 credit hours
Lecture: 1 hr, thrice a week
Discussion: 1 hr, once a week

3. Instructor's or course coordinator's name
Robert Gordon

4. Text book, title, author, and year
Engel and Reid. Physical Chemistry. 3rd Edition.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Kinetic and molecular theory of gases; introduction to the principles of quantum mechanics with application to model systems, multi-electron atoms, diatomic molecules, and bonding.

5b. Prerequisites or co-requisites
Grade of C or better in CHEM 342 and Grade of C or better in MATH 210.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction

7. Brief list of topics to be covered
- Historical Antecedents to Quantum Mechanics
- Waves and Wave Equations
- Complex Numbers; Eigenfunctions and Eigenvalues
- Normalization and Orthogonality of Functions
- Expansion of a Function in a Complete Basis Set
- Postulates of Quantum Mechanics
- The Free Particle and a Particle in a One-Dimensional Box
- Particle in a Multi-dimensional box; Determinants
- Potential Barriers and Wells
- The Uncertainty Principle
- The classical harmonic operator
- The quantum mechanical harmonic oscillator
- The rigid rotor
- Hydrogen atom

- Spin and the Pauli Exclusion Principle
- Slater Determinants
- The Variational Method
- Hartree-Fock Self-Consistent Field Method
- Molecular Orbitals
- Maxwell Boltzmann Distribution
- Collisions
- Flux and Effusion
- Transport Phenomena
- Chemical Kinetics: Temperature, Equilibrium, and Relaxation
- Rate Equations

Sequential and parallel reactions

PHYS 141

1. Course number and name
PHYS 141, General Physics I (Mechanics)

2. Credits and contact hours
4 credit hours, 6 contact hours/week (3 lecture, 3 laboratory)

3. Instructor's or course coordinator's name
Department of Physics staff

4. Text book, title, author, and year
Freedman and Young "Physics for Scientist and Engineers Vol. I" Special UIC edition.

4a. Other supplemental materials
None.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Kinematics; Newton's laws of motion; linear momentum and impulse; work and kinetic energy; potential energy; rotational dynamics; simple harmonic motion; gravitation.

5b. Prerequisites or co-requisites
Grade of C or better in MATH 180 or consent of the instructor.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction
The course objective for Physics 141 is to teach students the major ideas and methods in the physics of mechanics.

6b. Explicitly indicate which of the student outcomes are addressed by the course
a. an ability to apply knowledge of mathematics, science, and engineering
e. an ability to identify, formulate, and solve engineering problems

7. Brief list of topics to be covered

Introduction: physical quantities, physical laws, units	2
Kinematics of motion in one dimension	3
Newton's laws, one dimensional applications	4
Vectors, kinematics of motion in two and three dimensions, projectile motion, circular motion	4
Application of Newton's laws to two and three dimensional problems	4

Linear momentum: definition, impulse, conservation, rocket motion	3
Work, energy (kinetic and potential), work-energy theorem, conservation of mechanical energy, power	6
System of particles: center of mass, elastic and inelastic collisions	4
Equilibrium of extended bodies and the physics of rotation, angular momentum and its conservation	7
Oscillations: simple harmonic motion, object on spring, simple pendulum	3
Gravity: Kepler's laws and Newton's law of universal gravitation, gravitational field and gravitational potential, applications	5
Laboratory	45
Total Hours	90

PHYS 142

1. Course number and name
PHYS 142, General Physics II (Electricity and Magnetism)

2. Credits and contact hours
4 credit hours, 6 contact hours/week (3 lecture, 3 laboratory)

3. Instructor's or course coordinator's name
Department of Physics staff

4. Text book, title, author, and year
Young and Freedman, Physics for Scientist and Engineers Vol. 2 UIC special edition.

4a. Other supplemental materials
None.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Electrostatics; electric currents; d-c circuits; magnetic fields; magnetic media; elctromagnetic induction; a-c circuits; Maxwell's equations; electromagnetic waves; reflection and refraction; interference.

5b. Prerequisites or co-requisites
MATH 181; and grade of C or better in either Physics 141 or both Physics 105/106; or consent of the intructor.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction
The course objective for Physics 142 is to teach students the major ideas and methods in the physics of electricity and magnetism.

6b. Explicitly indicate which of the student outcomes are addressed by the course
a. an ability to apply knowledge of mathematics, science, and engineering
e. an ability to identify, formulate, and solve engineering problems

7. Brief list of topics to be covered

Electrostatics: electric charge, Coulomb's law, electric field	4
Gauss's law and applications	2
Electric potential and potential energy, capacitance	4
Electric currents: definitions, Ohm's law, classical model of	

conduction, d.c. circuits, transients	5
Magnetic fields: definition, action on magnets, current loops and point charges	3
Calculation of magnetic fields: Biot-Savart law and applications, Ampere's law and applications, magnetic field of a bar magnet	5
Electromagnetic induction: Faraday' and Lenz's laws, inductance, LR, LC, and LCR circuits	4
Magnetism in matter: ferromagnetic, paramagnetic and deamagnetic materials	2
Alternating-current circuits; the transformer	4
Mechanical waves: an introduction	2
Displacement current and Maxwell's equations; electromagnetic waves	4
Light: nature, reflection, refraction, polarization	3
Physical optics	3
Laboratory	45
Total Hours	90

A.4. English Courses.

ENGL 160

1. Course number and name
ENGL 160, Academic Writing I: Writing in Academic and Public Contexts

2. Credits and contact hours
3 credit hours, 3 contact hours/week

3. Instructor's or course coordinator's name
Department of English staff

4. Text book, title, author, and year
Feldman, Ann M., Nancy Downs, & Ellen McManus. In Context: Participating in Cultural Conversations. New York: Longman, 2003.

Anson, Chris and Robert Schweiger. The Longman Handbook for Writers and Readers. 3rd Edition. New York: Longman, 2003.

4a. Other supplemental materials
Sample Sources and Resources from a recently taught section titled, "Writing on Location: Living, Working, and Workplaces" in which students wrote about neighborhood issues, the working life, and people in their workplaces.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Students write in a variety of genres with an emphasis on argument and sentence-level grammar. Topics vary by section. This class may be taught in a blended format. When that is the case, internet access will be required. A high-speed connection is strongly suggested. Please check the online class schedule for blended-online sections.

5b. Prerequisites or co-requisites
Eligibility as determined by performance on the Department placement test.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course
6a. Specific outcomes of instruction
Joining an Academic Conversation: The ability to conceptualize, articulate, and craft a response or an argument in an appropriate genre.

Key Rhetorical Concepts: Students explore some version of these four terms:

situation--histories, cultures, communities, and individual experiences.
genre--the forms of reading and writing.
language--syntax, lexicon, organization, and design-shape texts; and
consequences of writing

Attention to Written English: Special focus on academic written English, including discourse and stylistic conventions.

Readings: Analyze texts related to an issue or set of issues from a wide variety of genres and contexts, considering how these texts contribute to academic work.

Writing Process: Effective approaches to planning, drafting, revising and editing.

Correctness: Focus on syntactic and rhetorical choices, standard English usage, and using the assigned handbook as a reference.

6b. Explicitly indicate which of the student outcomes are addressed by the course
g. (SO #12) an ability to communicate effectively

7. Brief list of topics to be covered
Students in Academic Writing I will produce at least 20 pages of finished writing and complete five writing projects in a variety of genres, including an argumentative essay. Although each section covers different topics, typical projects could include: argumentative essay, opinion piece/commentary, speech, interview, formal letter, or proposal. Class work is organized in the following way:

The projects listed below were assigned in a section titled, "Writing on Location: Living, Working, and Workplaces."

Introduction to Writing in Academic and Public Contexts	5
Opinion Piece/Commentary: 2 pages and a 1 page cover letter	8
Interview Paper: 3 pages	8
Dialogue/Symposium: 5 pages	8
Argumentative Essay: 4 pages and a 1 page cover letter	8
Feature Story: 4 pages and a 1 page cover letter	8
Total Hours=	45

ENGL 161

1. Course number and name
ENGL 161, Academic Writing II: Writing for Inquiry and Research

2. Credits and contact hours
3 credit hours, 3 contact hours/week

3. Instructor's or course coordinator's name
Department of English staff

4. Text book, title, author, and year
McGinty, Sue. 1999. Resilience, Gender, and Success at School. New York: Peter Lang.

Smith, Zadie. 2000. White Teeth. London: Hamish Hamilton.

Anson, Chris and Robert Schwegler. 2005. The Longman Handbook for Writers and Readers. 3rd Edition. New York: Longman.

Pitchford, Veronda and Tobi Jacobi. The Road to Research (Road), Stipes Publishing.

Feldman, Ann Merle. 1996. Writing and Learning in the Disciplines. New York: Addison-Wesley Longman.

4a. Other supplemental materials
Sample Sources and Resources: from a section titled, "Makin' It: Identity, Resilience, and School Culture," in which students explored the dynamic relationships that contribute to the construction of self and success in school. Students read ethnographic reports, cultural theory, fiction, documents related to school reform, and stories of students' lives.

5. Specific course information
5a. Brief description of the content of the course (catalog description)
Students learn about academic inquiry and complete several writing projects including a documented research paper. Topics vary by section.

5b. Prerequisites or co-requisites
ENGL 160 or the equivalent. All students take the Writing Placement Test. If students place into ESL 050, ESL 060, ENGL 150, ENGL 152 or ENGL 160, the student must take that course (or courses) prior to enrolling in ENGL 161. Students with an ACT English subscore of 27 or higher receive a waiver of ENGL 160 and permission to enroll in ENGL 161.

5c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course

6a. Specific outcomes of instruction

Objectives for English 161 fall into two categories that roughly coincide with the first and second parts of the semester.

Part 1: Meaning-Making in Intellectual Communities

Students examine a topic for depth and breadth in the following ways: exploring discipline-specific topics and contexts for research; examining discipline specific and public contexts; writing summaries; writing syntheses and analyses; writing arguments and developing research skills. Instruction includes quoting, paraphrasing as well as a focus on grammar and style.

Part 2: The Classroom as a Research Community

Students plan and write a research paper through the following process: proposing an inquiry for independent research; identifying the consequential nature of the topic; identifying source materials; working in small groups to solve ongoing research problems; applying and refining skills for summarizing, paraphrasing, quoting, and documenting sources.

6b. Explicitly indicate which of the student outcomes are addressed by the course

g. (SO #12) an ability to communicate effectively

7. Brief list of topics to be covered

Students complete 20 pages of writing through 5 projects that ask students to summarize, synthesize, develop an argument, propose a research project, and write a research paper. Each section covers different topics but adheres to the following framework:

Introduction to Academic Inquiry	5
Summarizing	8
Analyzing and Synthesizing Texts	8
Argument	8
Research Proposal	8
Research Paper/Portfolio	8
Total Hours =	45

Appendix B – Faculty Vitae

Faculty Vitae: Belinda S Akpa

1. Education

B.A. Chemical Engineering, University of Cambridge (UK), 2002
M.Eng. Chemical Engineering, University of Cambridge (UK), 2002
Ph.D. Chemical Engineering, University of Cambridge – Cambridge/MIT Institute, 2007

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago
Assistant Professor, 2009-present
Department of Chemical Engineering, University of Illinois-Chicago
Post Doctoral Research Fellow, August 2007-August 2009
University of Cambridge, Cambridge, UK
Post Doctoral Research Associate, April 2006-February 2007

3. Non-Academic Experience

General Mills Inc., Buffalo, NY and West Chicago, IL. MIT School of Chemical Engineering Practice Student Consultant. 9/03-11/03
Schlumberger Oilfield Services**,** Youngsville, LA. Drilling and Measurements Intern. 6/01-8/01

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers 1999-present
American Chemical Society 2009-present

6. Honors and awards

UIC College of Engineering Undergraduate Advising Award, 2012
UIC College of Engineering Teaching Award, 2012 & 2013
CAREER Award, National Science Foundation, July 2008 − June 2013
Cambridge Commonwealth Trust Graduate Fellowship 2002-2005

7. Service activities

American Institute of Chemical Engineers, Societal Impact Operating Council 2012-present
American Institute of Chemical Engineers, Midwest Regional Conference, Poster session Chair 2014
Graduate Committee, Department of Chemical Engineering, University of Illinois at Chicago, 2009-2011, 2012-2013
Faculty Search Committee, Department of Chemical Engineering, University of Illinois at Chicago, 2010-2013

8a. Selected Publications

10. Fettiplace M.R., Akpa B.S., Ripper R., Zider B., Lang J., Rubinstein I., Weinberg G.L. Resuscitation with lipid emulsion: dose-dependent recovery from cardiac pharmacotoxicity requires a cardiotonic effect. *Anesthesiology*. 120(0):00-00 (2014)

9. Kuo I and Akpa B.S. Validity of the lipid sink as a mechanism for reversal of local anesthetic systemic toxicity: A physiologically based pharmacokinetic study. *Anesthesiology*. 118(6):1350-1361 (2013)

8. Akpa B.S., D'Agostino C., Gladden L.F., Hindle K., Manyar H., McGregor J., Li R., Neurock M., Sinha N., Stitt E.H., Weber D., Zeitler J.A., Rooney D.W. Solvent effects in the hydrogenation of 2-butanone. *Journal of Catalysis*. 289:30-41 (2012)

7. Magin RL, Akpa B.S., et al. Fractional order analysis of Sephadex gel structures: NMR measurements reflecting anomalous diffusion. *Communications in Nonlinear Science and Numerical Simulatio*n. 16(12):4581-4587 (2011)

5. Parkar N.S., Akpa B.S., et al. Vesicle Formation and Endocytosis: Function, Machinery, Mechanisms, and Modeling. *Antioxidants & Redox Signaling*. 11(6):1301-1312 (2009)

4. Graf von der Schulenburg D.A., Akpa B.S., et al. Non-invasive mass transfer measurements in complex biofilm-coated structures. *Biotechnology and Bioengineering*. 101(3):602-608 (2008)

3. Akpa B.S., Matthews S.M., et al. Study of miscible and immiscible flow in a microfluidic device using magnetic resonance imaging. *Analytical Chemistry*. 79:6128-6134 (2007)

2. Akpa B.S., Holland D.J., et al. Enhanced 13C PFG-NMR for the study of dispersion in porous media. *Journal of Magnetic Resonance*. 186:160-165 (2007)

1. Akpa B.S., Mantle M.D., et al. In situ 13C DEPT MRI as a tool to spatially resolve chemical conversion and selectivity of a heterogeneous catalytic reaction occurring in a fixed bed reactor. *Chemical Communications*. 21:2741-2743 (2005)

8b. Selected Presentations

5. American Institute of Chemical Engineers Annual Meeting. Nov 2013. "Sequestering of toxins for the reversal of drug toxicity: A coarse grained molecular dynamics study" Manuela Ayee*. and Belinda Akpa

4. Illinois Institute of Technology. Invited talk. Department of Chemical and Biological Engineering. Feb 2013. "Exogenous lipid droplets: Multi-functional therapeutic agents for treating toxicity". Belinda Akpa

3. Chicago Area NMR Discussion group. Nov 2012. "Vapor-phase condensation at elevated temperature and pressure in a packed bed". Belinda Akpa*, Matthew Renshaw, Andrew Sederman, Michael Mantle, Lynn Gladden

2. Baxter Healthcare. Invited talk. May 2012. "Computational studies of the mechanism underlying lipid resuscitation". Belinda Akpa

1. UIC Water Research Forum. April 2012. "Multi-scale applications of magnetic resonance spectroscopy and imaging in reactive transport". Belinda Akpa

Faculty Vitae: Brian P. Chaplin

1. Education

B.S. Civil Engineering, University of Minnesota, 1999
M.S. Civil Engineering, University of Minnesota, 2003
Ph.D. Environmental Engineering, University of Illinois at Urbana-Champaign, 2007

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago
Assistant Professor, 2013-present
Department of Civil and Environmental Engineering, Villanova University
Assistant Professor, 2010-2013
Department of Chemical and Environmental Engineering, University of Arizona
Post Doctoral Researcher, Jan 2008-June 2010

3. Non-Academic Experience

U.S. Geological Survey, Mounds View, MN. Groundwater Hydrologist. 1/99-1/03

4. Certifications or professional registrations

EIT certified (Mn), since 1999

5. Current membership in professional organizations

Association for Environmental Engineering and Science Professors 2010-present
American Institute of Chemical Engineers 2013-present
American Chemical Society 2006-present

6. Honors and awards

Villanova Institute for Teaching and Learning (VITAL) Minigrant, (2012)
Summer Research Fellowship/Research Support Grant, Villanova University (2011)
Environmental Water Resources Institute's 2011 Samuel Arnold Greeley Award (Best Paper).
Water CAMPWS award for *Excellence in Mentoring Undergraduate Researchers*, University of Illinois at Urbana-Champaign Summer 2006 and 2007

University of Illinois Graduate School Fellowship 2003-2004

7. Service activities

Safety Committee Chair, Department of Chemical Engineering, University of Illinois at Chicago, 2013-present

Undergraduate Committee, Department of Chemical Engineering, University of Illinois at Chicago, 2013-present
Chair for *Villanova Center for the Advancement of Sustainable Engineering* (VCASE) Seminar Series, 2011-13
Committee member for *Sustainable Engineering Graduate Program*, Villanova University, 2010-2013

8. Selected Publications Last 5 Years

10. A. Donaghue, B.P. Chaplin (2013) Effect of Select Organic Compounds on Perchlorate Formation at Boron-doped Diamond Film Anodes. *Environmental Science and Technology*, 47(21), 12391-12399.

9. A. Zaky, B.P. Chaplin (2013) Porous Sub-stoichiometric TiO_2 Anodes as Reactive Electrochemical Membranes for Water Treatment. *Environmental Science and Technology*, 47(12), 6554-6563.

8. B.P. Chaplin, D. Hubler, J. Farrell (2013) Understanding Anodic Wear at Boron Doped Diamond Film Electrodes. *Electrochimica Acta*, 89, 122-131.

7. B.P. Chaplin, M. Reinhard, W.F. Schneider, C. Schüth, J.R. Shapley, T.J. Strathmann, C.J., Werth (2012). A Critical Review of Pd-Based Catalytic Treatment of Priority Contaminants in Water. *Environmental Science and Technology*, 46(7), 3655-3670.

6. O. Azizi, D. Hubler, G. Schrader, J. Farrell, B.P. Chaplin (2011). Mechanism of Perchlorate Formation on Boron-doped Diamond Anodes. *Environmental Science and Technology*, 45(24), 10582-10590.

5. B.P. Chaplin, I. Wylie, H. Zheng, J.A. Carlisle, J. Farrell (2011). Characterization of the Performance and Failure Mechanisms of Boron-doped Diamond Ultrananocrystalline Diamond Electrodes. *Journal of Applied Electrochemistry*, 41(11), **1329-1340**.

4. B.P. Chaplin, G. Schrader, J. Farrell (2010). Electrochemical Destruction of N-Nitrosodimethylamine in Reverse Osmosis Concentrates using Boron-doped Diamond Film Electrodes. *Environmental Science and Technology*, 44(11), **4264-4269**.

3. D. Shuai, B.P. Chaplin, J.R. Shapley, N.P. Menendez, D.C. McCalman, W.F. Schneider, C.J. Werth (2010). Enhancement of Oxyanion and Diatrizoate Reduction Kinetics Using Selected Azo Dyes on Pd-based Catalysts. *Environmental Science and Technology*, 44(5), 1773-1779.

2. B.P. Chaplin, G. Schrader, J. Farrell (2009). Electrochemical Oxidation of N-Nitrosodimethylamine with Boron-doped Diamond Film Electrodes. *Environmental Science and Technology*, 43(21), **8302-8307**.

1. B.P. Chaplin, J.R. Shapley, C.J. Werth (2009). Oxidative Regeneration of Sulfide-fouled Catalysts for Water Treatment. *Catalysis Letters*, **132(1-2)**, **174-181**.

Faculty Vitae: Ying Liu

1. Education

B.S. Engineering Mechanics, Tsinghua University, 2001
M.A. Mechanical and Aerospace Engineering, Princeton University, 2004
Ph.D. Mechanical and Aerospace Engineering, Princeton University, 2007

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago
Assistant Professor, 2008-present
Department of Biopharmaceutical Sciences, University of Illinois-Chicago
Adjunct professor, 2008-present
Department of Chemistry, University of Chicago
Post Doctoral Researcher, July 2007- July 2008

3. Non-Academic Experience

Not applicable.

4. Certifications or professional registrations

Not applicable.

5. Current membership in professional organizations

American Institute of Chemical Engineers
American Physical Society
American Association of Pharmaceutical Science

6. Honors and awards

CAREER Award, National Science Foundation, Jan. 2014 − Jan. 2019
College of Engineering Research Award, UIC, 2013
OTM POC Award, UIC, 2013
Chancellor Discovery Award, UIC, 2011
Technology for Developing Regions Fellowship, Princeton University, 2006
Howard Crathorne Phillips Graduate Fellowship, Princeton University, 2003
Engineering Fellowship, Princeton University, 2002
Rong Hong National Fellowship, Tsinghua University, 2000

7. Service activities

Member, Graduate Committee, Department of Chemical Engineering, University of Illinois at Chicago, 2008-present
Student Sponsor for Illinois Math and Science Academy, 2010-11, 2011-2012, 2012-2013
Mentor, Girls' Electronic Mentoring in Science, Engineering and Technology, 2009-present
Member, Editorial Board, Advances in Materials Science and Engineering, 2013-present
Member, Editorial Board, Journal of Oncology and Biomarker Research, 2013-present

8. Selected Publications Last 5 Years (Selected from 18 publications)

1. H. Shen, X. Hu, M. Szymusiak, Z. Wang, and Y. Liu, Orally Administrated Nano-curcumin to Attenuate Morphine Tolerance and Dependence: Comparison between Negatively Charged PLGA and Partially and Fully PEGylated Nanoparticles. Molecular Pharmaceutics, article ASAP, 2013.
2. A. Banerjee, H. Shen, M. Hautman, J. Anwer, S. Hong, I. Kapetanovic, Y. Liu, and A. Lyubimov, Polymeric nanoparticles enhance oral bioavailability of the hydrophobic chemopreventive agent (SR13668) in beagle dogs, Current Pharmaceutical Biotechnology, 14, 464-469, 2013.
3. S. Sunoqrot, Y. Liu, D. Kim and S. Hong, In vitro evaluation of dendrimer-polymer hybrid nanoparticles on their controlled cellular targeting kinetics, Molecular Pharmaceutics, 10 (6), 2157-2166, 2013.
4. H. Shen, A. Banerjee, P. Mlynarska, Mathew Hautman, Seungpyo Hong, Izet M. Kapetanovic, Alexander V. Lyubimov, and Y. Liu, Enhanced oral bioavailability of a cancer preventive agent (SR13668) by employing polymeric nanoparticles with high drug loading, Journal of Pharmaceutical Sciences, 101 (10) 3877-3885, 2012.
5. S. Sunoqrot, J. Bae, R. Pearson, K. Shyu, Y. Liu, D. Kim, and S. Hong, Temporal control over cellular targeting through hybridization of folate-targeted dendrimers and PEG-PLA nanoparticles, Biomacromolecules, 13 (4), 1223 – 1230, 2012.
6. V. Sharma, M. Szymusiak, H. Shen, L. Nitsche, and Y. Liu, Formation of Polymeric Toroidal-Spiral Particles, Langmuir, 28 (1), 729-735, 2012.
7. M. Szymusiak, V. Sharma, L. Nitsche, and Y. Liu, Interaction of sedimenting drops in a miscible solution – formation of heterogeneous toroidal-spiral particles, Soft Matter, 8 (29), 7556 – 7559, 2012.
8. H. Shen, S. Hong, R. Prud'homme, and Y. Liu, Self-assembling process of flash nanoprecipitation in a multi-inlet vortex mixer to produce drug-loaded polymeric nanopartiles, Journal of Nanoparticle Research, 12, 4109-4120, 2011.
9. S. Sunoqrot, J. Bae, S. Jin, R. Pearson, Y. Liu, and S. Hong, Kinetically Controlled Cellular Interactions of Polymer-Polymer and Polymer-Liposome Nanohybrid Systems, Bioconjugate Chemistry, 22, 466-474, 2011.
10. B. Russ, Y. Liu, and R. Prud'homme, Optimized Descriptive Model for Micromixing in a Vortex Mixer, Chemical Engineering Communications. 197, 1068-1075, 2010.

Faculty Vita: Randall Meyer

1. Education

B.S. Mechanical Engineering, Rice University, 1992
M.S. Environmental Engineering, University of California at Los Angeles, 1995
Ph.D. Chemical Engineering, University of Texas at Austin, 2001

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago
Associate Professor, 2011-present
Department of Chemical Engineering, University of Illinois-Chicago
Assistant Professor, 2006-2011
Department of Chemical Engineering, University of Virginia
Post Doctoral Researcher, April 2004-Dec 2005
Fritz Haber Institut der Max Plank Gesellschaft, Berlin, Germany
Alexander von Humboldt Post Doctoral Fellow, October 2001-March 2004

3. Non-Academic Experience

CDI Stubbs Overbeck, Houston, TX. Process Engineer. 9/95-1/96
Plant Process Equipment**,** League City, TX. Process Engineer. 2/93−8/94
Smith-Hamm Inc**.,** Galveston, TX. Engineering Assistant. 6/92−2/93

4. Certifications or professional registrations

Passed FOE, Fall 1992

5. Current membership in professional organizations

Chicago Catalysis Club 2006- present
American Institute of Chemical Engineers 2004-present
American Chemical Society 2007-present

6. Honors and awards

UIC Graduate Mentoring Award, 2011
UIC College of Engineering Undergraduate Advising Award, Fall 2009
UIC College of Engineering Research Award, 2009
CAREER Award, National Science Foundation, July 2008 − June 2013
Alexander von Humboldt Fellowship, Fritz Haber Institute, February 2002 − May 2003
Leaton Thomas Oliver Scholarship, University of Texas at Austin, September 2000 − August 2001
Moody Graduate Fellowship, University of Texas at Austin, Spring 1998

7. Service activities

Student Sponsor for the Illinois Math and Science Academy, 2007-08, 2008-09, 2009-10, 2010-11, 2012-13, 2013-2014

Advisory Board, Chicago Catalysis Club, 2008-09
President, Chicago Catalysis Club, 2007-08
Program Chair, Chicago Catalysis Club, 2006-07
Graduate Committee, Department of Chemical Engineering, University of Illinois at Chicago, 2006-2010, 2011-2012
Undergraduate Committee, Department of Chemical Engineering, University of Illinois at Chicago, 2010-present
Campus Research Board, University of Illinois at Chicago, 2012-13

8. Selected Publications Last 5 Years (10 of 36)

10. D. Childers, A. Saha, N. Schweitzer, R. M. Rioux, J. T. Miller, and R. J. Meyer, "Correlating Heat of Adsorption of CO to Reaction Selectivity: Geometric Effects vs. Electronic Effects in Neopentane Isomerization over Pt and Pd catalysts" *ACS Catalysis (2013)*
9. T. P. Senftle, R. J. Meyer, M. J. Janik, and A.C.T. van Duin, "Development of a ReaxFF Potential for Pd/O and Application to Palladium Oxide Formation" *Journal of Chemical Physics* **139** (2013) 044109
8. H. Wei, C. Gomez, J. Liu, N. Guo, T. Wu, R. Lobo, C. L. Marshall, J. T. Miller and R. J. Meyer "Selective Hydrogenation of Acrolein on Supported Silver Catalysts: A Kinetics Study of Particle Size Effects" *Journal of Catalysis* **298** (2013) 18-26
7. A. J. Schoenfelt, Z. Ni, A. W. Korinda, R. J. Meyer, J. M. Notestein, "Manganese Triazacyclononane Oxidation Catalysts Grafted Under Reaction Conditions on Solid Co-Catalytic Supports " *Journal of the American Chemical Society*, **133** (2011) 18684-18695
6. P.Tiruppathi, J. J. Low, A.S .Y. Chan, S. R. Bare, and R. J. Meyer, "Density Functional Theory Study of the Effect of Subsurface H, C and Ag on C_2H_2 Hydrogenation on Pd(111)", *Catalysis Today*, **165** (2011) 106-111
5. N. Guo, B. R. Fingland, W. D. Williams, V. F. Kispersky, J. Jelic, W. N. Delgass, F. H. Ribeiro, R. J. Meyer, and Jeffrey T. Miller, "Determination of CO, H_2O and H_2 Coverage by XANES and EXAFS on Pt and Au During Water Gas Shift Reaction", *Physical Chemistry Chemical Physics,* **12** (2010) 5678-5693
4. J. Jelic, K. Reuter, R.Meyer, "The Role of Surface Oxides in NOx Storage Reduction Catalysts", *ChemCatChem*, **2** (2010) 658-660
3. Y. Lei, A. Uhl, C. Becker, K. Wandelt, B. C. Gates, R. Meyer and M. Trenary, "Adsorption and Reaction of $Rh(CO)_2$(acac) on $Al_2O_3/Ni_3Al(111)$", *Physical Chemistry Chemical Physics,* **12** (2010) 1264-1270
2. Y. Lei, F. Mehmood, S. Lee, J. Greeley, B. Lee, S. Seifert, R. E. Winans, J. W. Elam, R. J. Meyer, P. C. Redfern, D. Teschner, R. Schlögl, M. J. Pellin, L. C. Curtiss, and S. Vajda, "Increased Silver Activity for Direct Propylene Epoxidation via Subnanometer Size Effects" *Science*, **328** (2010) 224-228
1. T. E. Feltes, L. Espinosa-Alonso, E. de Smit, L. D'Souza, R. J. Meyer, B.M. Weckhuysen and J. R. Regalbuto "Selective Adsorption of Manganese onto Cobalt for Optimized $Mn/Co/TiO_2$ Fischer-Tropsch Catalysts", *Journal of Catalysis*, **270** (2010) 95-102

Faculty Vitae: Sohail Murad

1. Education

Ph.D. Chemical Engineering, Cornell University, Ithaca, N.Y., 1979
M.S. Chemical Engineering, University of Florida, Gainesville, 1976
B.S. Chemical Engineering, University of Engineering, Lahore, Pakistan, 1974

2. Academic Experience

Assistant Professor, 1979 - 86; Associate Professor, 1986 - 91; Professor, 1991 - date; Department Head 2004- date; University of Illinois, Chicago

3. Non-Academic Experience

Senior Engineer, Exxon Research and Engineering Company, 1981 – 82 (During leave of absence)
Research Fellow, U. S. Army Ballistic Research Laboratory, Summer 1985
Expert Witness – Environmental Pollution (2012-)
.

4. Certifications or professional registrations

EIT (State of Illinois)

5. Current membership in professional organizations

American Institute of Chemical Engineers.
American Chemical Society
American Society for Engineering Education

6. Honors and awards

Fellow, American Institute of Chemical Engineers (AIChE)
College of Engineering Excellence in Advising Award, 2013.
Guest Professor, Nanjing University of Technology, Nanjing, China, 2012 –
Coleman Foundation Fellow, 2009
Research Excellence Award, College of Engineering (1998, 2004, 2006)
Harold Simon Award for Teaching Excellence, College of Engineering (2002)
University Award for Excellence in Teaching (1999)
University Teaching Recognition Award (1997)
Outstanding Teacher of the Year, Chemical Engineering Department (1995).

7. Selected service activities

Search Committee, Vice Provost for Faculty Affairs, 2012
Discussion Leader, "How to Keep Faculty Morale High During Difficult Financial Times", Department Heads Workshop, University of Illinois at Chicago, January 2009.
Panelist, Workshop on Leadership in Energy and Environmental Design (LEED), November 2008.
Strategic Thinking and Planning for Diversity, 2008-10.
University Master Plan Advisory Committee, 2008-2010.
Judge, Engineering Expo, April 2008.
Invited Talk, "Designing an Effective Research Program", McNair Scholars Program, March 2008.

UIC Senate Library Committee, 2005-7.
UIC Senate, 1993-1996, 2002-2005.
Senate Committee for Research, 2003-2008.
Senate Committee for Academic Freedom and Tenure, 2003-2008.
Search Committee for Dean of Engineering, 2004.
Chair, Search Committee for Computer Science Department Head, 2000-2001.
Senate Committee on Academic Computing, 1998-2001.
College of Engineering Executive Committee, 2000-2004.
Departmental Advisory Committee, 1997-2003.
Departmental Graduate Committee, 1999-2000, 2003-2005.
Honors College Executive Committee, 1993-1995.

8. Selected Publications and presentations from the last 5 years

1. S. Murad and I. K. Puri, "A Tractable Design for a Thermal Transistor", Journal of Chemical Physics, in press (2013)
2. S. Murad and I. K. Puri," A Thermal Logic Device Based on Fluid-Solid Interfaces", Applied Physics Letters, 102, 193109 (2013).
3. B. Song, H.Yuan, S. Pham, C. J. Jameson, and S. Murad, "Nanoparticle Permeation Induces Water Penetration, Ion Transport and Lipid Flip-flop", Langmuir, 28, 16989-17000 (2012).
4. S. Murad and I. K. Puri, "Thermal Rectification in Liquids by Manipulating the Solid-Liquid Interface", Journal of Chemical Physics, 137, 081101 (2012).
5. S. Murad and I. K. Puri, "Thermal Rectification In A Fluid Reservoir", Applied Physics Letters, 100, 121901 (2012)
6. S. Murad and I. K. Puri, ""Molecular Simulations of Thermal Transport Across Interfaces: Solid-vapor and Solid-Solid" Molecular Simulation, 38,642-652 (2012)
7. "A thermal transistor based on fluid-solid interfaces", Indian Institute of Science Alumni Second Global Conference, Chicago, IL, July 2013
8. "Molecular Modeling of Membrane Based Separation Processes for Water Purification and Gases", Keynote Talk, Arab Academy of Sciences Annual Meeting, Beirut, Lebanon, December 2012.
9. "Fluid and Nanoparticle Transport in Lipid Bilayer Membranes Using Coarse Grained Molecular Dynamics", Chemical Engineering Department, Purdue University, September 2012
10. "Molecular Modeling of Membrane Based Separation Processes", Nanjing University, PR China, June 2012
11. "Molecular Modeling of Membrane Based Separation Processes", Changzhou University, PR China, June 2012
    "Gas and Nanoparticle Permeation in Lipid Membranes', Keynote Address, Midwest Statistical Mechanics and Thermodynamic Conference", Minneapolis, MN, May 2012

9. Professional development activities

Attendance at the 2012 NSF S-STEM Projects Meeting, Arlington, VA, October 14-16, 2012.
Attendance of Workshop on Excellence Empowered by a Diverse Academic Workforce Chemists, Chemical Engineers, and Materials Scientists With DisabilitiesFebruary 9, 2009 Arlington, Virginia

Faculty Vita: Ludwig C. Nitsche

1. Education

B.S. Chemical Engineering, University of Minnesota, 1984.
B.S. Mathematics, University of Minnesota, 1984
Ph.D. Chemical Engineering, Massachusetts Institute of Technology, 1989.

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago
Associate Professor, 1996–present.
Department of Chemical Engineering, University of Illinois at Chicago
Assistant Professor, 1991–1996.
Department of Chemical Engineering, University of Illinois at Chicago
Visiting Assistant Professor, Jan. – Aug. 1991.
Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge, UK, NSF-NATO Postdoctoral Fellow, Sep. 1989 – Dec. 1990.
Department of Chemical Engineering, Massachusetts Institute of Technology, Postdoctoral Associate, Jun. – Aug. 1989.

3. Non-Academic Experience

Not applicable.

4. Certifications or professional registrations

Not applicable.

5. Current membership in professional organizations

American Institute of Chemical Engineers.

6. Honors and awards

UIC Award for Excellence in Teaching, 2013.
UIC College of Engineering Harold A. Simon Award for Excellence in Teaching, 2012.
UIC College of Engineering Faculty Teaching Award, 2008.
UIC CETL Teaching Recognition Program Award, 2007
UIC College of Engineering Faculty Teaching Award, 2006.
USIA Fulbright Senior Scholar Award #9498, Austria, Research, Oct. 1999 – Jan. 2000.
National Science Foundation Young Investigator Award, 1994–1999.
NSF-NATO Postdoctoral Fellowship, Sep. 1989 – Dec. 1990.
NSF Graduate Fellowship Award, academic years 1984-85, 1985-86, 1986-87.

7. Selected service activities

Director of Undergraduate Studies, Department of Chemical Engineering, University of Illinois at Chicago, 2005–2014.
ABET Assessment Coordinator, Department of Chemical Engineering, University of Illinois at Chicago, 2005–2014.
Member, Undergraduate Committee, Department of Chemical Engineering, University of Illinois at Chicago, 1991-1992, 1998-1999, 2000-2001, 2001-2002; Fall 2002.

Member, Advisory Committee, Department of Chemical Engineering, University of Illinois at Chicago, 1993–1994, 2006–2014.

Educational Policy Committee, College of Engineering, University of Illinois at Chicago: Secretary 1991–1993. Chair 2006–2008, 2010–2011. 1998–1999, 2005–2006, 2008–2010, 2011–2014.

Faculty advisor for AIChE student chapter, Department of Chemical Engineering, University of Illinois at Chicago, 2006-2007, 2007-2008, 2010-2011, 2012-2014.

Departmental Facilitator, WISE (Women in Science and Engineering Committee), University of Illinois at Chicago, 2005 – 2014.

UIC Affiliate Professor, Project Lead the Way (Secondary educational outreach), 2005-2006. www.pltw.org

8. Publications and presentations from the last 5 years

L. C. Nitsche and P. Parthasarathi, Stokes flow singularity at the junction between impermeable and porous walls. *J. Fluid Mech.*, **713** (2012) 183-215.

M. Szymusiak M, V. Sharma V, L. C. Nitsche LC and Y. Liu, Interaction of sedimenting drops in miscible solution – formation of heterogeneous toroidal-spiral particles. *Soft Matter,* **8** (2012) 7556-7559.

V. Sharma, M. Szymusiak, H. Shen, L. C. Nitsche, and Y. Liu, Formation of polymeric Toroidal-spiral particles, *Langmuir*, **28** (2012) 729–735.

L. C. Nitsche and Y. Liu, Self-Assembled Toroidal-Spiral Particles and Manufacture and Uses Thereof. US Patent Application (2012) 13/510,214.

Y. Lei, J. Jelic, L. C. Nitsche, R. Meyer and J. Miller, Effect of Particle Size and Adsorbates on the L3, L2 and L1 X-ray Absorption Near Edge Structure of Supported Pt Nanoparticles, *Topics in Catalysis*, **54**, (2011) 334–348.

N. S. Parkar, B. S. Akpa, L. C. Nitsche, L. E. Wedgewood, M. S. Sverdlov, O. Chaga and R. D. Minshall, Vesicle formation and Endocytosis: Function, machinery, Mechanisms, and Modeling (Forum Review Article), *Antioxidants & Redox Signaling*, **11** (2009) 1301-1312.

L. C. Nitsche and P. Parthasarathi, Cubically Regularized Stokeslets for Fast Particle Simulations of Low-Reynolds-Number Drop Flows, *Chem. Eng. Commun.*, **197** (2010) 18-38.

9. Professional development activities

Attendance at the 2013 ABET Symposium, Portland, OR, April 12-13, 2013.

Attendance at the 2012 NSF S-STEM Projects Meeting, Arlington, VA, October 14-16, 2012.

Faculty Vitae: Jeffery P. Perl

1. Education

B.S. Chemical Engineering, Illinois Institute of Technology, 1977.
M.S. Chemical Engineering, Illinois Institute of Technology, 1979
Ph.D. Chemical Engineering, Illinois Institute of Technology, 1984.

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago
- Adjunct Professor, 2008–present

Department of Chemical Engineering, Illinois Institute of Technology,
- Adjunct Professor, Jan. – May, 1989

Department of Chemistry, Brown University
- Postdoctoral Research Faculty Appointment Jun 1984 – Nov 1985

3. Non-Academic Experience

Consulting Chemical Engineer, 1986-Present
- Environmental Pollution Prevention and Cleanup
- Process Production Improvement and Troubleshooting
- Chemical Process Design
- Expert Witness, Process Failure, Federal Fraud, Waste and Abuse
- Forensic Engineering, Root Cause Accidents and Process Failure

Bioenvironmental Engineer Officer, United States Air Force Reserve, 1993-2004
- HQ The AF Civil Engineer, Brooks and Randolph AFB, San Antonio TX 1993-99
  - Air Force Base Restoration and Pollution Prevention
  - Environment, Safety and Occupational Health Team Leader
- HQ The AF Surgeon General, Bolling AFB, Washington DC, 1999-2004
  - Mobilization Coordination
  - Primary Care Optimization

4. Certifications or professional registrations

Registered Professional Engineer, Illinois and Kentucky
Certified Hazardous Materials Manager

5. Current membership in professional organizations

American Institute of Chemical Engineers
Academy of Certified Hazardous Materials Managers
American Chemical Society

6. Honors and awards

Fellow, American Institute of Chemical Engineers
Fellow, Institute of Hazardous Materials Management

7. Selected service activities

8. Publications and presentations from the last 5 years

- Professional Licensure and Academia – IL Society of Prof. Engineers, March 12, 2014
- Chair and Presenter – Teaching the Undergraduate ChE Capstone Design Course, AIChE Midwest Regional Conference, Illinois Institute of Technology, Jan 31, 2013
- Chair and Presenter – Undergraduate ChE Education – Academic, Industry and Professional Licensure Needs, AIChE Midwest Regional Conference, University of Illinois-Chicago, March 11, 2014
- Chair - All about Professional Licensure... Motivations for Attaining the PE, the Processes Involved, Today's Critical Issues, and How Different State Laws Could Affect You, AIChE Spring Meeting San Antonio TX, April 30, 2013
- Chemical Engineering Process Design Methodology at the University of Illinois - Chicago, AIChE Process Development Symposium, Oak Brook IL, June 2013
- Co-Chair, "Environmental, Health, and Safety Concerns", AIChE Process Development Symposium, Oak Brook IL, June 2013
- Fundamentals of Engineering Exam, Joseph Cramer and Jeffery Perl, AIChE Annual Conference, October 27, 2012, Pittsburgh, PA

9. Professional development activities

Past Chair AIChE Chicago Section
Past Chair AIChE Research and New Technology Committee (RANTC)
Past Treasurer, AIChE Environmental Division
National Council of Examiners of Engineering and Surveyors, Chemical Exam Committee 2007-Present, member 25 person team creating Chemical, PE Exam

Faculty Vitae: Vivek Sharma

1. Education

Ph.D., Polymer Science & Engineering, with a Minor in Physics.
Georgia Institute of Technology, Atlanta, GA. 2008

M. S., Chemical Engineering, with a Minor in Nonlinear Dynamics & Chaos.
Georgia Institute of Technology, Atlanta, GA. *2006*

M. S. in Polymer Science. University of Akron, OH. 2003

B. Tech. in Textile Technology. Indian Institute of Technology, Delhi, India. 2001

2. Academic Experience

Department of Chemical Engineering, University of Illinois, Chicago, IL.
Assistant Professor, Nov 2012-Present

Department of Mechanical Engineering, Massachusetts Institute of Technology, MA
Post Doctoral Researcher, Sept 2008-Aug 2012

3. Non-Academic Experience

Chemical Engineering Consultant, 1366 Technologies Inc, Lexington, MA. 04/11-09/11

4. Certifications or professional registrations

None

5. Current membership in professional organizations

American Institute of Chemical Engineers, American Chemical Society, American Physical Society, Society of Rheology

6. Honors and awards

National Talent Scholarship, India. 1995-2001
Selected to attend *APS Opportunities in Energy Research Workshop*, Portland, OR. 2010
Chair's travel award, *Gordon Research Conference - Polymer Physics*. 2004 & 2006
Fellowship for *ACS PRF Summer School on Nanoparticle Materials*, Ypsilanti, MI, 2004
National Talent Scholarship (NTSE), India. 1995-2001
Himachal Pradesh State Scholarship for Undergraduate Study in Engineering. 1997-2001

7. Service activities

Graduate Committee, Department of Chemical Engineering, University of Illinois at Chicago, 2012-Present
Conference technical program organizer for Interfacial and Nonlinear Flows, AIChE Annual Meeting, Atlanta, 2014
Conference technical program organizer for Non-Newtonian Flows, Society of Rheology (SOR), Annual Meeting, Montreal, Canada, 2013.

Conference technical program organizer for Interfacial and Nonlinear Flows, AIChE Annual Meeting, San Francisco, 2013
Session chair, Dynamics of Nanostructured Polymers, ACS Meeting, Denver CO 2011.
Session chair, Jets and wakes. APS DFD Meeting, Long Beach, CA. 2010.
Session chair, New experimental, theoretical and computational methods in polymer and soft matter physics. APS March Meeting, Portland, OR. 2010.
Reviewer for *Soft Matter, Journal of Fluid Mechanics, Physical Review Letters, Advanced Materials, Journal of Rheology, Physical Review E, International Journal of Multiphase Flow, Applied Rheology, Macromolecules, Lab Chip.*

8. Selected Publications Last 5 Years (10 of 13)

1. V. Sharma, M. Crne, J. O. Park and M. Srinivasarao "Structural Origin of Circularly Polarized Iridescence in Jeweled Beetles," *Science*, 325, 449-452 (2009).

2. V. Sharma, K. Park and M. Srinivasarao "Shape Separation of gold nanorods using centrifugation," *Proceedings of National Academy of Sciences*, 106(13), 4981-4985 (2009).

3. V. Sharma, K. Park and M. Srinivasarao "Colloidal dispersion of gold nanorods: Historical background, optical properties, synthesis, separation and self-assembly," *Material Science and Engineering Reports*, 65, 1-38 (2009).
4. V. Sharma, L. Song, R. L. Jones, P. R. Williams and M. Srinivasarao "Effect of solvent choice on breath-figure-templated assembly of 'holey' polymer films," *EPL*, 91, 38001 (2010).
5. R.O. Grigoriev, M.F. Schatz and V. Sharma "Chaotic mixing in microdroplets," *Lab Chip*, 6, 1369-1372 (2006).
6. A. M. Ardekani, V. Sharma and G. H. McKinley, "Dynamics of bead formation, filament thinning and breakup in weakly viscoelastic jets," *J. Fluid Mech.*, 665, 46-56 (2010).
7. S.J. Haward, V. Sharma and J. A. Odell, "Cross-slot Oscillatory Flow Extensional rheometer (COFER) for Low Volumes and Ultra-dilute Complex Fluids," *Soft Matter*, 7 (21), 9908-9921 (2011). Cover Art
8. V. Sharma, A. Jaishankar, Y. Wang and G. H. McKinley, "Apparent yield stress, interfacial viscosity and high shear rate viscosity of Bovine Serum Albumin Solutions," *Soft Matter*, 7 (11), 5150-5160 (2011).
9. V. Sharma and G. H. McKinley, "An intriguing empirical rule for estimating the first normal stress difference from steady shear viscosity data for polymer solutions and melts," *Rheologica Acta*, 51 (6), 487-495 (2012)
10. S. J. Haward, V. Sharma, G. H. McKinley and S. Rahatekar, "Shear and extensional rheology of cellulose in ionic liquids", *Biomacromolecules*, 13, 1688-1699 (2012).

Faculty Vita: Christos G. Takoudis

1. Education

National Technical University of Athens, Greece	Chemical Engineering	Diploma, 1977
University of Minnesota	Chemical Engineering	Ph.D., 1982

2. Academic Experience

Professor, University of Illinois at Chicago, Dept. of BioEngineering, 2/2005 – Present
Professor, University of Illinois at Chicago, Dept. of Chemical Engineering, 8/1996 – Present
Professor, Purdue University, School of Chemical Engineering, 8/1995 - 5/1997
Associate Professor, Purdue University, School of Chemical Engineering, 8/1987 - 8/1995
Assistant Professor, Purdue University, School of Chemical Engineering, 12/1981 - 8/1987
Instructor, University of Minnesota, Dept. of Chem. Eng. & Materials Science, 1/1981 - 12/1981
Visiting Professor, Institute of Biomedical Research and Technology, Greece 4/2008 - 7/2008
Adjunct Professor, U. of Illinois at Chicago, Department of Bioengineering, 1/2003 – 1/2005
Adjunct Professor, Purdue University, School of Chemical Engineering, 8/1997 - 5/1998
Visiting Professor, University of Leeds, England, School of Chemistry, 1/1990 - 2/1990
Visiting Scientist, Applied Materials, Santa Clara, Epi Applications Lab, 1/1988 - 2/1988

3. Non-Academic Experience

Not applicable.

4. Certifications or professional registrations

Not applicable.

5. Current membership in professional organizations

American Institute of Chemical Engineers.
The Electrochemical Society
American Chemical Society
American Physical Society
International Microelectronics and Packaging Society
American Society of Engineering Education
Technical Chamber of Greece

6. Honors and awards

Advisor of the Year Nominee, 2002-2003.
UIC CETL Teaching Recognition Program Award, 2000-2001
Member of the Editorial Board of the Journal: "*Chaos, Solitions and Fractals. Applications in Science and Engineering*," Pergamon Press, 1990 - present.
ARCO Faculty Fellow Award, 1982, Purdue University.
Institute of Technology Fellow, 1979, University of Minnesota.
Bush Foundation Fellow, 1978 –1981,University of Minnesota.
Ph.D. Dissertation Fellow, 1977, University of Minnesota.
Kondilion Award, Thomaidion Award, and Chryssovergion Award 1977, Greece
First Award of the Technical Chamber of Greece, 1977, Greece

7. Selected service activities

In Bioengineering Department

8. Publications and presentations from the last 5 years

Novel functionalization of Ti-V alloy and Ti-II using atomic layer deposition for improved surface wettability By: Patel, Sweetu; Butt, Arman; Tao, Qian; et al. COLLOIDS AND SURFACES B-BIOINTERFACES Volume: 115 Pages: 280-285 Published: MAR 1 2014

.

Growth behavior and properties of atomic layer deposited tin oxide on silicon from novel tin(II)acetylacetonate precursor and ozone By: Selvaraj, Sathees Kannan; Feinerman, Alan; Takoudis, Christos G. JOURNAL OF VACUUM SCIENCE & TECHNOLOGY A Volume: 32 Issue: 1 Article Number: 01A112 Published: JAN 2014

Selective atomic layer deposition of zirconia on copper patterned silicon substrates using ethanol as oxygen source as well as copper reductant By: Selvaraj, Sathees Kannan; Parulekar, Jaya; Takoudis, Christos G. JOURNAL OF VACUUM SCIENCE & TECHNOLOGY A Volume: 32 Issue: 1 Article Number: 010601 Published: JAN 2014

Design and implementation of a novel portable atomic layer deposition/chemical vapor deposition hybrid reactor By: Selvaraj, Sathees Kannan; Jursich, Gregory; Takoudis, Christos G. REVIEW OF SCIENTIFIC INSTRUMENTS Volume: 84 Issue: 9 Article Number: 095109 Published: SEP 2013

Atomic Layer Deposition and Characterization of Amorphous ErxTi1-xOy Dielectric Ultra-Thin Films By: Xu, Runshen; Takoudis, Christos G.ECS JOURNAL OF SOLID STATE SCIENCE AND TECHNOLOGY Volume: 1 Issue: 6 Pages: N107-N114 Published: 2012

9. Professional development activities

In Bioengineering Department

Faculty Vitae: Lewis E. Wedgewood

1. Education

Ph.D. Chemical Engineering, University of Wisconsin-Madison, 1988,
Major Professor: R. Byron Bird,
Research Area: Macromolecular Fluid Mechanics
Thesis Title: *A Theory of Chainlike Polymers: This thesis investigated the effects of hydrodynamic interaction on flow material properties*

B.S. Chemical Engineering, Cum Laude with highest distinction**,**University of Illinois-Urbana, 1982

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago
Associate Professor, 1998-Present

Department of Chemical Engineering, University of Wisconsin-Madison,
Visiting Faculty, 1997-1998

Department of Chemical Engineering, University of Washington,
Assistant Professor, 1990-1998

Department of Chemical Engineering, Wisconsin-Madison,
Research Associate, 1989-1990

NATO/NSF Postdoctoral Fellow, Albert-Ludwigs University, Freiburg, West Germany, 1988-1989

3. Non-Academic Experience

None

4. Certifications or professional registrations

None

5. Current membership in professional organizations

American Institute of Chemical Engineers 1982-present
The Society of Rheology 1982-present

6. Honors and awards

Award for Excellence in Teaching / University of Illinois at Chicago, 2010-2011

2009 Faculty Teaching Award, UIC College of Engineering
Teaching Recognition Program Award 2008-2007
2008 Faculty Advising Award, UIC College of Engineering
2007 Faculty Teaching Award, UIC College of Engineering
2007 Faculty Advising Award, UIC College of Engineering
Harold Simon Teaching Award, 2004
Teaching Recognition Program Award 2000-2001
Outstanding Instructor Award / University of Wisconsin, 1998
Excellence Award - Exceptional Teacher/ University of Washington, 1990-1998
3M Company Non-Tenured Faculty Award, 1997
3M Company Non-Tenured Faculty Award, 1996
NATO/NSF Postdoctoral Fellowship, 1987

7. Service activities

Director of Graduate Studies, UIC Department of Chemical Engineering, 2003-present

Member of Executive Committee, UIC College of Chemical Engineering, 2008-2010, 2011-present

Member of Advisory Committee, UIC Department of Chemical Engineering, 2003-present

Faculty Judge – UIC 2011 Student Research Forum

8. Selected Publications Last 5 Years

1. Ayee, MAA, Minshall, RD, Wedgewood, LE , Nitsche, LC, Levitan, I, and Akpa, BS "Meshless particulate model for the study of cell membrane deformations" *ABSTRACTS OF PAPERS OF THE AMERICAN CHEMICAL SOCIETY Vol. 243(2012) 723*

2. Parkar, Nihal S., Akpa, Belinda S., Nitsche, Ludwig C, "Vesicle Formation and Endocytosis: Function, Machinery, Mechanisms, and Modeling" ANTIOXIDANTS & REDOX SIGNALING *Vol. 11(2009) 1301*

3. "Blood rheology using a Brownian dynamics simulation of bead-spring rings with a constant area constraint ," Kim, KH, Lopez RH and Wedgewood LE, submitted for publication.

Faculty Vitae: Alan D. Zdunek

1. Education

B.A. Chemistry, Knox College, 1982
M.S. Chemical Engineering, Illinois Institute of Technology, 1986
Ph.D. Chemical Engineering, Illinois Institute of Technology, 1990

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago
Adjunct Professor, 2010-present
Department of Chemical Engineering, Illinois Institute of Technology
Research Associate Professor/Adjunct Faculty, 2007-2009

3. Non-Academic Experience

Advanced Diamond Technologies, Romeoville IL, Consultant. 5/12−9/12
HOH Water Technology**,** Palatine IL, Consultant. 6/08−1/09
Argonne National Laboratory, CSE Division, Lemont IL, Consultant. 7/08−11/09
American Air Liquide, Countryside, IL, Group Expert. 3/03−8/06
American Air Liquide, Countryside IL, Senior Scientist. 3/99−2/03
American Air Liquide, Countryside IL, Scientist, 3/95—2/99
BIRL Industrial Research Laboratory, Evanston IL, Scientist, 11/90−11/94

4. Certifications or professional registrations

Project Management Certificate, Loyola University, 12/09.
Hazards Material Technician, 11/97

5. Current membership in professional organizations

American Institute of Chemical Engineers 1990-present
The Electrochemical Society 1985- present
American Chemical Society 1982-present

6. Honors and awards

Keynote Speaker, Conference on International Exchange of Talents & Projects Discussion of Yellow River Delta High-Efficiency Economic Zone, Dongying City, China, November 2011.
Air Liquide Inventor Recognition Award, January 2006
Air Liquide Inventor Recognition Award, February 2005.
Air Liquide Group Expert Appointment, March 2003.
Outstanding Project Award, Air Liquide, December 1998.

7. Service activities

Senior Project Student Mentor; North Lawndale College Prep High School, Chicago, IL, 9/11-present.
Faculty Advisor, AIChE Student Chapter, UIC. 9/11- 8/12.

Director of Undergraduate Studies, Department of Chemical Engineering, University of Illinois at Chicago, 8/11- 12/11.
Judge, Illinois Junior Academy of Science Region 6 Fair, 3/10.
Project Review Committee Member MEDRC (Middle East Desalination Research Council), 2008-2010.
Technical Advisory Board Committee Member , IIT Chemical & Environmental Engineering Department (CEEDAC) (2004-2007).
Session Chair, Hydrogen Fueling for Vehicle Applications Session, IEEE Power & Propulsion Conference, IIT, September 2005.
Task Force Leader, Corrosion, SEMI International Standards Committee, 1997-2002.
Conference Chair, NACE Mid-America Corrosion Conference, October 1995 Schaumburg, IL.
NACE International Chicago Section: Chair, Vice Chair, Treasurer, Secretary 1992-95.

8. Selected Publications Last 5 Years

R. Sparks, G. Jursich, A. Zdunek, Jorge Ivan Rossero, C. Takoudis, "Tunability and Electrical Properties of Yttria-doped-ceria Films Fabricated via Atomic Layer Deposition", The Journal of Undergraduate Research at the University of Illinois at Chicago, Vol.7 No. 1, (2014).

Hanaa Gadallah, Alan Zdunek, K.M. El-Khatib, Fouad Teymour, Said Al-Hallaj, "Study of Electro De-swelling Properties of Super Absorbents Polymers", Australian Journal of Basic and Applied Sciences, 6(6): 282-291, (2012).

H.A. Becker, J.J. Cohen, A. D. Zdunek, "Electrochemical Cooling Water Treatment: A New Strategy for Control of Hardness, Scale, Sludge and Reducing Water Usage", CH-09-042, ASHRAE Transactions Vol. 115, Part 1 (2009).

M. Ferrandon, M. Lewis, D. Tatterson, A. Zdunek, "Overview of the Development and Status of the Thermochemical CuCl Cycle", Paper 328D, Advances in Electrolytic and Thermochemical Processes, AICHE Annual Meeting, Philadelphia, PA, November 2008.

H. Chen, S. Parulekar, A. Zdunek, "Interactions of Chloride and 3-Mercaptopropanesulfonic Acid Interactions in Acidic Copper Sulfate Electrolyte", J. Electrochem. Soc.155 D349 (2008).

H. Chen, S. Parulekar, A. Zdunek, "Interactions of Chloride and Polyethylene Glycol (PEG) Interactions in Acidic Copper Sulfate Electrolyte", J. Electrochem. Soc. 155 D341 (2008).

C. Gabrielli, P. Moçoteguy, H. Perrot, D. Nieto Sanz, A. Zdunek, "An Investigation of Copper Interconnect Deposition Bath Aging by Electrochemical Impedance Spectroscopy", J. Appl. Electrochem. 38: 457-468 (2008).

Appendix C – Equipment

Equipment List for Chemical Engineering Laboratory (ChE 381/382)
Experimental Demonstration Apparatus:

1. Distillation
    a. Distillation column (Corning Glass Works/Hubble & Price); 72L cap., 6 trays
    b. Constant Temperature Bath/Circulator: Haake Model NB22
    c. Refractometer; Bausch & Lomb Model ABBE-3L
2. Liquid-Liquid Extraction
    a. Liquid-liquid extractor; Denver Corp. Model SEM-SU
3. Fluid Flow in Pipe Systems
    a. Hampden Model H-6920 Pipe Friction Demonstrator
4. UltraFiltration
    a. Ultrafiltration column and apparatus; UIC Designed/Built w/ Koch Membrane Systems PM50 filter column
5. Membrane Separation
    a. Membrane Air Separation Apparatus; Perfected Experiments, Inc. Model ChE06
6. Filtration
    a. Ertel-Alsop sealed disc filtration unit Model SDF
7. Heat Transfer
    a. Constant Stirred Tank Reactor (CSTR); UIC Designed/Built
8. Viscosity of Liquids
    a. Capillary Tube Viscometer; UIC Designed/Built
    b. Falling ball viscometers, nine (9); Gilmont Instruments, Inc.
    c. Constant Temperature Stirred Baths (3 units); Lindberg/BlueM Model MW1172SSA1
9. Packed Bed
    a. Two-Column packed bed apparatus; UIC Designed/Built
10. Fluidized Bed
    a. Two-Column fluidized bed apparatus; UIC Designed/Built
11. Humidification
    a. Falling Liquid Film humidification apparatus; UIC Designed/Built
12. Evaporation
    a. Single-stage glass evaporator, vacuum pressure; UIC Designed/Built
13. Gas Absorption
    a. Armfield Model UOP7 Gas Absorption Column
14. Chemical Reaction
    a. Armfield Model CEX Chemical Reaction Unit
15. Boiling Heat Transfer apparatus; UIC Designed/Built

General Lab & Support Equipment

1. Classroom space/ computers
    a. 2^{nd}-level classroom area with 10 PC-based computers, Windows XP Pro
    b. Printers; two (2) Brother lasers
    c. Countertop table space/chairs for lab team collaboration and lab analysis
2. Chemical storage cabinets
    a. Three (3) flammable liquid cabinets
    b. Acids/base storage (under fume hoods)
3. Laboratory Utilities:
    a. Two horizontal sash fume hoods
    b. Three laboratory sinks (hot/cold water)
    c. Steam boiler; Lattner Model 48 BF electric boiler, 100 psi max
    d. Compressed air; house, several taps throughout lab, max pressure 60 psi
    e. Tap water; city water supply, various taps throughout lab
    f. Deionized water; house, several taps throughout lab
    g. Electrical; individual circuit breakers on specific equipment
    h. Refrigerator; Kenmore ~18 ft^3 fridge/freezer
4. Miscellaneous:
    a. Two Digital balances, Ohaus Scout (300g), Valor 3000 Xtreme (100 g)
    b. Three mechanical scales; Fairbanks & Morse, 100 lb capacity (2), Fisher Scientific, 5 kg (1)
    c. Fisher Scientific Isotemp drying oven
    d. pH meters (2)
    e. Magnetic stirrer/burette holder
    f. Drum handling equipment (55-gallon)
    g. Sterilizer, GMP General Purpose
    h. Press, 110,00 pound hydraulic, Ramco Model RP55
5. Safety equipment:
    a. One (1) Eyewash station
    b. One (1) Safety Shower
    c. Safety gloves; heat-resistant, chemical-resistant, sharp-resistant, general nitrile gloves.
    d. Safety glasses; 50+ count

UG Research in Faculty Laboratories

Below is a representative list of students who have participated in UG research during the last 6 years.

2013

Andrew McNamara ChE Sr

Thermal degradation of lithium Ion Batteries:

Andrew is using an accelerated rate calorimeter we research various different properties of lithium ion batteries as they become thermally unstable. This research reveals important data like the total energy released and peak temperatures during decomposition.

Nathan Liebmann ChE Jr

Designer fuel system:

Using a series of columns we convert a solution of hydrogen peroxide, methanol, and water into hydrogen gas to be used in a hydrogen fuel cell. This research will provide a proof of concept so that the system can be scaled to automotive use.

2012

Han Wang ChE

Seawater Desalination by Electric Field under Flow and Non-Flow Operation:

Mass flux rates were measured in a desalination membrane cell with and without an electric field and saltwater flow. Student used a custom-made test cell, PAR2273 potentiostat/galvanostat and electrochemical system, ion-selective electrodes, and power supply.

2011

Lipi Vanhanwala ChE Sr

Simulation of finitely extensible polymers in various two-dimensional flows:

Lipi simulated FENE (Finitely Extensible Nonlinear Elastic) chains as a model of a polymer solution in a variety of 2D flows in order to compare the rotational properties with predictions of a vorticity decomposition theory.

Vijeta Patel ChE Sr

Constraints for a simple model of a blood cell:

Vijeta worked out the constraints for a bead-spring tetrahedron model of a blood cell in which the volume and surface area are maintained constant. These results were applied in a Brownian dynamics simulation.

2010

Thinh Tang ChE

Corrosive Properties of Stainless Steel Oil Containers:

Thinh worked for me for two semesters on a project to characterize and improve the performance of stainless steel containers for oil storage and shipping. Thinh learned several laboratory methods and performed characterization experiments for the project.

2009

Magdalena Furczon ChE

Electrochemical modeling of the impedance versus temperature of lithium-ion positive electrode containing LiBOB salt:

Xin Qin ChE

Brownian dynamics simulations of blood flow:

Xin optimized model parameters for a Brownian dynamics simulation of red blood cells. His result fit the existing data to great accuracy.

2008

Ramune Otterson Meskyte ChE

Demonstration of a Gas Thermometer:

Ramune worked out the details of an ideal gas thermometer and built a working model, which she demonstrated.

Ruzica Djordjevic ChE

Critique of an Open vs. Closed Analysis in Thermodynamics:

Ruzica evaluated a paper which discussed the application of both closed and open mass and energy balances to a model system. She criticized the analysis presented in the paper.

Brandon C. Brown SROP (Norfolk State University)

A Brownian Dynamics Study of the Forces that Induce Endocytosis:

Brandon tested a model of endocytosis to determine the relation between charge on proteins attached to the lipid bilayer and the amount of force transmitted to the bilayer.

Appendix D – Institutional Summary

Programs are requested to provide the following information.

1. *The Institution*

a. University of Illinois at Chicago
2800 University Hall (m/c 102), 601 S. Morgan Street, Chicago, IL 60607-7128

b. Chief Executive Officer:

Paula Allen-Meares, Chancellor
University of Illinois at Chicago
2800 University Hall (m/c 102), Chicago, IL 60607-7128

c. Michael McNallan, Associate Dean for Undergraduate Affairs
University of Illinois at Chicago
851 S. Morgan St. (m/c 154), Chicago, IL 60607-7128

d. The University of Illinois at Chicago is accredited by the Commission on Institutions of Higher Education of the North Central Association of Colleges and Schools (NCA), 30 N. LaSalle St., Suite 2400, Chicago, Illinois 60602-2504, (312) 263-0456. The NCA is recognized by the Commission on Recognition of Postsecondary Accreditation. The initial accreditation of UIC was in 1970. In 2007, the North Central Association of Colleges and Schools granted continued accreditation of the University of Illinois at Chicago for the maximum period of 10 years. The next comprehensive evaluation of UIC is scheduled for 2016-17. Verification of accreditation status is available in the Office of the Chancellor (M/C 102), University of Illinois at Chicago, 601 South Morgan Street, Chicago, Illinois 60607-7128; (312) 413-3350. The undergraduate academic degree programs at UIC have been approved by the Illinois Board of Higher Education, 431 East Adams, Second Floor, Springfield, Illinois 62701-1418, (217) 782-2551.

History of Institution

In 1946, an undergraduate division of the University of Illinois was established at Navy Pier. This facility, renamed the University of Illinois at Chicago Circle, moved to its present location in 1965, when it opened its doors as a four-year university. By 1982, it had grown to include eight academic colleges offering degree programs at both the undergraduate and graduate levels.

The University of Illinois at Chicago was formed by the consolidation, in the fall of 1982, of the two Chicago campuses (formerly known as the University of Illinois at the Medical Center and the University of Illinois at Chicago Circle) into a single institution of higher learning. At that time, the University set as a goal to become a quality alternative to the Big Ten campuses. The

University's facilities for medical instruction date back to 1894, when the Chicago College of Pharmacy became the School of Pharmacy of the University of Illinois. In 1897, the independent College of Physicians and Surgeons of Chicago became the "Department of Medicine" of the University; in 1901, the Columbian Dental College became the University School of Dentistry; and in 1925 the University Hospital opened. Programs in nursing education under University auspices began in the 1940s, becoming the School of Nursing in 1951 and, in 1959, the College of Nursing. Other health sciences units of the University of Illinois at Chicago include the College of Applied Health Sciences, the School of Public Health, and over 50 clinics and research facilities. A new $60 million University of Illinois Hospital was completed in 1981.

Today the University of Illinois at Chicago has a total enrollment of approximately 28,000 students, including almost 11,000 graduate and professional students. Academic support services include eight libraries, extensive computer facilities with a 10,000-user network, and an instructional resources development office. The campus has a number of centers and institutes whose research activities complement classroom teaching. Other support services include tutoring programs; guidance in the improvement of reading, mathematics, and study skills; a writing center; academic and personal counseling; special instruction in English for international students; and financial aid.

The College of Engineering was one of four colleges at Navy Pier along with the Colleges of Commerce and Business Administration, The College of Liberal Arts and Science, and the College of Physical Education. The College of Engineering at Navy Pier was quite small, with about 400 students attending in an average year.

In 1965, when the University of Illinois at Chicago Circle was established, the College of Engineering reorganized into “functional” departments of Materials Engineering, Energy Engineering, Information Engineering, and Systems Engineering. Degree programs in the College were identified by functions that engineers perform, such as Mechanical Analysis and Design, Systems Analysis, or Communications Engineering. The College grew rapidly at Chicago Circle, reaching nearly 2,000 undergraduate students by 1968. In 1973, a number of the College’s programs received their first accreditation from the Engineers Council for Professional Development (now ABET).

Although the non-traditional engineering degree programs administered by the College in the 1970’s were accredited, they were not well understood by the public, and it was clear that our industrial constituents preferred to recruit students from traditional degree programs, so that the backgrounds and qualifications of their employees could be more easily predicted. In 1982, the University made the decision to reorganize the College into traditional departments and traditional degree programs, essentially as they are offered now.

Student Body

The nearly 28,000 students who study at the University of Illinois at Chicago come from the city of Chicago and its suburbs, and from all 50 states, three United States territories, and 100 foreign

countries. The student body is rich in its diversity, its youth and maturity, and its cultural heritage. Of the 16,600 undergraduate students, 50.3 percent are female and 49.7 percent are male. Minority students comprise 32.7 percent of the undergraduate enrollment. Many full-time students also hold part-time jobs, both on and off campus. In addition, a large number find time to participate in one or more of approximately 233 campus student organizations. With a growing residential population, UIC houses students on the east, west, and south sides of campus. Approximately 3,800 students live on campus, including more than half of first-year undergraduate students.
University of Illinois enrollment figures for the past 10 years are shown below. The overall campus student population has been stable in the range of 24,500 to 27,500 students over the past 10 years.

Total Enrollment Fall 2004-2013

	2004	**2005**	**2006**	**2007**	**2008**	**2009**	**2010**	**2011**	**2012**	**2013**
Undergraduate	15,448	15,148	14,999	15,672	15,648	16,036	16,790	16,911	16,671	16,660
Graduate	6,581	6,766	6,704	6,916	7,080	7,746	7,925	8,012	8,119	8,186
Professional	2,378	2,439	2,497	2,537	2,515	2,560	2,594	2,657	2,722	2,743
Total	24,407	24,353	24,200	25,125	25,243	26,342	27,309	27,580	27,512	27,589

Ethnic diversity among the undergraduate student population are shown below. UIC continues to have one of the most ethnically diverse student bodies of any American University. Apart for some small fluctuations from year to year, the overall numbers have been fairly stable over the past 10 years, with the Hispanic population showing significant growth since 2008.

Race/Ethnicity	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
Am. Indian	34	37	45	35	34	31	24	22	21	24
	0.2%	0.2%	0.3%	0.2%	0.2%	0.2%	0.1%	0.1%	0.1%	0.1%
Afr Amer	1,37	1,359	1,307	1,367	1,345	1,353	1,392	1,361	1,339	1,303
	8.9%	9.0%	8.7%	8.7%	8.6%	8.4%	8.3%	8.0%	8.0%	7.8%
Asian	3,84	3,679	3,590	3,633	3,572	3,584	3,576	3,605	3,656	3,737
	24.9	24.3%	23.9%	23.2%	22.8%	22.3%	21.3%	21.3%	21.9%	22.4%
Hispanic	2,77	2,782	2,765	2,695	2,669	2,960	3,386	3,711	3,950	4,120
	16.3	16.5%	16.3%	16.4%	17.0%	18.5%	20.2%	21.9%	23.7%	24.7%
Caucasian	6,64	6,561	6,602	7,006	7,001	7,101	7,451	7,173	6,664	6,304
	43.0	43.3%	44.0%	44.7%	44.7%	44.3%	44.4%	42.4%	40.0%	37.8%
Foreign	174	210	230	253	263	230	241	236	265	350
	1.1%	1.4%	1.5%	1.6%	1.7%	1.4%	1.4%	1.4%	1.6%	2.1%
Nat HI/Pac Isl	na	na	na	na	na	na	76	86	79	69
	na	na	na	na	na	na	0.5%	0.5%	0.5%	0.4%
Two or More	na	na	na	na	na	na	344	371	407	410
	na	na	na	na	na	na	2.0%	2.2%	2.4%	2.5%
Unknown	854	803	775	802	780	777	300	346	290	343
	5.5%	5.3%	5.2%	5.1%	5.0%	4.8%	1.8%	2.0%	1.7%	2.1%
UIC Total	15,4	15,148	14,999	15,672	15,664	16,036	16,790	16,911	16,671	16,660

*Beginning Fall 2010, new ethnicity and race categories were created in response to federal reporting requirements. The implementation of these new categories included a survey of all current students to verify ethnicity and race. Changes include the ability of respondents to indicate Hispanic ethnicity separately from racial origin and to identify more than one race category if applicable. The federal guidelines also require separating the category 'Asian or Pacific Islander' into two groups, one for 'Asian' and one for 'Native Hawaiian or Other Pacific Islander'. Students who are in the U.S. on a visa or temporary basis are defined as International.

UIC's undergraduate student body is largely drawn from local sources as shown in the following table showing the geographical origin of entering freshmen. Chicago and Cook County account for a majority of UIC's entering students, but have declined substantially as a percentage of entering students over the past 10 years, as UIC has become increasingly important as a statewide educational resource. The number of international students and non-resident U.S. students in the UIC undergraduate student body remains small.

The Geographical Location of entering Freshmen is shown in the following table:

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
Chicago	894	971	832	1,015	886	963	932	991	1,025	1,070
	32.9%	35.0%	29.2%	30.8%	29.9%	30.6%	29.1%	31.8%	32.8%	34.5%
Cook County	1,831	1,879	1,804	2,059	1,832	2,003	1,993	1,987	2,029	2,038
	67.4%	67.7%	63.3%	62.6%	61.8%	63.6%	62.2%	63.8%	65.0%	65.7%
Illinois	2,640	2,673	2,739	3,175	2,857	3,044	3,097	2,980	2,992	2,928
	97.2%	96.3%	96.0%	96.5%	96.4%	96.7%	96.7%	95.7%	95.8%	94.3%
Out of State	58	59	72	80	66	66	57	93	71	110
	2.1%	2.1%	2.5%	2.4%	2.2%	2.1%	1.8%	3.0%	2.3%	3.5%
Foreign	18	44	41	36	41	37	50	42	60	66
	0.7%	1.6%	1.4%	1.1%	1.4%	1.2%	1.6%	1.3%	1.9%	2.1%

Admission to the University of Illinois at Chicago is competitive, with the average enrolled freshman student being in the top 25% of his or her high school class, as shown in the following table for freshmen enrolled over the past 10 years. While the high school rank has been steady over this period, the average ACT score of entering freshmen has gradually increased by 1 point over this period, rising from 23.1 to 24.1.

The Mean ACT and High School Percentile Rank for New Freshmen for Fall 2003-2013 is show below:

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
ACT	23.1	23.3	23.7	23.6	23.9	23.9	23.8	23.7	24.0	24.1
HSPR	74.9	75.5	74.9	76.2	76.4	77.4	75.9	76.6	75.0	76.8

As shown below, College of Engineering enrollments have fluctuated more substantially over the ten year period than the overall university enrollment. From 2000 to 2005, the College of Engineering experienced a gradual decrease in undergraduate enrollment, as the College focused on student quality and graduate programs. Since 2005, the College has been implementing a recruiting drive aimed at increasing the undergraduate enrollments while maintaining the high student quality that had been obtained in the smaller classes.

Figure D-1 shows the College of Engineering enrollment since 2004 leading to a projected undergraduate enrollment of approximately 2650 for Fall 2014.

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
COE Undergrad	1,641	1,550	1,624	1,714	1,781	1,914	2,098	2,158	2,311	2524

Figure D-2 shows the distribution of the enrollments among the College's programs during that period. The Bioengineering, Chemical Engineering, Computer Science, Industrial Engineering and Mechanical Engineering programs have grown significantly during this time, while other programs have had stable or decreasing enrollment.

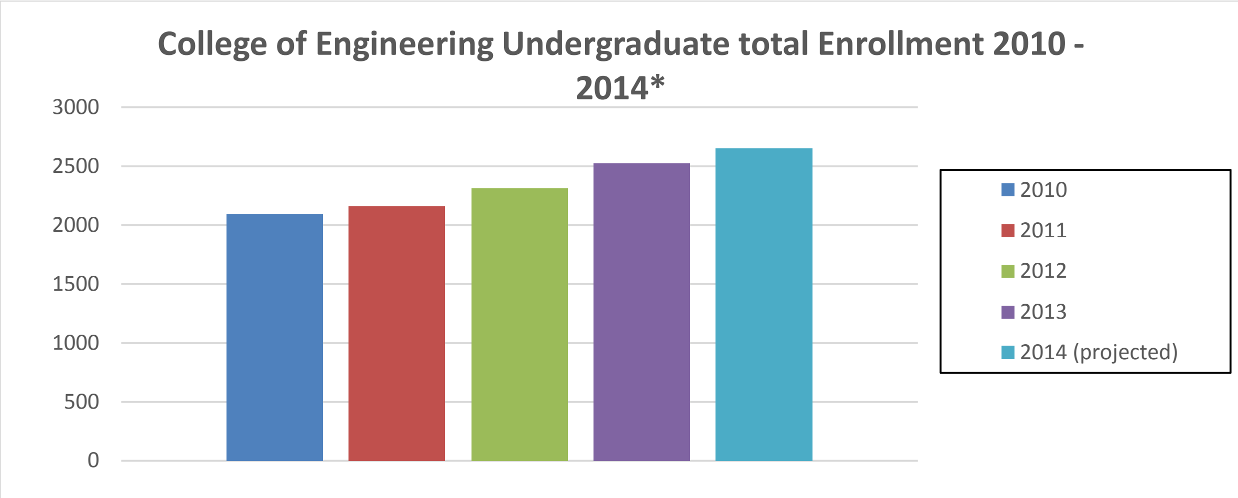

Figure D-1. College of Engineering Undergraduate Enrollment in Fall semester from 2010 to 2014 (projected).

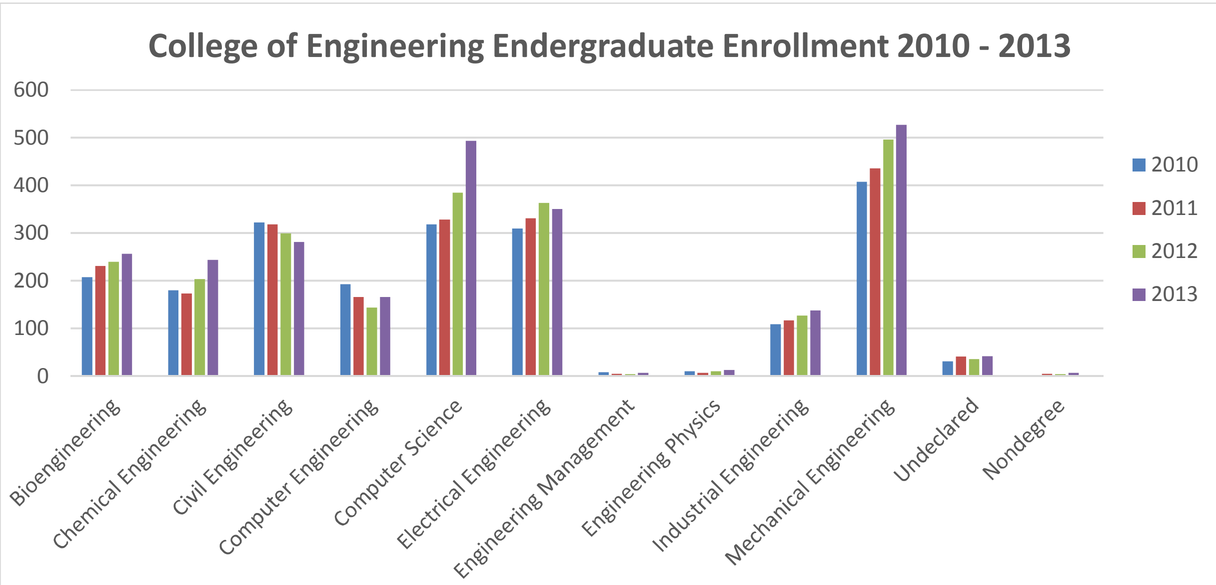

Figure D-2. College of Engineering Undergraduate Enrollment shown by program from 2010 to 2013.

COE Freshmen Demographics

As shown below for entering freshmen students in the 2004 to 2013 period, the ethnic diversity of the undergraduate engineering student body is generally consistent with the overall University statistics. There has been a small increase in the percentage of Caucasian students in the College as the recruiting drive has led to the enrollment of a larger number of majority students from suburban high schools. The percentage of women in the undergraduate student body has been in the range of 15% since the 1990's, but has also fallen slightly during the recruiting drive. Changes in the recruiting program to make the College more attractive to women and minority students are being implemented to reverse these trends and have shown some results in 2012 and 2013.

Ethnicity of Freshmen Engineering Students

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
Nat. Amer	1	0	5	0	0	1	1	0	0	2
	0.4%	1.5%	0%	0%	0.3%	0.3%	0	0	0.5%	1.5%
Afr. Amer	17	91	67	70	94	79	78	100	103	91
	7.0%	9.1%	7.5%	6.7%	5.9%	6.3%	5.0%	2.5%	3.1%	4.3%
Asian	74	51	91	67	70	94	79	78	100	103
	30.5%	22.2%	27.4%	21.3%	22.9%	23.9%	23.2%	23.9%	30.6%	27.8%
Hispanic	28	36	37	50	43	66	68	64	67	79
	11.5%	15.6%	11.1%	15.9%	14.1%	16.8%	19.9%	19.6%	20.5%	21.3%
Caucasian	106	107	161	160	148	184	157	152	125	140
	43.6%	46.5%	48.5%	50.8%	48.4%	46.7%	46.0%	46.6	38.2%	37.7%
Foreign	1	4	3	3	10	8	5	9	8	12
	0.4%	1.7%	0.9%	1.0%	3.3%	2.0%	1.5%	2.8%	2.4%	3.2%
Nat. HI/Pac Isl	na	na	na	na	na	na	2	2	2	2
	na	na	na	na	na	na	0.6%	0.6%	0.6%	0.5%
Two or More	na	na	na	na	na	na	11	8	11	14
	na	na	na	na	na	na	3.2%	2.5%	3.4%	3.8%
Unknown	16	11	10	14	17	16	1	5	4	3
	6.6%	4.8%	3.0%	4.4%	5.6%	4.1%	0.3%	1.5%	1.2%	0.8%
COE Total	243	230	332	315	306	394	341	326	327	371

*Beginning Fall 2010, new ethnicity and race categories were created in response to federal reporting requirements. The implementation of these new categories included a survey of all current students to verify ethnicity and race. Changes include the ability of respondents to indicate Hispanic ethnicity separately from racial origin and to identify more than one race category if applicable. The federal guidelines also require separating the category 'Asian or Pacific Islander' into two groups, one for 'Asian' and one for 'Native Hawaiian or Other Pacific Islander'. Students who are in the U.S. on a visa or temporary basis are defined as International.

Gender

	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014 (projected)
Female	35	47	37	51	51	42	34	53	61	63
Male	195	285	278	255	343	299	292	274	310	317
COE	230	332	315	306	394	341	326	327	371	380

Student quality, as measured by ACT and high school class rank, has increased every year since 2009 with ACT Composite score increasing by 1.9. Slight increase are also projected for Fall 2014

ACT / HSPR data for entering freshmen

	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014 (projected)

ACT	25.4	25.6	25.6	25.5	25.4	26.2	26.5	27.1	27.3	27.5
HSPR	73.8	69.3	73.8	71.3	75.1	75.2	75.5	78.1	79.7	80.0

2. *Type of Control*

The State of Illinois through a Board of Trustees has managerial control for the University of Illinois System. The University of Illinois at Chicago (UIC) is one of three campuses. The other two campuses are the University of Illinois at Urbana-Champaign and the University of Illinois at Springfield. Each campus has a Chancellor, and there is one President and one Board of Trustees for the three campus system.

3. *Educational Unit*

The organizational structure of the College of Engineering at the University of Illinois at Chicago is shown schematically in Figure 1-2. The Directors of Undergraduate Study responsible for the B.S. programs presented in this document, report to the Department Heads of their respective departments, who report, in turn, to the Dean of the College. The Dean is supported in his activity by Associate Deans with responsibilities for Research and Graduate Studies, Corporate Relations (also supervises the Engineering Career Center and Co-op Program), On-line Master of Engineering Program and International Programs (all graduate level programs), Administration (primarily budget and financial issues), and Advancement.

The authority lines for the Associate Dean for Undergraduate Affairs are shown in more detail in Figure 1-3, because these personnel address the programs reviewed in this report. The Associate Dean for Undergraduate Affairs supervises the Director of the Minority Recruiting and Retention Program (MERRP). The Assistant Dean is supported by Assistant Directors for both the recruiting program and the MERRP program. The recruiting activities are carried out throughout the Chicago region to continuously enhance the number and quality of students entering the College's programs. The MERRP program (described in more detail elsewhere in this report) is in place to recruit and support minority enrollments in the College of Engineering.

The Director of Engineering Admission and Records also reports to the Associate Dean for Undergraduate Affairs. The Director of Admission and Records reviews the records of student applicants to the College of Engineering and makes admission decisions. His office also maintains the records of continuing students and reviews them for compliance with College policies and graduation standards. In this, he is supported by three academic counselors who meet with individual students to advise them on policies and requirements, and also conduct orientation and advising sessions with new freshmen and transfer students. He is also supported by an Office Manager, who is usually the first responder for students with problems. Together with the Associate Dean for Undergraduate Affairs, he sets policies for special petitions and enforces probation and academic drop rules.

Organizational Structure

The authority lines for the Associate Dean for Undergraduate Affairs are shown in more detail below, because these personnel address the programs reviewed in this report. The Associate Dean for Undergraduate Affairs supervises the Assistant Dean for Undergraduate Recruiting, who is also the Director of the Minority Recruiting and Retention Program (MERRP). The Assistant Dean is supported by Assistant Directors for both the recruiting program and the MERRP program. The recruiting activities are carried out throughout the Chicago region to continuously enhance the number and quality of students entering the College's programs. The MERRP program (described in more detail elsewhere in this report) is in place to recruit and support minority enrollments in the College of Engineering. The following flowchart (Figure D3) shows the organizational relationships among the offices of the College of Engineering

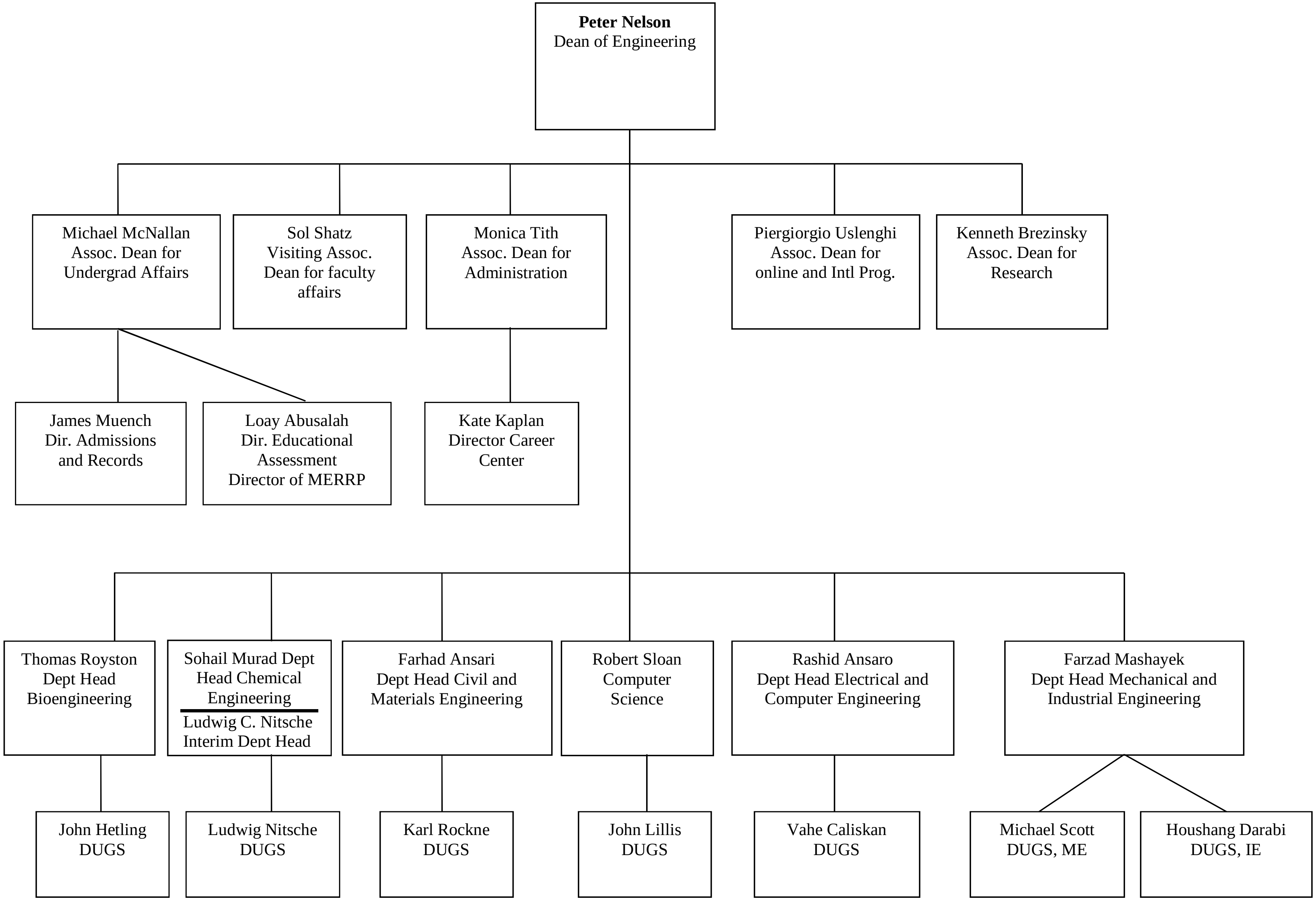

Figure D3. The flowchart shows the organizational relationships among the offices of the College of Engineering.

The Director of Engineering Admission and Records also reports to the Associate Dean for Undergraduate Affairs. The Director of Admission and Records reviews the records of student applicants to the College of Engineering and makes admission decisions. His office also maintains the records of continuing students and reviews them for compliance with College policies and graduation standards. In this, he is supported by three academic counselors who meet with individual students to advise them on policies and requirements, and also conduct orientation and advising sessions with new freshmen and transfer students. He is also supported by an Office Manager, who is usually the first responder for students with problems. Together with the Associate Dean for Undergraduate Affairs, he sets policies for special petitions and enforces probation and academic drop rules. The list of college of engineering associate deans and directors could also be found on our website http://engineering.uic.edu/bin/view/COE/Staff The organizational structure for the University of Illinois at Chicago is shown in Figure D4.

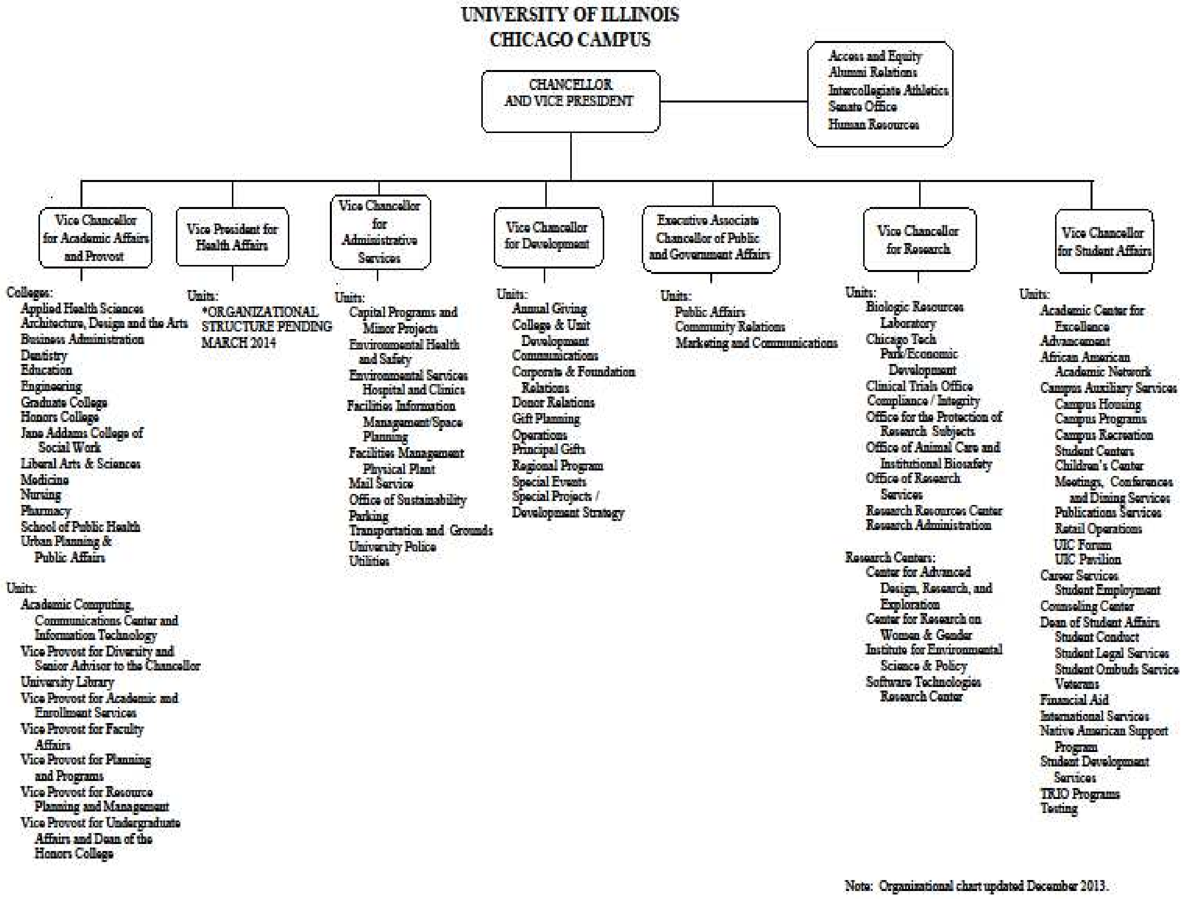

Figure D4. The organizational structure for the University of Illinois at Chicago.

4. *Academic Support Units*

Required courses for College of Engineering programs are taught by faculty from several departments in the College of Liberal Arts and Sciences. These include:

Department of Mathematics, Statistics, and Computer Science

Department of Mathematics, Statistics, and Computer Science which teaches required mathematics courses. The MSCS Department is headquartered at:

Department of Mathematics, Statistics, and Computer Science
University of Illinois at Chicago
322 Science and Engineering Offices (M/C 249)
851 S. Morgan Street
Chicago, IL 60607-7045

Department Head: Lawrence Ein, 419 SEO; (312) 996-2372; ein@uic.edu

The MSCS Department houses 85 faculty and 89 teaching assistants, who are responsible for offering mathematics courses taken by students in the College of Engineering and in other programs which rely on mathematics as a component of their knowledge base. Additional information is available at: http://www.math.uic.edu/

Department of Chemistry

The Department of Chemistry, which teaches required chemistry courses. The Chemistry Department is headquartered at:

Department of Chemistry MC 111
4500 SES (Science and Engineering South)
University of Illinois at Chicago
845 West Taylor Street
Chicago, IL 60607

Department Head: Luke Hanley, 5105 SES; (312) 996-3161; lhanley@uic.edu

The Department of Chemistry houses 25 faculty and a number of graduate teaching assistants who are responsible for offering chemistry courses taken by students in the College of Engineering and in other programs which rely on chemistry as a component of their knowledge base. Additional information is available at: http://www.chem.uic.edu

Department of Physics

The Department of Physics, which teaches the required physics courses. The Department of Physics is headquartered at:

Department of Physics
2244 SES (Science and Engineering South)
University of Illinois at Chicago
845 W. Taylor St. M/C 273
Chicago IL 60607-7059

Department Head: David J. Hofman, 2244 SES; (312)-413-2798 ; hofman@uic.edu

The Department of Physics houses 21 faculty and 52 graduate teaching assistants who are responsible for offering physics courses taken by students in the College of Engineering and in other programs which rely on physics as a component of their knowledge base. Additional information is available at: http://www.phy.uic.edu

Department of English

The Department of English, which teaches the required English composition courses. The Department of English is headquartered at:

Department of English
College of Liberal Arts and Sciences
2027 University Hall
601 South Morgan Street
Chicago, Illinois 60607

Department Head: John Huntington, 2027 University Hall; (312) 413-2200; huntingj@uic.edu

The Department of English houses 49 faculty and 52 lecturers, who are responsible for offering the English composition courses taken by students in the College of Engineering and by other undergraduate students at the University of Illinois at Chicago. Additional Information is available at: http://www.uic.edu/depts/engl

In addition, several other academic departments, including the Department of Biological Sciences, http://www.uic.edu/depts/bios/ and the Department of Earth and Environmental Sciences, http://www.uic.edu/depts/geos/ offer courses which are used in specific College of Engineering programs. Courses that satisfy the Humanities and Social Sciences requirements for College of Engineering programs are selected from a list of courses that satisfy the General Education requirements of the University of Illinois at Chicago, http://www.uic.edu/ucat/catalog/GE.shtml and are offered by a variety of departments within the University.

5. *Non-academic Support Units*

Library

The University of Illinois at Chicago is supported by an extensive library system with sites at several locations around the campus. The Library is administered by:

University Librarian: Mary Case E-mail: marycase@uic.edu
Richard J. Daley Library
801 S. Morgan, M/C 234
Chicago, IL 60607

Reference materials relevant to College of engineering courses are most commonly found at either:

The RICHARD J. DALEY LIBRARY, 801 S. Morgan Street, Chicago
The SCIENCE LIBRARY, 3500 SES; 845 W. Taylor Street
Or
The LIBRARY OF THE HEALTH SCIENCES, 1750 W. Polk Street

In addition, the library maintains extensive collections of electronic resources and access to web based journals and abstracting services that are frequently used by College of Engineering students and faculty. More information is available at: http://library.uic.edu

Computing facilities

Computing facilities at the University of Illinois at Chicago are provided by: the Academic Computing and Communications Center (ACCC). The ACCC is directed by:

Cynthia E. Herrera Lindstrom, Chief Information Officer and Executive Director, Academic Computing and Communications Center
114 BGRC MC 135
1940 West Taylor St
Chicago IL 60612-7352
cynthiar@uic.edu

The ACCC provides accounts for all students, faculty and staff at UIC. They also maintain an extensive set of public computer laboratories for student use, maintain software licenses for the University, and provide consulting services on computing issues. The ACCC coordinates frequent training courses and seminars on various aspects of computer usage and software. ACCC supports the Instructional Technology Laboratory (ITL), which provides computing services to support classroom use and teaching through electronic means. In addition to the ACCC, the College of Engineering departments all maintain individual computing facilities for their students, containing equipment and software appropriate to their specific programs. More information about the ACCC and computing resources at UIC is available at: http://accc.uic.edu

Tutoring and academic resources: Tutoring and other academic support is available to College of Engineering students through a number of organizations on campus. These include:

Academic Center for Excellence

The Academic Center for Excellence (ACE) is a multifaceted academic support program administered by the Office of the Vice Chancellor for Student Affairs designed to help UIC students accomplish their academic goals. ACE is open to all UIC students, from freshman through graduate level and provides assistance with learning styles, time management, and tutoring support. More information about ACE is available at: http://www.uic.edu/depts/ace/index.shtml

The Honors College:

After they have been admitted to UIC, all incoming freshmen with an ACT composite score of at least 28 or an SAT total score of 1240, and who rank in the top 15 percent of their high school class, are invited to join the Honors College (membership is not automatic; students must submit an application). Members are required to carry out honors activity every semester.

One of the honors activity services is to provide tutoring to underclass students in a variety of subjects, including the math and science activities relevant to engineering. More information about Honors College tutoring is available at: http://www.uic.edu/honors

The Mathematical Sciences Learning Center (MSLC):

The Center hosts several activities focused on improving mathematics instruction and student performance. A major initiative of the Department will be the opening of the Mathematical Sciences Learning Center (MSLC) dedicated to supporting the study of the mathematical sciences for all UIC students. MSLC will offer a program of peer-led workshops and tutoring, as well as more traditional assistance from graduate students. One successful support activity, the Emerging Scholars Program (ESP), is a long-standing UIC effort derived from the work of Uri Treisman (Treisman 1985). Treisman showed the importance of creating opportunities for students, particularly those from underrepresented groups, to work in teams while mastering mathematics. ESP offers students the opportunity to work in groups on more challenging math problems for an additional 2.5 hours each week, supervised by TAs. An internal study has demonstrated the effectiveness of the program: >1/2 of ESP students receive an A or B in pre-Calculus compared to just over 1/3 of non-ESP students; 65% of ESP students in multivariable calculus receiving an A or B while only about half of non-ESP students do this well. More information about the Mathematical Sciences Learning Center and the Emerging Scholars Program is available at: **http://www.math.uic.edu/undergrad/mslc/**

The Science Learning Center (SLC)

The Science Learning Center is a wonderfully designed common space at UIC shared by the sciences (biology, chemistry, earth and environmental science, and physics) and dedicated to enhancing the educational experience of undergraduates enrolled in UIC's introductory science courses. Although the main task of the learning center is to provide tutoring in the sciences, secondary goals are to demonstrate the interdisciplinary nature of science, to provide students with a nurturing learning environment, and to encourage team problem-solving. Two types of structured tutoring occur in the SLC: tutoring by TAs from the natural sciences and peer-led study groups for 100-level science courses, led by students who have recently completed the course. Peer leaders are trained in group dynamics and serve to promote a group's ability to work together toward the solution of a problem. More information about the Science Learning Center is available at: http://www.chem.uic.edu/slc/

The Writing Center

Students who are having difficulties with written communication can obtain tutoring and support services through UIC's Writing Center. The Writing Center at the University of Illinois at Chicago is committed to the improvement of writing across the campus through peer-centered education and tutor development. The Writing Center complements classroom instruction by providing tutorials, resources, workshops, and publication opportunities for writers, educators, and writing groups from all disciplines and experience levels. More information about the Writing Center is available at:
http://www.uic.edu/depts/engl/writing/

In addition to these programs, the College of Engineering and the campus also provide specific support programs aimed at underrepresented and disadvantaged students.

Minority Engineering Recruitment and Retention Program (MERRP)

The Minority Engineering Recruitment and Retention Program (MERRP), housed in the College of Engineering, has, as its major effort, to increase in the number of African American, Latino, and Native American students earning engineering degrees. The Program staff is composed of professionals with backgrounds in higher education, engineering, and engineering education as follows.

Gerald A. Smith, Director
1232 SEO
gasmith@uic.edu
Elsa Soto, Assistant Director
1237 SEO
esoto3@uic.edu

MERRP is the foundation of the minority engineering student experience at UIC. Most minority engineering students are introduced to MERRP through the college orientation program and the engineering student orientation course, ENGR 100.

MERRP has supported the student chapters of the Society of Hispanic Professional Engineers (SHPE) and the National Society of Black Engineers (NSBE) and has been a source of experience and advice for both groups. The hallmark of MERRP is Supplemental Instruction (SI) and the "academic family" atmosphere.

As part of its minority student recruitment efforts, MERRP maintains summer programs for high school students including.

Preparation for Majoring in Engineering (PREP-ME)

This program is for UIC freshmen and community college students in engineering. Prep-ME Plus is a six week program designed to introduce new UIC engineering students to college-level math and engineering as a discipline and profession. The primary focus is math instruction. Students also attend seminars on goal setting, study skills, and time management. Students gain an understanding of college requirements and expectations. Qualified students enroll in a freshman English course and a math workshop.

Women in Science & Engineering Program (WISE)

845 West Taylor Street (MC 180)
Office 205D Science Learning Center, SES
Chicago, IL 60607

The goal of the UIC Women in Science and Engineering Program, WISE, is to increase the number of women students pursuing and graduating in science, technology, engineering and math, STEM, disciplines, and to promote the recruitment, retention and advancement of women who have chosen academic careers. Nationally, retention rates for women STEM students fall far behind those of their male counterparts, and women consistently make up less than 20% of tenured STEM faculty. WISE supports women undergrad/graduate students and faculty in STEM by sponsoring activities that foster a positive educational and professional environment, and enable excellence in scholarship, teaching and service. More information about the WISE program is available at: http://www.uicwise.org/

African American Academic Network

Student Services 1200 West Harrison
Suite 2800
Chicago, IL 60607

AAAN's mission is to increase the recruitment, retention, and graduation rates of African American students. In keeping with that focus, AAAN is also committed to establishing an inclusive and supportive campus environment. AAAN sponsors social and cultural activities to encourage student involvement, and advocates for the interests of its participants. During the 2006-2007 academic year, the African American Academic Network advised 65% of African American first time freshmen. More information about AAAN is available at: http://www.uic.edu/depts/aaan/index.shtml

Latin American Recruitment and Educational Services Program (LARES)

Suite 2640, Student Services Building
1200 West Harrison Street
Chicago, Illinois 60607

LARES is a recruitment and academic assistance program at the University of Illinois at Chicago. It was established to recruit, advise and provide educational assistance to Latino students at both the high school and college levels. LARES was primarily designed to assist Latino students who are interested in pursuing higher education and who are in need of guidance and support. More information about LARES is available at: http://lares.uic.edu

Native American Support Program (NASP)

The Native American Support Program (NASP) strives to increase the enrollment, retention and graduation rates of Native American students. NASP fosters a climate supportive of positive academic experiences for Native American students at the University of Illinois at Chicago. More information about NASP is available at: http://www.uic.edu/depts/nasp

Disability Resource Center

As reflected in the University of Illinois' Nondiscrimination Statement and the UIC Chancellor's Statement of Commitment to Persons with Disabilities, UIC strives to maintain a barrier-free environment so that students with disabilities can fully access classes, programs, services and other campus activities. The Disability Resource Center facilitates access for students through consultation with faculty and campus departments, and the provision of reasonable accommodations. The Disability Resource Center recognizes various environments in which people function: physical, programmatic, informational and attitudinal. Some modifications to these environments are readily-achievable through direct consultation with faculty or staff. To be eligible for accommodations, students should apply for services through the Disability Resource Center. More information about the Disability Resource Center is available at: http://www.uic.edu/depts/oaa/disability_resources/index.html

Counseling Center

Counseling Services offices are located in Suite 2010, Student Services Building, on the corner of Racine and Harrison (1200 West Harrison Street). As one of the two units of the Counseling Center, the Services are staffed by licensed and board certified psychologists and a psychiatrist, clinical therapists, advanced doctoral psychology trainees, psychiatric residents, and undergraduate paraprofessional volunteers, all of whom are trained to help students with a wide range of personal problems, emotional and psychological difficulties, career questions, and relationship issues. The Center is accredited by the International Association of Counseling Services, and its internship is accredited by the American Psychological Association. The Counseling Services staff adheres to all relevant ethical codes of conduct and state and federal laws and regulations. The staff is committed to the highest standards of competency in meeting the needs of individuals from diverse backgrounds, including differences of culture, race, ethnicity, national origin, class, gender, ability, age, and sexual orientation. More information about the Counseling Center is available at: http://www.uic.edu/depts/counseling

Student career placement and co-op activities are supported by office at both the College of Engineering and campus level.

The Cooperative Education and Internship Program

College of Engineering (M/C 160)
University of Illinois at Chicago
818 Science & Engineering Offices
851 South Morgan Street
Chicago, Illinois 60607-7050
Kate Kaplan, Director, Engineering Career Center (312.996.2238, engrjobs@uic.edu)

The Engineering Cooperative and Internship Program gives UIC engineering students the opportunity to gain practical work experience while they are still students. The Engineering Co-op/Internship Office helps students find full time or part time engineering employment before graduation. There are a substantial number of opportunities for qualified engineering students interested in obtaining relevant work experience. The Co-op Office matches employer requirements with student skills, abilities, interests, and background. More information about the Engineering Career Center and the Cooperative Education and Internship Program is available at: http://www.ecc.uic.edu/ECC/WebHome

UIC Office of Career Services

1200 W. Harrison
Room 3050 SSB (MC 099)
Chicago, IL 60607

The mission of the UIC Office of Career Services is to provide personalized services that assist UIC students and recent graduates in a process of self-assessment, career planning and preparation in order to facilitate lifetime career development and success.

Information is also provided about part-time employment opportunities to help make the connection between the academic world and real world work experiences while at UIC; full-time employment opportunities to ensure a successful transition from UIC to the world of work; and graduate/professional school catalogues are available to assist students interested in continuing their education.

In addition, the office maintains and continues to develop mutually beneficial relationships with regional employers of all sizes to promote the employment of UIC students and graduates. More information about the UIC Office of Career Services is available at:
http://www.uic.edu/depts/ocs

6. Credit Unit

A semester hour is the University's unit of academic credit. During the fall and spring semesters, a University semester hour represents one classroom period of fifty minutes weekly for one semester in lecture or discussion or a longer period of time in laboratory, studio, or other work. For example, a three-semester-hour lecture/discussion course could meet 3 times a week for 50 minutes each period or 2 times a week for 75 minutes each period. In either case, a student attends the lecture/discussion course for an equivalent amount of time each week during a 15-week semester. A minimum of two 50-minute periods each week per credit hour is required for lab, practicum, or clinical activity. During the eight-week summer session, the classroom period is 100 minutes for lecture/discussion. It is expected that students will spend at least the equivalent of two classroom periods of outside preparation for one classroom period per week of lecture or discussion. Those courses in which semester hours exceed contact hours may require additional readings, assigned papers, or other course work.

Instructional Modes

All of the programs in the College of Engineering are offered as traditional on-campus instruction. Students in the Co-op program may spend one or more semesters off campus in engineering internships, but their course work is all taken on campus in traditional classes.

Grade-Point Average

In order to calculate the grade point average: Take the grades for each course taken and determine the grade points per hour: A=4, B=3, C=2, D=1, F=0. Multiply the grade points per hour for each course to get the grade points for each course. Add the points for each course to get the total number of grade points for the semester. Add the semester hours taken for each course to get the total number of semester hours. Divide the total number of grade points for the semester by the total number of semester hours taken to get the grade point average.

In order to receive a degree from the College of Engineering at the University of Illinois at Chicago, a student must present a minimum grade point average of 2.00/4.00 in all work taken in the major. In addition, the student must satisfy the University requirement of a 2.00/4.00 grade point average in two categories: (1) all work taken at UIC; (2) all work taken at UIC and all other two- and four-year institutions combined.

Faculty Workload

The teaching load in most departments in the College of Engineering is four classroom courses per year or faculty who have some research activity (some graduate students and some research publications). Faculty who are active in research have reduced loads of three courses per year. Faculty who have no M.S. or PhD. Students should teach one additional course per semester in exchange for their lack of individual graduate instruction. Hence, faculty who have no research activities, (no graduate students, no publications, no funding) will be expected to teach six classroom courses per year. Departments have buy-out policies for reducing the teaching loads. A common model is for faculty to pay 1/9 of their academic year salary for every course reduction. In addition, faculty members receive course buy-outs for significant administrative loads, such as Department Heads, Director of Graduate Studies, Director of Undergraduate Studies, Associate Deans, etc.

laboratory hours per week. One academic year normally represents at least 28 weeks of classes, exclusive of final examinations. If other standards are used for this program, the differences should be indicated.

7. Tables

Complete the following tables for the program undergoing evaluation.

Table D-1. Program Enrollment and Degree Data

Department of Chemical Engineering – Chemical Engineering Program

	Academic Year		Enrollment Year					Total Undergrad	Total Grad	Degrees Awarded[1,2]			
			1st	2nd	3rd	4th	5th			Associates	Bachelors	Masters	Doctorates
Current 2013-2014		FT	36	53	53	80	-	222	49	0	3	4	3
		PT	0	1	5	16	-	22					
1 2012-2013		FT	38	31	45	69	-	183	46	0	39	5	4
		PT	1	4	4	11	-	20					
2 2011-2012		FT	26	31	37	59	-	153	52	0	33	7	3
		PT	1	1	1	17	-	20					
3 2010-2011		FT	34	29	37	70	-	170	49	0	40	2	3
		PT	0	1	2	7	-	10					
4 2009-2010		FT	37	30	28	61	-	156	45	0	30	3	5
		PT	1	1	3	10	-	15					

Give official fall term enrollment figures (head count) for the current and preceding four academic years and undergraduate and graduate degrees conferred during each of those years. The "current" year means the academic year preceding the on-site visit.

[1]Degree data are from Summer XX, Fall XX, Spring XX+1 terms (Summer 2012, Fall 2012, Spring 2013)
[2]Current 2013-2014 only includes Summer 2013 and Fall 2013 data. Spring 2014 data available after June 13, 2014

FT--full time
PT--part time

Table D-2. Personnel

Name of the Program

Year[1]: Fall 2013

	HEAD COUNT		FTE[2]
	FT	PT	
Administrative[2]	0.80		0.80
Faculty (tenure-track)[3]	7	0.50	7.50
Other Faculty (excluding student Assistants)		0.75	0.75
Student Teaching Assistants[4]		7.00	7.00
Technicians/Specialists	1	0.40	1.40
Office/Clerical Employees	2		2.00
Others[5]			

Report data for the program being evaluated.

1. Data on this table should be for the fall term immediately preceding the visit. Updated tables for the fall term when the ABET team is visiting are to be prepared and presented to the team when they arrive.

2. Persons holding joint administrative/faculty positions or other combined assignments should be allocated to each category according to the fraction of the appointment assigned to that category.

3. For faculty members, 1 FTE equals what your institution defines as a full-time load

4. For student teaching assistants, 1 FTE equals 20 hours per week of work (or service). For undergraduate and graduate students, 1 FTE equals 15 semester credit-hours (or 24 quarter credit-hours) per term of institutional course work, meaning all courses — science, humanities and social sciences, etc.

5. Specify any other category considered appropriate, or leave blank.

Appendix E – Sample Grade Distributions in Chemical Engineering Courses

SPRING 2014

COURSE	A	%	B	%	C	%	D	%	F	%	NR	%	IN	%	DR	%	WD	%	STUD
CHE 201	9	16.4	22	40.0	10	18.2	4	7.3	5	9.1	0	0.0	0	0.0	0	0.0	5	9.1	55
CHE 205	21	32.8	19	29.7	15	23.4	1	1.6	3	4.7	0	0.0	0	0.0	0	0.0	5	7.8	64
CHE 210	10	18.9	13	24.5	15	28.3	8	15.1	7	13.2	0	0.0	0	0.0	0	0.0	0	0.0	53
CHE 312	11	19.3	20	35.1	20	35.1	4	7.0	0	0.0	0	0.0	0	0.0	0	0.0	2	3.5	57
CHE 313	19	33.9	18	32.1	12	21.4	6	10.7	1	1.8	0	0.0	0	0.0	0	0.0	0	0.0	56
CHE 321	11	19.3	13	22.8	17	29.8	8	14.0	8	14.0	0	0.0	0	0.0	0	0.0	0	0.0	57
CHE 341	7	16.7	15	35.7	14	33.3	5	11.9	0	0.0	0	0.0	0	0.0	0	0.0	1	2.4	42
CHE 382	27	62.8	14	32.6	2	4.7	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	43
CHE 397	20	46.5	20	46.5	3	7.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	43
CHE 494-RET_UG	3	75.0	1	25.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	4
CHE 494-RET_G	2	33.3	4	66.7	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	6
CHE 494-FE_UG	0	0.0	0	0.0	1	33.3	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	2	66.7	3
CHE 494-FE_G	0	0.0	6	75.0	1	12.5	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	1	12.5	8

FALL 2013

COURSE	A	%	B	%	C	%	D	%	F	%	NR	%	IN	%	DR	%	WD	%	STUD
CHE 201	16	27.1	12	20.3	10	16.9	7	11.9	4	6.8	0	0.0	0	0.0	0	0.0	10	16.9	59
CHE 205	7	17.1	11	26.8	5	12.2	11	26.8	0	0.0	2	4.9	0	0.0	0	0.0	5	12.2	41
CHE 210	10	25.0	7	17.5	10	25.0	8	20.0	1	2.5	0	0.0	0	0.0	0	0.0	4	10.0	40
CHE 301	10	15.9	15	23.8	15	23.8	15	23.8	4	6.3	0	0.0	0	0.0	0	0.0	4	6.3	63
CHE 311	9	13.8	21	32.3	18	27.7	10	15.4	3	4.6	0	0.0	0	0.0	0	0.0	4	6.2	65
CHE 381	21	48.8	20	46.5	2	4.7	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	43
CHE 396	12	27.9	18	41.9	13	30.2	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	43
CHE 410	6	35.3	9	52.9	1	5.9	0	0.0	0	0.0	0	0.0	1	5.9	0	0.0	0	0.0	17
CHE 422-UG	1	20.0	1	20.0	2	40.0	1	20.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	5
CHE 422-G	2	66.7	0	0.0	1	33.3	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	3
CHE 445	6	25.0	10	41.7	8	33.3	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	24
CHE 494-FIZZICS_UG	5	83.3	1	16.7	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	6
CHE 494-FIZZICS_G	6	66.7	3	33.3	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	9
CHE 494-ASPEN_UG	9	52.9	6	35.3	0	0.0	0	0.0	1	5.9	0	0.0	0	0.0	0	0.0	1	5.9	17

SPRING 2013

COURSE	A	%	B	%	C	%	D	%	F	%	NR	%	IN	%	DR	%	WD	%	STUD
CHE 201	13	26.5	13	26.5	8	16.3	8	16.3	0	0.0	0	0.0	0	0.0	0	0.0	7	14.3	49
CHE 205	12	24.5	19	38.8	8	16.3	6	12.2	2	4.1	0	0.0	1	2.0	0	0.0	1	2.0	49
CHE 210	7	20.0	6	17.1	12	34.3	5	14.3	2	5.7	0	0.0	0	0.0	0	0.0	3	8.6	35
CHE 312	8	17.4	16	34.8	12	26.1	9	19.6	1	2.2	0	0.0	0	0.0	0	0.0	0	0.0	46
CHE 313	7	14.9	28	59.6	11	23.4	1	2.1	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	47
CHE 321	6	12.2	8	16.3	16	32.7	15	30.6	3	6.1	0	0.0	0	0.0	0	0.0	1	2.0	49
CHE 341	6	15.8	11	28.9	17	44.7	3	7.9	0	0.0	0	0.0	0	0.0	0	0.0	1	2.6	38
CHE 382	19	52.8	17	47.2	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	36
CHE 397	23	63.9	13	36.1	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	36
CHE 494-SYS_UG	2	20.0	5	50.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	3	30.0	10
CHE 494-SYS_G	7	100.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	7
CHE 511	4	28.6	5	35.7	4	28.6	0	0.0	0	0.0	0	0.0	1	7.1	0	0.0	0	0.0	14
CHE 527	4	33.3	6	50.0	1	8.3	0	0.0	0	0.0	0	0.0	1	8.3	0	0.0	0	0.0	12

FALL 2012

COURSE	A	%	B	%	C	%	D	%	F	%	NR	%	IN	%	DR	%	WD	%	STUD
CHE 201	11	19.3	20	35.1	14	24.6	7	12.3	2	3.5	0	0.0	0	0.0	0	0.0	3	5.3	57
CHE 210	4	14.3	10	35.7	5	17.9	4	14.3	0	0.0	0	0.0	0	0.0	0	0.0	5	17.9	28
CHE 301	5	9.6	15	28.8	22	42.3	5	9.6	0	0.0	0	0.0	0	0.0	0	0.0	5	9.6	52
CHE 311	8	14.5	13	23.6	21	38.2	6	10.9	0	0.0	0	0.0	0	0.0	0	0.0	7	12.7	55
CHE 381	20	54.1	17	45.9	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	37
CHE 396	9	24.3	17	45.9	11	29.7	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	37
CHE 410	6	46.2	7	53.8	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	13
CHE 422-UG	1	10.0	5	50.0	3	30.0	1	10.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	10
CHE 422-G	9	75.0	3	25.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	12
CHE 431	7	53.8	5	38.5	1	7.7	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	13
CHE 494-ASPEN_UG	11	47.8	9	39.1	3	13.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	23
CHE 502	11	100.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0	11

Signature Attesting to Compliance

By signing below, I attest to the following:

That Chemical Engineering has conducted an honest assessment of compliance and has provided a complete and accurate disclosure of timely information regarding compliance with ABET's *Criteria for Accrediting Engineering Programs* to include the General Criteria and any applicable Program Criteria, and the *ABET Accreditation Policy and Procedure Manual.*

Peter Nelson

Dean's Name (As indicated on the RFE)

Signature

June 30, 2014

Date